National Space Science Data Center (Source: http://nssdc.gsfc.nasa.gov/planetary...cometfact.html), the perihelion distance for Halley's comet is 0.587 A.U. To calculate the aphelion distance, we add

$$P = a + c = 17.96 + 17.37 = 35.33 \text{ A.U.}$$

This is approximately 3.3×10^9 mi. The average distance from Pluto to the Sun is 39.5 A.U. (Source: http://www.oarval.org/furthest.htm), so it would appear that Halley's Comet stays just within the orbit of Pluto.

NAVIGATING A BANKED TURN

How fast can a racecar travel through a circular turn without skidding and hitting the wall? The answer could depend on several factors:

- The weight of the car;
- The friction between the tires and the road;
- The radius of the circle;
- The "steepness" of the turn.

In this project we investigate this question for NASCAR racecars at the Bristol Motor Speedway in Tennessee. Before considering this track in particular, we use vector functions to develop the mathematics and physics necessary for answering questions such as this.

A car of mass m moves with constant angular speed ω around a circular curve of radius R (Figure 13.4.9). The curve is banked at an angle θ. If the height of the car off the ground is h, then the position of the car at time t is given by the function $\vec{\mathbf{r}}(t) =< R\cos(\omega t), R\sin(\omega t), h >$.

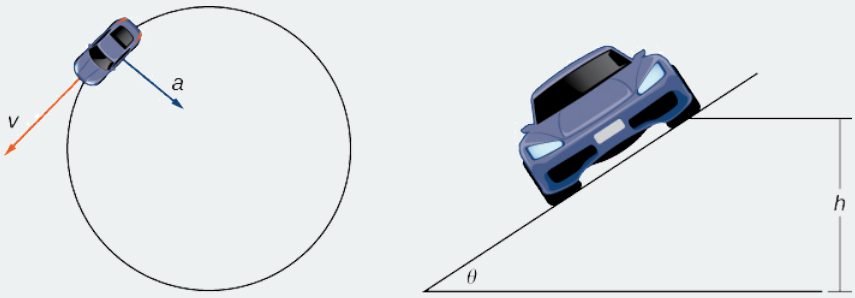


Figure 13.4.9: Views of a race car moving around a track.

1. Find the velocity function $\vec{\mathbf{v}}(t)$ of the car. Show that $\vec{\mathbf{v}}$ is tangent to the circular curve. This means that, without a force to keep the car on the curve, the car will shoot off of it.
2. Show that the speed of the car is ωR. Use this to show that $(2\pi 4)/\|\vec{\mathbf{v}}\| = (2\pi)/\omega$.
3. Find the acceleration $\vec{\mathbf{a}}$. Show that this vector points toward the center of the circle and that $\|\vec{\mathbf{a}}\| = R\omega^2$.
4. The force required to produce this circular motion is called the *centripetal force*, and it is denoted $\vec{\mathbf{F}}_{cent}$. This force points toward the center of the circle (not toward the ground). Show that $\|\vec{\mathbf{F}}_{cent}\| = (m|\vec{\mathbf{v}}|^2)/R$.

As the car moves around the curve, three forces act on it: gravity, the force exerted by the road (this force is perpendicular to the ground), and the friction force (Figure 13.4.10). Because describing the frictional force generated by the tires and the road is complex, we use a standard approximation for the frictional force. Assume that $\vec{\mathbf{f}} = \mu\vec{\mathbf{N}}$ for some positive constant μ. The constant μ is called the *coefficient of friction*.

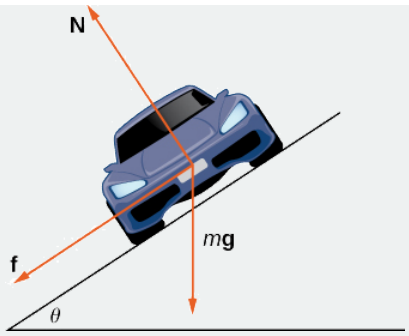


Figure 13.4.10: The car has three forces acting on it: gravity (denoted by $m\vec{\mathbf{g}}$), *the friction force* $\vec{\mathbf{f}}$, *and the force exerted by the road* $\vec{\mathbf{N}}$.

Let v_{max} denote the maximum speed the car can attain through the curve without skidding. In other words, v_{max} is the fastest speed at which the car can navigate the turn. When the car is traveling at this speed, the magnitude of the centripetal force is

$$\|\vec{\mathbf{F}}_{cent}\| = \frac{m(v_{max})^2}{R}.$$

The next three questions deal with developing a formula that relates the speed v_{max} to the banking angle θ.

5. Show that $\vec{\mathbf{N}}\cos\theta = m\vec{\mathbf{g}} + \vec{\mathbf{f}}\sin\theta$. Conclude that $\vec{\mathbf{N}} = (m\vec{\mathbf{g}})/(\cos\theta - \mu\sin\theta)$.
6. The centripetal force is the sum of the forces in the horizontal direction, since the centripetal force points toward the center of the circular curve. Show that

   $$\vec{\mathbf{F}}_{cent} = \vec{\mathbf{N}}\sin\theta + \vec{\mathbf{f}}\cos\theta.$$

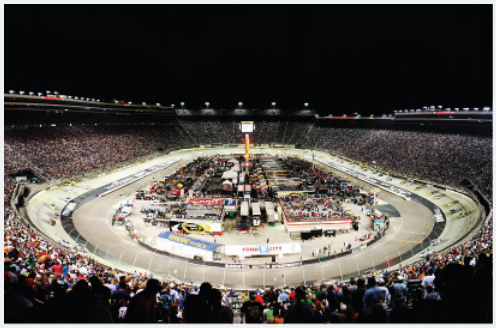
   Conclude that

   $$\vec{\mathbf{F}}_{cent} = \frac{\sin\theta + \mu\cos\theta}{cos\theta - \mu\sin\theta} m\vec{\mathbf{g}}.$$

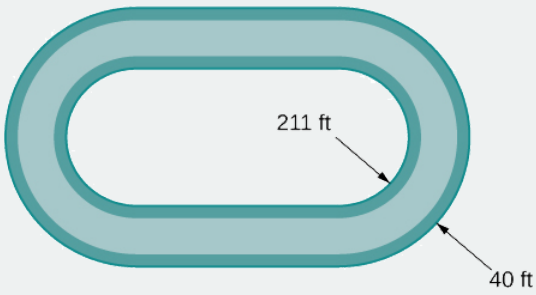
7. Show that $(v_{\max})^2 = ((\sin\theta + \mu\ cos\theta)/(\cos\theta - \mu\sin\theta))gR$. Conclude that the maximum speed does not actually depend on the mass of the car.

   Now that we have a formula relating the maximum speed of the car and the banking angle, we are in a position to answer the questions like the one posed at the beginning of the project.

   The Bristol Motor Speedway is a NASCAR short track in Bristol, Tennessee. The track has the approximate shape shown in Figure 13.4.11. Each end of the track is approximately semicircular, so when cars make turns they are traveling along an approximately circular curve. If a car takes the inside track and speeds along the bottom of turn 1, the car travels along a semicircle of radius approximately 211 ft with a banking angle of 24°. If the car decides to take the outside track and speeds along the top of turn 1, then the car travels along a semicircle with a banking angle of 28°. (The track has variable angle banking.)



(a)



(b)

Figure 13.4.11: At the Bristol Motor Speedway, Bristol, Tennessee (a), the turns have an inner radius of about 211 ft and a width of 40 ft (b). (credit: part (a) photo by Raniel Diaz, Flickr)

The coefficient of friction for a normal tire in dry conditions is approximately 0.7. Therefore, we assume the coefficient for a NASCAR tire in dry conditions is approximately 0.98.

Before answering the following questions, note that it is easier to do computations in terms of feet and seconds, and then convert the answers to miles per hour as a final step.

8. In dry conditions, how fast can the car travel through the bottom of the turn without skidding?
9. In dry conditions, how fast can the car travel through the top of the turn without skidding?
10. In wet conditions, the coefficient of friction can become as low as 0.1. If this is the case, how fast can the car travel through the bottom of the turn without skidding?
11. Suppose the measured speed of a car going along the outside edge of the turn is 105 mph. Estimate the coefficient of friction for the car's tires.

Key Concepts

- If $\vec{\mathbf{r}}(t)$ represents the position of an object at time t, then $\vec{\mathbf{r}}\,'(t)$ represents the velocity and $\vec{\mathbf{r}}\,''(t)$ represents the acceleration of the object at time t. The magnitude of the velocity vector is speed.
- The acceleration vector always points toward the concave side of the curve defined by $\vec{\mathbf{r}}(t)$. The tangential and normal components of acceleration $a_{\vec{\mathbf{T}}}$ and $a_{\vec{\mathbf{N}}}$ are the projections of the acceleration vector onto the unit tangent and unit normal vectors to the curve.
- Kepler's three laws of planetary motion describe the motion of objects in orbit around the Sun. His third law can be modified to describe motion of objects in orbit around other celestial objects as well.
- Newton was able to use his law of universal gravitation in conjunction with his second law of motion and calculus to prove Kepler's three laws.

Key Equations

- **Velocity**

$$\vec{\mathbf{v}}(t) = \vec{\mathbf{r}}\,'(t)$$

- **Acceleration**

$$\vec{\mathbf{a}}(t) = \vec{\mathbf{v}}\,'(t) = \vec{\mathbf{r}}\,''(t)$$

- **Speed**

$$v(t) = ||\vec{\mathbf{v}}(t)|| = ||\vec{\mathbf{r}}\,'(t)|| = \frac{ds}{dt}$$

- **Tangential component of acceleration**

$$a_{\vec{\mathbf{T}}} = \vec{\mathbf{a}} \cdot \vec{\mathbf{T}} = \frac{\vec{\mathbf{v}} \cdot \vec{\mathbf{a}}}{||\vec{\mathbf{v}}||}$$

- **Normal component of acceleration**

$$a_{\vec{\mathbf{N}}} = \vec{\mathbf{a}} \cdot \vec{\mathbf{N}} = \frac{||\vec{\mathbf{v}} \times \vec{\mathbf{a}}||}{||\vec{\mathbf{v}}||} = \sqrt{||\vec{\mathbf{a}}||^2 - \left(a_{\vec{\mathbf{T}}}\right)^2}$$

Glossary

acceleration vector
the second derivative of the position vector

Kepler's laws of planetary motion
three laws governing the motion of planets, asteroids, and comets in orbit around the Sun

normal component of acceleration

the coefficient of the unit normal vector $\vec{\mathbf{N}}$ when the acceleration vector is written as a linear combination of $\vec{\mathbf{T}}$ and $\vec{\mathbf{N}}$

projectile motion
motion of an object with an initial velocity but no force acting on it other than gravity

tangential component of acceleration
the coefficient of the unit tangent vector $\vec{\mathbf{T}}$ when the acceleration vector is written as a linear combination of $\vec{\mathbf{T}}$ and $\vec{\mathbf{N}}$

velocity vector
the derivative of the position vector

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- Edited by Paul Seeburger
  Paul Seeburger added finding point $(1, 2)$ when $t = 1$ in Example 13.4.1.
  He also created Figure 13.4.1.

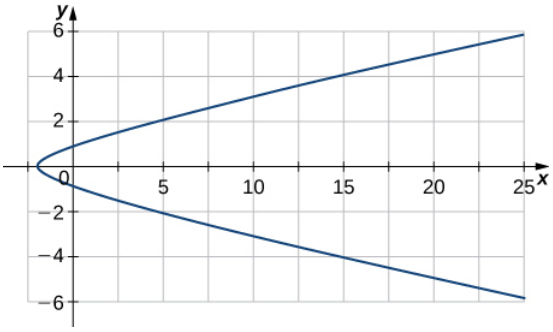
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13.4E: Exercises for Section 13.4

1) Given $\vec{\mathbf{r}}(t) = (3t^2 - 2)\,\hat{\mathbf{i}} + (2t - \sin t)\,\hat{\mathbf{j}}$,

a. find the velocity of a particle moving along this curve.

b. find the acceleration of a particle moving along this curve.



Answer

a. $\vec{\mathbf{v}}(t) = 6t\,\hat{\mathbf{i}} + (2 - \cos t)\,\hat{\mathbf{i}}$

b. $\vec{\mathbf{a}}(t) = 6\,\hat{\mathbf{i}} + \sin t\,\hat{\mathbf{i}}$

In questions 2 - 5, given the position function, find the velocity, acceleration, and speed in terms of the parameter t.

2) $\vec{\mathbf{r}}(t) = e^{-t}\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + \tan t\,\hat{\mathbf{k}}$

3) $\vec{\mathbf{r}}(t) = \langle 3\cos t,\ 3\sin t,\ t^2 \rangle$

Answer

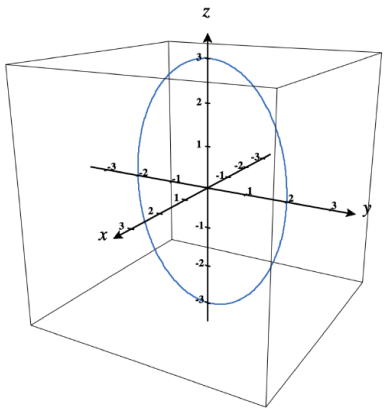
$\vec{\mathbf{v}}(t) = -3\sin t\,\hat{\mathbf{i}} + 3\cos t\,\hat{\mathbf{j}} + 2t\,\hat{\mathbf{k}}$

$\vec{\mathbf{a}}(t) = -3\cos t\,\hat{\mathbf{i}} - 3\sin t\,\hat{\mathbf{j}} + 2\,\hat{\mathbf{k}}$

$\text{Speed}(t) = \|\vec{\mathbf{v}}(t)\| = \sqrt{9 + 4t^2}$

4) $\vec{\mathbf{r}}(t) = t^5\,\hat{\mathbf{i}} + (3t^2 + 2t - 5)\,\hat{\mathbf{j}} + (3t - 1)\,\hat{\mathbf{k}}$

5) $\vec{\mathbf{r}}(t) = 2\cos t\,\hat{\mathbf{j}} + 3\sin t\,\hat{\mathbf{k}}$. The graph is shown here:



Answer

$\vec{\mathbf{v}}(t) = -2\sin t\,\hat{\mathbf{j}} + 3\cos t\,\hat{\mathbf{k}}$

$\vec{\mathbf{a}}(t) = -2\cos t\,\hat{\mathbf{j}} - 3\sin t\,\hat{\mathbf{k}}$

$\text{Speed}(t) = \|\vec{\mathbf{v}}(t)\| = \sqrt{4\sin^2 t + 9\cos^2 t} = \sqrt{4 + 5\cos^2 t}$

In questions 6 - 8, find the velocity, acceleration, and speed of a particle with the given position function.

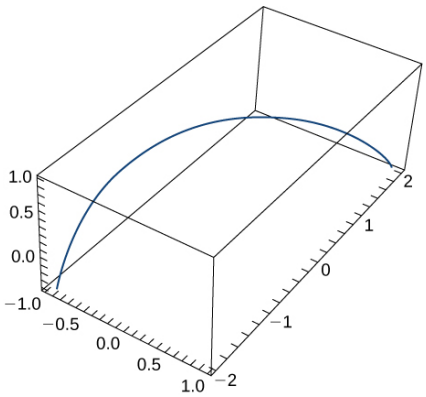
6) $\vec{\mathbf{r}}(t) = \langle t^2 - 1, t \rangle$

7) $\vec{\mathbf{r}}(t) = \langle e^t, e^{-t} \rangle$

Answer

$\vec{\mathbf{v}}(t) = \langle e^t, -e^{-t} \rangle,$
$\vec{\mathbf{a}}(t) = \langle e^t, e^{-t} \rangle,$
$\|\vec{\mathbf{v}}(t)\| = \sqrt{e^{2t} + e^{-2t}}$

8) $\vec{\mathbf{r}}(t) = \langle \sin t, t, \cos t \rangle$. The graph is shown here:



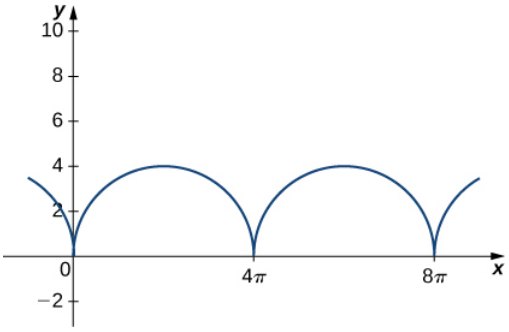
9) The position function of an object is given by $\vec{\mathbf{r}}(t) = \langle t^2, 5t, t^2 - 16t \rangle$. At what time is the speed a minimum?

Answer

$t = 4$

10) Let $\vec{\mathbf{r}}(t) = r\cosh(\omega t)\,\hat{\mathbf{i}} + r\sinh(\omega t)\,\hat{\mathbf{j}}$. Find the velocity and acceleration vectors and show that the acceleration is proportional to $\vec{\mathbf{r}}(t)$.

11) Consider the motion of a point on the circumference of a rolling circle. As the circle rolls, it generates the cycloid $\vec{\mathbf{r}}(t) = (\omega t - \sin(\omega t))\,\hat{\mathbf{i}} + (1 - \cos(\omega t))\,\hat{\mathbf{j}}$, where ω is the angular velocity of the circle and b is the radius of the circle:

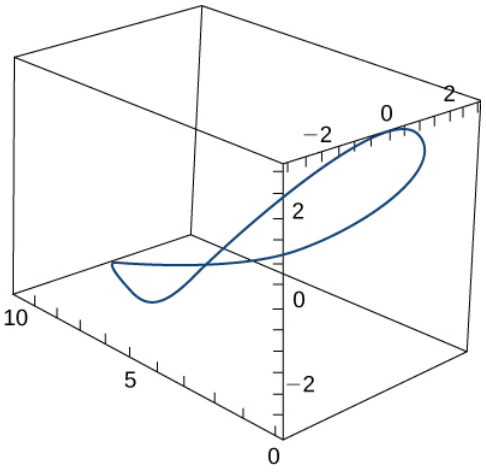


Find the equations for the velocity, acceleration, and speed of the particle at any time.

Answer

$\vec{\mathbf{v}}(t) = (\omega - \omega\cos(\omega t))\,\hat{\mathbf{i}} + (\omega\sin(\omega t))\,\hat{\mathbf{j}}$
$\vec{\mathbf{a}}(t) = (\omega^2\sin(\omega t))\,\hat{\mathbf{i}} + (\omega^2\cos(\omega t))\,\hat{\mathbf{j}}$
$\text{speed}(t) = \sqrt{(\omega - \omega\cos(\omega t))^2 + (\omega\sin(\omega t))^2}$
$= \sqrt{\omega^2 - 2\omega^2\cos(\omega t) + \omega^2\cos^2(\omega t) + \omega^2\sin^2(\omega t)}$
$= \sqrt{2\omega^2(1 - \cos(\omega t))}$

12) A person on a hang glider is spiraling upward as a result of the rapidly rising air on a path having position vector $\vec{\mathbf{r}}(t) = (3\cos t)\,\hat{\mathbf{i}} + (3\sin t)\,\hat{\mathbf{j}} + t^2\,\hat{\mathbf{k}}$. The path is similar to that of a helix, although it is not a helix. The graph is shown here:



Find the following quantities:

a. The velocity and acceleration vectors

b. The glider's speed at any time

Answer

$\|\vec{\mathbf{v}}(t)\| = \sqrt{9 + 4t^2}$

c. The times, if any, at which the glider's acceleration is orthogonal to its velocity

13) Given that $\vec{\mathbf{r}}(t) = \langle e^{-5t}\sin t,\ e^{-5t}\cos t,\ 4e^{-5t}\rangle$ is the position vector of a moving particle, find the following quantities:

a. The velocity of the particle

Answer

$\vec{\mathbf{v}}(t) = \langle e^{-5t}(\cos t - 5\sin t),\ -e^{-5t}(\sin t + 5\cos t),\ -20e^{-5t}\rangle$

b. The speed of the particle

c. The acceleration of the particle

Answer

$\vec{\mathbf{a}}(t) = \langle e^{-5t}(-\sin t - 5\cos t) - 5e^{-5t}(\cos t - 5\sin t),\ -e^{-5t}(\cos t - 5\sin t) + 5e^{-5t}(\sin t + 5\cos t),\ 100e^{-5t}\rangle$

14) Find the maximum speed of a point on the circumference of an automobile tire of radius 1 ft when the automobile is traveling at 55 mph.

15) Find the position vector-valued function $\vec{\mathbf{r}}(t)$, given that $\vec{\mathbf{a}}(t) = \hat{\mathbf{i}} + e^t\,\hat{\mathbf{j}}$, $\vec{\mathbf{v}}(0) = 2\,\hat{\mathbf{j}}$, and $\vec{\mathbf{r}}(0) = 2\,\hat{\mathbf{i}}$.

16) Find $\vec{\mathbf{r}}(t)$ given that $\vec{\mathbf{a}}(t) = -32\,\hat{\mathbf{j}}$, $\vec{\mathbf{v}}(0) = 600\sqrt{3}\,\hat{\mathbf{i}} + 600\,\hat{\mathbf{j}}$, and $\vec{\mathbf{r}}(0) = \vec{\mathbf{0}}$.

17) The acceleration of an object is given by $\vec{\mathbf{a}}(t) = t\,\hat{\mathbf{j}} + t\,\hat{\mathbf{k}}$. The velocity at $t = 1$ sec is $\vec{\mathbf{v}}(1) = 5\,\hat{\mathbf{j}}$ and the position of the object at $t = 1$ sec is $\vec{\mathbf{r}}(1) = 0\,\hat{\mathbf{i}} + 0\,\hat{\mathbf{j}} + 0\,\hat{\mathbf{k}}$. Find the object's position at any time.

Answer

$\vec{\mathbf{r}}(t) = 0\,\hat{\mathbf{i}} + \left(\frac{1}{6}t^3 + 4.5t - \frac{14}{3}\right)\hat{\mathbf{j}} + \left(\frac{1}{6}t^3 - \frac{1}{2}t + \frac{1}{3}\right)\hat{\mathbf{k}}$

Projectile Motion

18) A projectile is shot in the air from ground level with an initial velocity of 500 m/sec at an angle of 60° with the horizontal.

a. At what time does the projectile reach maximum height?

Answer

44.185 sec

b. What is the approximate maximum height of the projectile?

c. At what time is the maximum range of the projectile attained?

Answer

$t = 88.37$ sec

d. What is the maximum range?

e. What is the total flight time of the projectile?

Answer

$t = 88.37$ sec

19) A projectile is fired at a height of 1.5 m above the ground with an initial velocity of 100 m/sec and at an angle of 30° above the horizontal. Use this information to answer the following questions:

a. Determine the maximum height of the projectile.

b. Determine the range of the projectile.

Answer

The range is approximately 886.29 m.

20) A golf ball is hit in a horizontal direction off the top edge of a building that is 100 ft tall. How fast must the ball be launched to land 450 ft away?

21) A projectile is fired from ground level at an angle of 8° with the horizontal. The projectile is to have a range of 50 m. Find the minimum velocity (speed) necessary to achieve this range.

Answer

$v = 42.16$ m/sec

22) Prove that an object moving in a straight line at a constant speed has an acceleration of zero.

Finding Components of Acceleration & Kepler's Laws

23) Find the tangential and normal components of acceleration for $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + 2t\,\hat{\mathbf{j}}$ when $t = 1$.

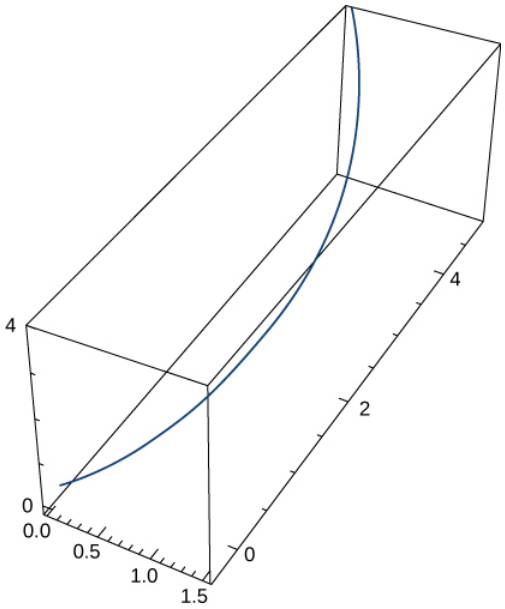
Answer

$a_{\vec{\mathbf{T}}} = \sqrt{2}, \quad a_{\vec{\mathbf{N}}} = \sqrt{2}$

In questions 24 - 30, find the tangential and normal components of acceleration.

24) $\vec{\mathbf{r}}(t) = \langle \cos(2t),\ \sin(2t),\ 1 \rangle$

25) $\vec{\mathbf{r}}(t) = \langle e^t \cos t,\ e^t \sin t,\ e^t \rangle$. The graph is shown here:



Answer

$a_{\vec{\mathbf{T}}} = \sqrt{3}e^t, \quad a_{\vec{\mathbf{N}}} = \sqrt{2}e^t$

26) $\vec{\mathbf{r}}(t) = \langle \frac{2}{3}(1+t)^{3/2}, \frac{2}{3}(1-t)^{3/2}, \sqrt{2}t \rangle$

27) $\vec{\mathbf{r}}(t) = \left\langle 2t, t^2, \frac{t^3}{3} \right\rangle$

Answer

$a_{\vec{\mathbf{T}}} = 2t, \quad a_{\vec{\mathbf{N}}} = 2$

28) $\vec{\mathbf{r}}(t) = t^2\,\hat{\mathbf{i}} + t^2\,\hat{\mathbf{j}} + t^3\,\hat{\mathbf{k}}$

29) $\vec{\mathbf{r}}(t) = \langle 6t, 3t^2, 2t^3 \rangle$

Answer

$a_{\vec{\mathbf{T}}} = \dfrac{6t + 12t^3}{\sqrt{1 + t^2 + t^4}}, \quad a_{\vec{\mathbf{N}}} = 6\sqrt{\dfrac{1 + 4t^2 + t^4}{1 + t^2 + t^4}}$

30) $\vec{\mathbf{r}}(t) = 3\cos(2\pi t)\,\hat{\mathbf{i}} + 3\sin(2\pi t)\,\hat{\mathbf{j}}$

Answer

$a_{\vec{\mathbf{T}}} = 0, \quad a_{\vec{\mathbf{N}}} = 12\pi^2$

31) Find the tangential and normal components of acceleration for $\vec{\mathbf{r}}(t) = a\cos(\omega t)\,\hat{\mathbf{i}} + b\sin(\omega t)\,\hat{\mathbf{j}}$ at $t = 0$.

Answer

$a_{\vec{\mathbf{T}}} = 0, \quad a_{\vec{\mathbf{N}}} = a\omega^2$

32) Suppose that the position function for an object in three dimensions is given by the equation $\vec{\mathbf{r}}(t) = t\cos(t)\,\hat{\mathbf{i}} + t\sin(t)\,\hat{\mathbf{j}} + 3t\,\hat{\mathbf{k}}$.

a. Show that the particle moves on a circular cone.

b. Find the angle between the velocity and acceleration vectors when $t = 1.5$.

c. Find the tangential and normal components of acceleration when $t = 1.5$.

Answer

c. $a_{\vec{\mathbf{T}}} = 0.43 \text{ m/sec}^2, \quad a_{\vec{\mathbf{N}}} = 2.46 \text{ m/sec}^2$

33) The force on a particle is given by $\vec{\mathbf{f}}(t) = (\cos t)\,\hat{\mathbf{i}} + (\sin t)\,\hat{\mathbf{j}}$. The particle is located at point $(c, 0)$ at $t = 0$. The initial velocity of the particle is given by $\vec{\mathbf{v}}(0) = v_0\,\hat{\mathbf{j}}$. Find the path of the particle of mass m. (Recall, $\vec{\mathbf{F}} = m\vec{\mathbf{a}}$.)

Answer

$$\vec{\mathbf{r}}(t) = \left(\frac{-\cos t}{m} + c + \tfrac{1}{m}\right)\hat{\mathbf{i}} + \left(\frac{-\sin t}{m} + \left(v_0 + \tfrac{1}{m}\right)t\right)\hat{\mathbf{j}}$$

34) An automobile that weighs 2700 lb makes a turn on a flat road while traveling at 56 ft/sec. If the radius of the turn is 70 ft, what is the required frictional force to keep the car from skidding?

35) Using Kepler's laws, it can be shown that $v_0 = \sqrt{\dfrac{2GM}{r_0}}$ is the minimum speed needed when $\theta = 0$ so that an object will escape from the pull of a central force resulting from mass M. Use this result to find the minimum speed when $\theta = 0$ for a space capsule to escape from the gravitational pull of Earth if the probe is at an altitude of 300 km above Earth's surface.

Answer

10.94 km/sec

36) Find the time in years it takes the dwarf planet Pluto to make one orbit about the Sun given that $a = 39.5$ A.U.

Contributors

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13.R: Chapter 13 Review Exercises

***True or False*? Justify your answer with a proof or a counterexample.**

1. A parametric equation that passes through points P and Q can be given by $\vec{\mathbf{r}}(t) = \langle t^2,\ 3t+1,\ t-2\rangle$, where $P(1,4,-1)$ and $Q(16,11,2)$.

2. $\dfrac{d}{dt}\left[\vec{\mathbf{u}}(t)\times\vec{\mathbf{u}}(t)\right] = 2\vec{\mathbf{u}}'(t)\times\vec{\mathbf{u}}(t)$

Answer

False, $\dfrac{d}{dt}\left[\vec{\mathbf{u}}(t)\times\vec{\mathbf{u}}(t)\right] = \vec{\mathbf{0}}$.

3. The curvature of a circle of radius r is constant everywhere. Furthermore, the curvature is equal to $1/r$.

4. The speed of a particle with a position function $\vec{\mathbf{r}}(t)$ is $\dfrac{\vec{\mathbf{r}}'(t)}{\|\vec{\mathbf{r}}'(t)\|}$.

Answer

False, it is $\|\vec{\mathbf{r}}'(t)\|$

Find the domains of the vector-valued functions.

5. $\vec{\mathbf{r}}(t) = \langle \sin(t),\ \ln(t),\ \sqrt{t}\rangle$

6. $\vec{\mathbf{r}}(t) = \left\langle e^t,\ \dfrac{1}{\sqrt{4-t}},\ \sec t\right\rangle$

Answer

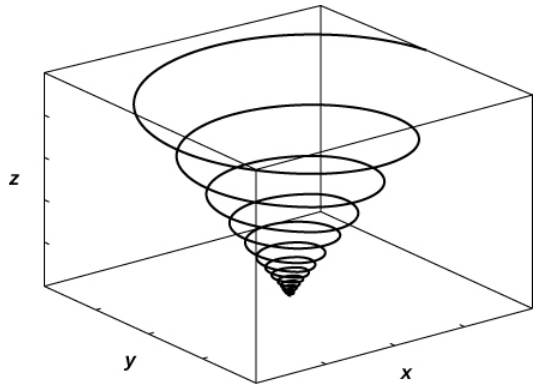
$t<4,\ t\neq\dfrac{n\pi}{2}$

Sketch the curves for the following vector equations. Use a calculator if needed.

7. **[T]** $\vec{\mathbf{r}}(t) = \langle t^2,\ t^3\rangle$

8. **[T]** $\vec{\mathbf{r}}(t) = \langle \sin(20t)e^{-t},\ \cos(20t)e^{-t},\ e^{-t}\rangle$

Answer



Find a vector function that describes the following curves.

9. Intersection of the cylinder $x^2+y^2=4$ with the plane $x+z=6$

10. Intersection of the cone $z=\sqrt{x^2+y^2}$ and plane $z=y-4$

Answer

$\vec{\mathbf{r}}(t) = \left\langle t,\ 2-\frac{t^2}{8},\ -2-\frac{t^2}{8}\right\rangle$

Find the derivatives of $\vec{\mathbf{u}}(t)$, $\vec{\mathbf{u}}'(t)$, $\vec{\mathbf{u}}'(t) \times \vec{\mathbf{u}}(t)$, $\vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}'(t)$, and $\vec{\mathbf{u}}(t) \cdot \vec{\mathbf{u}}'(t)$. Find the unit tangent vector.

11. $\vec{\mathbf{u}}(t) = \langle e^t,\, e^{-t} \rangle$

12. $\vec{\mathbf{u}}(t) = \langle t^2,\, 2t+6,\, 4t^5 - 12 \rangle$

Answer

$\vec{\mathbf{u}}'(t) = \langle 2t,\, 2,\, 20t^4 \rangle$,
$\vec{\mathbf{u}}''(t) = \langle 2,\, 0,\, 80t^3 \rangle$,
$\frac{d}{dt}\left[\vec{\mathbf{u}}'(t) \times \vec{\mathbf{u}}(t)\right] = \langle -480t^3 - 160t^4,\, 24 + 75t^2,\, 12 + 4t \rangle$,
$\frac{d}{dt}\left[\vec{\mathbf{u}}(t) \times \vec{\mathbf{u}}'(t)\right] = \langle 480t^3 + 160t^4,\, -24 - 75t^2,\, -12 - 4t \rangle$,
$\frac{d}{dt}\left[\vec{\mathbf{u}}(t) \cdot \vec{\mathbf{u}}'(t)\right] = 720t^8 - 9600t^3 + 6t^2 + 4$,
unit tangent vector: $\vec{\mathbf{T}}(t) = \dfrac{2t}{\sqrt{400t^8 + 4t^2 + 4}}\,\hat{\mathbf{i}} + \dfrac{2}{\sqrt{400t^8 + 4t^2 + 4}}\,\hat{\mathbf{j}} + \dfrac{20t^4}{\sqrt{400t^8 + 4t^2 + 4}}\,\hat{\mathbf{k}}$

Evaluate the following integrals.

13. $\displaystyle\int \left(\tan(t)\sec(t)\,\hat{\mathbf{i}} - te^{3t}\,\hat{\mathbf{j}}\right) dt$

14. $\displaystyle\int_1^4 \vec{\mathbf{u}}(t)\,dt$, with $\vec{\mathbf{u}}(t) = \left\langle \frac{\ln t}{t},\, \frac{1}{\sqrt{t}},\, \sin\left(\frac{t\pi}{4}\right) \right\rangle$

Answer

$\dfrac{\ln(4^2)}{2}\,\hat{\mathbf{i}} + 2\,\hat{\mathbf{j}} + \dfrac{2(2+\sqrt{2})}{\pi}\,\hat{\mathbf{k}}$

Find the length for the following curves.

15. $\vec{\mathbf{r}}(t) = \langle 3t,\, 4\cos t,\, 4\sin t \rangle$ for $1 \le t \le 4$

16. $\vec{\mathbf{r}}(t) = 2\,\hat{\mathbf{i}} + t\,\hat{\mathbf{j}} + 3t^2\,\hat{\mathbf{k}}$ for $0 \le t \le 1$

Answer

$\dfrac{\sqrt{37}}{2} + \frac{1}{12}\sinh^{-1} 6$

Reparameterize the following functions with respect to their arc length measured from $t = 0$ in direction of increasing t.

17. $\vec{\mathbf{r}}(t) = 2t\,\hat{\mathbf{i}} + (4t-5)\,\hat{\mathbf{j}} + (1-3t)\,\hat{\mathbf{k}}$

18. $\vec{\mathbf{r}}(t) = \cos(2t)\,\hat{\mathbf{i}} + 8t\,\hat{\mathbf{j}} - \sin(2t)\,\hat{\mathbf{k}}$

Answer

$\vec{\mathbf{r}}(t(s)) = \cos\left(\frac{2s}{\sqrt{65}}\right)\,\hat{\mathbf{i}} + \frac{8s}{\sqrt{65}}\,\hat{\mathbf{j}} - \sin\left(\frac{2s}{\sqrt{65}}\right)\,\hat{\mathbf{k}}$

Find the curvature for the following vector functions.

19. $\vec{\mathbf{r}}(t) = (2\sin t)\,\hat{\mathbf{i}} - 4t\,\hat{\mathbf{j}} + (2\cos t)\,\hat{\mathbf{k}}$

20. $\vec{\mathbf{r}}(t) = \sqrt{2}e^t\,\hat{\mathbf{i}} + \sqrt{2}e^{-t}\,\hat{\mathbf{j}} + 2t\,\hat{\mathbf{k}}$

Answer

$\dfrac{e^{2t}}{\left(e^{2t}+1\right)^2}$

21. Find the unit tangent vector, the unit normal vector, and the binormal vector for $\vec{\mathbf{r}}(t) = 2\cos t\,\hat{\mathbf{i}} + 3t\,\hat{\mathbf{j}} + 2sint\,\hat{\mathbf{k}}$.

22. Find the tangential and normal acceleration components with the position vector $\vec{\mathbf{r}}(t) = \langle \cos t,\, \sin t,\, e^t \rangle$.

Answer

$$a_T = \frac{e^{2t}}{1+e^{2t}},$$

$$a_N = \frac{\sqrt{2e^{2t}+4e^{2t}\sin t\cos t+1}}{1+e^{2t}}$$

23. A Ferris wheel car is moving at a constant speed v and has a constant radius r. Find the tangential and normal acceleration of the Ferris wheel car.

24. The position of a particle is given by $\vec{\mathbf{r}}(t) = \langle t^2, \ln t, \sin(\pi t)\rangle$, where t is measured in seconds and r is measured in meters. Find the velocity, acceleration, and speed functions. What are the position, velocity, speed, and acceleration of the particle at 1 sec?

Answer

$\vec{\mathbf{v}}(t) = \left\langle 2t, \frac{1}{t}, \pi\cos(\pi t)\right\rangle$ m/sec,
$\vec{\mathbf{a}}(t) = \left\langle 2, -\frac{1}{t^2}, -\pi^2\sin(\pi t)\right\rangle$ m/sec^2,
$\text{speed}(t) = \sqrt{4t^2+\frac{1}{t^2}+\pi^2\cos^2(\pi t)}$ m/sec;
At $t=1$, $\vec{\mathbf{r}}(1) = \langle 1,0,0\rangle$ m, $\vec{\mathbf{v}}(1) = \langle 2,-1,\pi\rangle$ m/sec, $\vec{\mathbf{a}}(1) = \langle 2,-1,0\rangle$ m/sec^2, and $\text{speed}(1) = \sqrt{5+\pi^2}$ m/sec

The following problems consider launching a cannonball out of a cannon. The cannonball is shot out of the cannon with an angle θ and initial velocity $\vec{\mathbf{v}}_0$. The only force acting on the cannonball is gravity, so we begin with a constant acceleration $\vec{\mathbf{a}}(t) = -g\,\hat{\mathbf{j}}$.

25. Find the velocity vector function $\vec{\mathbf{v}}(t)$.

26. Find the position vector $\vec{\mathbf{r}}(t)$ and the parametric representation for the position.

Answer

$\vec{\mathbf{r}}(t) = \vec{\mathbf{v}}_0 t - \frac{gt^2}{2}\hat{\mathbf{j}}$,
$\vec{\mathbf{r}}(t) = \langle v_0(\cos\theta)t,\ v_0(\sin\theta)t, -\frac{gt^2}{2}\rangle$ where $v_0 = \|\vec{\mathbf{v}}_0\|$.

27. At what angle do you need to fire the cannonball for the horizontal distance to be greatest? What is the total distance it would travel?

Contributors

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CHAPTER OVERVIEW

14: Differentiation of Functions of Several Variables

When dealing with a function of more than one independent variable, several questions naturally arise. For example, how do we calculate limits of functions of more than one variable? The definition of derivative we used before involved a limit. Does the new definition of derivative involve limits as well? Do the rules of differentiation apply in this context? Can we find relative extrema of functions using derivatives? All these questions are answered in this chapter.

14.0: Prelude to Differentiation of Functions of Several Variables

14.1: Functions of Several Variables

14.1E: Exercises for Section 14.1

14.2: Limits and Continuity

14.2E: Exercises for Section 14.2

14.3: Partial Derivatives

14.3E: Exercises for Section 14.3

14.4: Tangent Planes and Linear Approximations

14.4E: Exercises for Section 14.4

14.5: The Chain Rule for Multivariable Functions

14.5E: Exercises for Section 14.5

14.6: Directional Derivatives and the Gradient

14.6E: Exercises for Section 14.6

14.7: Maxima/Minima Problems

14.7E: Exercises for Section 14.7

14.8: Lagrange Multipliers

14.8E: Exercises for Section 14.8

14.R: Chapter 14 Review Exercises

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14.0: Prelude to Differentiation of Functions of Several Variables

In Introduction to Applications of Derivatives, we studied how to determine the maximum and minimum of a function of one variable over a closed interval. This function might represent the temperature over a given time interval, the position of a car as a function of time, or the altitude of a jet plane as it travels from New York to San Francisco. In each of these examples, the function has one independent variable.



Figure 14.0.1: Americans use (and lose) millions of golf balls a year, which keeps golf ball manufacturers in business. In this chapter, we study a profit model and learn methods for calculating optimal production levels for a typical golf ball manufacturing company. (credit: modification of work by oatsy40, Flickr)

Suppose, however, that we have a quantity that depends on more than one variable. For example, temperature can depend on location and the time of day, or a company's profit model might depend on the number of units sold and the amount of money spent on advertising. In this chapter, we look at a company that produces golf balls. We develop a profit model and, under various restrictions, we find that the optimal level of production and advertising dollars spent determines the maximum possible profit. Depending on the nature of the restrictions, both the method of solution and the solution itself changes.

When dealing with a function of more than one independent variable, several questions naturally arise. For example, how do we calculate limits of functions of more than one variable? The definition of derivative we used before involved a limit. Does the new definition of derivative involve limits as well? Do the rules of differentiation apply in this context? Can we find relative extrema of functions using derivatives? All these questions are answered in this chapter.

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14.1: Functions of Several Variables

Learning Objectives

- Recognize a function of two variables and identify its domain and range.
- Sketch a graph of a function of two variables.
- Sketch several traces or level curves of a function of two variables.
- Recognize a function of three or more variables and identify its level surfaces.

Our first step is to explain what a function of more than one variable is, starting with functions of two independent variables. This step includes identifying the domain and range of such functions and learning how to graph them. We also examine ways to relate the graphs of functions in three dimensions to graphs of more familiar planar functions.

Functions of Two Variables

The definition of a function of two variables is very similar to the definition for a function of one variable. The main difference is that, instead of mapping values of one variable to values of another variable, we map ordered pairs of variables to another variable.

Definition: Function of Two Variables

A *function of two variables* $z = f(x, y)$ maps each ordered pair (x, y) in a subset D of the real plane R^2 to a unique real number z. The set D is called the domain of the function. The range of f is the set of all real numbers z that has at least one ordered pair $(x, y) \in D$ such that $f(x, y) = z$ as shown in Figure 14.1.1.

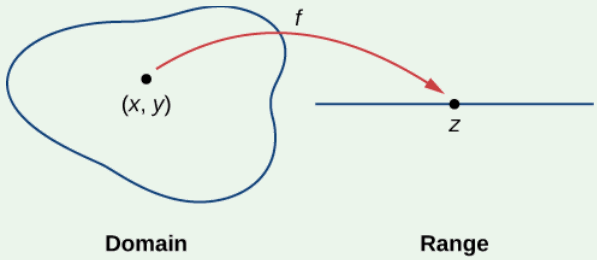


Figure 14.1.1: The domain of a function of two variables consists of ordered pairs (x, y).

Determining the domain of a function of two variables involves taking into account any domain restrictions that may exist. Let's take a look.

Example 14.1.1: Domains and Ranges for Functions of Two Variables

Find the domain and range of each of the following functions:

a. $f(x, y) = 3x + 5y + 2$
b. $g(x, y) = \sqrt{9 - x^2 - y^2}$

Solution

a. This is an example of a linear function in two variables. There are no values or combinations of x and y that cause $f(x, y)$ to be undefined, so the domain of f is R^2. To determine the range, first pick a value for z. We need to find a solution to the equation $f(x, y) = z$, or $3x - 5y + 2 = z$. One such solution can be obtained by first setting $y = 0$, which yields the equation $3x + 2 = z$. The solution to this equation is $x = \frac{z-2}{3}$, which gives the ordered pair $\left(\frac{z-2}{3}, 0\right)$ as a solution to the equation $f(x, y) = z$ for any value of z. Therefore, the range of the function is all real numbers, or R.

b. For the function $g(x, y)$ to have a real value, the quantity under the square root must be nonnegative:

$$9 - x^2 - y^2 \geq 0.$$

This inequality can be written in the form

$$x^2 + y^2 \leq 9.$$

Therefore, the domain of $g(x, y)$ is $\{(x, y) \in R^2 \mid x^2 + y^2 \leq 9\}$. The graph of this set of points can be described as a disk of radius 3 centered at the origin. The domain includes the boundary circle as shown in the following graph.

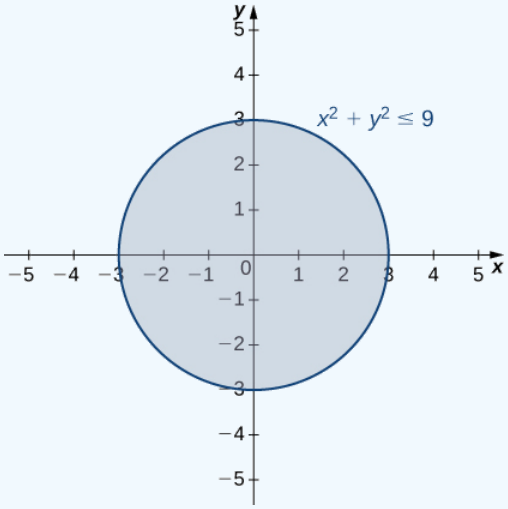


Figure 14.1.2: The domain of the function $g(x, y) = \sqrt{9 - x^2 - y^2}$ is a closed disk of radius 3.

To determine the range of $g(x, y) = \sqrt{9 - x^2 - y^2}$ we start with a point (x_0, y_0) on the boundary of the domain, which is defined by the relation $x^2 + y^2 = 9$. It follows that $x_0^2 + y_0^2 = 9$ and

$$\begin{aligned} g(x_0, y_0) &= \sqrt{9 - x_0^2 - y_0^2} \\ &= \sqrt{9 - (x_0^2 + y_0^2)} \\ &= \sqrt{9 - 9} \\ &= 0. \end{aligned}$$

If $x_0^2 + y_0^2 = 0$ (in other words, $x_0 = y_0 = 0$), then

$$\begin{aligned} g(x_0, y_0) &= \sqrt{9 - x_0^2 - y_0^2} \\ &= \sqrt{9 - (x_0^2 + y_0^2)} \\ &= \sqrt{9 - 0} = 3. \end{aligned}$$

This is the maximum value of the function. Given any value c between 0 and 3, we can find an entire set of points inside the domain of g such that $g(x, y) = c$:

$$\begin{aligned} \sqrt{9 - x^2 - y^2} &= c \\ 9 - x^2 - y^2 &= c^2 \\ x^2 + y^2 &= 9 - c^2. \end{aligned}$$

Since $9 - c^2 > 0$, this describes a circle of radius $\sqrt{9 - c^2}$ centered at the origin. Any point on this circle satisfies the equation $g(x, y) = c$. Therefore, the range of this function can be written in interval notation as $[0, 3]$.

? Exercise 14.1.1

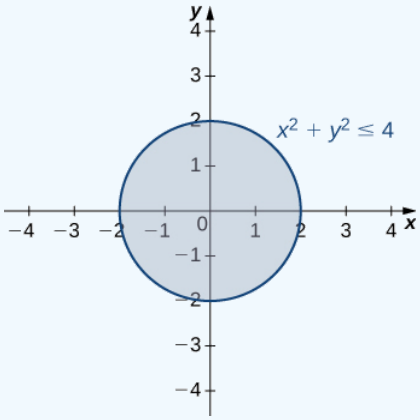
Find the domain and range of the function $f(x, y) = \sqrt{36 - 9x^2 - 9y^2}$.

Hint

Determine the set of ordered pairs that do not make the radicand negative.

Solution

The domain is $\{(x,y)|x^2+y^2 \leq 4\}$ the shaded circle defined by the inequality $x^2+y^2 \leq 4$, which has a circle of radius 2 as its boundary. The range is $[0,6]$.



Graphing Functions of Two Variables

Suppose we wish to graph the function $z=f(x,y)$. This function has two independent variables (x and y) and one dependent variable (z). When graphing a function $y=f(x)$ of one variable, we use the Cartesian plane. We are able to graph any ordered pair (x,y) in the plane, and every point in the plane has an ordered pair (x,y) associated with it. With a function of two variables, each ordered pair (x,y) in the domain of the function is mapped to a real number z. Therefore, the graph of the function f consists of ordered triples (x,y,z). The graph of a function $z=f(x,y)$ of two variables is called a surface.

To understand more completely the concept of plotting a set of ordered triples to obtain a surface in three-dimensional space, imagine the (x,y) coordinate system laying flat. Then, every point in the domain of the function f has a unique z-value associated with it. If z is positive, then the graphed point is located above the xy-plane, if z is negative, then the graphed point is located below the xy-plane. The set of all the graphed points becomes the two-dimensional surface that is the graph of the function f.

Example 14.1.2: Graphing Functions of Two Variables

Create a graph of each of the following functions:

a. $g(x,y)=\sqrt{9-x^2-y^2}$
b. $f(x,y)=x^2+y^2$

Solution

a. In Example 14.1.2, we determined that the domain of $g(x,y)=\sqrt{9-x^2-y^2}$ is $\{(x,y) \in R^2 \mid x^2+y^2 \leq 9\}$ and the range is $\{z \in R^2 \mid 0 \leq z \leq 3\}$. When $x^2+y^2=9$ we have $g(x,y)=0$. Therefore any point on the circle of radius 3 centered at the origin in the xy-plane maps to $z=0$ in R^3. If $x^2+y^2=8$, then $g(x,y)=1$, so any point on the circle of radius $2\sqrt{2}$ centered at the origin in the xy-plane maps to $z=1$ in R^3. As x^2+y^2 gets closer to zero, the value of z approaches 3. When $x^2+y^2=0$, then $g(x,y)=3$. This is the origin in the xy-plane If x^2+y^2 is equal to any other value between 0 and 9, then $g(x,y)$ equals some other constant between 0 and 3. The surface described by this function is a hemisphere centered at the origin with radius 3 as shown in the following graph.

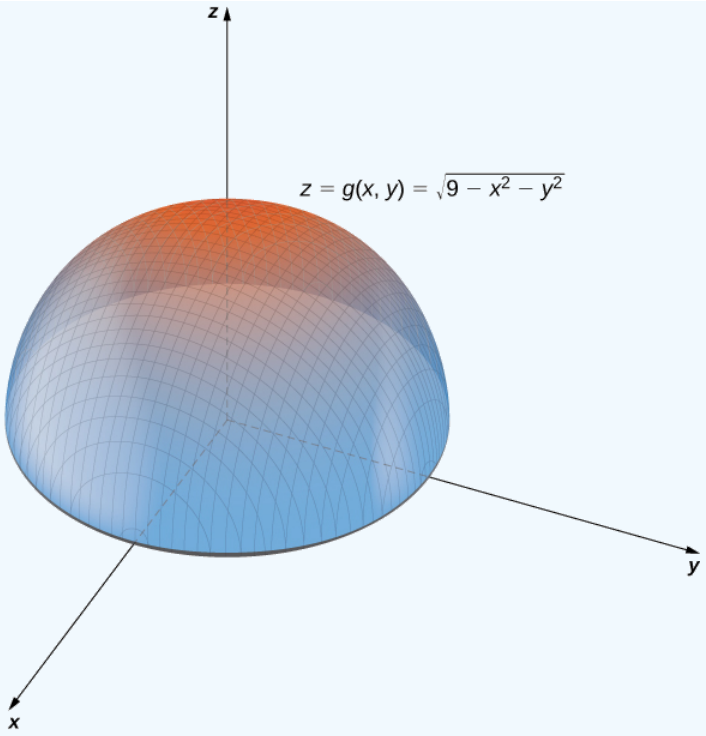


Figure 14.1.3: Graph of the hemisphere represented by the given function of two variables.

b. This function also contains the expression $x^2 + y^2$. Setting this expression equal to various values starting at zero, we obtain circles of increasing radius. The minimum value of $f(x, y) = x^2 + y^2$ is zero (attained when $x = y = 0$.. When $x = 0$, the function becomes $z = y^2$, and when $y = 0$, then the function becomes $z = x^2$. These are cross-sections of the graph, and are parabolas. Recall from Introduction to Vectors in Space that the name of the graph of $f(x, y) = x^2 + y^2$ is a **paraboloid**. The graph of f appears in the following graph.

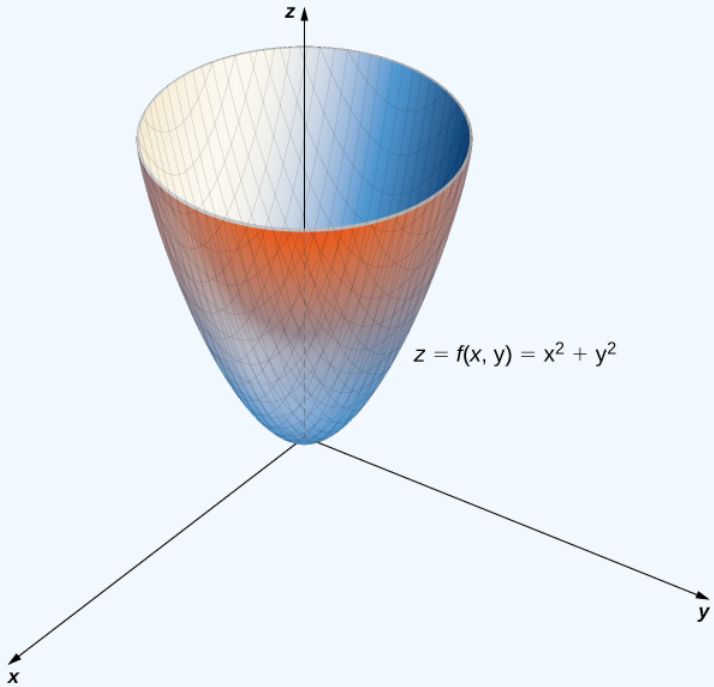


Figure 14.1.4: A paraboloid is the graph of the given function of two variables.

Example 14.1.3: Nuts and Bolts

A profit function for a hardware manufacturer is given by

$$f(x,y) = 16 - (x-3)^2 - (y-2)^2,$$

where x is the number of nuts sold per month (measured in thousands) and y represents the number of bolts sold per month (measured in thousands). Profit is measured in thousands of dollars. Sketch a graph of this function.

Solution

This function is a polynomial function in two variables. The domain of f consists of (x,y) coordinate pairs that yield a nonnegative profit:

$$16 - (x-3)^2 - (y-2)^2 \geq 0$$
$$(x-3)^2 + (y-2)^2 \leq 16.$$

This is a disk of radius 4 centered at $(3,2)$. A further restriction is that both x and y must be nonnegative. When $x = 3$ and $y = 2, f(x,y) = 16$. Note that it is possible for either value to be a noninteger; for example, it is possible to sell 2.5 thousand nuts in a month. The domain, therefore, contains thousands of points, so we can consider all points within the disk. For any $z < 16$, we can solve the equation $f(x,y) = 16$:

$$16 - (x-3)^2 - (y-2)^2 = z$$
$$(x-3)^2 + (y-2)^2 = 16 - z.$$

Since $z < 16$, we know that $16 - z > 0$, so the previous equation describes a circle with radius $\sqrt{16-z}$ centered at the point $(3,2)$. Therefore. the range of $f(x,y)$ is $\{z \in \mathbb{R} | z \leq 16\}$. The graph of $f(x,y)$ is also a paraboloid, and this paraboloid points downward as shown.

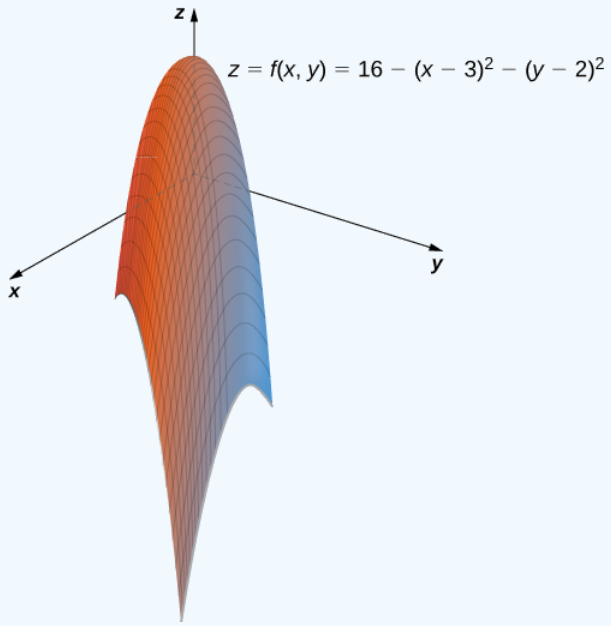
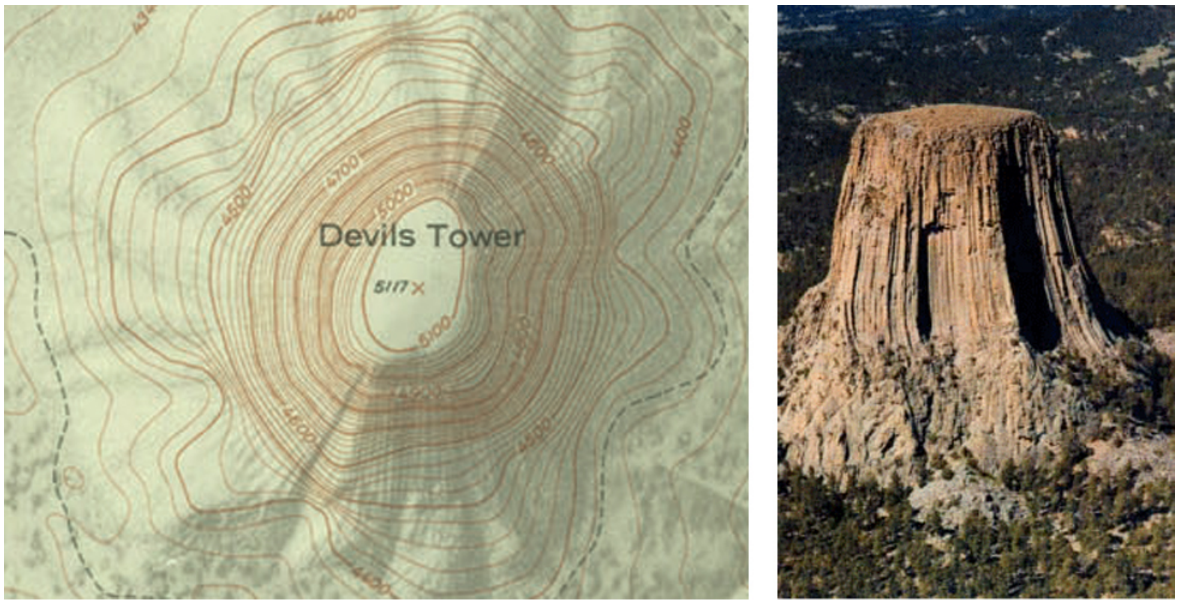


Figure 14.1.5: The graph of the given function of two variables is also a paraboloid.

Level Curves

If hikers walk along rugged trails, they might use a topographical map that shows how steeply the trails change. A *topographical map* contains curved lines called contour lines. Each contour line corresponds to the points on the map that have equal elevation (Figure 14.1.6). A level curve of a function of two variables $f(x,y)$ is completely analogous to a contour line on a topographical map.



(a) (b)

Figure 14.1.6: (a) A topographical map of Devil's Tower, Wyoming. Lines that are close together indicate very steep terrain. (b) A perspective photo of Devil's Tower shows just how steep its sides are. Notice the top of the tower has the same shape as the center of the topographical map.

Definition: Level Curves

Given a function $f(x, y)$ and a number c in the range of f, a level curve of a function of two variables for the value c is defined to be the set of points satisfying the equation $f(x, y) = c$.

Returning to the function $g(x, y) = \sqrt{9 - x^2 - y^2}$, we can determine the level curves of this function. The range of g is the closed interval $[0, 3]$. First, we choose any number in this closed interval—say, $c = 2$. The level curve corresponding to $c = 2$ is described by the equation

$$\sqrt{9 - x^2 - y^2} = 2.$$

To simplify, square both sides of this equation:

$$9 - x^2 - y^2 = 4.$$

Now, multiply both sides of the equation by -1 and add 9 to each side:

$$x^2 + y^2 = 5.$$

This equation describes a circle centered at the origin with radius $\sqrt{5}$. Using values of c between 0 and 3 yields other circles also centered at the origin. If $c = 3$, then the circle has radius 0, so it consists solely of the origin. Figure 14.1.7 is a graph of the level curves of this function corresponding to $c = 0, 1, 2$, and 3. Note that in the previous derivation it may be possible that we introduced extra solutions by squaring both sides. This is not the case here because the range of the square root function is nonnegative.

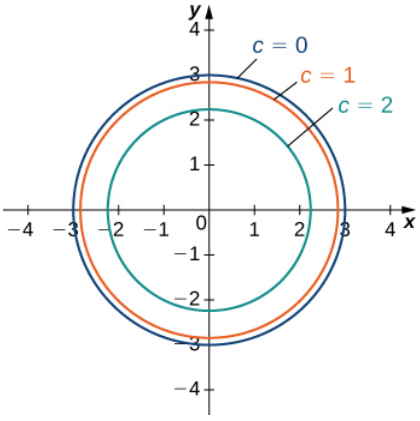


Figure 14.1.7: Level curves of the function $g(x,y)=\sqrt{9-x^2-y^2}$, using $c=0,1,2$, and $3(c=3$ corresponds to the origin).

A graph of the various level curves of a function is called a *contour map*.

Example 14.1.4: Making a Contour Map

Given the function $f(x,y)=\sqrt{8+8x-4y-4x^2-y^2}$, find the level curve corresponding to $c=0$. Then create a contour map for this function. What are the domain and range of f?

Solution

To find the level curve for $c=0$, we set $f(x,y)=0$ and solve. This gives

$$0=\sqrt{8+8x-4y-4x^2-y^2}.$$

We then square both sides and multiply both sides of the equation by -1:

$$4x^2+y^2-8x+4y-8=0.$$

Now, we rearrange the terms, putting the x terms together and the y terms together, and add 8 to each side:

$$4x^2-8x+y^2+4y=8.$$

Next, we group the pairs of terms containing the same variable in parentheses, and factor 4 from the first pair:

$$4(x^2-2x)+(y^2+4y)=8.$$

Then we complete the square in each pair of parentheses and add the correct value to the right-hand side:

$$4(x^2-2x+1)+(y^2+4y+4)=8+4(1)+4.$$

Next, we factor the left-hand side and simplify the right-hand side:

$$4(x-1)^2+(y+2)^2=16.$$

Last, we divide both sides by 16 :

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{16}=1. \tag{14.1.1}$$

This equation describes an ellipse centered at $(1,-2)$. The graph of this ellipse appears in the following graph.

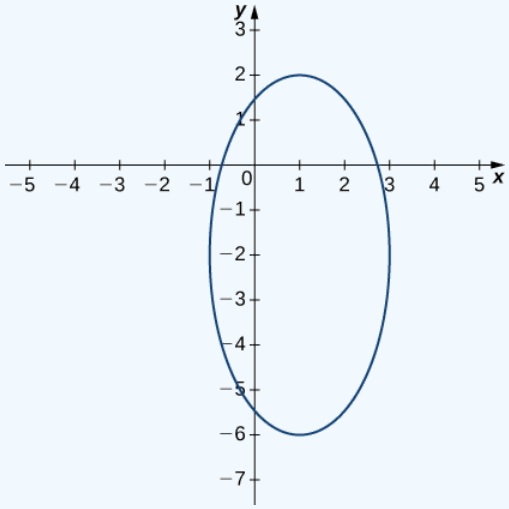


Figure 14.1.8: Level curve of the function $f(x,y)=\sqrt{8+8x-4y-4x^2-y^2}$ corresponding to $c=0$

We can repeat the same derivation for values of c less than 4. Then, Equation 14.1.1 becomes

$$\frac{4(x-1)^2}{16-c^2}+\frac{(y+2)^2}{16-c^2}=1$$

for an arbitrary value of c. Figure 14.1.9 shows a contour map for $f(x,y)$ using the values $c=0,1,2,$ and 3. When $c=4$, the level curve is the point $(-1,2)$.

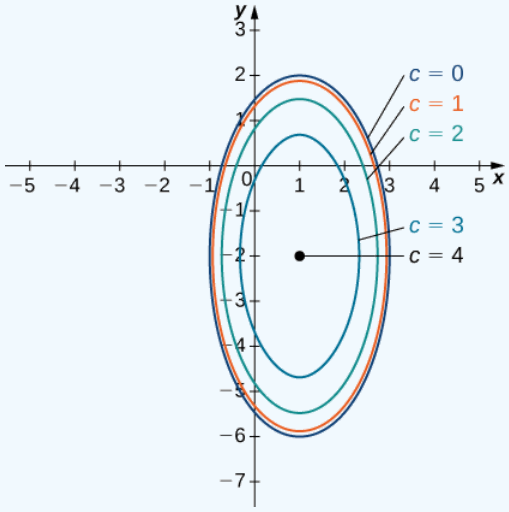


Figure 14.1.9: Contour map for the function $f(x,y)=\sqrt{8+8x-4y-4x^2-y^2}$ using the values $c=0,1,2,3,$ and 4.

Finding the Domain & Range

Since this is a square root function, the radicand must not be negative. So we have

$$8+8x-4y-4x^2-y^2\geq 0$$

Recognizing that the boundary of the domain is an ellipse, we repeat the steps we showed above to obtain

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{16}\leq 1$$

So the domain of f can be written: $\left\{(x,y)\mid \frac{(x-1)^2}{4}+\frac{(y+2)^2}{16}\leq 1\right\}$.

To find the range of f, we need to consider the possible outputs of this square root function. We know the output cannot be negative, so we need to next check if its output is ever 0. From the work we completed above to find the level curve for $c=0$, we know the value of f is 0 for any point on that level curve (on the ellipse, $\frac{(x-1)^2}{4}+\frac{(y+2)^2}{16}=1$). So we know the lower bound of the range of this function is 0.

To determine the upper bound for the range of the function in this problem, it's easier if we first complete the square under the radical.

$$\begin{aligned} f(x,y) &= \sqrt{8+8x-4y-4x^2-y^2} \\ &= \sqrt{8-4(x^2-2x\quad)-(y^2+4y\quad)} \\ &= \sqrt{8-4(x^2-2x+1-1)-(y^2+4y+4-4)} \\ &= \sqrt{8-4(x^2-2x+1)+4-(y^2+4y+4)+4} \\ &= \sqrt{16-4(x-1)^2-(y+2)^2} \end{aligned}$$

Now that we have f in this form, we can see how large the radicand can be. Since we are subtracting two perfect squares from 16, we know that the value of the radicand cannot be greater than 16. At the point $(1,-2)$, we can see the radicand will be 16 (since we will be subtracting 0 from 16 at this point. This gives us the maximum value of f, that is $f(1,-2)=\sqrt{16}=4$.

So the range of this function is $[0,4]$.

Exercise 14.1.2

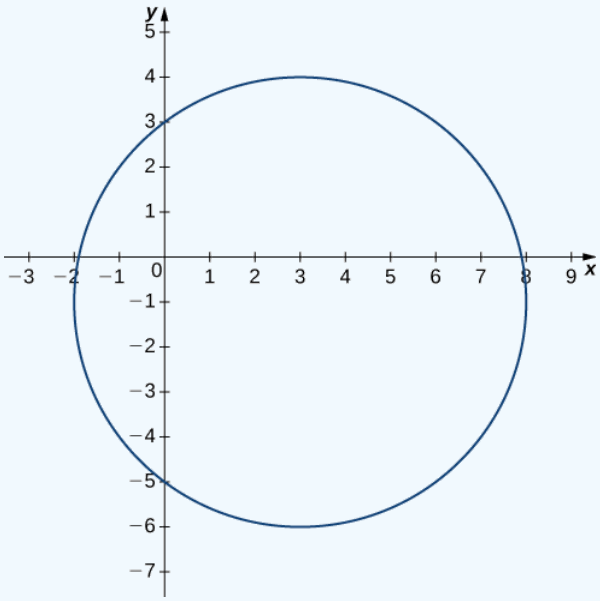
Find and graph the level curve of the function $g(x,y)=x^2+y^2-6x+2y$ corresponding to $c=15$.

Hint

First, set $g(x,y)=15$ and then complete the square.

Solution

The equation of the level curve can be written as $(x-3)^2+(y+1)^2=25$, which is a circle with radius 5 centered at $(3,-1)$.



Another useful tool for understanding the **graph of a function of two variables** is called a vertical trace. Level curves are always graphed in the xy-plane, but as their name implies, vertical traces are graphed in the xz- or yz-planes.

Definition: Vertical Traces

Consider a function $z=f(x,y)$ with domain $D\subseteq\mathbb{R}^2$. A *vertical trace* of the function can be either the set of points that solves the equation $f(a,y)=z$ for a given constant $x=a$ or $f(x,b)=z$ for a given constant $y=b$.

https://math.libretexts.org/@go/page/2600

Example 14.1.5: Finding Vertical Traces

Find vertical traces for the function $f(x,y) = \sin x \cos y$ corresponding to $x = -\frac{\pi}{4}, 0$, and $\frac{\pi}{4}$, and $y = -\frac{\pi}{4}, 0$, and $\frac{\pi}{4}$.

Solution

First set $x = -\frac{\pi}{4}$ in the equation $z = \sin x \cos y$:

$$z = \sin\left(-\frac{\pi}{4}\right)\cos y = -\frac{\sqrt{2}\cos y}{2} \approx -0.7071 \cos y.$$

This describes a cosine graph in the plane $x = -\frac{\pi}{4}$. The other values of z appear in the following table.

Vertical Traces Parallel to the xz-Plane for the Function $f(x,y) = \sin x \cos y$

c	Vertical Trace for $x = c$
$-\frac{\pi}{4}$	$z = -\frac{\sqrt{2}\cos y}{2}$
0	$z = 0$
$\frac{\pi}{4}$	$z = \frac{\sqrt{2}\cos y}{2}$

In a similar fashion, we can substitute the y-values in the equation $f(x,y)$ to obtain the traces in the yz-plane, as listed in the following table.

Vertical Traces Parallel to the yz-Plane for the Function $f(x,y) = \sin x \cos y$

d	Vertical Trace for $y = d$
$\frac{\pi}{4}$	$z = \frac{\sqrt{2}\sin x}{2}$
0	$z = \sin x$
$-\frac{\pi}{4}$	$z = \frac{\sqrt{2}\sin x}{2}$

The three traces in the xz-plane are cosine functions; the three traces in the yz-plane are sine functions. These curves appear in the intersections of the surface with the planes $x = -\frac{\pi}{4}, x = 0, x = \frac{\pi}{4}$ and $y = -\frac{\pi}{4}, y = 0, y = \frac{\pi}{4}$ as shown in the following figure.

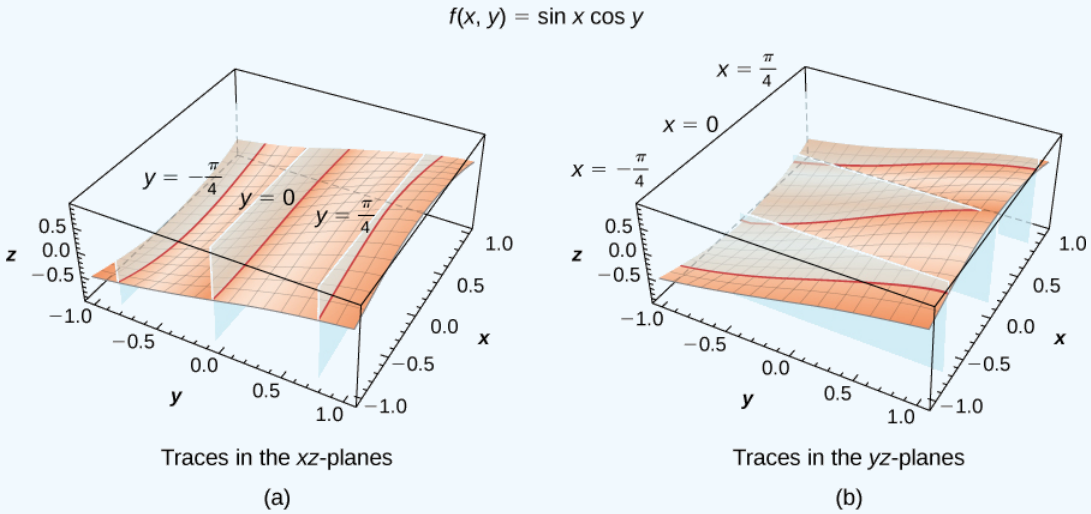


Figure 14.1.10: Vertical traces of the function $f(x,y)$ are cosine curves in the $xz-planes$ (a) and sine curves in the $yz-planes$ (b).

Exercise 14.1.3

Determine the equation of the vertical trace of the function $g(x,y) = -x^2 - y^2 + 2x + 4y - 1$ corresponding to $y = 3$, and describe its graph.

Hint

Set $y = 3$ in the equation $z = -x^2 - y^2 + 2x + 4y - 1$ and complete the square.

Solution

$z = 3 - (x-1)^2$. This function describes a parabola opening downward in the plane $y = 3$.

Functions of two variables can produce some striking-looking surfaces. Figure 14.1.11 shows two examples.

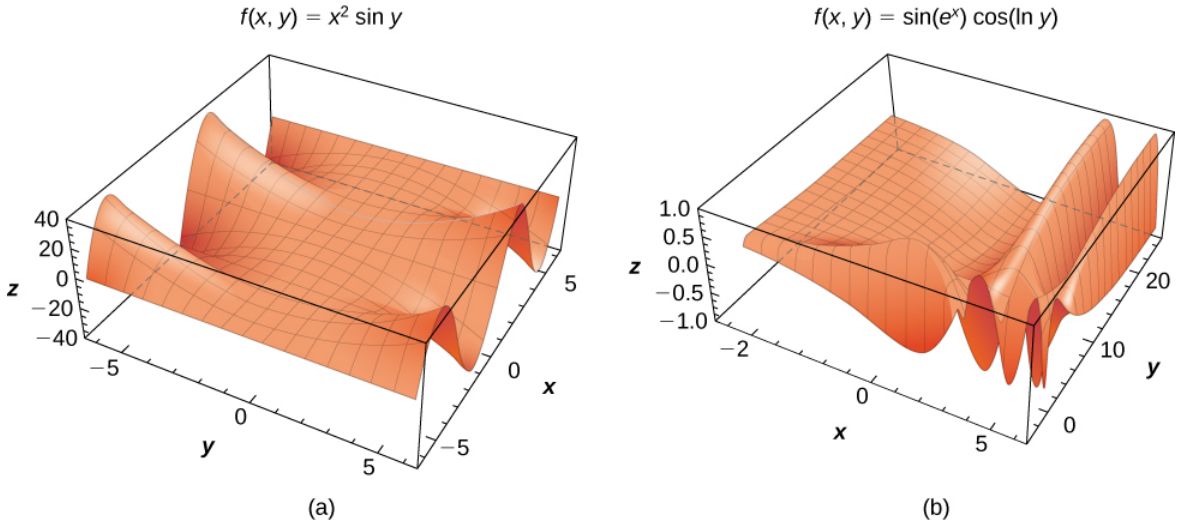


Figure 14.1.11: Examples of surfaces representing functions of two variables: (a) a combination of a power function and a sine function and (b) a combination of trigonometric, exponential, and logarithmic functions.

Functions of More Than Two Variables

So far, we have examined only functions of two variables. However, it is useful to take a brief look at functions of more than two variables. Two such examples are

$$\underbrace{f(x,y,z) = x^2 - 2xy + y^2 + 3yz - z^2 + 4x - 2y + 3x - 6}_{\text{a polynomial in three variables}}$$

and

$$g(x,y,t) = (x^2 - 4xy + y^2)\sin t - (3x + 5y)\cos t.$$

In the first function, (x,y,z) represents a point in space, and the function f maps each point in space to a fourth quantity, such as temperature or wind speed. In the second function, (x,y) can represent a point in the plane, and t can represent time. The function might map a point in the plane to a third quantity (for example, pressure) at a given time t. The method for finding the domain of a function of more than two variables is analogous to the method for functions of one or two variables.

Example 14.1.6: Domains for Functions of Three Variables

Find the domain of each of the following functions:

a. $f(x,y,z) = \dfrac{3x - 4y + 2z}{\sqrt{9 - x^2 - y^2 - z^2}}$

b. $g(x,y,t) = \dfrac{\sqrt{2t - 4}}{x^2 - y^2}$

Solution:

a. For the function $f(x,y,z)=\dfrac{3x-4y+2z}{\sqrt{9-x^2-y^2-z^2}}$ to be defined (and be a real value), two conditions must hold:

1. The denominator **cannot** be zero.
2. The radicand **cannot** be negative.

Combining these conditions leads to the inequality

$$9-x^2-y^2-z^2>0.$$

Moving the variables to the other side and reversing the inequality gives the domain as

$$domain(f)=\{(x,y,z)\in R^3 \mid x^2+y^2+z^2<9\},$$

which describes a ball of radius 3 centered at the origin. (Note: The surface of the ball is not included in this domain.)

b. For the function $g(x,y,t)=\dfrac{\sqrt{2t-4}}{x^2-y^2}$ to be defined (and be a real value), two conditions must hold:

1. The radicand cannot be negative.
2. The denominator cannot be zero.

Since the radicand cannot be negative, this implies $2t-4\geq 0$, and therefore that $t\geq 2$. Since the denominator cannot be zero, $x^2-y^2\neq 0$, or $x^2\neq y^2$, Which can be rewritten as $y=\pm x$, which are the equations of two lines passing through the origin. Therefore, the domain of g is

$$domain(g)=\{(x,y,t)|y\neq\pm x, t\geq 2\}.$$

? Exercise 14.1.4

Find the domain of the function $h(x,y,t)=(3t-6)\sqrt{y-4x^2+4}$.

Hint

Check for values that make radicands negative or denominators equal to zero.

Solution

$$domain(h)=\{(x,y,t)\in\mathbb{R}^3 \mid y\geq 4x^2-4\}$$

Functions of two variables have level curves, which are shown as curves in the $xy-plane$. However, when the function has three variables, the curves become surfaces, so we can define level surfaces for functions of three variables.

Definition: Level Surface of a Function of Three Variables

Given a function $f(x,y,z)$ and a number c in the range of f, a *level surface of a function of three variables* is defined to be the set of points satisfying the equation $f(x,y,z)=c$.

Example 14.1.7: Finding a Level Surface

Find the level surface for the function $f(x,y,z)=4x^2+9y^2-z^2$ corresponding to $c=1$.

Solution

The level surface is defined by the equation $4x^2+9y^2-z^2=1$. This equation describes a hyperboloid of one sheet as shown in Figure 14.1.12.

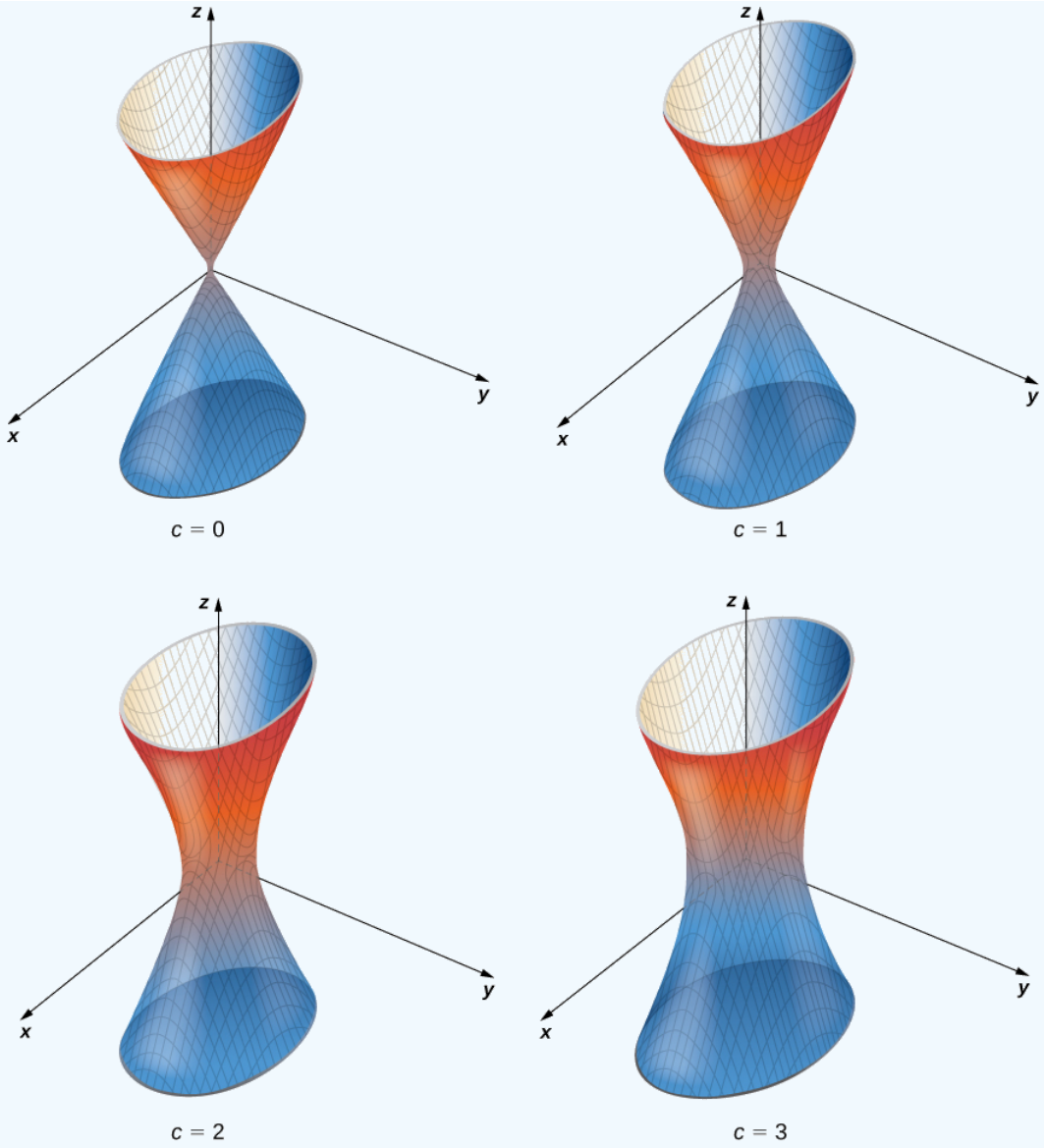


Figure 14.1.12: A hyperboloid of one sheet with some of its level surfaces.

? Exercise 14.1.5

Find the equation of the level surface of the function

$$g(x, y, z) = x^2 + y^2 + z^2 - 2x + 4y - 6z$$

corresponding to $c = 2$, and describe the surface, if possible.

Hint

Set $g(x, y, z) = c$ and complete the square.

Solution

$(x-1)^2 + (y+2)^2 + (z-3)^2 = 16$ describes a sphere of radius 4 centered at the point $(1, -2, 3)$.

Summary

- The graph of a function of two variables is a surface in $\mathbb{R}^3$ and can be studied using level curves and vertical traces.
- A set of level curves is called a contour map.

Key Equations

- **Vertical trace**

  $f(a, y) = z$ for $x = a$ or $f(x, b) = z$ for $y = b$

- **Level surface of a function of three variables**

  $f(x, y, z) = c$

Glossary

contour map
a plot of the various level curves of a given function $f(x, y)$

function of two variables
a function $z = f(x, y)$ that maps each ordered pair (x, y) in a subset D of R^2 to a unique real number z

graph of a function of two variables
a set of ordered triples (x, y, z) that satisfies the equation $z = f(x, y)$ plotted in three-dimensional Cartesian space

level curve of a function of two variables
the set of points satisfying the equation $f(x, y) = c$ for some real number c in the range of f

level surface of a function of three variables
the set of points satisfying the equation $f(x, y, z) = c$ for some real number c in the range of f

surface
the graph of a function of two variables, $z = f(x, y)$

vertical trace
the set of ordered triples (c, y, z) that solves the equation $f(c, y) = z$ for a given constant $x = c$ or the set of ordered triples (x, d, z) that solves the equation $f(x, d) = z$ for a given constant $y = d$

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14.1E: EXERCISES FOR SECTION 14.1

For the following exercises, evaluate each function at the indicated values.

1) $W(x,y) = 4x^2 + y^2$. Find $W(2,-1), W(-3,6)$.

Answer

$W(2,-1) = 17, \quad W(-3,6) = 72$

2) $W(x,y) = 4x^2 + y^2$. Find $W(2+h, 3+h)$.

3) The volume of a right circular cylinder is calculated by a function of two variables, $V(x,y) = \pi x^2 y$, where x is the radius of the right circular cylinder and y represents the height of the cylinder. Evaluate $V(2,5)$ and explain what this means.

Answer

$V(2,5) = 20\pi \text{ units}^3$ This is the volume when the radius is 2 and the height is 5.

4) An oxygen tank is constructed of a right cylinder of height y and radius x with two hemispheres of radius x mounted on the top and bottom of the cylinder. Express the volume of the cylinder as a function of two variables, x and y, find $V(10,2)$, and explain what this means.

For exercises 5 - 10, find the domain and range of the given function. State the domain in set-builder notation and the range in interval notation.

5) $V(x,y) = 4x^2 + y^2$

Answer

Domain: $\{(x,y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}$ That is, all points in the xy-plane

Range: $[0, \infty)$

6) $f(x,y) = \sqrt{x^2 + y^2 - 4}$

Answer

Domain: $\{(x,y) \mid x^2 + y^2 \geq 4\}$

Range: $[0, \infty)$

7) $f(x,y) = 4\ln(y^2 - x)$

Answer

Domain: $\{(x,y) \mid x < y^2\}$

Range: $(-\infty, \infty)$

8) $g(x,y) = \sqrt{16 - 4x^2 - y^2}$

Answer

Domain: $\{(x,y) \mid \frac{x^2}{4} + \frac{y^2}{16} \leq 1\}$

Range: $[0,4]$

9) $z = \arccos(y - x)$

Answer

Domain: $\{(x,y) \mid x - 1 \leq y \leq x + 1\}$ That is, all points between the graphs of $y = x - 1$ and $y = x + 1$.

Range: $[0, \pi]$

10) $f(x,y) = \frac{y+2}{x^2}$

Answer

Domain: $\{(x,y) \mid x \neq 0\}$

Range: $(-\infty, \infty)$

Find the range of the functions.

11) $g(x,y)=\sqrt{16-4x^2-y^2}$

Answer

$\{z \mid 0 \le z \le 4\}$ or in interval notation: $[0,4]$

12) $V(x,y)=4x^2+y^2$

13) $z=y^2-x^2$

Answer

The set $\mathbb{R}$

In exercises 14 - 29, find the level curves of each function at the indicated values of c to visualize the given function. Sketch a contour plot for those exercises where you are asked for more than 3 values of c.

14) $z(x,y)=y^2-x^2, \quad c=1$

15) $z(x,y)=y^2-x^2, \quad c=4$

Answer

$y^2-x^2=4$, a hyperbola

16) $g(x,y)=x^2+y^2; \quad c=0,1,2,3,4,9$

17) $g(x,y)=4-x-y; \quad c=0,1,2,3,4$

Answer

Level curves are lines with $y=-x+(4-c)$.

For each value of c these are:

$c=0: \ y=-x+4,$

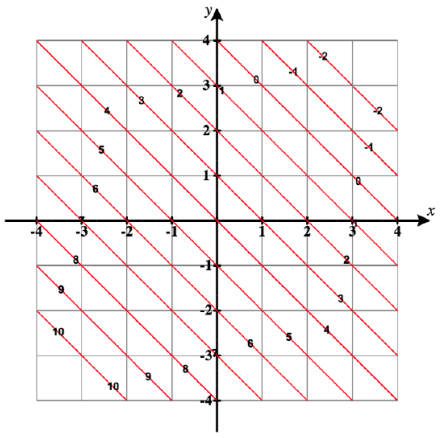
$c=1: \ y=-x+3,$

$c=2: \ y=-x+2,$

$c=3: \ y=-x+1,$

$c=4: \ y=-x.$

The contour plot consists of a series of parallel lines.



18) $f(x,y)=xy; c=1; \quad c=-1$

19) $h(x,y)=2x-y; \quad c=-2,0,2$

Answer

$2x-y=0, 2x-y=-2, 2x-y=2$; three lines

20) $f(x,y)=x^2-y; \quad c=1,2$

21) $g(x,y)=\dfrac{x}{x+y}; c=-1,0,1,2$

Answer

Level curves are lines with the form $y=x\left(\dfrac{1-c}{c}\right)$. At $c=0$, we solve it directly from the equation $\dfrac{x}{x+y}=0$ to get $x=0$.

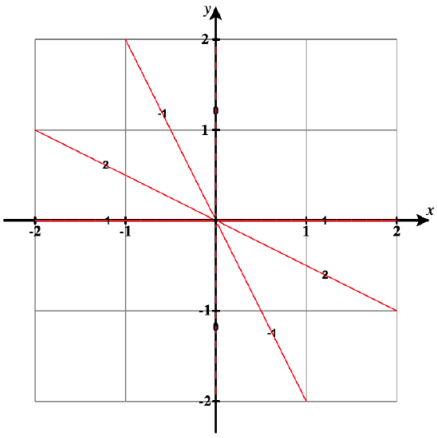
For each value of c these are:

$c=-1: y=-2x$,

$c=0: x=0,$ with $y\neq 0$,

$c=1: y=0,$ with $x\neq 0$,

$c=2: y=-\frac{1}{2}x$.



22) $g(x,y)=x^3-y; \quad c=-1,0,2$

23) $g(x,y)=e^{xy}; \quad c=\frac{1}{2},3$

Answer

The level curves have the form, $y=\dfrac{\ln c}{x}$.

For each value of c these are:

$c=\frac{1}{2}: y=\dfrac{\ln\frac{1}{2}}{x}$ that can be rewritten as, $y=-\dfrac{\ln 2}{x}$

$c=3: y=\dfrac{\ln 3}{x}$.

24) $f(x,y)=x^2; \quad c=4,9$

25) $f(x,y)=xy-x; \quad c=-2,0,2$

Answer

Level curves have the form: $y=\dfrac{c}{x}+1$.

Here $y=\dfrac{-2}{x}+1, \quad y=1, \quad y=\dfrac{2}{x}+1$ or $xy-x=-2$, $xy-x=0$, $xy-x=2$

26) $h(x,y)=\ln(x^2+y^2); \quad c=-1,0,1$

27) $g(x,y)=\ln\left(\dfrac{y}{x^2}\right); \quad c=-2,0,2$

Answer

The level curves have the form, $y=e^c x^2$.

For each value of c these are:

$c=-2: y=e^{-2}x^2$,

$c=0: y=x^2$,

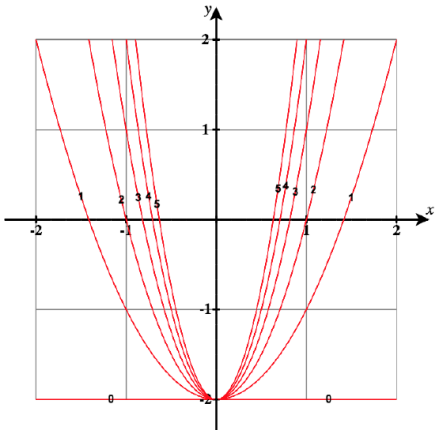
$c=2: y=e^2x^2$.

28) $z=f(x,y)=\sqrt{x^2+y^2}, \quad c=3$

29) $f(x,y)=\dfrac{y+2}{x^2}$, $c=$ any constant

Answer

The level curves are parabolas of the form $y=cx^2-2,$ **with** $x\neq 0$.



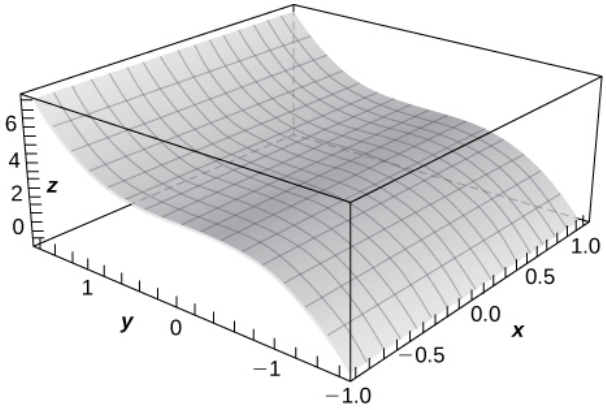
In exercises 30-32, find the vertical traces of the functions at the indicated values of x and y, and plot the traces.

30) $z=4-x-y, \quad x=2$

31) $f(x,y)=3x+y^3, \quad x=1$

Answer

$z=3+y^3$, a curve in the zy-**plane** with rulings parallel to the x-axis



32) $z=\cos\sqrt{x^2+y^2}, \quad x=1$

In exercises 33 - 38, find the domain and range of each function.

33) $z=\sqrt{100-4x^2-25y^2}$

Answer

Domain: $\left\{(x,y)\,|\,\dfrac{x^2}{25}+\dfrac{y^2}{4}\leq 1\right\}$

Range: $[0,10]$

34) $z=\ln(x-y^2)$

35) $f(x,y,z)=\dfrac{1}{\sqrt{36-4x^2-9y^2-z^2}}$

Answer

Domain: $\left\{(x,y,z)\,|\,\dfrac{x^2}{9}+\dfrac{y^2}{4}+\dfrac{z^2}{36}<1\right\}$

Range: $\left[\frac{1}{6},\infty\right)$

36) $f(x,y,z) = \sqrt{49 - x^2 - y^2 - z^2}$

37) $f(x,y,z) = \sqrt[3]{16 - x^2 - y^2 - z^2}$

Answer

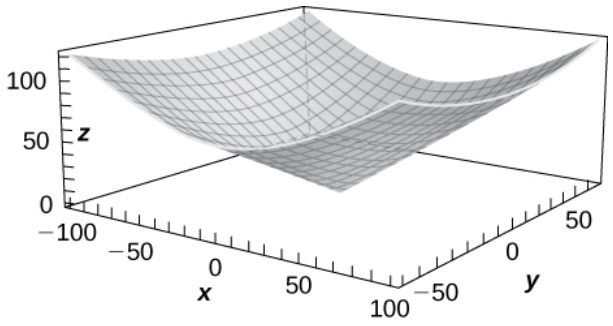
Domain: All points in xyz-space
Range: $\left(-\infty, \sqrt[3]{16}\right]$

38) $f(x,y) = \cos\sqrt{x^2 + y^2}$

In exercises 39 - 40, plot a graph of the function.

39) $z = f(x,y) = \sqrt{x^2 + y^2}$

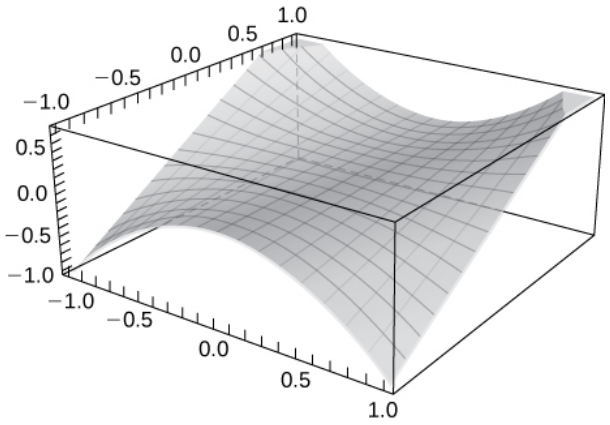
Answer



40) $z = x^2 + y^2$

41) Use technology to graph $z = x^2 y$.

Answer

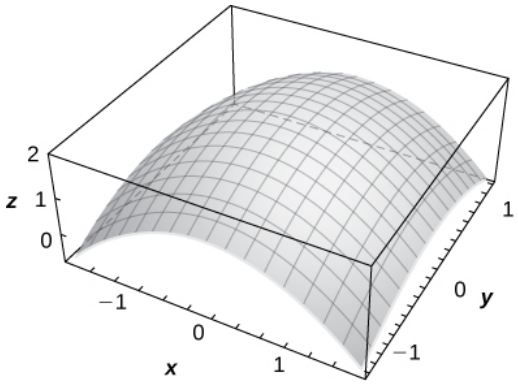


In exercises 42 - 46, sketch the function by finding its level curves. Verify the graph using technology, such as CalcPlot3D.

42) $f(x,y) = \sqrt{4 - x^2 - y^2}$

43) $f(x,y) = 2 - \sqrt{x^2 + y^2}$

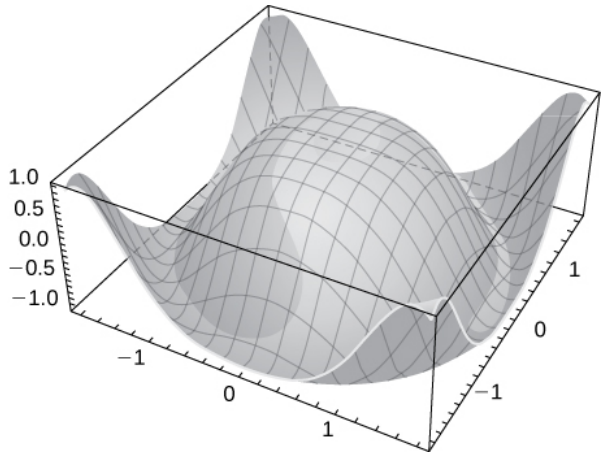
Answer



44) $z = 1 + e^{-x^2-y^2}$

45) $z = \cos\sqrt{x^2+y^2}$

Answer



46) $z = y^2 - x^2$

47) Describe the contour lines for several values of c for $z = x^2 + y^2 - 2x - 2y$.

Answer

The contour lines are concentric circles centered at the point, $(1,1)$.
You can see this by completing the square after setting this function equal to c.
That is, we write $x^2 - 2x + 1 + y^2 - 2y + 1 = c + 2$ which can be rewritten as, $(x-1)^2 + (y-1)^2 = c + 2$.
This gives us circles centered at the point, $(1,1)$, each with a radius of $\sqrt{c+2}$.

In exercises, 48 - 52, find the level surface for the given value of c for each function of three variables and describe it.

48) $w(x,y,z) = x - 2y + z, \quad c = 4$

49) $w(x,y,z) = x^2 + y^2 + z^2, \quad c = 9$

Answer

$x^2 + y^2 + z^2 = 9$, a sphere of radius 3

50) $w(x,y,z) = x^2 + y^2 - z^2, \quad c = -4$

51) $w(x,y,z) = x^2 + y^2 - z^2, \quad c = 4$

Answer

$x^2 + y^2 - z^2 = 4$, a hyperboloid of one sheet

52) $w(x,y,z) = 9x^2 - 4y^2 + 36z^2, \quad c = 0$

In exercises 53 - 55, find an equation of the level curve of f that contains the point P.

53) $f(x,y) = 1 - 4x^2 - y^2, \quad P(0,1)$

Answer

$4x^2 + y^2 = 1,$

54) $g(x,y) = y^2 \arctan x, \quad P(1,2)$

55) $g(x,y) = e^{xy}(x^2 + y^2), \quad P(1,0)$

Answer

$1 = e^{xy}(x^2 + y^2)$

56) The strength E of an electric field at point (x,y,z) resulting from an infinitely long charged wire lying along the y-axis is given by $E(x,y,z) = k/\sqrt{x^2 + y^2}$, where k is a positive constant. For simplicity, let $k = 1$ and find the equations of the level surfaces for $E = 10$ and $E = 100$.

57) A thin plate made of iron is located in the xy-plane The temperature T in degrees Celsius at a point $P(x,y)$ is inversely proportional to the square of its distance from the origin. Express T as a function of x and y.

Answer

$T(x,y) = \dfrac{k}{x^2 + y^2}$

58) Refer to the preceding problem. Using the temperature function found there, determine the proportionality constant if the temperature at point $P(1,2)$ is $50°C$. Use this constant to determine the temperature at point $Q(3,4)$.

59) Refer to the preceding problem. Find the level curves for $T = 40°C$ and $T = 100°C$, and describe what the level curves represent.

Answer

$x^2 + y^2 = \dfrac{k}{40}, \quad x^2 + y^2 = \dfrac{k}{100}$. The level curves represent circles of radii $\sqrt{10k}/20$ and $\sqrt{k}/10$

CONTRIBUTORS

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) added the contour plots to answers for problems 17, 21 and 29.

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14.2: Limits and Continuity

Learning Objectives

- Calculate the limit of a function of two variables.
- Learn how a function of two variables can approach different values at a boundary point, depending on the path of approach.
- State the conditions for continuity of a function of two variables.
- Verify the continuity of a function of two variables at a point.
- Calculate the limit of a function of three or more variables and verify the continuity of the function at a point.

We have now examined functions of more than one variable and seen how to graph them. In this section, we see how to take the limit of a function of more than one variable, and what it means for a function of more than one variable to be continuous at a point in its domain. It turns out these concepts have aspects that just don't occur with functions of one variable.

Limit of a Function of Two Variables

Recall from Section 2.5 that the definition of a limit of a function of one variable:

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a. Let L be a real number. Then

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$, such that if $0 < |x - a| < \delta$ for all x in the domain of f, then

$$|f(x) - L| < \varepsilon.$$

Before we can adapt this definition to define a limit of a function of two variables, we first need to see how to extend the idea of an open interval in one variable to an open interval in two variables.

Definition: δ Disks

Consider a point $(a, b) \in \mathbb{R}^2$. A δ **disk** centered at point (a, b) is defined to be an open disk of radius δ centered at point (a, b) — that is,

$$\{(x, y) \in \mathbb{R}^2 \mid (x - a)^2 + (y - b)^2 < \delta^2\}$$

as shown in Figure 14.2.1.

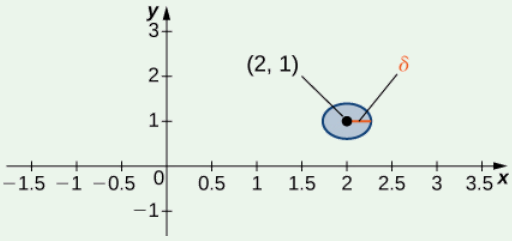


Figure 14.2.1: A δ disk centered around the point $(2, 1)$.

The idea of a δ disk appears in the definition of the limit of a function of two variables. If δ is small, then all the points (x, y) in the δ disk are close to (a, b). This is completely analogous to x being close to a in the definition of a limit of a function of one variable. In one dimension, we express this restriction as

$$a - \delta < x < a + \delta.$$

In more than one dimension, we use a δ disk.

Definition: limit of a function of two variables

Let f be a function of two variables, x and y. The limit of $f(x,y)$ as (x,y) approaches (a,b) is L, written

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

if for each $\varepsilon > 0$ there exists a small enough $\delta > 0$ such that for all points (x,y) in a δ disk around (a,b), except possibly for (a,b) itself, the value of $f(x,y)$ is no more than ε away from L (Figure 14.2.2).

Using symbols, we write the following: For any $\varepsilon > 0$, there exists a number $\delta > 0$ such that

$$|f(x,y) - L| < \varepsilon$$

whenever

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

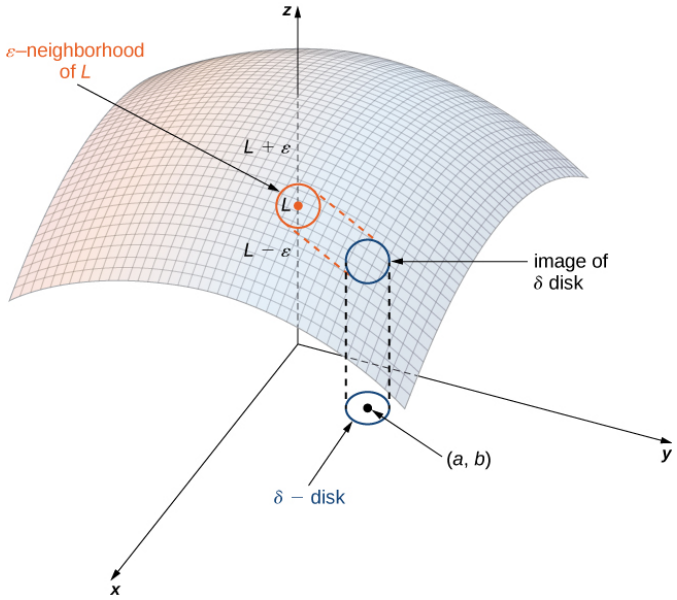


Figure 14.2.2: The limit of a function involving two variables requires that $f(x,y)$ be within ε of L whenever (x,y) is within δ of (a,b). The smaller the value of ε, the smaller the value of δ.

Proving that a limit exists using the definition of a **limit of a function of two variables** can be challenging. Instead, we use the following theorem, which gives us shortcuts to finding limits. The formulas in this theorem are an extension of the formulas in the limit laws theorem in the section titled The Limit Laws in single-variable calculus.

Theorem 14.2.1: Limit Laws for Functions of Two Variables

Let $f(x,y)$ and $g(x,y)$ be defined for all $(x,y) \neq (a,b)$ in a neighborhood around (a,b), and assume the neighborhood is contained completely inside the domain of f. Assume that L and M are real numbers such that

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

and

$$\lim_{(x,y)\to(a,b)} g(x,y) = M,$$

and let c be a constant. Then each of the following statements holds:

Constant Law:

$$\lim_{(x,y)\to(a,b)} c = c$$

Identity Laws:

$$\lim_{(x,y)\to(a,b)} x = a$$

$$\lim_{(x,y)\to(a,b)} y = b$$

Sum Law:

$$\lim_{(x,y)\to(a,b)} (f(x,y) + g(x,y)) = L + M$$

Difference Law:

$$\lim_{(x,y)\to(a,b)} (f(x,y) - g(x,y)) = L - M$$

Constant Multiple Law:

$$\lim_{(x,y)\to(a,b)} (cf(x,y)) = cL$$

Product Law:

$$\lim_{(x,y)\to(a,b)} (f(x,y)g(x,y)) = LM$$

Quotient Law:

$$\lim_{(x,y)\to(a,b)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M} \text{ for } M \neq 0$$

Power Law:

$$\lim_{(x,y)\to(a,b)} (f(x,y))^n = L^n$$

for any positive integer n.

Root Law:

$$\lim_{(x,y)\to(a,b)} \sqrt[n]{f(x,y)} = \sqrt[n]{L}$$

for all L if n is odd and positive, and for $L \geq 0$ if n is even and positive.

The proofs of these properties are similar to those for the limits of functions of one variable. We can apply these laws to finding limits of various functions.

Example 14.2.1: Finding the Limit of a Function of Two Variables

Find each of the following limits:

a. $\lim\limits_{(x,y)\to(2,-1)} (x^2 - 2xy + 3y^2 - 4x + 3y - 6)$

b. $\lim\limits_{(x,y)\to(2,-1)} \dfrac{2x + 3y}{4x - 3y}$

Solution

a. First use the sum and difference laws to separate the terms:

$$\begin{aligned}\lim_{(x,y)\to(2,-1)} (x^2 - 2xy + 3y^2 - 4x + 3y - 6) \\ = \left(\lim_{(x,y)\to(2,-1)} x^2\right) - \left(\lim_{(x,y)\to(2,-1)} 2xy\right) + \left(\lim_{(x,y)\to(2,-1)} 3y^2\right) - \left(\lim_{(x,y)\to(2,-1)} 4x\right) \\ + \left(\lim_{(x,y)\to(2,-1)} 3y\right) - \left(\lim_{(x,y)\to(2,-1)} 6\right).\end{aligned}$$

Next, use the constant multiple law on the second, third, fourth, and fifth limits:

$$= (\lim_{(x,y)\to(2,-1)} x^2) - 2(\lim_{(x,y)\to(2,-1)} xy) + 3(\lim_{(x,y)\to(2,-1)} y^2) - 4(\lim_{(x,y)\to(2,-1)} x)$$
$$+3(\lim_{(x,y)\to(2,-1)} y) - \lim_{(x,y)\to(2,-1)} 6.$$

Now, use the power law on the first and third limits, and the product law on the second limit:

$$\left(\lim_{(x,y)\to(2,-1)} x\right)^2 - 2\left(\lim_{(x,y)\to(2,-1)} x\right)\left(\lim_{(x,y)\to(2,-1)} y\right) + 3\left(\lim_{(x,y)\to(2,-1)} y\right)^2$$
$$-4\left(\lim_{(x,y)\to(2,-1)} x\right) + 3\left(\lim_{(x,y)\to(2,-1)} y\right) - \lim_{(x,y)\to(2,-1)} 6.$$

Last, use the identity laws on the first six limits and the constant law on the last limit:

$$\lim_{(x,y)\to(2,-1)} (x^2 - 2xy + 3y^2 - 4x + 3y - 6) = (2)^2 - 2(2)(-1) + 3(-1)^2 - 4(2) + 3(-1) - 6$$
$$= -6.$$

b. Before applying the quotient law, we need to verify that the limit of the denominator is nonzero. Using the difference law, constant multiple law, and identity law,

$$\lim_{(x,y)\to(2,-1)} (4x - 3y) = \lim_{(x,y)\to(2,-1)} 4x - \lim_{(x,y)\to(2,-1)} 3y$$
$$= 4(\lim_{(x,y)\to(2,-1)} x) - 3(\lim_{(x,y)\to(2,-1)} y)$$
$$= 4(2) - 3(-1) = 11.$$

Since the limit of the denominator is nonzero, the quotient law applies. We now calculate the limit of the numerator using the difference law, constant multiple law, and identity law:

$$\lim_{(x,y)\to(2,-1)} (2x + 3y) = \lim_{(x,y)\to(2,-1)} 2x + \lim_{(x,y)\to(2,-1)} 3y$$
$$= 2(\lim_{(x,y)\to(2,-1)} x) + 3(\lim_{(x,y)\to(2,-1)} y)$$
$$= 2(2) + 3(-1) = 1.$$

Therefore, according to the quotient law we have

$$\lim_{(x,y)\to(2,-1)} \frac{2x + 3y}{4x - 3y} = \frac{\lim_{(x,y)\to(2,-1)} (2x + 3y)}{\lim_{(x,y)\to(2,-1)} (4x - 3y)}$$
$$= \frac{1}{11}.$$

? Exercise 14.2.1:

Evaluate the following limit:

$$\lim_{(x,y)\to(5,-2)} \sqrt[3]{\frac{x^2 - y}{y^2 + x - 1}}.$$

Hint

Use the limit laws.

Answer

$$\lim_{(x,y)\to(5,-2)} \sqrt[3]{\frac{x^2 - y}{y^2 + x - 1}} = \frac{3}{2}$$

Since we are taking the limit of a function of two variables, the point (a, b) is in $\mathbb{R}^2$, and it is possible to approach this point from an infinite number of directions. Sometimes when calculating a limit, the answer varies depending on the path taken toward (a, b). If

this is the case, then the limit fails to exist. In other words, the limit must be unique, regardless of path taken.

Example 14.2.2: Limits That Fail to Exist

Show that neither of the following limits exist:

a. $\lim\limits_{(x,y)\to(0,0)} \dfrac{2xy}{3x^2+y^2}$

b. $\lim\limits_{(x,y)\to(0,0)} \dfrac{4xy^2}{x^2+3y^4}$

Solution

a. The domain of the function $f(x,y)=\dfrac{2xy}{3x^2+y^2}$ consists of all points in the xy-plane except for the point $(0,0)$ (Figure 14.2.3). To show that the limit does not exist as (x,y) approaches $(0,0)$, we note that it is impossible to satisfy the definition of a limit of a function of two variables because of the fact that the function takes different values along different lines passing through point $(0,0)$. First, consider the line $y=0$ in the xy-plane. Substituting $y=0$ into $f(x,y)$ gives

$$f(x,0)=\frac{2x(0)}{3x^2+0^2}=0$$

for any value of x. Therefore the value of f remains constant for any point on the x-axis, and as y approaches zero, the function remains fixed at zero.

Next, consider the line $y=x$. Substituting $y=x$ into $f(x,y)$ gives

$$f(x,x)=\frac{2x(x)}{3x^2+x^2}=\frac{2x^2}{4x^2}=\tfrac{1}{2}.$$

This is true for any point on the line $y=x$. If we let x approach zero while staying on this line, the value of the function remains fixed at $\frac{1}{2}$, regardless of how small x is.

Choose a value for ε that is less than $1/2$—say, $1/4$. Then, no matter how small a δ disk we draw around $(0,0)$, the values of $f(x,y)$ for points inside that δ disk will include both 0 and $\frac{1}{2}$. Therefore, the definition of limit at a point is never satisfied and the limit fails to exist.

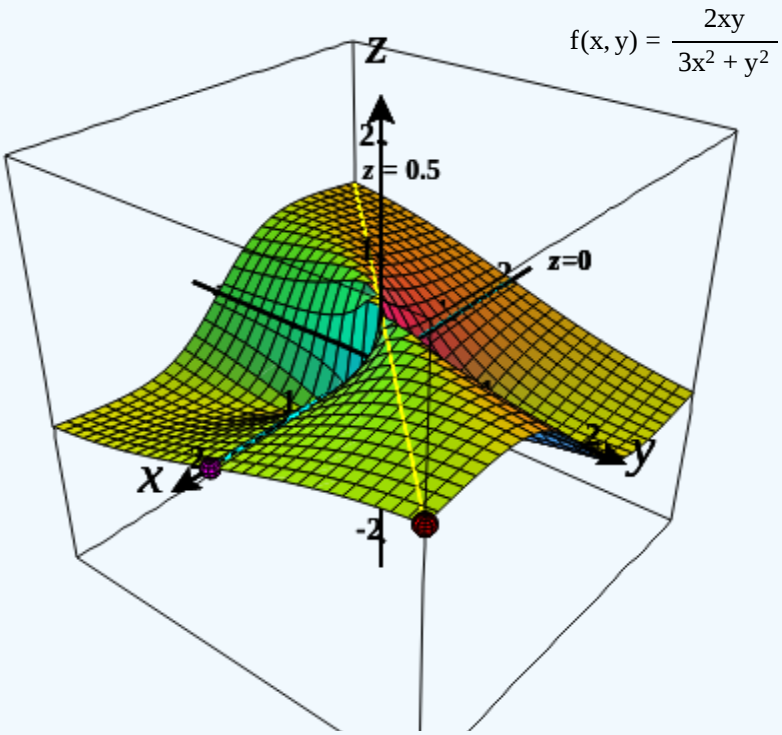


Figure 14.2.3: Graph of the function $f(x,y)=\dfrac{2xy}{3x^2+y^2}$. Along the line $y=0$, the function is equal to zero; along the line $y=x$, the function is equal to $\frac{1}{2}$.

b. In a similar fashion to a., we can approach the origin along any straight line passing through the origin. If we try the x-axis (i.e., $y=0$), then the function remains fixed at zero. The same is true for the y-axis. Suppose we approach the origin along a straight line of slope k. The equation of this line is $y=kx$. Then the limit becomes

$$\begin{aligned}\lim_{(x,y)\to(0,0)}\frac{4xy^2}{x^2+3y^4} &= \lim_{(x,y)\to(0,0)}\frac{4x(kx)^2}{x^2+3(kx)^4}\\ &= \lim_{(x,y)\to(0,0)}\frac{4k^2x^3}{x^2+3k^4x^4}\\ &= \lim_{(x,y)\to(0,0)}\frac{4k^2x}{1+3k^4x^2}\\ &= \frac{\lim\limits_{(x,y)\to(0,0)}(4k^2x)}{\lim\limits_{(x,y)\to(0,0)}(1+3k^4x^2)}\\ &= 0.\end{aligned}$$

regardless of the value of k. It would seem that the limit is equal to zero. What if we chose a curve passing through the origin instead? For example, we can consider the parabola given by the equation $x=y^2$. Substituting y^2 in place of x in $f(x,y)$ gives

$$\begin{aligned}\lim_{(x,y)\to(0,0)}\frac{4xy^2}{x^2+3y^4} &= \lim_{(x,y)\to(0,0)}\frac{4(y^2)y^2}{(y^2)^2+3y^4}\\ &= \lim_{(x,y)\to(0,0)}\frac{4y^4}{y^4+3y^4}\\ &= \lim_{(x,y)\to(0,0)}1\\ &= 1.\end{aligned}$$

By the same logic in part a, it is impossible to find a δ disk around the origin that satisfies the definition of the limit for any value of $\varepsilon<1$. Therefore,

$$\lim_{(x,y)\to(0,0)} \frac{4xy^2}{x^2+3y^4}$$

does **not** exist.

Exercise 14.2.2:

Show that

$$\lim_{(x,y)\to(2,1)} \frac{(x-2)(y-1)}{(x-2)^2+(y-1)^2}$$

does not exist.

Hint

Pick a line with slope k passing through point $(2,1)$.

Answer

If $y=k(x-2)+1$, then $\lim_{(x,y)\to(2,1)} \frac{(x-2)(y-1)}{(x-2)^2+(y-1)^2} = \frac{k}{1+k^2}$. Since the answer depends on k, the limit fails to exist.

Interior Points and Boundary Points

To study continuity and differentiability of a function of two or more variables, we first need to learn some new terminology.

Definition: interior and boundary points

Let S be a subset of $\mathbb{R}^2$ (Figure 14.2.4).

1. A point P_0 is called an *interior point* of S if there is a δ disk centered around P_0 contained completely in S.
2. A point P_0 is called a *boundary point* of S if every δ disk centered around P_0 contains points both inside and outside S.

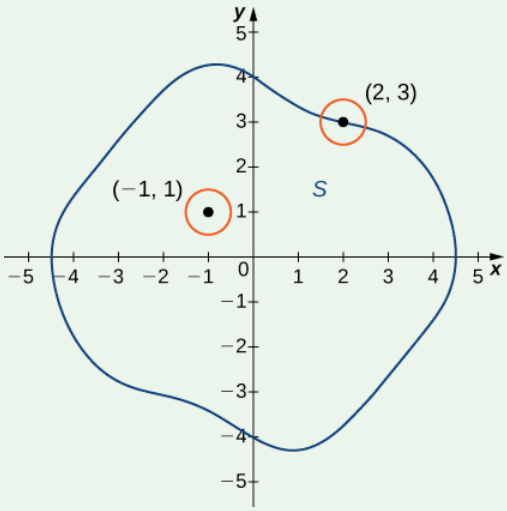


Figure 14.2.4: In the set S shown, $(-1,1)$ is an interior point and $(2,3)$ is a boundary point.

Definition: Open and closed sets

Let S be a subset of $\mathbb{R}^2$ (Figure 14.2.4).

1. S is called an *open set* if every point of S is an interior point.
2. S is called a *closed set* if it contains all its boundary points.

An example of an open set is a δ disk. If we include the boundary of the disk, then it becomes a closed set. A set that contains some, but not all, of its boundary points is neither open nor closed. For example if we include half the boundary of a δ disk but not the other half, then the set is neither open nor closed.

Definition: connected sets and Regions

Let S be a subset of $\mathbb{R}^2$ (Figure 14.2.4).

1. An open set S is a *connected set* if it cannot be represented as the union of two or more disjoint, nonempty open subsets.
2. A set S is a *region* if it is open, connected, and nonempty.

The definition of a limit of a function of two variables requires the δ disk to be contained inside the domain of the function. However, if we wish to find the limit of a function at a boundary point of the domain, the δ **disk** is not contained inside the domain. By definition, some of the points of the δ **disk** are inside the domain and some are outside. Therefore, we need only consider points that are inside both the δ disk and the domain of the function. This leads to the definition of the limit of a function at a boundary point.

Definition

Let f be a function of two variables, x and y, and suppose (a,b) is on the boundary of the domain of f. Then, the limit of $f(x,y)$ as (x,y) approaches (a,b) is L, written

$$\lim_{(x,y)\to(a,b)} f(x,y) = L,$$

if for any $\varepsilon > 0$, there exists a number $\delta > 0$ such that for any point (x,y) inside the domain of f and within a suitably small distance positive δ of (a,b), the value of $f(x,y)$ is no more than ε away from L (Figure 14.2.2). Using symbols, we can write: For any $\varepsilon > 0$, there exists a number $\delta > 0$ such that

$$|f(x,y) - L| < \varepsilon \text{ whenever } 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

Example 14.2.3: Limit of a Function at a Boundary Point

Prove

$$\lim_{(x,y)\to(4,3)} \sqrt{25 - x^2 - y^2} = 0.$$

Solution

The domain of the function $f(x,y) = \sqrt{25 - x^2 - y^2}$ is $\{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 25\}$, which is a circle of radius 5 centered at the origin, along with its interior as shown in Figure 14.2.5.

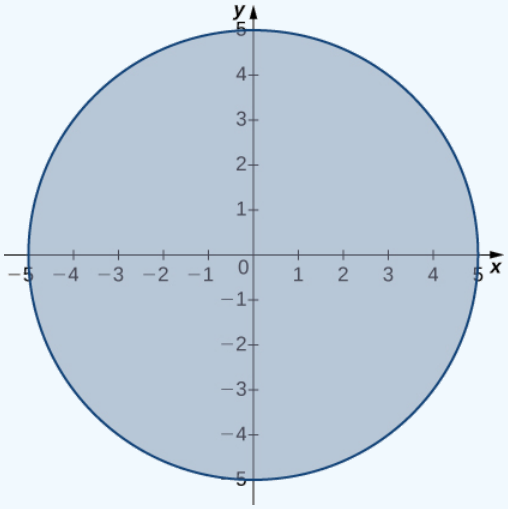


Figure 14.2.5: Domain of the function $f(x,y)=\sqrt{25-x^2-y^2}$.

We can use the limit laws, which apply to limits at the boundary of domains as well as interior points:

$$\begin{aligned}\lim_{(x,y)\to(4,3)}\sqrt{25-x^2-y^2} &= \sqrt{\lim_{(x,y)\to(4,3)}(25-x^2-y^2)}\\ &= \sqrt{\lim_{(x,y)\to(4,3)}25-\lim_{(x,y)\to(4,3)}x^2-\lim_{(x,y)\to(4,3)}y^2}\\ &= \sqrt{25-4^2-3^2}\\ &= 0\end{aligned}$$

See the following graph.

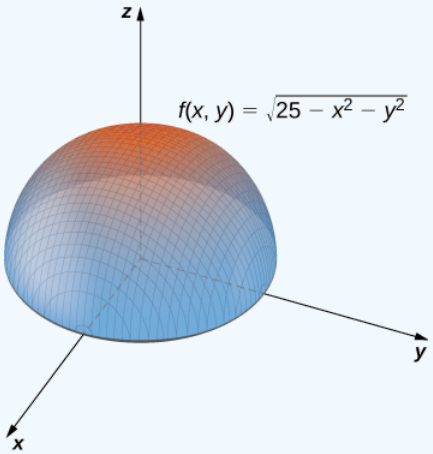


Figure 14.2.6: Graph of the function $f(x,y)=\sqrt{25-x^2-y^2}$.

? Exercise 14.2.3

Evaluate the following limit:

$$\lim_{(x,y)\to(5,-2)}\sqrt{29-x^2-y^2}.$$

Hint

Determine the domain of $f(x,y)=\sqrt{29-x^2-y^2}$.

Answer

$$\lim_{(x,y)\to(5,-2)} \sqrt{29-x^2-y^2} = \sqrt{29-5^2-(-2)^2} = \sqrt{29-25-4} = \sqrt{0} = 0$$

Continuity of Functions of Two Variables

In Continuity, we defined the continuity of a function of one variable and saw how it relied on the limit of a function of one variable. In particular, three conditions are necessary for $f(x)$ to be continuous at point $x = a$

1. $f(a)$ exists.
2. $\lim_{x\to a} f(x)$ exists.
3. $\lim_{x\to a} f(x) = f(a)$.

These three conditions are necessary for continuity of a function of two variables as well.

Definition: continuous Functions

A function $f(x, y)$ is continuous at a point (a, b) in its domain if the following conditions are satisfied:

1. $f(a, b)$ exists.
2. $\lim_{(x,y)\to(a,b)} f(x, y)$ exists.
3. $\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$.

Example 14.2.4: Demonstrating Continuity for a Function of Two Variables

Show that the function

$$f(x, y) = \frac{3x + 2y}{x + y + 1}$$

is continuous at point $(5, -3)$.

Solution

There are three conditions to be satisfied, per the definition of continuity. In this example, $a = 5$ and $b = -3$.

1. $f(a, b)$ exists. This is true because the domain of the function f consists of those ordered pairs for which the denominator is nonzero (i.e., $x + y + 1 \neq 0$). Point $(5, -3)$ satisfies this condition. Furthermore,

$$f(a, b) = f(5, -3) = \frac{3(5) + 2(-3)}{5 + (-3) + 1} = \frac{15 - 6}{2 + 1} = 3.$$

2. $\lim_{(x,y)\to(a,b)} f(x, y)$ exists. This is also true:

$$\begin{aligned}\lim_{(x,y)\to(a,b)} f(x, y) &= \lim_{(x,y)\to(5,-3)} \frac{3x + 2y}{x + y + 1} \\ &= \frac{\lim_{(x,y)\to(5,-3)} (3x + 2y)}{\lim_{(x,y)\to(5,-3)} (x + y + 1)} \\ &= \frac{15 - 6}{5 - 3 + 1} \\ &= 3.\end{aligned}$$

3. $\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$. This is true because we have just shown that both sides of this equation equal three.

Exercise 14.2.4

Show that the function

$$f(x, y) = \sqrt{26 - 2x^2 - y^2}$$

is continuous at point $(2, -3)$.

Hint

Use the three-part definition of continuity.

Answer

1. The domain of f contains the ordered pair $(2, -3)$ because $f(a, b) = f(2, -3) = \sqrt{16 - 2(2)^2 - (-3)^2} = 3$
2. $\lim\limits_{(x,y)\to(a,b)} f(x, y) = 3$
3. $\lim\limits_{(x,y)\to(a,b)} f(x, y) = f(a, b) = 3$

Continuity of a function of any number of variables can also be defined in terms of delta and epsilon. A function of two variables is continuous at a point (x_0, y_0) in its domain if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that, whenever $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$ it is true, $|f(x, y) - f(a, b)| < \varepsilon$. This definition can be combined with the formal definition (that is, the *epsilon–delta definition*) of continuity of a function of one variable to prove the following theorems:

Theorem 14.2.2: The Sum of Continuous Functions is Continuous

If $f(x, y)$ is continuous at (x_0, y_0), and $g(x, y)$ is continuous at (x_0, y_0), then $f(x, y) + g(x, y)$ is continuous at (x_0, y_0).

Theorem 14.2.3: The Product of Continuous Functions is Continuous

If $g(x)$ is continuous at x_0 and $h(y)$ is continuous at y_0, then $f(x, y) = g(x)h(y)$ is continuous at (x_0, y_0).

Theorem 14.2.4: The Composition of Continuous Functions is Continuous

Let g be a function of two variables from a domain $D \subseteq \mathbb{R}^2$ to a range $R \subseteq R$. Suppose g is continuous at some point $(x_0, y_0) \in D$ and define $z_0 = g(x_0, y_0)$. Let f be a function that maps R to R such that z_0 is in the domain of f. Last, assume f is continuous at z_0. Then $f \circ g$ is continuous at (x_0, y_0) as shown in Figure 14.2.7.

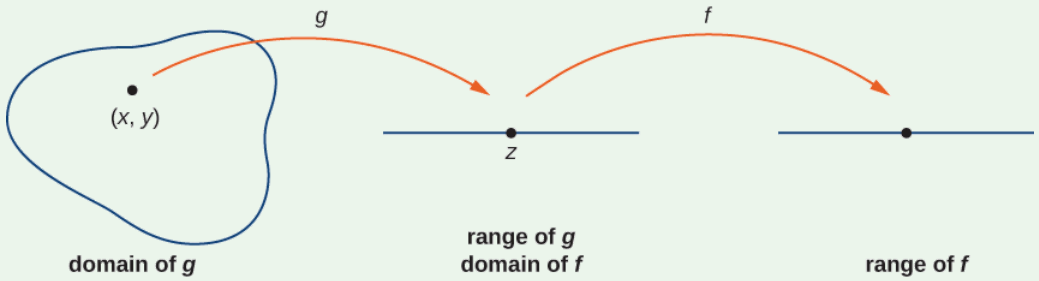


Figure 14.2.7: The composition of two continuous functions is continuous.

Let's now use the previous theorems to show continuity of functions in the following examples.

Example 14.2.5: More Examples of Continuity of a Function of Two Variables

Show that the functions $f(x, y) = 4x^3y^2$ and $g(x, y) = \cos(4x^3y^2)$ are continuous everywhere.

Solution

The polynomials $g(x) = 4x^3$ and $h(y) = y^2$ are continuous at every real number, and therefore by the product of continuous functions theorem, $f(x, y) = 4x^3y^2$ is continuous at every point (x, y) in the xy-plane. Since $f(x, y) = 4x^3y^2$ is continuous at every point (x, y) in the xy-plane and $g(x) = \cos x$ is continuous at every real number x, the continuity of the composition of functions tells us that $g(x, y) = \cos(4x^3y^2)$ is continuous at every point (x, y) in the xy-plane.

Exercise 14.2.5

Show that the functions $f(x,y) = 2x^2y^3 + 3$ and $g(x,y) = (2x^2y^3 + 3)^4$ are continuous everywhere.

Hint

Use the continuity of the sum, product, and composition of two functions.

Answer

The polynomials $g(x) = 2x^2$ and $h(y) = y^3$ are continuous at every real number; therefore, by the product of continuous functions theorem, $f(x,y) = 2x^2y^3$ is continuous at every point (x,y) in the xy-plane. Furthermore, any constant function is continuous everywhere, so $g(x,y) = 3$ is continuous at every point (x,y) in the xy-plane. Therefore, $f(x,y) = 2x^2y^3 + 3$ is continuous at every point (x,y) in the xy-plane. Last, $h(x) = x^4$ is continuous at every real number x, so by the continuity of composite functions theorem $g(x,y) = (2x^2y^3 + 3)^4$ is continuous at every point (x,y) in the xy-plane.

Functions of Three or More Variables

The limit of a function of three or more variables occurs readily in applications. For example, suppose we have a function $f(x,y,z)$ that gives the temperature at a physical location (x,y,z) in three dimensions. Or perhaps a function $g(x,y,z,t)$ can indicate air pressure at a location (x,y,z) at time t. How can we take a limit at a point in $\mathbb{R}^3$? What does it mean to be continuous at a point in four dimensions?

The answers to these questions rely on extending the concept of a δ disk into more than two dimensions. Then, the ideas of the limit of a function of three or more variables and the continuity of a function of three or more variables are very similar to the definitions given earlier for a function of two variables.

Definition: δ-balls

Let (x_0,y_0,z_0) be a point in $\mathbb{R}^3$. Then, a δ-ball in three dimensions consists of all points in $\mathbb{R}^3$ lying at a distance of less than δ from (x_0,y_0,z_0) —that is,

$$\{(x,y,z) \in \mathbb{R}^3 \mid \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2} < \delta\}.$$

To define a δ-ball in higher dimensions, add additional terms under the radical to correspond to each additional dimension. For example, given a point $P = (w_0,x_0,y_0,z_0)$ in $\mathbb{R}^4$, a δ ball around P can be described by

$$\{(w,x,y,z) \in \mathbb{R}^4 \mid \sqrt{(w-w_0)^2 + (x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2} < \delta\}.$$

To show that a limit of a function of three variables exists at a point (x_0,y_0,z_0), it suffices to show that for any point in a δ ball centered at (x_0,y_0,z_0), the value of the function at that point is arbitrarily close to a fixed value (the limit value). All the limit laws for functions of two variables hold for functions of more than two variables as well.

Example 14.2.6: Finding the Limit of a Function of Three Variables

Find

$$\lim_{(x,y,z)\to(4,1,-3)} \frac{x^2y - 3z}{2x + 5y - z}.$$

Solution

Before we can apply the quotient law, we need to verify that the limit of the denominator is nonzero. Using the difference law, the identity law, and the constant law,

$$\begin{aligned}\lim_{(x,y,z)\to(4,1,-3)} (2x + 5y - z) &= 2\left(\lim_{(x,y,z)\to(4,1,-3)} x\right) + 5\left(\lim_{(x,y,z)\to(4,1,-3)} y\right) - \left(\lim_{(x,y,z)\to(4,1,-3)} z\right) \\ &= 2(4) + 5(1) - (-3) \\ &= 16.\end{aligned}$$

Since this is nonzero, we next find the limit of the numerator. Using the product law, power law, difference law, constant multiple law, and identity law,

$$\begin{aligned}\lim_{(x,y,z)\to(4,1,-3)}(x^2y-3z) &= (\lim_{(x,y,z)\to(4,1,-3)}x)^2(\lim_{(x,y,z)\to(4,1,-3)}y)-3\lim_{(x,y,z)\to(4,1,-3)}z\\ &= (4^2)(1)-3(-3)\\ &= 16+9\\ &= 25\end{aligned}$$

Last, applying the quotient law:

$$\lim_{(x,y,z)\to(4,1,-3)}\frac{x^2y-3z}{2x+5y-z}=\frac{\lim\limits_{(x,y,z)\to(4,1,-3)}(x^2y-3z)}{\lim\limits_{(x,y,z)\to(4,1,-3)}(2x+5y-z)}=\frac{25}{16}$$

? Exercise 14.2.6

Find

$$\lim_{(x,y,z)\to(4,-1,3)}\sqrt{13-x^2-2y^2+z^2}$$

Hint

Use the limit laws and the continuity of the composition of functions.

Answer

$$\lim_{(x,y,z)\to(4,-1,3)}\sqrt{13-x^2-2y^2+z^2}=2$$

Key Concepts

- To study limits and continuity for functions of two variables, we use a δ disk centered around a given point.
- A function of several variables has a limit if for any point in a δ ball centered at a point P, the value of the function at that point is arbitrarily close to a fixed value (the limit value).
- The limit laws established for a function of one variable have natural extensions to functions of more than one variable.
- A function of two variables is continuous at a point if the limit exists at that point, the function exists at that point, and the limit and function are equal at that point.

Glossary

boundary point

a point P_0 of R is a boundary point if every δ disk centered around P_0 contains points both inside and outside R

closed set

a set S that contains all its boundary points

connected set

an open set S that cannot be represented as the union of two or more disjoint, nonempty open subsets

δ disk

an open disk of radius δ centered at point (a,b)

δ ball

all points in $\mathbb{R}^3$ lying at a distance of less than δ from (x_0,y_0,z_0)

interior point

a point P_0 of $\mathbb{R}$ is a boundary point if there is a δ disk centered around P_0 contained completely in $\mathbb{R}$

open set

a set S that contains none of its boundary points

region

an open, connected, nonempty subset of $\mathbb{R}^2$

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14.2E: EXERCISES FOR SECTION 14.2

1) Use the limit laws for functions of two variables to evaluate each limit below, given that $\lim_{(x,y)\to(a,b)} f(x,y)=5$ and $\lim_{(x,y)\to(a,b)} g(x,y)=2$.

a. $\lim_{(x,y)\to(a,b)} [f(x,y)+g(x,y)]$

b. $\lim_{(x,y)\to(a,b)} [f(x,y)g(x,y)]$

c. $\lim_{(x,y)\to(a,b)} \left[\dfrac{7f(x,y)}{g(x,y)}\right]$

d. $\lim_{(x,y)\to(a,b)} \left[\dfrac{2f(x,y)-4g(x,y)}{f(x,y)-g(x,y)}\right]$

Answer

a. $\lim_{(x,y)\to(a,b)} [f(x,y)+g(x,y)] = \lim_{(x,y)\to(a,b)} f(x,y) + \lim_{(x,y)\to(a,b)} g(x,y) = 5+2=7$

b. $\lim_{(x,y)\to(a,b)} [f(x,y)g(x,y)] = \left(\lim_{(x,y)\to(a,b)} f(x,y)\right)\left(\lim_{(x,y)\to(a,b)} g(x,y)\right) = 5(2) = 10$

c. $\lim_{(x,y)\to(a,b)} \left[\dfrac{7f(x,y)}{g(x,y)}\right] = \dfrac{7\left(\lim_{(x,y)\to(a,b)} f(x,y)\right)}{\lim_{(x,y)\to(a,b)} g(x,y)} = \dfrac{7(5)}{2} = 17.5$

d. $\lim_{(x,y)\to(a,b)} \left[\dfrac{2f(x,y)-4g(x,y)}{f(x,y)-g(x,y)}\right] = \dfrac{2\left(\lim_{(x,y)\to(a,b)} f(x,y)\right)-4\left(\lim_{(x,y)\to(a,b)} g(x,y)\right)}{\lim_{(x,y)\to(a,b)} f(x,y)-\lim_{(x,y)\to(a,b)} g(x,y)} = \dfrac{2(5)-4(2)}{5-2} = \dfrac{2}{3}$

In exercises 2 - 4, find the limit of the function.

2) $\lim_{(x,y)\to(1,2)} x$

3) $\lim_{(x,y)\to(1,2)} \dfrac{5x^2y}{x^2+y^2}$

Answer

$\lim_{(x,y)\to(1,2)} \dfrac{5x^2y}{x^2+y^2} = 2$

4) Show that the limit $\lim_{(x,y)\to(0,0)} \dfrac{5x^2y}{x^2+y^2}$ exists and is the same along the paths: y-axis and x-axis, and along $y=x$.

In exercises 5 - 19, evaluate the limits at the indicated values of x and y. If the limit does not exist, state this and explain why the limit does not exist.

5) $\lim_{(x,y)\to(0,0)} \dfrac{4x^2+10y^2+4}{4x^2-10y^2+6}$

Answer

$\lim_{(x,y)\to(0,0)} \dfrac{4x^2+10y^2+4}{4x^2-10y^2+6} = \dfrac{2}{3}$

6) $\lim_{(x,y)\to(11,13)} \sqrt{\dfrac{1}{xy}}$

7) $\lim_{(x,y)\to(0,1)} \dfrac{y^2\sin x}{x}$

Answer

$\lim_{(x,y)\to(0,1)} \dfrac{y^2\sin x}{x} = 1$

8) $\lim_{(x,y)\to(0,0)} \sin\left(\dfrac{x^8+y^7}{x-y+10}\right)$

9) $\lim_{(x,y)\to(\pi/4,1)} \dfrac{y\tan x}{y+1}$

Answer

$\lim_{(x,y)\to(\pi/4,1)} \dfrac{y\tan x}{y+1} = \dfrac{1}{2}$

10) $\lim_{(x,y)\to(0,\pi/4)} \dfrac{\sec x + 2}{3x - \tan y}$

11) $\lim_{(x,y)\to(2,5)} \left(\dfrac{1}{x} - \dfrac{5}{y}\right)$

Answer

$$\lim_{(x,y)\to(2,5)} \left(\frac{1}{x} - \frac{5}{y}\right) = -\frac{1}{2}$$

12) $\lim_{(x,y)\to(4,4)} x \ln y$

13) $\lim_{(x,y)\to(4,4)} e^{-x^2-y^2}$

Answer

$$\lim_{(x,y)\to(4,4)} e^{-x^2-y^2} = e^{-32}$$

14) $\lim_{(x,y)\to(0,0)} \sqrt{9 - x^2 - y^2}$

15) $\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y)$

Answer

$$\lim_{(x,y)\to(1,2)} (x^2y^3 - x^3y^2 + 3x + 2y) = 11$$

16) $\lim_{(x,y)\to(\pi,\pi)} x \sin\left(\dfrac{x+y}{4}\right)$

17) $\lim_{(x,y)\to(0,0)} \dfrac{xy+1}{x^2+y^2+1}$

Answer

$$\lim_{(x,y)\to(0,0)} \frac{xy+1}{x^2+y^2+1} = 1$$

18) $\lim_{(x,y)\to(0,0)} \dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}-1}$

19) $\lim_{(x,y)\to(0,0)} \ln(x^2+y^2)$

Answer

The limit does not exist because when x and y both approach zero, the function approaches $\ln 0$, which is undefined (approaches negative infinity).

In exercises 20 - 21, complete the statement.

20) A point (x_0, y_0) in a plane region R is an interior point of R if _____.

21) A point (x_0, y_0) in a plane region R is called a boundary point of R if _____.

Answer

Every open disk centered at (x_0, y_0) contains points inside R and outside R.

In exercises 22 - 25, use algebraic techniques to evaluate the limit.

22) $\lim_{(x,y)\to(2,1)} \dfrac{x-y-1}{\sqrt{x-y}-1}$

23) $\lim_{(x,y)\to(0,0)} \dfrac{x^4-4y^4}{x^2+2y^2}$

Answer

$$\lim_{(x,y)\to(0,0)} \frac{x^4-4y^4}{x^2+2y^2} = 0$$

24) $\lim_{(x,y)\to(0,0)} \dfrac{x^3-y^3}{x-y}$

25) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2-xy}{\sqrt{x}-\sqrt{y}}$

Answer

$$\lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}} = 0$$

In exercises 26 - 27, evaluate the limits of the functions of three variables.

26) $\lim\limits_{(x,y,z)\to(1,2,3)} \dfrac{xz^2-y^2z}{xyz-1}$

27) $\lim\limits_{(x,y,z)\to(0,0,0)} \dfrac{x^2-y^2-z^2}{x^2+y^2-z^2}$

Answer

The limit does not exist.

In exercises 28 - 31, evaluate the limit of the function by determining the value the function approaches along the indicated paths. If the limit does not exist, explain why not.

28) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy+y^3}{x^2+y^2}$

a. Along the x-axis $(y=0)$

b. Along the y-axis $(x=0)$

c. Along the path $y=2x$

29) Evaluate $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy+y^3}{x^2+y^2}$ using the results of previous problem.

Answer

The limit does not exist. The function approaches two different values along different paths.

30) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2y}{x^4+y^2}$

a. Along the x-axis $(y=0)$

b. Along the y-axis $(x=0)$

c. Along the path $y=x^2$

31) Evaluate $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2y}{x^4+y^2}$ using the results of previous problem.

Answer

The limit does not exist because the function approaches two different values along the paths.

In exercises 32 - 35, discuss the continuity of each function. Find the largest region in the xy-plane in which each function is continuous.

32) $f(x,y)=\sin(xy)$

33) $f(x,y)=\ln(x+y)$

Answer

The function f is continuous in the region $y>-x$.

34) $f(x,y)=e^{3xy}$

35) $f(x,y)=\dfrac{1}{xy}$

Answer

The function f is continuous at all points in the xy-plane except at points on the x- and y-axes.

In exercises 36 - 38, determine the region in which the function is continuous. Explain your answer.

36) $f(x,y)=\dfrac{x^2y}{x^2+y^2}$

37) $f(x,y)=\begin{cases}\dfrac{x^2y}{x^2+y^2} & if (x,y)\neq(0,0)\\ 0 & if (x,y)=(0,0)\end{cases}$

Hint:

Show that the function approaches different values along two different paths.

Answer

The function is continuous at $(0,0)$ since the limit of the function at $(0,0)$ is 0, the same value of $f(0,0)$.

38) $f(x,y)=\dfrac{\sin(x^2+y^2)}{x^2+y^2}$

39) Determine whether $g(x,y)=\dfrac{x^2-y^2}{x^2+y^2}$ is continuous at $(0,0)$.

Answer

The function is discontinuous at $(0,0)$. The limit at $(0,0)$ fails to exist and $g(0,0)$ does not exist.

40) Create a plot using graphing software to determine where the limit does not exist. Determine the region of the coordinate plane in which $f(x,y)=\dfrac{1}{x^2-y}$ is continuous.

41) Determine the region of the xy-plane in which the composite function $g(x,y)=\arctan(\frac{xy^2}{x+y})$ is continuous. Use technology to support your conclusion.

Answer

Since the function $\arctan x$ is continuous over $(-\infty,\infty)$, $g(x,y)=\arctan(\frac{xy^2}{x+y})$ is continuous where $z=\dfrac{xy^2}{x+y}$ is continuous. The inner function z is continuous on all points of the xy-plane except where $y=-x$. Thus, $g(x,y)=\arctan(\frac{xy^2}{x+y})$ is continuous on all points of the coordinate plane except at points at which $y=-x$.

42) Determine the region of the xy-plane in which $f(x,y)=\ln(x^2+y^2-1)$ is continuous. Use technology to support your conclusion. (Hint: Choose the range of values for x and y carefully!)

43) At what points in space is $g(x,y,z)=x^2+y^2-2z^2$ continuous?

Answer

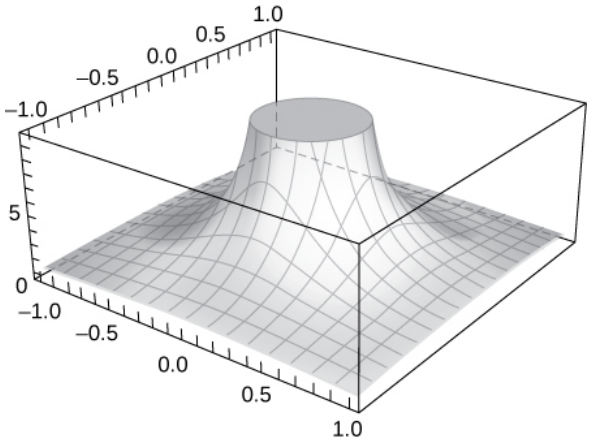
All points $P(x,y,z)$ in space

44) At what points in space is $g(x,y,z)=\dfrac{1}{x^2+z^2-1}$ continuous?

45) Show that $\lim\limits_{(x,y)\to(0,0)}\dfrac{1}{x^2+y^2}$ does not exist at $(0,0)$ by plotting the graph of the function.

Answer

The graph increases without bound as x and y both approach zero.



46) [T] Evaluate $\lim_{(x,y)\to(0,0)} \frac{-xy^2}{x^2+y^4}$ by plotting the function using a CAS. Determine analytically the limit along the path $x = y^2$.

47) [T]

a. Use a CAS to draw a contour map of $z = \sqrt{9-x^2-y^2}$.

b. What is the name of the geometric shape of the level curves?

c. Give the general equation of the level curves.

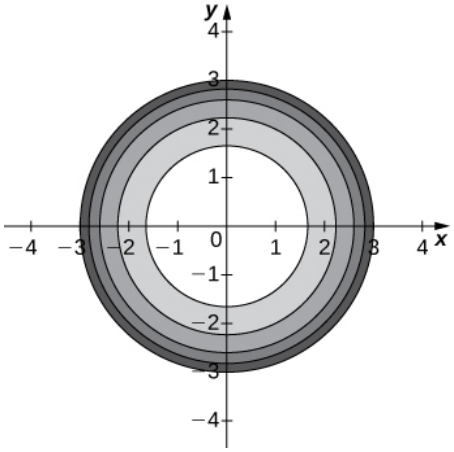
d. What is the maximum value of z?

e. What is the domain of the function?

f. What is the range of the function?

Answer

a.



b. The level curves are circles centered at $(0,0)$ with radius $9-c$.

c. $x^2+y^2 = 9-c$

d. $z = 3$

e. $\{(x,y) \in R^2 \mid x^2+y^2 \le 9\}$

f. $\{z | 0 \le z \le 3\}$

48) True or False: If we evaluate $\lim_{(x,y)\to(0,0)} f(x)$ along several paths and each time the limit is 1, we can conclude that $\lim_{(x,y)\to(0,0)} f(x) = 1$.

49) Use polar coordinates to find $\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}}$. You can also find the limit using L'Hôpital's rule.

Answer

$$\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} = 1$$

50) Use polar coordinates to find $\lim_{(x,y)\to(0,0)} \cos(x^2+y^2)$.

51) Discuss the continuity of $f(g(x,y))$ where $f(t) = 1/t$ and $g(x,y) = 2x - 5y$.

Answer

$f(g(x,y))$ is continuous at all points (x,y) that are not on the line $2x - 5y = 0$.

52) Given $f(x,y) = x^2 - 4y$, find $\lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}$.

53) Given $f(x,y) = x^2 - 4y$, find $\lim_{h\to 0} \frac{f(1+h,y) - f(1,y)}{h}$.

Answer

$$\lim_{h\to 0} \frac{f(1+h,y) - f(1,y)}{h} = 2$$

CONTRIBUTORS

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14.3: Partial Derivatives

Learning Objectives

- Calculate the partial derivatives of a function of two variables.
- Calculate the partial derivatives of a function of more than two variables.
- Determine the higher-order derivatives of a function of two variables.
- Explain the meaning of a partial differential equation and give an example.

Now that we have examined limits and continuity of functions of two variables, we can proceed to study derivatives. Finding derivatives of functions of two variables is the key concept in this chapter, with as many applications in mathematics, science, and engineering as differentiation of single-variable functions. However, we have already seen that limits and continuity of multivariable functions have new issues and require new terminology and ideas to deal with them. This carries over into differentiation as well.

Derivatives of a Function of Two Variables

When studying derivatives of functions of one variable, we found that one interpretation of the derivative is an instantaneous rate of change of y as a function of x. Leibniz notation for the derivative is dy/dx, which implies that y is the dependent variable and x is the independent variable. For a function $z = f(x, y)$ of two variables, x and y are the independent variables and z is the dependent variable. This raises two questions right away: How do we adapt Leibniz notation for functions of two variables? Also, what is an interpretation of the derivative? The answer lies in partial derivatives.

Definition: Partial Derivatives

Let $f(x, y)$ be a function of two variables. Then the *partial derivative* of f with respect to x, written as $\partial f/\partial x$, or f_x, is defined as

$$\frac{\partial f}{\partial x} = f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h} \tag{14.3.1}$$

The partial derivative of f with respect to y, written as $\partial f/\partial y$, or f_y, is defined as

$$\frac{\partial f}{\partial y} = f_y(x, y) = \lim_{k \to 0} \frac{f(x, y + k) - f(x, y)}{k}. \tag{14.3.2}$$

This definition shows two differences already. First, the notation changes, in the sense that we still use a version of Leibniz notation, but the d in the original notation is replaced with the symbol ∂. (This rounded "d" is usually called "partial," so $\partial f/\partial x$ is spoken as the "partial of f with respect to x.") This is the first hint that we are dealing with partial derivatives. Second, we now have two different derivatives we can take, since there are two different independent variables. Depending on which variable we choose, we can come up with different partial derivatives altogether, and often do.

Example 14.3.1: Calculating Partial Derivatives from the Definition

Use the definition of the partial derivative as a limit to calculate $\partial f/\partial x$ and $\partial f/\partial y$ for the function

$$f(x, y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12.$$

Solution

First, calculate $f(x + h, y)$.

$$\begin{aligned} f(x + h, y) &= (x + h)^2 - 3(x + h)y + 2y^2 - 4(x + h) + 5y - 12 \\ &= x^2 + 2xh + h^2 - 3xy - 3hy + 2y^2 - 4x - 4h + 5y - 12. \end{aligned}$$

Next, substitute this into Equation 14.3.1 and simplify:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}\\ &= \lim_{h\to 0}\frac{(x^2+2xh+h^2-3xy-3hy+2y^2-4x-4h+5y-12)-(x^2-3xy+2y^2-4x+5y-12)}{h}\\ &= \lim_{h\to 0}\frac{x^2+2xh+h^2-3xy-3hy+2y^2-4x-4h+5y-12-x^2+3xy-2y^2+4x-5y+12}{h}\\ &= \lim_{h\to 0}\frac{2xh+h^2-3hy-4h}{h}\\ &= \lim_{h\to 0}\frac{h(2x+h-3y-4)}{h}\\ &= \lim_{h\to 0}(2x+h-3y-4)\\ &= 2x-3y-4.\end{aligned}$$

To calculate $\frac{\partial f}{\partial y}$, first calculate $f(x,y+h)$:

$$\begin{aligned}f(x,y+h) &= x^2-3x(y+h)+2(y+h)^2-4x+5(y+h)-12\\ &= x^2-3xy-3xh+2y^2+4yh+2h^2-4x+5y+5h-12.\end{aligned}$$

Next, substitute this into Equation 14.3.2 and simplify:

$$\begin{aligned}\frac{\partial f}{\partial y} &= \lim_{h\to 0}\frac{f(x,y+h)-f(x,y)}{h}\\ &= \lim_{h\to 0}\frac{(x^2-3xy-3xh+2y^2+4yh+2h^2-4x+5y+5h-12)-(x^2-3xy+2y^2-4x+5y-12)}{h}\\ &= \lim_{h\to 0}\frac{x^2-3xy-3xh+2y^2+4yh+2h^2-4x+5y+5h-12-x^2+3xy-2y^2+4x-5y+12}{h}\\ &= \lim_{h\to 0}\frac{-3xh+4yh+2h^2+5h}{h}\\ &= \lim_{h\to 0}\frac{h(-3x+4y+2h+5)}{h}\\ &= \lim_{h\to 0}(-3x+4y+2h+5)\\ &= -3x+4y+5\end{aligned}$$

? Exercise 14.3.1

Use the definition of the partial derivative as a limit to calculate $\partial f/\partial x$ and $\partial f/\partial y$ for the function

$$f(x,y)=4x^2+2xy-y^2+3x-2y+5.$$

Hint

Use Equations 14.3.1 and 14.3.2 from the definition of partial derivatives.

Answer

$$\frac{\partial f}{\partial x}=8x+2y+3$$
$$\frac{\partial f}{\partial y}=2x-2y-2$$

The idea to keep in mind when calculating partial derivatives is to treat all independent variables, other than the variable with respect to which we are differentiating, as constants. Then proceed to differentiate as with a function of a single variable. To see why this is true, first fix y and define $g(x)=f(x,y)$ as a function of x. Then

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$
$$= \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$
$$= \frac{\partial f}{\partial x}.$$

The same is true for calculating the partial derivative of f with respect to y. This time, fix x and define $h(y) = f(x, y)$ as a function of y. Then

$$h'(x) = \lim_{k \to 0} \frac{h(x+k) - h(x)}{k}$$
$$= \lim_{k \to 0} \frac{f(x, y+k) - f(x, y)}{k}$$
$$= \frac{\partial f}{\partial y}.$$

All differentiation rules apply.

Example 14.3.2: Calculating Partial Derivatives

Calculate $\partial f/\partial x$ and $\partial f/\partial y$ for the following functions by holding the opposite variable constant then differentiating:

a. $f(x, y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12$
b. $g(x, y) = \sin(x^2 y - 2x + 4)$

Solution:

a. To calculate $\partial f/\partial x$, treat the variable y as a constant. Then differentiate $f(x, y)$ with respect to x using the sum, difference, and power rules:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left[x^2 - 3xy + 2y^2 - 4x + 5y - 12\right]$$
$$= \frac{\partial}{\partial x}[x^2] - \frac{\partial}{\partial x}[3xy] + \frac{\partial}{\partial x}[2y^2] - \frac{\partial}{\partial x}[4x] + \frac{\partial}{\partial x}[5y] - \frac{\partial}{\partial x}[12]$$
$$= 2x - 3y + 0 - 4 + 0 - 0$$
$$= 2x - 3y - 4.$$

The derivatives of the third, fifth, and sixth terms are all zero because they do not contain the variable x, so they are treated as constant terms. The derivative of the second term is equal to the coefficient of x, which is $-3y$. Calculating $\partial f/\partial y$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left[x^2 - 3xy + 2y^2 - 4x + 5y - 12\right]$$
$$= \frac{\partial}{\partial y}[x^2] - \frac{\partial}{\partial y}[3xy] + \frac{\partial}{\partial y}[2y^2] - \frac{\partial}{\partial y}[4x] + \frac{\partial}{\partial y}[5y] - \frac{\partial}{\partial y}[12]$$
$$= 0 - 3x + 4y - 0 + 5 - 0$$
$$= -3x + 4y + 5.$$

These are the same answers obtained in Example 14.3.1.

b. To calculate $\partial g/\partial x$, treat the variable y as a constant. Then differentiate $g(x, y)$ with respect to x using the chain rule and power rule:

$$\frac{\partial g}{\partial x} = \frac{\partial}{\partial x}\left[\sin(x^2 y - 2x + 4)\right]$$
$$= \cos(x^2 y - 2x + 4)\frac{\partial}{\partial x}[x^2 y - 2x + 4]$$
$$= (2xy - 2)\cos(x^2 y - 2x + 4).$$

To calculate $\partial g/\partial y$, treat the variable x as a constant. Then differentiate $g(x,y)$ with respect to y using the chain rule and power rule:

$$\begin{aligned}\frac{\partial g}{\partial y} &= \frac{\partial}{\partial y}\left[\sin(x^2y-2x+4)\right] \\ &= \cos(x^2y-2x+4)\frac{\partial}{\partial y}[x^2y-2x+4] \\ &= x^2\cos(x^2y-2x+4).\end{aligned}$$

? Exercise 14.3.2

Calculate $\partial f/\partial x$ and $\partial f/\partial y$ for the function

$$f(x,y)=\tan(x^3-3x^2y^2+2y^4)$$

by holding the opposite variable constant, then differentiating.

Hint

Use Equations 14.3.1 and 14.3.1 from the definition of partial derivatives.

Answer

$$\frac{\partial f}{\partial x}=(3x^2-6xy^2)\sec^2(x^3-3x^2y^2+2y^4)$$
$$\frac{\partial f}{\partial y}=(-6x^2y+8y^3)\sec^2(x^3-3x^2y^2+2y^4)$$

How can we interpret these partial derivatives? Recall that the graph of a function of two variables is a surface in R^3. If we remove the limit from the definition of the partial derivative with respect to x, the difference quotient remains:

$$\frac{f(x+h,y)-f(x,y)}{h}.$$

This resembles the difference quotient for the derivative of a function of one variable, except for the presence of the y variable. Figure 14.3.1 illustrates a surface described by an arbitrary function $z=f(x,y)$.

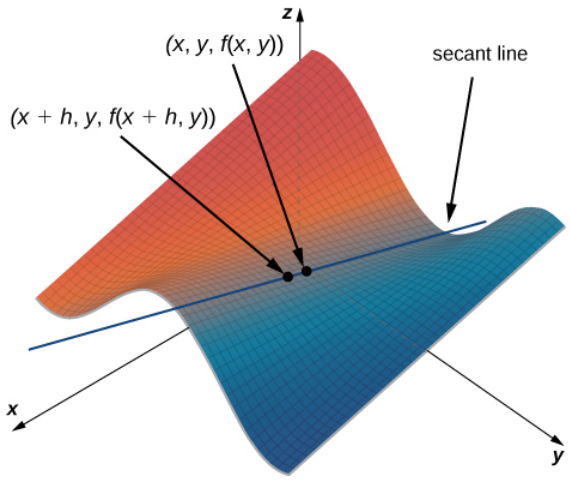


Figure 14.3.1: Secant line passing through the points $(x,y,f(x,y))$ and $(x+h,y,f(x+h,y))$.

In Figure 14.3.1, the value of h is positive. If we graph $f(x,y)$ and $f(x+h,y)$ for an arbitrary point (x,y), then the slope of the secant line passing through these two points is given by

$$\frac{f(x+h,y)-f(x,y)}{h}.$$

This line is parallel to the x-axis. Therefore, the slope of the secant line represents an average rate of change of the function f as we travel parallel to the x-axis. As h approaches zero, the slope of the secant line approaches the slope of the tangent line.

If we choose to change y instead of x by the same incremental value h, then the secant line is parallel to the y-axis and so is the tangent line. Therefore, $\partial f/\partial x$ represents the slope of the tangent line passing through the point $(x, y, f(x, y))$ parallel to the x-axis and $\partial f/\partial y$ represents the slope of the tangent line passing through the point $(x, y, f(x, y))$ parallel to the y-axis. If we wish to find the slope of a tangent line passing through the same point in any other direction, then we need what are called directional derivatives.

We now return to the idea of contour maps, which we introduced in Functions of Several Variables. We can use a **contour map** to estimate partial derivatives of a function $g(x, y)$.

Example 14.3.3: Partial Derivatives from a Contour Map

Use a contour map to estimate $\partial g/\partial x$ at the point $(\sqrt{5}, 0)$ for the function

$$g(x, y) = \sqrt{9 - x^2 - y^2}.$$

Solution

Figure 14.3.2 represents a contour map for the function $g(x, y)$.

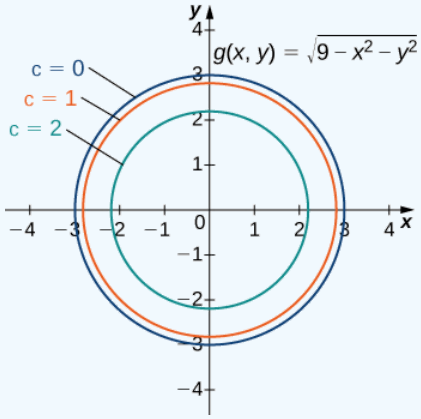


Figure 14.3.2: Contour map for the function $g(x, y) = \sqrt{9 - x^2 - y^2}$, using $c = 0, 1, 2,$ and $3 (c = 3$ corresponds to the origin).

The inner circle on the contour map corresponds to $c = 2$ and the next circle out corresponds to $c = 1$. The first circle is given by the equation $2 = \sqrt{9 - x^2 - y^2}$; the second circle is given by the equation $1 = \sqrt{9 - x^2 - y^2}$. The first equation simplifies to $x^2 + y^2 = 5$ and the second equation simplifies to $x^2 + y^2 = 8$. The x-intercept of the first circle is $(\sqrt{5}, 0)$ and the x-intercept of the second circle is $(2\sqrt{2}, 0)$. We can estimate the value of $\partial g/\partial x$ evaluated at the point $(\sqrt{5}, 0)$ using the slope formula:

$$\begin{aligned} \frac{\partial g}{\partial x}\Big|_{(x,y)=(\sqrt{5},0)} &\approx \frac{g(\sqrt{5}, 0) - g(2\sqrt{2}, 0)}{\sqrt{5} - 2\sqrt{2}} \\ &= \frac{2 - 1}{\sqrt{5} - 2\sqrt{2}} \\ &= \frac{1}{\sqrt{5} - 2\sqrt{2}} \approx -1.688. \end{aligned}$$

To calculate the exact value of $\partial g/\partial x$ evaluated at the point $(\sqrt{5}, 0)$, we start by finding $\partial g/\partial x$ using the chain rule. First, we rewrite the function as

$$g(x, y) = \sqrt{9 - x^2 - y^2} = (9 - x^2 - y^2)^{1/2}$$

and then differentiate with respect to x while holding y constant:

$$\begin{aligned} \frac{\partial g}{\partial x} &= \frac{1}{2}(9 - x^2 - y^2)^{-1/2}(-2x) \\ &= -\frac{x}{\sqrt{9 - x^2 - y^2}}. \end{aligned}$$

Next, we evaluate this expression using $x = \sqrt{5}$ and $y = 0$:

$$\begin{aligned}\frac{\partial g}{\partial x}\Big|_{(x,y)=(\sqrt{5},0)} &= -\frac{\sqrt{5}}{\sqrt{9-(\sqrt{5})^2-(0)^2}} \\ &= -\frac{\sqrt{5}}{\sqrt{4}} \\ &= -\frac{\sqrt{5}}{2} \approx -1.118.\end{aligned}$$

The estimate for the partial derivative corresponds to the slope of the secant line passing through the points $(\sqrt{5}, 0, g(\sqrt{5}, 0))$ and $(2\sqrt{2}, 0, g(2\sqrt{2}, 0))$. It represents an approximation to the slope of the tangent line to the surface through the point $(\sqrt{5}, 0, g(\sqrt{5}, 0))$, which is parallel to the x-axis.

? Exercise 14.3.3

Use a contour map to estimate $\partial f/\partial y$ at point $(0, \sqrt{2})$ for the function

$$f(x, y) = x^2 - y^2.$$

Compare this with the exact answer.

Hint

Create a contour map for f using values of c from -3 to 3. Which of these curves passes through point $(0, \sqrt{2})$?

Answer

Using the curves corresponding to $c = -2$ and $c = -3$, we obtain

$$\begin{aligned}\frac{\partial f}{\partial y}\Big|_{(x,y)=(0,\sqrt{2})} &\approx \frac{f(0,\sqrt{3}) - f(0,\sqrt{2})}{\sqrt{3}-\sqrt{2}} \\ &= \frac{-3-(-2)}{\sqrt{3}-\sqrt{2}} \cdot \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}} \\ &= -\sqrt{3}-\sqrt{2} \approx -3.146.\end{aligned}$$

The exact answer is

$$\frac{\partial f}{\partial y}\Big|_{(x,y)=(0,\sqrt{2})} = (-2y|_{(x,y)=(0,\sqrt{2})} = -2\sqrt{2} \approx -2.828.$$

Functions of More Than Two Variables

Suppose we have a function of three variables, such as $w = f(x, y, z)$. We can calculate partial derivatives of w with respect to any of the independent variables, simply as extensions of the definitions for partial derivatives of functions of two variables.

Definition: Partial Derivatives

Let $f(x, y, z)$ be a function of three variables. Then, the partial derivative of f with respect to x, written as $\partial f/\partial x$, or f_x, is defined to be

$$\frac{\partial f}{\partial x} = f_x(x, y, z) = \lim_{h\to 0} \frac{f(x+h, y, z) - f(x, y, z)}{h}. \tag{14.3.3}$$

The *partial derivative* of f with *respect* to y, written as $\partial f/\partial y$, or f_y, is defined to be

$$\frac{\partial f}{\partial y} = f_y(x, y, z) = \lim_{k\to 0} \frac{f(x, y+k, z) - f(x, y, z)}{k.} \tag{14.3.4}$$

The *partial derivative* of f with respect to z, written as $\partial f/\partial z$, or f_z, is defined to be

$$\frac{\partial f}{\partial z} = f_z(x,y,z) = \lim_{m\to 0} \frac{f(x,y,z+m) - f(x,y,z)}{m}. \tag{14.3.5}$$

We can calculate a partial derivative of a function of three variables using the same idea we used for a function of two variables. For example, if we have a function f of x, y, and z, and we wish to calculate $\partial f/\partial x$, then we treat the other two independent variables as if they are constants, then differentiate with respect to x.

Example 14.3.4: Calculating Partial Derivatives for a Function of Three Variables

Use the limit definition of partial derivatives to calculate $\partial f/\partial x$ for the function

$$f(x,y,z) = x^2 - 3xy + 2y^2 - 4xz + 5yz^2 - 12x + 4y - 3z.$$

Then, find $\partial f/\partial y$ and $\partial f/\partial z$ by setting the other two variables constant and differentiating accordingly.

Solution:

We first calculate $\partial f/\partial x$ using Equation 14.3.3, then we calculate the other two partial derivatives by holding the remaining variables constant. To use the equation to find $\partial f/\partial x$, we first need to calculate $f(x+h,y,z)$:

$$\begin{aligned} f(x+h,y,z) &= (x+h)^2 - 3(x+h)y + 2y^2 - 4(x+h)z + 5yz^2 - 12(x+h) + 4y - 3z \\ &= x^2 + 2xh + h^2 - 3xy - 3xh + 2y^2 - 4xz - 4hz + 5yz^2 - 12x - 12h + 4y - 3z \end{aligned}$$

and recall that $f(x,y,z) = x^2 - 3xy + 2y^2 - 4zx + 5yz^2 - 12x + 4y - 3z$. Next, we substitute these two expressions into the equation:

$$\begin{aligned} \frac{\partial f}{\partial x} &= \lim_{h\to 0} \\ &\quad \left[\frac{\begin{array}{r} x^2 + 2xh + h^2 - 3xy - 3hy + 2y^2 - 4xz - 4hz + 5yz^2 - 12x - 12h + 4y - 3zh - x^2 - 3xy \\ + 2y^2 - 4xz + 5yz^2 - 12x + 4y - 3z \end{array}}{h}\right] \\ &= \lim_{h\to 0}\left[\frac{2xh + h^2 - 3hy - 4hz - 12h}{h}\right] \\ &= \lim_{h\to 0}\left[\frac{h(2x + h - 3y - 4z - 12)}{h}\right] \\ &= \lim_{h\to 0}(2x + h - 3y - 4z - 12) \\ &= 2x - 3y - 4z - 12. \end{aligned}$$

Then we find $\partial f/\partial y$ by holding x and z constant. Therefore, any term that does not include the variable y is constant, and its derivative is zero. We can apply the sum, difference, and power rules for functions of one variable:

$$\begin{aligned} &\frac{\partial}{\partial y}\left[x^2 - 3xy + 2y^2 - 4xz + 5yz^2 - 12x + 4y - 3z\right] \\ &= \frac{\partial}{\partial y}[x^2] - \frac{\partial}{\partial y}[3xy] + \frac{\partial}{\partial y}[2y^2] - \frac{\partial}{\partial y}[4xz] + \frac{\partial}{\partial y}[5yz^2] - \frac{\partial}{\partial y}[12x] + \frac{\partial}{\partial y}[4y] - \frac{\partial}{\partial z}[3z] \\ &= 0 - 3x + 4y - 0 + 5z^2 - 0 + 4 - 0 \\ &= -3x + 4y + 5z^2 + 4. \end{aligned}$$

To calculate $\partial f/\partial z$, we hold x and y constant and apply the sum, difference, and power rules for functions of one variable:

$$\begin{aligned} &\frac{\partial}{\partial z}[x^2 - 3xy + 2y^2 - 4xz + 5yz^2 - 12x + 4y - 3z] \\ &= \frac{\partial}{\partial z}[x^2] - \frac{\partial}{\partial z}[3xy] + \frac{\partial}{\partial z}[2y^2] - \frac{\partial}{\partial z}[4xz] + \frac{\partial}{\partial z}[5yz^2] - \frac{\partial}{\partial z}[12x] + \frac{\partial}{\partial z}[4y] - \frac{\partial}{\partial z}[3z] \\ &= 0 - 0 + 0 - 4x + 10yz - 0 + 0 - 3 \\ &= -4x + 10yz - 3 \end{aligned}$$

Exercise 14.3.4

Use the limit definition of partial derivatives to calculate $\partial f/\partial x$ for the function

$$f(x,y,z) = 2x^2 - 4x^2y + 2y^2 + 5xz^2 - 6x + 3z - 8.$$

Then find $\partial f/\partial y$ and $\partial f/\partial z$ by setting the other two variables constant and differentiating accordingly.

Hint

Use the strategy in the preceding example.

Answer

$\dfrac{\partial f}{\partial x} = 4x - 8xy + 5z^2 - 6, \dfrac{\partial f}{\partial y} = -4x^2 + 4y, \dfrac{\partial f}{\partial z} = 10xz + 3$

Example 14.3.5: Calculating Partial Derivatives for a Function of Three Variables

Calculate the three partial derivatives of the following functions.

a. $f(x,y,z) = \dfrac{x^2y - 4xz + y^2}{x - 3yz}$

b. $g(x,y,z) = \sin(x^2y - z) + \cos(x^2 - yz)$

Solution

In each case, treat all variables as constants except the one whose partial derivative you are calculating.

a.

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}\left[\frac{x^2y - 4xz + y^2}{x - 3yz}\right] \\ &= \frac{\frac{\partial}{\partial x}(x^2y - 4xz + y^2)(x - 3yz) - (x^2y - 4xz + y^2)\frac{\partial}{\partial x}(x - 3yz)}{(x - 3yz)^2} \\ &= \frac{(2xy - 4z)(x - 3yz) - (x^2y - 4xz + y^2)(1)}{(x - 3yz)^2} \\ &= \frac{2x^2y - 6xy^2z - 4xz + 12yz^2 - x^2y + 4xz - y^2}{(x - 3yz)^2} \\ &= \frac{x^2y - 6xy^2z - 4xz + 12yz^2 + 4xz - y^2}{(x - 3yz)^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}\left[\frac{x^2y - 4xz + y^2}{x - 3yz}\right] \\ &= \frac{\frac{\partial}{\partial y}(x^2y - 4xz + y^2)(x - 3yz) - (x^2y - 4xz + y^2)\frac{\partial}{\partial y}(x - 3yz)}{(x - 3yz)^2} \\ &= \frac{(x^2 + 2y)(x - 3yz) - (x^2y - 4xz + y^2)(-3z)}{(x - 3yz)^2} \\ &= \frac{x^3 - 3x^2yz + 2xy - 6y^2z + 3x^2yz - 12xz^2 + 3y^2z}{(x - 3yz)^2} \\ &= \frac{x^3 + 2xy - 3y^2z - 12xz^2}{(x - 3yz)^2}\end{aligned}$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}\left[\frac{x^2y - 4xz + y^2}{x - 3yz}\right]$$

$$= \frac{\frac{\partial}{\partial z}(x^2y - 4xz + y^2)(x - 3yz) - (x^2y - 4xz + y^2)\frac{\partial}{\partial z}(x - 3yz)}{(x - 3yz)^2}$$

$$= \frac{(-4x)(x - 3yz) - (x^2y - 4xz + y^2)(-3y)}{(x - 3yz)^2}$$

$$= \frac{-4x^2 + 12xyz + 3x^2y^2 - 12xyz + 3y^3}{(x - 3yz)^2}$$

$$= \frac{-4x^2 + 3x^2y^2 + 3y^3}{(x - 3yz)^2}$$

b.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left[\sin(x^2y - z) + \cos(x^2 - yz)\right]$$

$$= (\cos(x^2y - z))\frac{\partial}{\partial x}(x^2y - z) - (\sin(x^2 - yz))\frac{\partial}{\partial x}(x^2 - yz)$$

$$= 2xy\cos(x^2y - z) - 2x\sin(x^2 - yz)$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}[\sin(x^2y - z) + \cos(x^2 - yz)]$$

$$= (\cos(x^2y - z))\frac{\partial}{\partial y}(x^2y - z) - (\sin(x^2 - yz))\frac{\partial}{\partial y}(x^2 - yz)$$

$$= x^2\cos(x^2y - z) + z\sin(x^2 - yz)$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}[\sin(x^2y - z) + \cos(x^2 - yz)]$$

$$= (\cos(x^2y - z))\frac{\partial}{\partial z}(x^2y - z) - (\sin(x^2 - yz))\frac{\partial}{\partial z}(x^2 - yz)$$

$$= -\cos(x^2y - z) + y\sin(x^2 - yz)$$

? Exercise 14.3.5

Calculate $\partial f/\partial x, \partial f/\partial y$, and $\partial f/\partial z$ for the function

$$f(x, y, z) = \sec(x^2y) - \tan(x^3yz^2).$$

Hint

Use the strategy in the preceding example.

Answer

$$\frac{\partial f}{\partial x} = 2xy\sec(x^2y)\tan(x^2y) - 3x^2yz^2\sec^2(x^3yz^2)$$

$$\frac{\partial f}{\partial y} = x^2\sec(x^2y)\tan(x^2y) - x^3z^2\sec^2(x^3yz^2)$$

$$\frac{\partial f}{\partial z} = -2x^3yz\sec^2(x^3yz^2)$$

Higher-Order Partial Derivatives

Consider the function

$$f(x, y) = 2x^3 - 4xy^2 + 5y^3 - 6xy + 5x - 4y + 12.$$

Its partial derivatives are

$$\frac{\partial f}{\partial x} = 6x^2 - 4y^2 - 6y + 5$$

and

$$\frac{\partial f}{\partial y} = -8xy + 15y^2 - 6x - 4.$$

Each of these partial derivatives is a function of two variables, so we can calculate partial derivatives of these functions. Just as with derivatives of single-variable functions, we can call these *second-order derivatives*, third-order derivatives, and so on. In general, they are referred to as **higher-order partial derivatives.** There are four second-order partial derivatives for any function (provided they all exist):

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right]$$
$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right]$$
$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right]$$
$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right].$$

An alternative notation for each is $f_{xx}, f_{xy}, f_{yx},$ and f_{yy}, respectively. Higher-order partial derivatives calculated with respect to different variables, such as f_{xy} and f_{yx}, are commonly called **mixed partial derivatives.**

Example 14.3.6: Calculating Second Partial Derivatives

Calculate all four second partial derivatives for the function

$$f(x, y) = xe^{-3y} + \sin(2x - 5y). \tag{14.3.6}$$

Solution:

To calculate $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y \partial x}$, we first calculate $\partial f/\partial x$:

$$\frac{\partial f}{\partial x} = e^{-3y} + 2\cos(2x - 5y). \tag{14.3.7}$$

To calculate $\frac{\partial^2 f}{\partial x^2}$, differentiate $\partial f/\partial x$ (Equation 14.3.7) with respect to x:

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial x}\right] \\ &= \frac{\partial}{\partial x}[e^{-3y} + 2\cos(2x - 5y)] \\ &= -4\sin(2x - 5y).\end{aligned}$$

To calculate $\frac{\partial^2 f}{\partial y \partial x}$, differentiate $\partial f/\partial x$ (Equation 14.3.7) with respect to y:

$$\begin{aligned}\frac{\partial^2 f}{\partial y\, \partial x} &= \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial x}\right] \\ &= \frac{\partial}{\partial y}[e^{-3y} + 2\cos(2x - 5y)] \\ &= -3e^{-3y} + 10\sin(2x - 5y).\end{aligned}$$

To calculate $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y^2}$, first calculate $\partial f/\partial y$:

$$\frac{\partial f}{\partial y} = -3xe^{-3y} - 5\cos(2x - 5y). \tag{14.3.8}$$

To calculate $\frac{\partial^2 f}{\partial x \partial y}$, differentiate $\partial f/\partial y$ (Equation 14.3.8) with respect to x:

$$\begin{aligned}\frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x}\left[\frac{\partial f}{\partial y}\right] \\ &= \frac{\partial}{\partial x}[-3xe^{-3y} - 5\cos(2x-5y)] \\ &= -3e^{-3y} + 10\sin(2x-5y).\end{aligned}$$

To calculate $\frac{\partial^2 f}{\partial y^2}$, differentiate $\partial f/\partial y$ (Equation 14.3.8) with respect to y:

$$\begin{aligned}\frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y}\left[\frac{\partial f}{\partial y}\right] \\ &= \frac{\partial}{\partial y}[-3xe^{-3y} - 5\cos(2x-5y)] \\ &= 9xe^{-3y} - 25\sin(2x-5y).\end{aligned}$$

? Exercise 14.3.6

Calculate all four second partial derivatives for the function

$$f(x,y) = \sin(3x-2y) + \cos(x+4y).$$

Hint

Follow the same steps as in the previous example.

Answer

$\frac{\partial^2 f}{\partial x^2} = -9\sin(3x-2y) - \cos(x+4y)$

$\frac{\partial^2 f}{\partial y \partial x} = 6\sin(3x-2y) - 4\cos(x+4y)$

$\frac{\partial^2 f}{\partial x \partial y} = 6\sin(3x-2y) - 4\cos(x+4y)$

$\frac{\partial^2 f}{\partial y^2} = -4\sin(3x-2y) - 16\cos(x+4y)$

At this point we should notice that, in both Example 14.3.6 and the checkpoint, it was true that $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$. Under certain conditions, this is always true. In fact, it is a direct consequence of the following theorem.

Theorem 14.3.1: Equality of Mixed Partial Derivatives (Clairaut's Theorem)

Suppose that $f(x,y)$ is defined on an open disk D that contains the point (a,b). If the functions f_{xy} and f_{yx} are continuous on D, then $f_{xy}(a,b) = f_{yx}(a,b)$.

Clairaut's theorem guarantees that as long as mixed second-order derivatives are continuous, the order in which we choose to differentiate the functions (i.e., which variable goes first, then second, and so on) does not matter. It can be extended to higher-order derivatives as well. The proof of Clairaut's theorem can be found in most advanced calculus books.

Two other second-order partial derivatives can be calculated for any function $f(x,y)$. The partial derivative f_{xx} is equal to the partial derivative of f_x with respect to x, and f_{yy} is equal to the partial derivative of f_y with respect to y.

Partial Differential Equations

Previously, we studied differential equations in which the unknown function had one independent variable. A partial differential equation is an equation that involves an unknown function of more than one independent variable and one or more of its partial

derivatives. Examples of partial differential equations are

$$u_t = c^2(u_{xx} + u_{yy})$$
heat equation in two dimensions

$$u_{tt} = c^2(u_{xx} + u_{yy})$$
wave equation in two dimensions

$$u_{xx} + u_{yy} = 0$$
Laplace's equation in two dimensions

In the heat and wave equations, the unknown function u has three independent variables: t, x, and y with c is an arbitrary constant. The independent variables x and y are considered to be spatial variables, and the variable t represents time. In Laplace's equation, the unknown function u has two independent variables x and y.

Example 14.3.7: A Solution to the Wave Equation

Verify that

$$u(x, y, t) = 5\sin(3\pi x)\sin(4\pi y)\cos(10\pi t)$$

is a solution to the wave equation

$$u_{tt} = 4(u_{xx} + u_{yy}). \tag{14.3.9}$$

Solution

First, we calculate u_{tt}, u_{xx}, and u_{yy} :

$$\begin{aligned} u_{tt}(x, y, t) &= \frac{\partial}{\partial t}\left[\frac{\partial u}{\partial t}\right] \\ &= \frac{\partial}{\partial t}[5\sin(3\pi x)\sin(4\pi y)(-10\pi\sin(10\pi t))] \\ &= \frac{\partial}{\partial t}[-50\pi\sin(3\pi x)\sin(4\pi y)\sin(10\pi t)] \\ &= -500\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t) \end{aligned}$$

$$\begin{aligned} u_{xx}(x, y, t) &= \frac{\partial}{\partial x}\left[\frac{\partial u}{\partial x}\right] \\ &= \frac{\partial}{\partial x}[15\pi\cos(3\pi x)\sin(4\pi y)\cos(10\pi t)] \\ &= -45\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t) \end{aligned}$$

$$\begin{aligned} u_{yy}(x, y, t) &= \frac{\partial}{\partial y}\left[\frac{\partial u}{\partial y}\right] \\ &= \frac{\partial}{\partial y}[5\sin(3\pi x)(4\pi\cos(4\pi y))\cos(10\pi t)] \\ &= \frac{\partial}{\partial y}[20\pi\sin(3\pi x)\cos(4\pi y)\cos(10\pi t)] \\ &= -80\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t). \end{aligned}$$

Next, we substitute each of these into the right-hand side of Equation 14.3.9 and simplify:

$$\begin{aligned} 4(u_{xx} + u_{yy}) &= 4(-45\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t) + -80\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t)) \\ &= 4(-125\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t)) \\ &= -500\pi^2\sin(3\pi x)\sin(4\pi y)\cos(10\pi t) \\ &= u_{tt}. \end{aligned}$$

This verifies the solution.

Exercise 14.3.7: A Solution to the Heat Equation

Verify that

$$u(x,y,t) = 2\sin\left(\frac{x}{3}\right)\sin\left(\frac{y}{4}\right)e^{-25t/16}$$

is a solution to the heat equation

$$u_t = 9(u_{xx} + u_{yy}).$$

Hint

Calculate the partial derivatives and substitute into the right-hand side.

Answer

TBA

Since the solution to the two-dimensional heat equation is a function of three variables, it is not easy to create a visual representation of the solution. We can graph the solution for fixed values of t, which amounts to snapshots of the heat distributions at fixed times. These snapshots show how the heat is distributed over a two-dimensional surface as time progresses. The graph of the preceding solution at time $t = 0$ appears in Figure 14.3.3. As time progresses, the extremes level out, approaching zero as t approaches infinity.

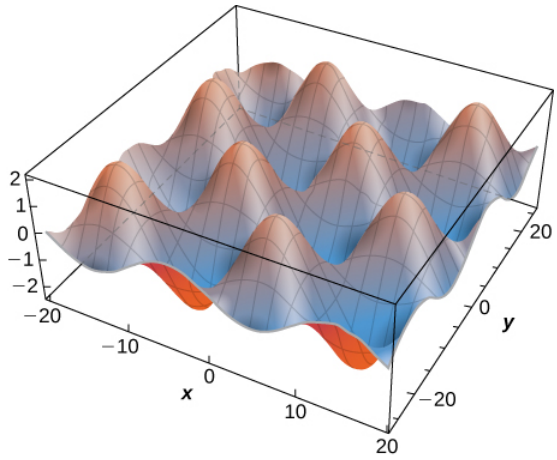


Figure 14.3.3

If we consider the heat equation in one dimension, then it is possible to graph the solution over time. The heat equation in one dimension becomes

$$u_t = c^2 u_{xx},$$

where c^2 represents the thermal diffusivity of the material in question. A solution of this differential equation can be written in the form

$$u_m(x,t) = e^{-\pi^2 m^2 c^2 t}\sin(m\pi x)$$

where m is any positive integer. A graph of this solution using $m = 1$ appears in Figure 14.3.4, where the initial temperature distribution over a wire of length 1 is given by $u(x,0) = \sin \pi x$. Notice that as time progresses, the wire cools off. This is seen because, from left to right, the highest temperature (which occurs in the middle of the wire) decreases and changes color from red to blue.

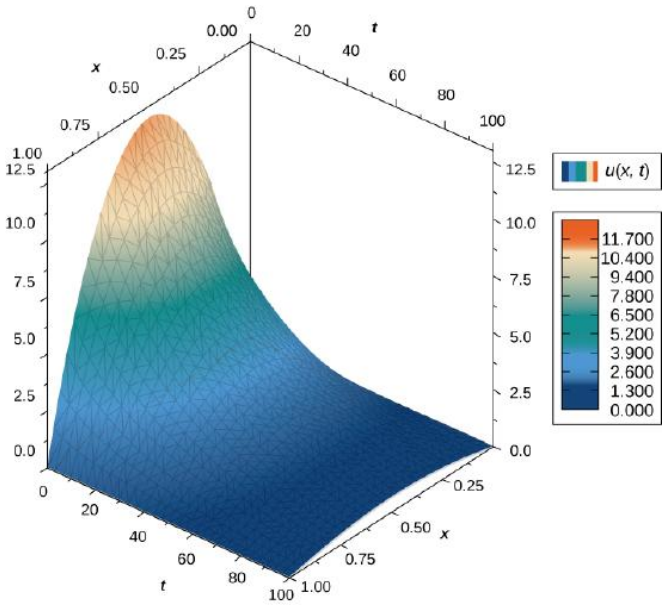
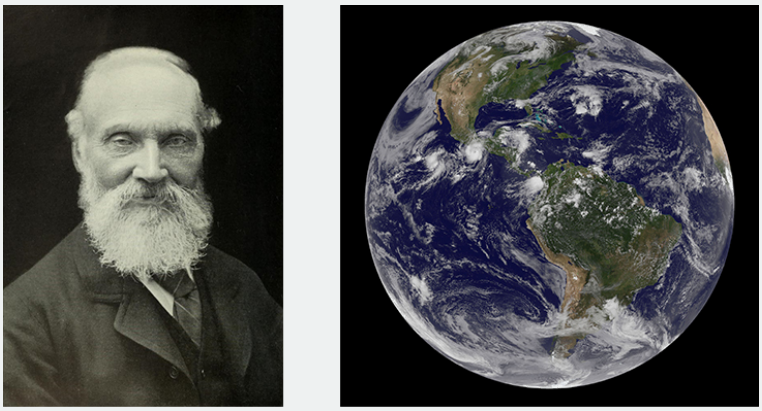


Figure 14.3.4: Graph of a solution of the heat equation in one dimension over time.

Lord Kelvin and the Age of Earth

During the late 1800s, the scientists of the new field of geology were coming to the conclusion that Earth must be “millions and millions” of years old. At about the same time, Charles Darwin had published his treatise on evolution. Darwin’s view was that evolution needed many millions of years to take place, and he made a bold claim that the Weald chalk fields, where important fossils were found, were the result of 300 million years of erosion.



(a) (b)

Figure 14.3.5: (a) William Thomson (Lord Kelvin), 1824-1907, was a British physicist and electrical engineer; (b) Kelvin used the heat diffusion equation to estimate the age of Earth (credit: modification of work by NASA).

At that time, eminent physicist William Thomson (Lord Kelvin) used an important partial differential equation, known as the heat diffusion equation, to estimate the age of Earth by determining how long it would take Earth to cool from molten rock to what we had at that time. His conclusion was a range of **20** to **400** million years, but most likely about **50** million years. For many decades, the proclamations of this irrefutable icon of science did not sit well with geologists or with Darwin.

- Read Kelvin’s paper on estimating the age of the Earth.

Kelvin made reasonable assumptions based on what was known in his time, but he also made several assumptions that turned out to be wrong. One incorrect assumption was that Earth is solid and that the cooling was therefore via conduction only, hence justifying the use of the diffusion equation. But the most serious error was a forgivable one—omission of the fact that Earth contains radioactive elements that continually supply heat beneath Earth’s mantle. The discovery of radioactivity came near the end of Kelvin’s life and he acknowledged that his calculation would have to be modified.

Kelvin used the simple one-dimensional model applied only to Earth's outer shell, and derived the age from graphs and the roughly known temperature gradient near Earth's surface. Let's take a look at a more appropriate version of the diffusion equation in radial coordinates, which has the form

$$\frac{\partial T}{\partial t} = K\left[\frac{\partial^2 T}{\partial^2 r} + \frac{2}{r}\frac{\partial T}{\partial r}\right] \tag{14.3.10}$$

.

Here, $T(r,t)$ is temperature as a function of r (measured from the center of Earth) and time t.K is the heat conductivity—for molten rock, in this case. The standard method of solving such a partial differential equation is by separation of variables, where we express the solution as the product of functions containing each variable separately. In this case, we would write the temperature as

$$T(r,t) = R(r)f(t).$$

1. Substitute this form into Equation 14.3.10 and, noting that $f(t)$ is constant with respect to distance (r) and $R(r)$ is constant with respect to time (t), show that

$$\frac{1}{f}\frac{\partial f}{\partial t} = \frac{K}{R}\left[\frac{\partial^2 R}{\partial r^2} + \frac{2}{r}\frac{\partial R}{\partial r}\right].$$

2. This equation represents the separation of variables we want. The left-hand side is only a function of t and the right-hand side is only a function of r, and they must be equal for all values of r and t. Therefore, they both must be equal to a constant. Let's call that constant $-\lambda^2$. (The convenience of this choice is seen on substitution.) So, we have

$$\frac{1}{f}\frac{\partial f}{\partial t} = -\lambda^2 \text{and} \frac{K}{R}\left[\frac{\partial^2 R}{\partial r^2} + \frac{2}{r}\frac{\partial R}{\partial r}\right] = -\lambda^2.$$

3. Now, we can verify through direct substitution for each equation that the solutions are $f(t) = Ae^{-\lambda^2 t}$ and $R(r) = B\left(\frac{\sin \alpha r}{r}\right) + C\left(\frac{\cos \alpha r}{r}\right)$, where $\alpha = \lambda/\sqrt{K}$. Note that $f(t) = Ae^{+\lambda n^2 t}$ is also a valid solution, so we could have chosen $+\lambda^2$for our constant. Can you see why it would not be valid for this case as time increases?
4. Let's now apply boundary conditions.
   a. The temperature must be finite at the center of Earth, $r = 0$. Which of the two constants, B or C, must therefore be zero to keep R finite at $r = 0$? (Recall that $\sin(\alpha r)/r \to \alpha =$ as $r \to 0$, but $\cos(\alpha r)/r$ behaves very differently.)
   b. Kelvin argued that when magma reaches Earth's surface, it cools very rapidly. A person can often touch the surface within weeks of the flow. Therefore, the surface reached a moderate temperature very early and remained nearly constant at a surface temperature T_s. For simplicity, let's set $T = 0$ at $r = R_E$ and find α such that this is the temperature there for all time t. (Kelvin took the value to be $300K \approx 80°F$. We can add this $300K$ constant to our solution later.) For this to be true, the sine argument must be zero at $r = R_E$. Note that α has an infinite series of values that satisfies this condition. Each value of α represents a valid solution (each with its own value for A). The total or general solution is the sum of all these solutions.
   c. At $t = 0$, we assume that all of Earth was at an initial hot temperature T_0 (Kelvin took this to be about $7000K$.) The application of this boundary condition involves the more advanced application of Fourier coefficients. As noted in part b. each value of α_n represents a valid solution, and the general solution is a sum of all these solutions. This results in a series solution:

$$T(r,t) = \left(\frac{T_0 R_E}{\pi}\right)\sum_n \frac{(-1)^{n-1}}{n} e^{-\lambda n^2 t}\frac{\sin(\alpha_n r)}{r}$$

   where $\alpha_n = n\pi/R_E$.

Note how the values of α_n come from the boundary condition applied in part b. The term $\frac{-1^{n-1}}{n}$ is the constant A_n for each term in the series, determined from applying the Fourier method. Letting $\beta = \frac{\pi}{R_E}$, examine the first few terms of this solution shown here and note how λ^2 in the exponential causes the higher terms to decrease quickly as time progresses:

$$T(r,t) = \frac{T_0 R_E}{\pi r}\left(e^{-K\beta^2 t}(\sin\beta r) - \frac{1}{2}e^{-4K\beta^2 t}(\sin 2\beta r) + \frac{1}{3}e^{-9K\beta^2 t}(\sin 3\beta r) - \frac{1}{4}e^{-16K\beta^2 t}(\sin 4\beta r) + \frac{1}{5}e^{-25K\beta^2 t}(\sin 5\beta r)\ldots\right).$$

Near time $t = 0$, many terms of the solution are needed for accuracy. Inserting values for the conductivity K and $\beta = \pi/R_E$ for time approaching merely thousands of years, only the first few terms make a significant contribution. Kelvin only needed to look at the solution near Earth's surface (Figure 14.3.6) and, after a long time, determine what time best yielded the estimated temperature gradient known during his era ($1°F$ increase per $50ft$). He simply chose a range of times with a gradient close to this value. In Figure 14.3.6, the solutions are plotted and scaled, with the $300 - K$ surface temperature added. Note that the center of Earth would be relatively cool. At the time, it was thought Earth must be solid.

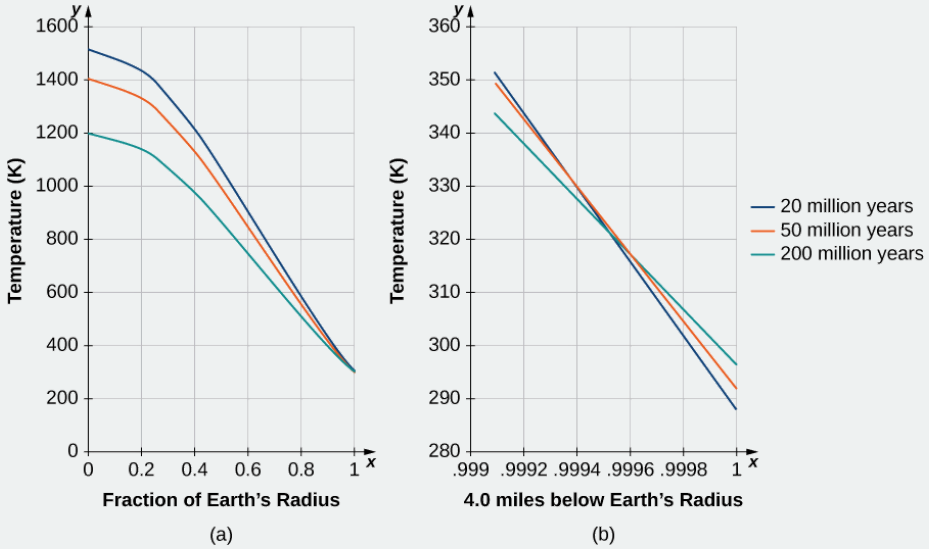


Figure 14.3.6: Temperature versus radial distance from the center of Earth. (a) Kelvin's results, plotted to scale. (b) A close-up of the results at a depth of 4.0 miles below Earth's surface.

Epilog

On May **20, 1904,** physicist **Ernest Rutherford** spoke at the Royal Institution to announce a revised calculation that included the contribution of radioactivity as a source of Earth's heat. In Rutherford's own words:

> "I came into the room, which was half-dark, and presently spotted Lord Kelvin in the audience, and realized that I was in for trouble at the last part of my speech dealing with the age of the Earth, where my views conflicted with his. To my relief, Kelvin fell fast asleep, but as I came to the important point, I saw the old bird sit up, open an eye and cock a baleful glance at me.
>
> Then a sudden inspiration came, and I said Lord Kelvin had limited the age of the Earth, provided no new source [of heat] was discovered. That prophetic utterance referred to what we are now considering tonight, radium! Behold! The old boy beamed upon me."

Rutherford calculated an age for Earth of about **500** million years. Today's accepted value of Earth's age is about **4.6** billion years.

Key Concepts

- A partial derivative is a derivative involving a function of more than one independent variable.
- To calculate a partial derivative with respect to a given variable, treat all the other variables as constants and use the usual differentiation rules.
- Higher-order partial derivatives can be calculated in the same way as higher-order derivatives.

Key Equations

Partial derivative of f with respect to x

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

Partial derivative of f with respect to y

$$\frac{\partial f}{\partial y} = \lim_{k \to 0} \frac{f(x, y+k) - f(x, y)}{k}$$

Glossary

higher-order partial derivatives

second-order or higher partial derivatives, regardless of whether they are mixed partial derivatives

mixed partial derivatives

second-order or higher partial derivatives, in which at least two of the differentiations are with respect to different variables

partial derivative

a derivative of a function of more than one independent variable in which all the variables but one are held constant

partial differential equation

an equation that involves an unknown function of more than one independent variable and one or more of its partial derivatives

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14.3E: Exercises for Section 14.3

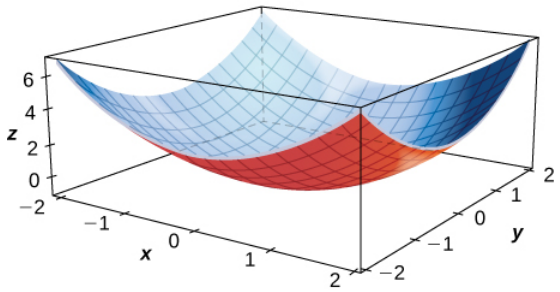
In the following exercise, calculate the partial derivative using the limit definitions only.

1) $\frac{\partial z}{\partial y}$ for $z = x^2 - 3xy + y^2$

Answer

$\frac{\partial z}{\partial y} = -3x + 2y$

For exercises 2 - 5, calculate the sign of the partial derivative using the graph of the surface.



2) $f_x(1,1)$

3) $f_x(-1,1)$

Answer

The sign is negative.

4) $f_y(1,1)$

5) $f_x(0,0)$

Answer

The partial derivative is zero at the origin.

In exercises 6 - 16, calculate the requested partial derivatives.

6) $\frac{\partial z}{\partial x}$ for $z = \sin(3x)\cos(3y)$

7) $\frac{\partial z}{\partial y}$ for $z = \sin(3x)\cos(3y)$

Answer

$\frac{\partial z}{\partial y} = -3\sin(3x)\sin(3y)$

8) $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = x^8 e^3 y$

9) $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $z = \ln(x^6 + y^4)$

Answer

$\frac{\partial z}{\partial x} = \frac{6x^5}{x^6+y^4}$; $\frac{\partial z}{\partial y} = \frac{4y^3}{x^6+y^4}$

10) Find $f_y(x,y)$ for $f(x,y) = e^{xy}\cos(x)\sin(y)$.

11) Let $z = e^{xy}$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

-

Answer

$\frac{\partial z}{\partial x} = ye^{xy}; \quad \frac{\partial z}{\partial y} = xe^{xy}$

12) Let $z = \ln(\frac{x}{y})$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

13) Let $z = \tan(2x - y)$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Answer

$\frac{\partial z}{\partial x} = 2\sec^2(2x - y), \quad \frac{\partial z}{\partial y} = -\sec^2(2x - y)$

14) Let $z = \sinh(2x + 3y)$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

15) Let $f(x, y) = \arctan(\frac{y}{x})$. Evaluate $f_x(2, -2)$ and $f_y(2, -2)$.

Answer

$f_x(2, -2) = \frac{1}{4} = f_y(2, -2)$

16) Let $f(x, y) = \frac{xy}{x - y}$. Find $f_x(2, -2)$ and $f_y(2, -2)$.

17) Find $\frac{\partial z}{\partial x}$ at $(0, 1)$ for $z = e^{-x}cos(y)$.

Answer

$\frac{\partial z}{\partial x} = -\cos(1)$

18) Given $f(x, y, z) = x^3yz^2$, find $\frac{\partial^2 f}{\partial x \partial y}$ and $f_z(1, 1, 1)$.

19) Given $f(x, y, z) = 2\sin(x + y)$, find $f_x(0, \frac{\pi}{2}, -4)$, $f_y(0, \frac{\pi}{2}, -4)$, and $f_z(0, \frac{\pi}{2}, -4)$.

Answer

$f_x(x, y, z) = 0, \quad f_y(x, y, z) = 0, \quad f_z(x, y, z) = 0$

20) The area of a parallelogram with adjacent side lengths that are a and b, and in which the angle between these two sides is θ, is given by the function $A(a, b, \theta) = ba\sin(\theta)$.Find the rate of change of the area of the parallelogram with respect to the following:

a. Side a

b. Side b

c. Angle θ

21) Express the volume of a right circular cylinder as a function of two variables:

a. its radius r and its height h.

b. Show that the rate of change of the volume of the cylinder with respect to its radius is the product of its circumference multiplied by its height.

c. Show that the rate of change of the volume of the cylinder with respect to its height is equal to the area of the circular base.

Answer

$a. V(r, h) = \pi r^2 h$
$b. \frac{\partial V}{\partial r} = 2\pi rh$
$c. \frac{\partial V}{\partial h} = \pi r^2$

22) Calculate $\frac{\partial w}{\partial z}$ for $w = z\sin(xy^2 + 2z)$.

In exercises 23 - 39, find the indicated higher-order partial derivatives.

23) $f_{xy}(x,y)$ for $f(x,y)=\ln(x-y)$

Answer

$f_{xy}(x,y)=\frac{1}{(x-y)^2}$

24) $f_{yx}(x,y)$ for $f(x,y)=\ln(x-y)$

25) Let $z=x^2+3xy+2y^2$. Find $\dfrac{\partial^2 z}{\partial x^2}$ and $\dfrac{\partial^2 z}{\partial y^2}$.

Answer

$\dfrac{\partial^2 z}{\partial x^2}=2, \quad \dfrac{\partial^2 z}{\partial y^2}=4$

26) Given $z=e^x\tan y$, find $\dfrac{\partial^2 z}{\partial x\partial y}$ and $\dfrac{\partial^2 z}{\partial y\partial x}$.

27) Given $f(x,y,z)=xyz$, find $f_{xyy}(x,y,z)$, $f_{yxy}(x,y,z)$, and $f_{yyx}(x,y,z)$.

Answer

$f_{xyy}(x,y,z)=f_{yxy}(x,y,z)=f_{yyx}(x,y,z)=0$

28) Given $f(x,y,z)=e^{-2x}\sin(z^2y)$, show that $f_{xyy}(x,y,z)=f_{yxy}(x,y,z)$.

29) Show that $z=\frac{1}{2}(e^y-e^{-y})\sin x$ is a solution of the differential equation $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=0$.

Answer

$\dfrac{d^2z}{dx^2}=-\frac{1}{2}(e^y-e^{-y})\sin x$
$\dfrac{d^2z}{dy^2}=\frac{1}{2}(e^y-e^{-y})\sin x$
$\dfrac{d^2z}{dx^2}+\dfrac{d^2z}{dy^2}=0$

30) Find $f_{xx}(x,y)$ for $f(x,y)=\frac{4x^2}{y}+\frac{y^2}{2x}$.

31) Let $f(x,y,z)=x^2y^3z-3xy^2z^3+5x^2z-y^3z$. Find f_{xyz}.

Answer

$f_{xyz}(x,y,z)=6y^2x-18yz^2$

32) Let $F(x,y,z)=x^3yz^2-2x^2yz+3xz-2y^3z$. Find $F_{xyz}(x,y,z)$.

33) Given $f(x,y)=x^2+x-3xy+y^3-5$, find all points at which $f_x(x,y)=f_y(x,y)=0$ simultaneously.

Answer

$(\frac{1}{4},\frac{1}{2}), \quad (1,1)$

34) Given $f(x,y)=2x^2+2xy+y^2+2x-3$, find all points at which $\dfrac{\partial f}{\partial x}=0$ and $\dfrac{\partial f}{\partial y}=0$ simultaneously.

35) Given $f(x,y)=y^3-3yx^2-3y^2-3x^2+1$, find all points on f at which $f_x(x,y)=f_y(x,y)=0$ simultaneously.

Answer

$(0,0), \quad (0,2), \quad (\sqrt{3},-1), \quad (-\sqrt{3},-1)$

36) Given $f(x,y)=15x^3-3xy+15y^3$, find all points at which $f_x(x,y)=f_y(x,y)=0$ simultaneously.

37) Show that $z=e^x\sin y$ satisfies the equation $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=0$.

Answer

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^x \sin y - e^x \sin y = 0$$

38) Show that $f(x, y) = \ln(x^2 + y^2)$ solves Laplace's equation $\dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial y^2} = 0$.

39) Show that $z = e^{-t}\cos(\frac{x}{c})$ satisfies the heat equation $\dfrac{\partial z}{\partial t} = -e^{-t}\cos(\frac{x}{c})$.

Answer

$$c^2 \frac{\partial^2 z}{\partial x^2} = e^{-t}\cos(\tfrac{x}{c})$$

40) Find $\lim\limits_{\Delta x \to 0} \dfrac{f(x + \Delta x) - f(x, y)}{\Delta x}$ for $f(x, y) = -7x - 2xy + 7y$.

41) Find $\lim\limits_{\Delta y \to 0} \dfrac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$ for $f(x, y) = -7x - 2xy + 7y$.

Answer

$$\frac{\partial f}{\partial y} = -2x + 7$$

42) Find $\lim\limits_{\Delta x \to 0} \dfrac{\Delta f}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$ for $f(x, y) = x^2 y^2 + xy + y$.

43) Find $\lim\limits_{\Delta x \to 0} \dfrac{\Delta f}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$ for $f(x, y) = \sin(xy)$.

Answer

$$\frac{\partial f}{\partial x} = y \cos xy$$

44) The function $P(T, V) = \dfrac{nRT}{V}$ gives the pressure at a point in a gas as a function of temperature T and volume V. The letters n and R are constants. Find $\dfrac{\partial P}{\partial V}$ and $\dfrac{\partial P}{\partial T}$, and explain what these quantities represent.

45) The equation for heat flow in the xy-plane is $\dfrac{\partial f}{\partial t} = \dfrac{\partial^2 f}{\partial x^2} + \dfrac{\partial^2 f}{\partial y^2}$. Show that $f(x, y, t) = e^{-2t} \sin x \sin y$ is a solution.

46) The basic wave equation is $f_{tt} = f_{xx}$. Verify that $f(x, t) = \sin(x + t)$ and $f(x, t) = \sin(x - t)$ are solutions.

47) The law of cosines can be thought of as a function of three variables. Let x, y, and θ be two sides of any triangle where the angle θ is the included angle between the two sides. Then, $F(x, y, \theta) = x^2 + y^2 - 2xy\cos\theta$ gives the square of the third side of the triangle. Find $\dfrac{\partial F}{\partial \theta}$ and $\dfrac{\partial F}{\partial x}$ when $x = 2$, $y = 3$, and $\theta = \frac{\pi}{6}$.

Answer

$$\frac{\partial F}{\partial \theta} = 6, \quad \frac{\partial F}{\partial x} = 4 - 3\sqrt{3}$$

48) Suppose the sides of a rectangle are changing with respect to time. The first side is changing at a rate of 2in./sec whereas the second side is changing at the rate of 4 in/sec. How fast is the diagonal of the rectangle changing when the first side measures 16 in. and the second side measures 20 in.? (Round answer to three decimal places.)

49) A **Cobb-Douglas production function** is $f(x, y) = 200x^{0.7}y^{0.3}$, where x and y represent the amount of labor and capital available. Let $x = 500$ and $y = 1000$. Find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ at these values, which represent the marginal productivity of labor and capital, respectively.

Answer

$$\frac{\partial f}{\partial x} \text{ at } (500, 1000) = 172.36, \quad \frac{\partial f}{\partial y} \text{ at } (500, 1000) = 36.93$$

50) The apparent temperature index is a measure of how the temperature feels, and it is based on two variables: h, which is relative humidity, and t, which is the air temperature.

$A = 0.885t - 22.4h + 1.20th - 0.544$. Find $\frac{\partial A}{\partial t}$ and $\frac{\partial A}{\partial h}$ when $t = 20°F$ and $h = 0.90$.

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14.4: Tangent Planes and Linear Approximations

Learning Objectives

- Determine the equation of a plane tangent to a given surface at a point.
- Use the tangent plane to approximate a function of two variables at a point.
- Explain when a function of two variables is differentiable.
- Use the total differential to approximate the change in a function of two variables.

In this section, we consider the problem of finding the tangent plane to a surface, which is analogous to finding the equation of a tangent line to a curve when the curve is defined by the graph of a function of one variable, $y = f(x)$. The slope of the tangent line at the point $x = a$ is given by $m = f'(a)$; what is the slope of a tangent plane? We learned about the equation of a plane in Equations of Lines and Planes in Space; in this section, we see how it can be applied to the problem at hand.

Tangent Planes

Intuitively, it seems clear that, in a plane, only one line can be tangent to a curve at a point. However, in three-dimensional space, many lines can be tangent to a given point. If these lines lie in the same plane, they determine the tangent plane at that point. A more intuitive way to think of a tangent plane is to assume the surface is smooth at that point (no corners). Then, a tangent line to the surface at that point in any direction does not have any abrupt changes in slope because the direction changes smoothly. Therefore, in a small-enough neighborhood around the point, a tangent plane touches the surface at that point only.

Definition: Tangent Lines

Let $P_0 = (x_0, y_0, z_0)$ be a point on a surface S, and let C be any curve passing through P_0 and lying entirely in S. If the tangent lines to all such curves C at P_0 lie in the same plane, then this plane is called the *tangent plane* to S at P_0 (Figure 14.4.1).

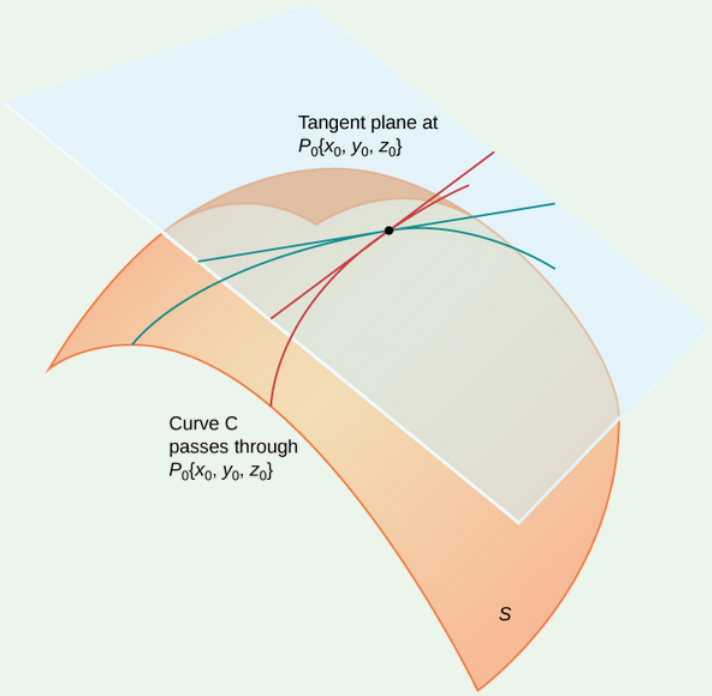


Figure 14.4.1: The tangent plane to a surface S at a point P_0 contains all the tangent lines to curves in S *that pass through* P_0.

For a tangent plane to a surface to exist at a point on that surface, it is sufficient for the function that defines the surface to be differentiable at that point. We define the term tangent plane here and then explore the idea intuitively.

Definition: Tangent Planes

Let S be a surface defined by a differentiable function $z = f(x, y)$, and let $P_0 = (x_0, y_0)$ be a point in the domain of f. Then, the equation of the tangent plane to S at P_0 is given by

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0). \tag{14.4.1}$$

To see why this formula is correct, let's first find two tangent lines to the surface S. The equation of the tangent line to the curve that is represented by the intersection of S with the vertical trace given by $x = x_0$ is $z = f(x_0, y_0) + f_y(x_0, y_0)(y - y_0)$. Similarly, the equation of the tangent line to the curve that is represented by the intersection of S with the vertical trace given by $y = y_0$ is $z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0)$. A parallel vector to the first tangent line is $\vec{\mathbf{a}} = \hat{\mathbf{j}} + f_y(x_0, y_0)\,\hat{\mathbf{k}}$; a parallel vector to the second tangent line is $\vec{\mathbf{b}} = \hat{\mathbf{i}} + f_x(x_0, y_0)\,\hat{\mathbf{k}}$. We can take the cross product of these two vectors:

$$\begin{aligned}
\vec{\mathbf{a}} \times \vec{\mathbf{b}} &= (\hat{\mathbf{j}} + f_y(x_0, y_0)\,\hat{\mathbf{k}}) \times (\hat{\mathbf{i}} + f_x(x_0, y_0)\,\hat{\mathbf{k}}) \\
&= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 1 & f_y(x_0, y_0) \\ 1 & 0 & f_x(x_0, y_0) \end{vmatrix} \\
&= f_x(x_0, y_0)\,\hat{\mathbf{i}} + f_y(x_0, y_0)\,\hat{\mathbf{j}} - \hat{\mathbf{k}}.
\end{aligned}$$

This vector is perpendicular to both lines and is therefore perpendicular to the tangent plane. We can use this vector as a normal vector to the tangent plane, along with the point $P_0 = (x_0, y_0, f(x_0, y_0))$ in the equation for a plane:

$$\begin{aligned}
\vec{\mathbf{n}} \cdot ((x - x_0)\,\hat{\mathbf{i}} + (y - y_0)\,\hat{\mathbf{j}} + (z - f(x_0, y_0))\,\hat{\mathbf{k}}) &= 0 \\
(f_x(x_0, y_0)\,\hat{\mathbf{i}} + f_y(x_0, y_0)\,\hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot ((x - x_0)\,\hat{\mathbf{i}} + (y - y_0)\,\hat{\mathbf{j}} + (z - f(x_0, y_0))\,\hat{\mathbf{k}}) &= 0 \\
f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - f(x_0, y_0)) &= 0.
\end{aligned}$$

Solving this equation for z gives Equation 14.4.1.

Example 14.4.1: Finding a Tangent Plane

Find the equation of the tangent plane to the surface defined by the function $f(x, y) = 2x^2 - 3xy + 8y^2 + 2x - 4y + 4$ at point $(2, -1)$.

Solution

First, we must calculate $f_x(x, y)$ and $f_y(x, y)$, then use Equation with $x_0 = 2$ and $y_0 = -1$:

$$\begin{aligned}
f_x(x, y) &= 4x - 3y + 2 \\
f_y(x, y) &= -3x + 16y - 4 \\
f(2, -1) &= 2(2)^2 - 3(2)(-1) + 8(-1)^2 + 2(2) - 4(-1) + 4 = 34 \\
f_x(2, -1) &= 4(2) - 3(-1) + 2 = 13 \\
f_y(2, -1) &= -3(2) + 16(-1) - 4 = -26.
\end{aligned}$$

Then Equation 14.4.1 becomes

$$\begin{aligned}
z &= f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \\
z &= 34 + 13(x - 2) - 26(y - (-1)) \\
z &= 34 + 13x - 26 - 26y - 26 \\
z &= 13x - 26y - 18.
\end{aligned}$$

(See the following figure).

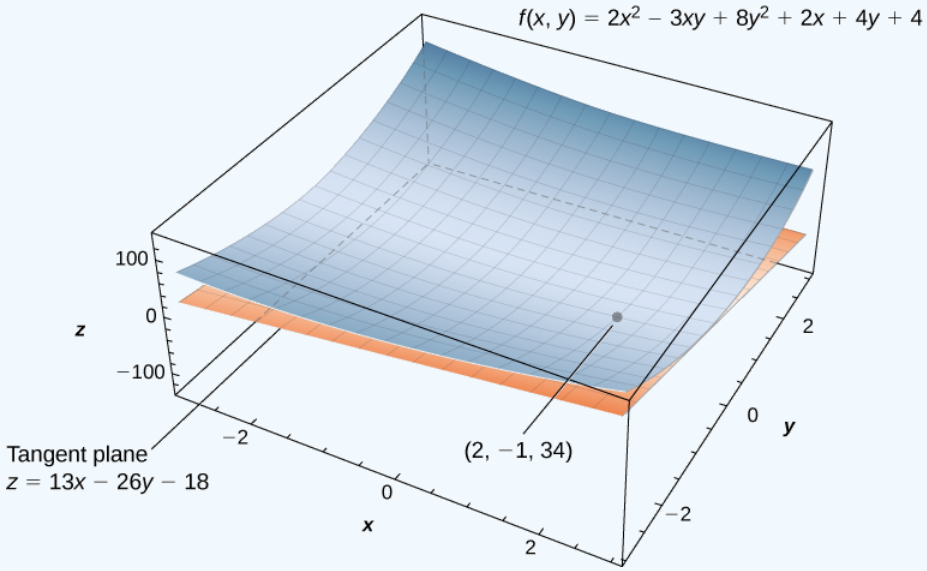


Figure 14.4.2: Calculating the equation of a tangent plane to a given surface at a given point.

Exercise 14.4.1

Find the equation of the tangent plane to the surface defined by the function $f(x,y) = x^3 - x^2y + y^2 - 2x + 3y - 2$ at point $(-1,3)$.

Hint

First, calculate $f_x(x,y)$ and $f_y(x,y)$, then use Equation 14.4.1.

Answer

$z = 7x + 8y - 3$

Example 14.4.2: Finding Another Tangent Plane

Find the equation of the tangent plane to the surface defined by the function $f(x,y) = \sin(2x)\cos(3y)$ at the point $(\pi/3, \pi/4)$.

Solution

First, calculate $f_x(x,y)$ and $f_y(x,y)$, then use Equation 14.4.1 with $x_0 = \pi/3$ and $y_0 = \pi/4$:

$$f_x(x,y) = 2\cos(2x)\cos(3y)$$
$$f_y(x,y) = -3\sin(2x)\sin(3y)$$
$$f\left(\frac{\pi}{3},\frac{\pi}{4}\right) = \sin\left(2\left(\frac{\pi}{3}\right)\right)\cos\left(3\left(\frac{\pi}{4}\right)\right) = \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{6}}{4}$$
$$f_x\left(\frac{\pi}{3},\frac{\pi}{4}\right) = 2\cos\left(2\left(\frac{\pi}{3}\right)\right)\cos\left(3\left(\frac{\pi}{4}\right)\right) = 2\left(-\frac{1}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2}$$
$$f_y\left(\frac{\pi}{3},\frac{\pi}{4}\right) = -3\sin\left(2\left(\frac{\pi}{3}\right)\right)\sin\left(3\left(\frac{\pi}{4}\right)\right) = -3\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = -\frac{3\sqrt{6}}{4}.$$

Then Equation 14.4.1 becomes

$$
\begin{aligned}
z &= f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \\
&= -\frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{2}\left(x - \frac{\pi}{3}\right) - \frac{3\sqrt{6}}{4}\left(y - \frac{\pi}{4}\right) \\
&= \frac{\sqrt{2}}{2}x - \frac{3\sqrt{6}}{4}y - \frac{\sqrt{6}}{4} - \frac{\pi\sqrt{2}}{6} + \frac{3\pi\sqrt{6}}{16}
\end{aligned}
$$

A tangent plane to a surface does not always exist at every point on the surface. Consider the piecewise function

$$
f(x, y) = \begin{cases} \dfrac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}. \tag{14.4.2}
$$

The graph of this function follows.

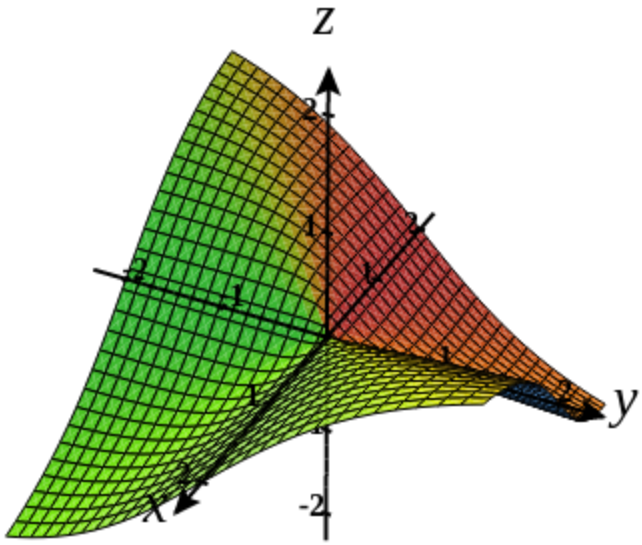


Figure 14.4.3***:*** *Graph of a function that does not have a tangent plane at the origin. Dynamic figure powered by CalcPlot3D.*

If either $x = 0$ or $y = 0$, then $f(x, y) = 0$, so the value of the function does not change on either the x- or y-axis. Therefore, $f_x(x, 0) = f_y(0, y) = 0$, so as either x or y approach zero, these partial derivatives stay equal to zero. Substituting them into Equation gives $z = 0$ as the equation of the tangent line. However, if we approach the origin from a different direction, we get a different story. For example, suppose we approach the origin along the line $y = x$. If we put $y = x$ into the original function, it becomes

$$
f(x, x) = \frac{x(x)}{\sqrt{x^2 + (x)^2}} = \frac{x^2}{\sqrt{2x^2}} = \frac{|x|}{\sqrt{2}}.
$$

When $x > 0$, the slope of this curve is equal to $\sqrt{2}/2$; when $x < 0$, the slope of this curve is equal to $-(\sqrt{2}/2)$. This presents a problem. In the definition of tangent plane, we presumed that all tangent lines through point P (in this case, the origin) lay in the

same plane. This is clearly not the case here. When we study differentiable functions, we will see that this function is not differentiable at the origin.

Linear Approximations

Recall from Linear Approximations and Differentials that the formula for the linear approximation of a function $f(x)$ at the point $x = a$ is given by

$$y \approx f(a) + f'(a)(x - a).$$

The diagram for the linear approximation of a function of one variable appears in the following graph.

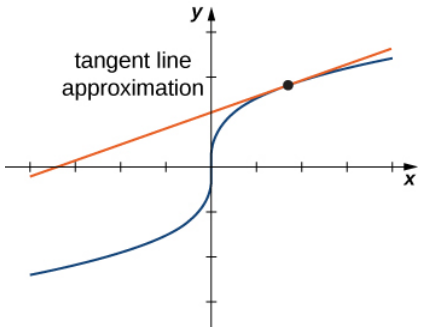


Figure 14.4.4: Linear approximation of a function in one variable.

The tangent line can be used as an approximation to the function $f(x)$ for values of x reasonably close to $x = a$. When working with a function of two variables, the tangent line is replaced by a tangent plane, but the approximation idea is much the same.

Definition: Linear Approximation

Given a function $z = f(x, y)$ with continuous partial derivatives that exist at the point (x_0, y_0), the linear approximation of f at the point (x_0, y_0) is given by the equation

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0). \tag{14.4.3}$$

Notice that this equation also represents the tangent plane to the surface defined by $z = f(x, y)$ at the point (x_0, y_0). The idea behind using a linear approximation is that, if there is a point (x_0, y_0) at which the precise value of $f(x, y)$ is known, then for values of (x, y) reasonably close to (x_0, y_0), the linear approximation (i.e., tangent plane) yields a value that is also reasonably close to the exact value of $f(x, y)$ (Figure 14.4.5). Furthermore the plane that is used to find the linear approximation is also the tangent plane to the surface at the point (x_0, y_0).

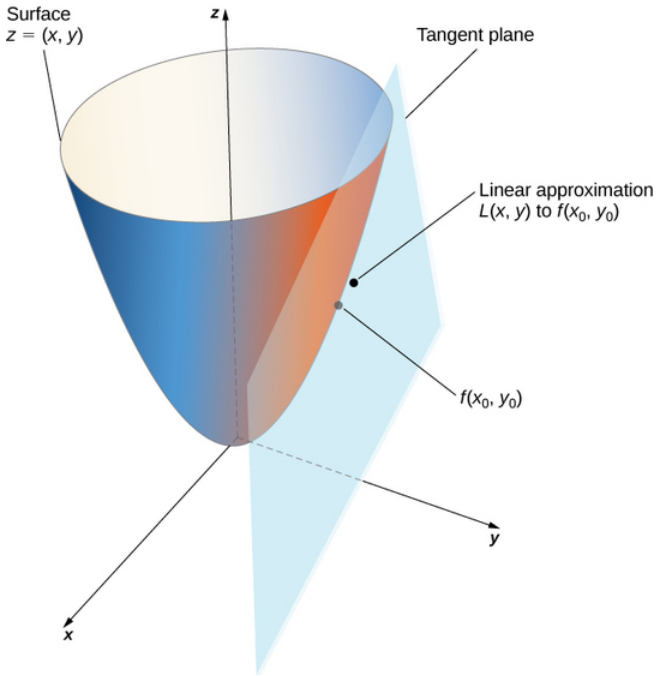


Figure 14.4.5: Using a tangent plane for linear approximation at a point.

Example 14.4.3: Using a Tangent Plane Approximation

Given the function $f(x,y) = \sqrt{41-4x^2-y^2}$, approximate $f(2.1, 2.9)$ using point $(2,3)$ for (x_0, y_0). What is the approximate value of $f(2.1, 2.9)$ to four decimal places?

Solution

To apply Equation 14.4.3, we first must calculate $f(x_0,y_0)$, $f_x(x_0,y_0)$, and $f_y(x_0,y_0)$ using $x_0 = 2$ and $y_0 = 3$:

$$f(x_0,y_0) = f(2,3) = \sqrt{41-4(2)^2-(3)^2} = \sqrt{41-16-9} = \sqrt{16} = 4$$

$$f_x(x,y) = -\frac{4x}{\sqrt{41-4x^2-y^2}} \text{ so } f_x(x_0,y_0) = -\frac{4(2)}{\sqrt{41-4(2)^2-(3)^2}} = -2$$

$$f_y(x,y) = -\frac{y}{\sqrt{41-4x^2-y^2}} \text{ so } f_y(x_0,y_0) = -\frac{3}{\sqrt{41-4(2)^2-(3)^2}} = -\frac{3}{4}.$$

Now we substitute these values into Equation 14.4.3:

$$\begin{aligned} L(x,y) &= f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) \\ &= 4 - 2(x-2) - \frac{3}{4}(y-3) \\ &= \frac{41}{4} - 2x - \frac{3}{4}y. \end{aligned}$$

Last, we substitute $x = 2.1$ and $y = 2.9$ into $L(x,y)$:

$$L(2.1, 2.9) = \frac{41}{4} - 2(2.1) - \frac{3}{4}(2.9) = 10.25 - 4.2 - 2.175 = 3.875.$$

The approximate value of $f(2.1, 2.9)$ to four decimal places is

$$f(2.1, 2.9) = \sqrt{41-4(2.1)^2-(2.9)^2} = \sqrt{14.95} \approx 3.8665,$$

which corresponds to a 0.2 error in approximation.

Exercise 14.4.2

Given the function $f(x,y) = e^{5-2x+3y}$, approximate $f(4.1, 0.9)$ using point $(4,1)$ for (x_0, y_0). Also state the approximate value of $f(4.1, 0.9)$ to four decimal places to compare the two values.

Hint

First calculate $f(x_0, y_0)$, $f_x(x_0, y_0)$, and $f_y(x_0, y_0)$ using $x_0 = 4$ and $y_0 = 1$, then use Equation 14.4.3.

Answer

$L(x,y) = 6 - 2x + 3y$, so $L(4.1, 0.9) = 6 - 2(4.1) + 3(0.9) = 0.5$
Note that $f(4.1, 0.9) = e^{5-2(4.1)+3(0.9)} = e^{-0.5} \approx 0.6065$.

Differentiability

When working with a function $y = f(x)$ of one variable, the function is said to be differentiable at a point $x = a$ if $f'(a)$ exists. Furthermore, if a function of one variable is differentiable at a point, the graph is "smooth" at that point (i.e., no corners exist) and a tangent line is well-defined at that point.

The idea behind differentiability of a function of two variables is connected to the idea of smoothness at that point. In this case, a surface is considered to be smooth at point P if a tangent plane to the surface exists at that point. If a function is differentiable at a point, then a tangent plane to the surface exists at that point. Recall the formula (Equation 14.4.1) for a tangent plane at a point (x_0, y_0) is given by

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

For a tangent plane to exist at the point (x_0, y_0), the partial derivatives must therefore exist at that point. However, this is not a sufficient condition for smoothness, as was illustrated in Figure 14.4.3. In that case, the partial derivatives existed at the origin, but the function also had a corner on the graph at the origin.

Definition: Differentiable Functions

A function $f(x,y)$ is *differentiable* at a point $P(x_0, y_0)$ if, for all points (x,y) in a δ disk around P, we can write

$$f(x,y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x,y), \tag{14.4.4}$$

where the error term E satisfies

$$\lim_{(x,y)\to(x_0,y_0)} \frac{E(x,y)}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} = 0. \tag{14.4.5}$$

The last term in Equation 14.4.4 is referred to as the **error term.** It represents how closely the tangent plane comes to the surface in a small neighborhood (δ disk) of point P. For the function f to be differentiable at P, the function must be smooth—that is, the graph of f must be close to the tangent plane for points near P.

Example 14.4.4: Demonstrating Differentiability

Show that the function $f(x,y) = 2x^2 - 4y$ is differentiable at point $(2, -3)$.

Solution

First, we calculate $f(x_0, y_0)$, $f_x(x_0, y_0)$, and $f_y(x_0, y_0)$ using $x_0 = 2$ and $y_0 = -3$, then we use Equation 14.4.4:

$$\begin{aligned} f(2,-3) &= 2(2)^2 - 4(-3) = 8 + 12 = 20 \\ f_x(2,-3) &= 4(2) = 8 \\ f_y(2,-3) &= -4. \end{aligned}$$

Therefore $m_1 = 8$ and $m_2 = -4$, and Equation 14.4.4 becomes

$$
\begin{aligned}
f(x,y) &= f(2,-3) + f_x(2,-3)(x-2) + f_y(2,-3)(y+3) + E(x,y)\\
2x^2 - 4y &= 20 + 8(x-2) - 4(y+3) + E(x,y)\\
2x^2 - 4y &= 20 + 8x - 16 - 4y - 12 + E(x,y)\\
2x^2 - 4y &= 8x - 4y - 8 + E(x,y)\\
E(x,y) &= 2x^2 - 8x + 8.
\end{aligned}
$$

Next, we calculate the limit in Equation 14.4.5:

$$
\begin{aligned}
\lim_{(x,y)\to(x_0,y_0)} \frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}} &= \lim_{(x,y)\to(2,-3)} \frac{2x^2-8x+8}{\sqrt{(x-2)^2+(y+3)^2}}\\
&= \lim_{(x,y)\to(2,-3)} \frac{2(x^2-4x+4)}{\sqrt{(x-2)^2+(y+3)^2}}\\
&= \lim_{(x,y)\to(2,-3)} \frac{2(x-2)^2}{\sqrt{(x-2)^2+(y+3)^2}}\\
&\le \lim_{(x,y)\to(2,-3)} \frac{2((x-2)^2+(y+3)^2)}{\sqrt{(x-2)^2+(y+3)^2}}\\
&= \lim_{(x,y)\to(2,-3)} 2\sqrt{(x-2)^2+(y+3)^2}\\
&= 0.
\end{aligned}
$$

Since $E(x,y) \ge 0$ for any value of x or y, the original limit must be equal to zero. Therefore, $f(x,y) = 2x^2 - 4y$ is differentiable at point $(2,-3)$.

? Exercise 14.4.3

Show that the function $f(x,y) = 3x - 4y^2$ is differentiable at point $(-1,2)$.

Hint

First, calculate $f(x_0,y_0)$, $f_x(x_0,y_0)$, and $f_y(x_0,y_0)$ using $x_0 = -1$ and $y_0 = 2$, then use Equation 14.4.5 to find $E(x,y)$.

Since you will find that $E(x,y) \le 0$, show

$$
\lim_{(x,y)\to(x_0,y_0)} \frac{|E(x,y)|}{\sqrt{(x-x_0)^2+(y-y_0)^2}} = 0
$$

Answer

$$
f(-1,2) = -19, \quad f_x(-1,2) = 3, \quad f_y(-1,2) = -16, \quad E(x,y) = -4(y-2)^2.
$$

$$
\begin{aligned}
\lim_{(x,y)\to(x_0,y_0)} \frac{|E(x,y)|}{\sqrt{(x-x_0)^2+(y-y_0)^2}} &= \lim_{(x,y)\to(-1,2)} \frac{|-4(y-2)^2|}{\sqrt{(x+1)^2+(y-2)^2}}\\
&= \lim_{(x,y)\to(-1,2)} \frac{4(y-2)^2}{\sqrt{(x+1)^2+(y-2)^2}}\\
&\le \lim_{(x,y)\to(-1,2)} \frac{4((x+1)^2+(y-2)^2)}{\sqrt{(x+1)^2+(y-2)^2}}\\
&= \lim_{(x,y)\to(-1,2)} 4\sqrt{(x+1)^2+(y-2)^2}\\
&= 0.
\end{aligned}
$$

Since $\lim_{(x,y)\to(x_0,y_0)} \frac{|E(x,y)|}{\sqrt{(x-x_0)^2+(y-y_0)^2}} = 0$, we know it's also true that

$$
\lim_{(x,y)\to(x_0,y_0)} \frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}} = 0
$$

Therefore, $f(x,y) = 3x - 4y^2$ is differentiable at point $(-1, 2)$.

This function from (Equation 14.4.2)

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

is not differentiable at the origin (Figure 14.4.3). We can see this by calculating the partial derivatives. This function appeared earlier in the section, where we showed that $f_x(0,0) = f_y(0,0) = 0$. Substituting this information into Equations 14.4.4 and 14.4.5 using $x_0 = 0$ and $y_0 = 0$, we get

$$f(x,y) = f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0) + E(x,y)$$
$$E(x,y) = \frac{xy}{\sqrt{x^2+y^2}}.$$

Calculating

$$\lim_{(x,y)\to(x_0,y_0)} \frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$$

gives

$$\lim_{(x,y)\to(x_0,y_0)} \frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}} = \lim_{(x,y)\to(0,0)} \frac{\dfrac{xy}{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}$$
$$= \lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}.$$

Depending on the path taken toward the origin, this limit takes different values. Therefore, the limit does not exist and the function f is not differentiable at the origin as shown in the following figure.

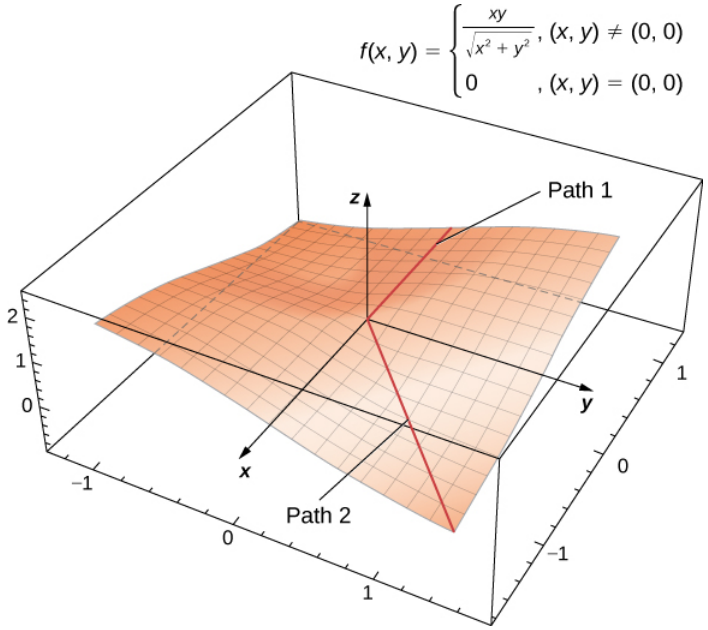


Figure 14.4.6: This function $f(x,y)$ (Equation 14.4.2) is not differentiable at the origin.

Differentiability and continuity for functions of two or more variables are connected, the same as for functions of one variable. In fact, with some adjustments of notation, the basic theorem is the same.

Theorem 14.4.1: Differentiability Implies Continuity

Let $z = f(x, y)$ be a function of two variables with (x_0, y_0) in the domain of f. If $f(x, y)$ is differentiable at (x_0, y_0), then $f(x, y)$ is continuous at (x_0, y_0).

Theorem 14.4.1 shows that if a function is differentiable at a point, then it is continuous there. However, if a function is continuous at a point, then it is not necessarily differentiable at that point. For example, the function discussed above (Equation 14.4.2)

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

is *continuous* at the origin, but it is *not differentiable* at the origin. This observation is also similar to the situation in single-variable calculus.

We can further explore the connection between continuity and differentiability at a point. This next theorem says that if the function and its partial derivatives are continuous at a point, the function is differentiable.

Theorem 14.4.2: Continuity of First Partials Implies Differentiability

Let $z = f(x, y)$ be a function of two variables with (x_0, y_0) in the domain of f. If $f(x, y)$, $f_x(x, y)$, and $f_y(x, y)$ all exist in a neighborhood of (x_0, y_0) and are continuous at (x_0, y_0), then $f(x, y)$ is differentiable there.

Recall that earlier we showed that the function in Equation 14.4.2 was not differentiable at the origin. Let's calculate the partial derivatives f_x and f_y:

$$\frac{\partial f}{\partial x} = \frac{y^3}{(x^2+y^2)^{3/2}}$$

and

$$\frac{\partial f}{\partial y} = \frac{x^3}{(x^2+y^2)^{3/2}}.$$

The contrapositive of the preceding theorem states that if a function is not differentiable, then at least one of the hypotheses must be false. Let's explore the condition that $f_x(0, 0)$ must be continuous. For this to be true, it must be true that

$$\lim_{(x,y)\to(0,0)} f_x(x,y) = f_x(0,0)$$

therefor

$$\lim_{(x,y)\to(0,0)} f_x(x,y) = \lim_{(x,y)\to(0,0)} \frac{y^3}{(x^2+y^2)^{3/2}}.$$

Let $x = ky$. Then

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{y^3}{(x^2+y^2)^{3/2}} &= \lim_{y\to 0} \frac{y^3}{((ky)^2+y^2)^{3/2}} \\ &= \lim_{y\to 0} \frac{y^3}{(k^2y^2+y^2)^{3/2}} \\ &= \lim_{y\to 0} \frac{y^3}{|y|^3(k^2+1)^{3/2}} \\ &= \frac{1}{(k^2+1)^{3/2}} \lim_{y\to 0} \frac{|y|}{y}. \end{aligned}$$

If $y > 0$, then this expression equals $1/(k^2+1)^{3/2}$; if $y < 0$, then it equals $-(1/(k^2+1)^{3/2})$. In either case, the value depends on k, so the limit fails to exist.

Differentials

In Linear Approximations and Differentials we first studied the concept of differentials. The differential of y, written dy, is defined as $f'(x)dx$. The differential is used to approximate $\Delta y = f(x + \Delta x) - f(x)$, where $\Delta x = dx$. Extending this idea to the linear approximation of a function of two variables at the point (x_0, y_0) yields the formula for the total differential for a function of two variables.

Definition: Total Differential

Let $z = f(x, y)$ be a function of two variables with (x_0, y_0) in the domain of f, and let Δx and Δy be chosen so that $(x_0 + \Delta x, y_0 + \Delta y)$ is also in the domain of f. If f is differentiable at the point (x_0, y_0), then the differentials dx and dy are defined as

$$dx = \Delta x$$

and

$$dy = \Delta y.$$

The differential dz, also called the ***total differential*** of $z = f(x, y)$ at (x_0, y_0), is defined as

$$dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy. \tag{14.4.6}$$

Notice that the symbol ∂ is not used to denote the total differential; rather, d appears in front of z. Now, let's define $\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$. We use dz to approximate Δz, so

$$\Delta z \approx dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy.$$

Therefore, the differential is used to approximate the change in the function $z = f(x_0, y_0)$ at the point (x_0, y_0) for given values of Δx and Δy. Since $\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$, this can be used further to approximate $f(x + \Delta x, y + \Delta y)$:

$$f(x + \Delta x, y + \Delta y) = f(x, y) + \Delta z \approx f(x, y) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y.$$

See the following figure.

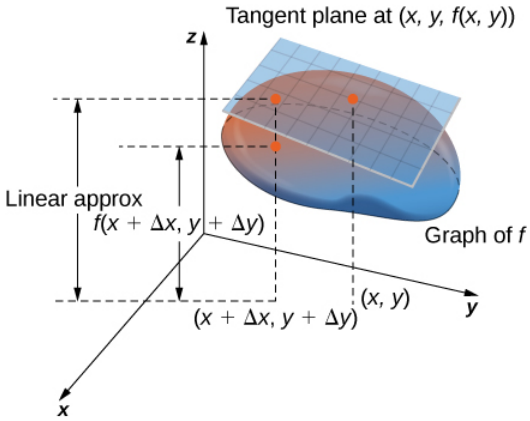


Figure 14.4.7: The linear approximation is calculated via the formula $f(x + \Delta x, y + \Delta y) \approx f(x, y) + f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y$.

One such application of this idea is to determine error propagation. For example, if we are manufacturing a gadget and are off by a certain amount in measuring a given quantity, the differential can be used to estimate the error in the total volume of the gadget.

Example 14.4.5: Approximation by Differentials

Find the differential dz of the function $f(x, y) = 3x^2 - 2xy + y^2$ and use it to approximate Δz at point $(2, -3)$. Use $\Delta x = 0.1$ and $\Delta y = -0.05$. What is the exact value of Δz?

Solution

First, we must calculate $f(x_0, y_0)$, $f_x(x_0, y_0)$, and $f_y(x_0, y_0)$ using $x_0 = 2$ and $y_0 = -3$:

$$f(x_0, y_0) = f(2, -3) = 3(2)^2 - 2(2)(-3) + (-3)^2 = 12 + 12 + 9 = 33$$
$$f_x(x, y) = 6x - 2y$$

$$f_y(x, y) = -2x + 2y$$
$$f_x(x_0, y_0) = fx(2, -3)$$
$$= 6(2) - 2(-3) = 12 + 6 = 18$$

$$f_y(x_0, y_0) = f_y(2, -3)$$
$$= -2(2) + 2(-3)$$
$$= -4 - 6 = -10.$$

Then, we substitute these quantities into Equation 14.4.6:

$$dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy$$
$$dz = 18(0.1) - 10(-0.05) = 1.8 + 0.5 = 2.3.$$

This is the approximation to $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$. The exact value of Δz is given by

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$
$$= f(2 + 0.1, -3 - 0.05) - f(2, -3)$$
$$= f(2.1, -3.05) - f(2, -3)$$
$$= 2.3425.$$

? Exercise 14.4.4

Find the differential dz of the function $f(x, y) = 4y^2 + x^2y - 2xy$ and use it to approximate Δz at point $(1, -1)$. Use $\Delta x = 0.03$ and $\Delta y = -0.02$. What is the exact value of Δz?

Hint

First, calculate $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ using $x_0 = 1$ and $y_0 = -1$, then use Equation 14.4.6.

Answer

$dz = 0.18$

$\Delta z = f(1.03, -1.02) - f(1, -1) = 0.180682$

Differentiability of a Function of Three Variables

All of the preceding results for differentiability of functions of two variables can be generalized to functions of three variables. First, the definition:

Definition: Differentiability at a Point

A function $f(x, y, z)$ is differentiable at a point $P(x_0, y_0, z_0)$ if for all points (x, y, z) in a δ disk around P we can write

$$f(x, y) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0)$$
$$+ f_z(x_0, y_0, z_0)(z - z_0) + E(x, y, z),$$

where the error term E satisfies

$$\lim_{(x,y,z)\to(x_0,y_0,z_0)} \frac{E(x, y, z)}{\sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}} = 0.$$

If a function of three variables is differentiable at a point (x_0, y_0, z_0), then it is continuous there. Furthermore, continuity of first partial derivatives at that point guarantees differentiability.

Key Concepts

- The analog of a tangent line to a curve is a tangent plane to a surface for functions of two variables.
- Tangent planes can be used to approximate values of functions near known values.
- A function is differentiable at a point if it is ”smooth” at that point (i.e., no corners or discontinuities exist at that point).
- The total differential can be used to approximate the change in a function $z = f(x_0, y_0)$ at the point (x_0, y_0) for given values of Δx and Δy.

Key Equations

- **Tangent plane**

  $$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- **Linear approximation**

  $$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- **Total differential**

  $$dz = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy.$$

- **Differentiability (two variables)**

  $$f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x, y),$$

  where the error term E satisfies

  $$\lim_{(x,y)\to(x_0,y_0)} \frac{E(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0.$$

- **Differentiability (three variables)**

  $$f(x, y) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) + E(x, y, z),$$

  where the error term E satisfies

  $$\lim_{(x,y,z)\to(x_0,y_0,z_0)} \frac{E(x, y, z)}{\sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2}} = 0.$$

Glossary

differentiable

a function $f(x, y)$ is differentiable at (x_0, y_0) if $f(x, y)$ can be expressed in the form $f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x, y)$, where the error term $E(x, y)$ satisfies $\lim_{(x,y)\to(x_0,y_0)} \frac{E(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0$

linear approximation

given a function $f(x, y)$ and a tangent plane to the function at a point (x_0, y_0), we can approximate $f(x, y)$ for points near (x_0, y_0) using the tangent plane formula

tangent plane

given a function $f(x, y)$ that is differentiable at a point (x_0, y_0), the equation of the tangent plane to the surface $z = f(x, y)$ is given by $z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

total differential

the total differential of the function $f(x, y)$ at (x_0, y_0) is given by the formula $dz = f_x(x_0, y_0)dx + fy(x_0, y_0)dy$

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14.4E: Exercises for Section 14.4

In exercises 1 - 2, find a unit normal vector to the surface at the indicated point.

1) $f(x,y) = x^3, \quad (2,-1,8)$

Answer

$(\frac{\sqrt{145}}{145})(12\hat{\mathbf{i}} - \hat{\mathbf{k}})$

2) $\ln\left(\dfrac{x}{y-z}\right) = 0$ when $x = y = 1$

In exercises 3 - 7, find a normal vector and a tangent vector at point P.

3) $x^2 + xy + y^2 = 3, \quad P(-1,-1)$

Answer

Normal vector: $\hat{\mathbf{i}} + \hat{\mathbf{j}}$, tangent vector: $\hat{\mathbf{i}} - \hat{\mathbf{j}}$

4) $(x^2+y^2)^2 = 9(x^2-y^2), \quad P(\sqrt{2},1)$

5) $xy^2 - 2x^2 + y + 5x = 6, \quad P(4,2)$

Answer

Normal vector: $7\hat{\mathbf{i}} - 17\hat{\mathbf{j}}$, tangent vector: $17\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$

6) $2x^3 - x^2y^2 = 3x - y - 7, \quad P(1,-2)$

7) $ze^{x^2-y^2} - 3 = 0, \quad P(2,2,3)$

Answer

Normal vector: $-12\hat{\mathbf{i}} + 12\hat{\mathbf{j}} - \hat{\mathbf{k}}$ or $12\hat{\mathbf{i}} - 12\hat{\mathbf{j}} + \hat{\mathbf{k}}$,
Tangent vector: $0\hat{\mathbf{i}} + 1\hat{\mathbf{j}} + 12\hat{\mathbf{k}}$ or $1\hat{\mathbf{i}} + 0\hat{\mathbf{j}} - 12\hat{\mathbf{k}}$

In exercises 8 - 19, find the equation for the tangent plane to the surface at the indicated point. (Hint: If the given function is not already solved for z, start by solving it for z in terms of x and y.)

8) $-8x - 3y - 7z = -19, \quad P(1,-1,2)$

9) $z = -9x^2 - 3y^2, \quad P(2,1,-39)$

Answer

$-36x - 6y - z = -39$

10) $x^2 + 10xyz + y^2 + 8z^2 = 0, \quad P(-1,-1,-1)$

11) $z = \ln(10x^2 + 2y^2 + 1), \quad P(0,0,0)$

Answer

$z = 0$

12) $z = e^{7x^2+4y^2}, \quad P(0,0,1)$

13) $xy + yz + zx = 11, \quad P(1,2,3)$

Answer

$5x + 4y + 3z - 22 = 0$

14) $x^2 + 4y^2 = z^2, \quad P(3,2,5)$

15) $x^3 + y^3 = 3xyz, \quad P(1, 2, \frac{3}{2})$

Answer

$4x - 5y + 4z = 0$

16) $z = axy, \quad P(1, \frac{1}{a}, 1)$

17) $z = \sin x + \sin y + \sin(x + y), \quad P(0, 0, 0)$

Answer

$2x + 2y - z = 0$

18) $h(x, y) = \ln \sqrt{x^2 + y^2}, \quad P(3, 4)$

19) $z = x^2 - 2xy + y^2, \quad P(1, 2, 1)$

Answer

$-2(x - 1) + 2(y - 2) - (z - 1) = 0$

In exercises 20 - 25, find parametric equations for the normal line to the surface at the indicated point. (Recall that to find the equation of a line in space, you need a point on the line, $P_0(x_0, y_0, z_0)$, and a vector $\vec{v} = \langle a, b, c \rangle$ that is parallel to the line. Then the equations of the line are: $x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$)

20) $-3x + 9y + 4z = -4, \quad P(1, -1, 2)$

21) $z = 5x^2 - 2y^2, \quad P(2, 1, 18)$

Answer

$x = 20t + 2, y = -4t + 1, z = -t + 18$

22) $x^2 - 8xyz + y^2 + 6z^2 = 0, \quad P(1, 1, 1)$

23) $z = \ln(3x^2 + 7y^2 + 1), \quad P(0, 0, 0)$

Answer

$x = 0, y = 0, z = t$

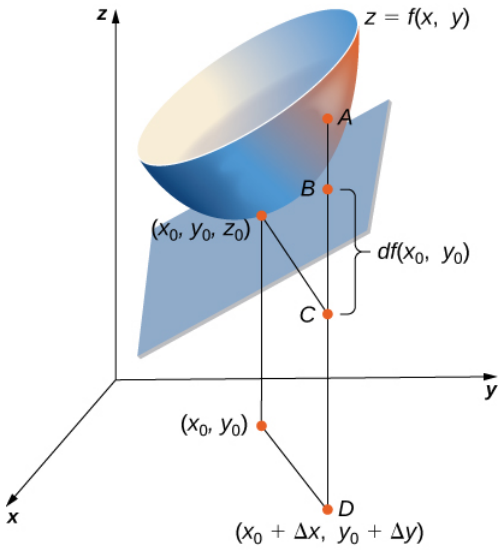
24) $z = e^{4x^2 + 6y^2}, \quad P(0, 0, 1)$

25) $z = x^2 - 2xy + y^2$ at point $P(1, 2, 1)$

Answer

$x - 1 = 2t; y - 2 = -2t; z - 1 = t$

In exercises 26 - 28, use the figure shown here.



26) The length of line segment AC is equal to what mathematical expression?

27) The length of line segment BC is equal to what mathematical expression?

Answer

The differential of the function $z(x,y) = dz = f_x dx + f_y dy$

28) Using the figure, explain what the length of line segment AB represents.

29) Show that $f(x,y) = e^{xy}x$ is differentiable at point $(1,0)$.

Answer

Using the definition of differentiability, we have $e^{xy}x \approx x + y$.

30) Show that $f(x,y) = x^2 + 3y$ is differentiable at every point. In other words, show that $\Delta z = f(x + \Delta x, y + \Delta y) - f(x,y) = f_x \Delta x + f_y \Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y$, where both ε_1 and ε_2 approach zero as $(\Delta x, \Delta y)$ approaches $(0,0)$.

Answer

$\Delta z = 2x\Delta x + 3\Delta y + (\Delta x)^2.(\Delta x)^2 \to 0$ for small Δx and z satisfies the definition of differentiability.

31) Find the total differential of each function:

a. $z = x^3 + y^3 - 5$
b. $z = e^{xy}$
c. $z = y\cos x + \sin y$
d. $P = t^2 + 3t + tu^3$
e. $w = e^y \cos(x) + z^2$

Answers:

a. $dz = 3x^2\,dx + 3y^2\,dy$
b. $dz = ye^{xy}\,dx + xe^{xy}\,dy$
c. $dz = -y\sin x\,dx + (\cos x + \cos y)\,dy$
d. $dP = (2t + 3 + u^3)\,dt + 3tu^2\,du$
e. $dw = -e^y \sin(x)\,dx + e^y \cos(x)\,dy + 2z\,dz$

32) a. Find the total differential dz of the function $z = \dfrac{xy}{y + x}$ and then

b. State its value where x changes from 10 to 10.5 and y changes from 15 to 13.

Answer

a. $dz = \dfrac{y^2}{(x+y)^2}\,dx + \dfrac{x^2}{(x+y)^2}\,dy$

b. $dx = 0.5$ and $dy = -2$ so

$$\begin{aligned} dz &= f_x(10,15)\,dx + f_y(10,15)\,dy \\ &= \frac{15^2}{25^2}\,dx + \frac{10^2}{25^2}\,dy \\ &= \frac{225}{625}\,(0.5) + \frac{100}{625}(-2) \\ &= \frac{9}{25}\left(\frac{1}{2}\right) + \frac{4}{25}(-2) \\ &= \frac{18}{100} - \frac{32}{100} \\ &= .18 - .32 = -0.14 \end{aligned}$$

33) Let $z = f(x,y) = xe^y$. State its total differential. Then compute Δz from $P(1,2)$ to $Q(1.05, 2.1)$ and then find the approximate change in z, dz, from point P to point Q. Recall $\Delta z = f(x+\Delta x, y+\Delta y) - f(x,y)$, and dz and Δz should be approximately equal, if dx and dy are both reasonably small.

Answer

Total Differential: $dz = e^y\,dx + xe^y\,dy$

$\Delta z \approx 1.185422$ and $dz \approx 1.108$. Note that they are relatively close.

34) The volume of a right circular cylinder is given by $V(r,h) = \pi r^2 h$. Find the differential dV. Interpret the formula geometrically.

Answer

$dV = 2\pi rh\,dr + \pi r^2\,dh$

35) See the preceding problem. Use differentials to estimate the amount of aluminum in an enclosed aluminum can with diameter $8.0cm$ and height $12cm$ if the aluminum is 0.04 cm thick.

Answer

$16\,\text{cm}^3$

36) Use the differential dz to approximate the change in $z = \sqrt{4 - x^2 - y^2}$ as (x,y) moves from point $(1,1)$ to point $(1.01, 0.97)$. Compare this approximation with the actual change in the function.

37) Let $z = f(x,y) = x^2 + 3xy - y^2$. Find the exact change in the function and the approximate change in the function as x changes from 2.00 to 2.05 and y changes from 3.00 to 2.96.

Answer

$\Delta z =$ exact change $= 0.6449$, approximate change is $dz = 0.65$. The two values are close.

38) The centripetal acceleration of a particle moving in a circle is given by $a(r,v) = \frac{v^2}{r}$, where v is the velocity and r is the radius of the circle. Approximate the maximum percent error in measuring the acceleration resulting from errors of 3% in v and 2% in r. (Recall that the percentage error is the ratio of the amount of error over the original amount. So, in this case, the percentage error in a is given by $\frac{da}{a}$.)

39) The radius r and height h of a right circular cylinder are measured with possible errors of 4% and 5%, respectively. Approximate the maximum possible percentage error in measuring the volume (Recall that the percentage error is the ratio of the amount of error over the original amount. So, in this case, the percentage error in V is given by $\frac{dV}{V}$.)

Answer

13% or 0.13

40) The base radius and height of a right circular cone are measured as 10 in. and 25 in., respectively, with a possible error in measurement of as much as 0.1 in. each. Use differentials to estimate the maximum error in the calculated volume of the cone.

41) The **electrical resistance** R produced by wiring resistors R_1 and R_2 in parallel can be calculated from the formula $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$. If R_1 and R_2 are measured to be 7Ω and 6Ω, respectively, and if these measurements are accurate to within 0.05Ω, estimate the maximum possible error in computing R. (The symbol Ω represents an ohm, the unit of electrical resistance.)

Answer

0.025

42) The area of an ellipse with axes of length $2a$ and $2b$ is given by the formula $A = \pi ab$. Approximate the percent change in the area when a increases by 2% and b increases by 1.5%.

43) The period T of a **simple pendulum** with small oscillations is calculated from the formula $T = 2\pi\sqrt{\frac{L}{g}}$, where L is the length of the pendulum and g is the acceleration resulting from gravity. Suppose that L and g have errors of, at most, 0.5% and 0.1%, respectively. Use differentials to approximate the maximum percentage error in the calculated value of T.

Answer

0.3%

44) **Electrical power** P is given by $P = \frac{V^2}{R}$, where V is the voltage and R is the resistance. Approximate the maximum percentage error in calculating power if $120V$ is applied to a $2000 - \Omega$ resistor and the possible percent errors in measuring V and R are 3% and 4%, respectively.

For exercises 45 - 49, find the linear approximation of each function at the indicated point.

45) $f(x, y) = x\sqrt{y}, \quad P(1, 4)$

Answer

$L(x, y) = 2x + \frac{1}{4}y - 1$

46) $f(x, y) = e^x \cos y; \quad P(0, 0)$

47) $f(x, y) = \arctan(x + 2y), \quad P(1, 0)$

Answer

$L(x, y) = \frac{1}{2}x + y + \frac{1}{4}\pi - \frac{1}{2}$

48) $f(x, y) = \sqrt{20 - x^2 - 7y^2}, \quad P(2, 1)$

49) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}, \quad P(3, 2, 6)$

Answer

$L(x, y, z) = \frac{3}{7}x + \frac{2}{7}y + \frac{6}{7}z$

50) [T] Find the equation of the tangent plane to the surface $f(x, y) = x^2 + y^2$ at point $(1, 2, 5)$, and graph the surface and the tangent plane at the point.

51) [T] Find the equation for the tangent plane to the surface at the indicated point, and graph the surface and the tangent plane: $z = \ln(10x^2 + 2y^2 + 1), \quad P(0, 0, 0)$.

Answer

$z = 0$

52) [T] Find the equation of the tangent plane to the surface $z = f(x, y) = \sin(x + y^2)$ at point $\left(\frac{\pi}{4}, 0, \frac{\sqrt{2}}{2}\right)$, and graph the surface and the tangent plane.

Contributors

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14.5: The Chain Rule for Multivariable Functions

Learning Objectives

- State the chain rules for one or two independent variables.
- Use tree diagrams as an aid to understanding the chain rule for several independent and intermediate variables.
- Perform implicit differentiation of a function of two or more variables.

In single-variable calculus, we found that one of the most useful differentiation rules is the chain rule, which allows us to find the derivative of the composition of two functions. The same thing is true for multivariable calculus, but this time we have to deal with more than one form of the chain rule. In this section, we study extensions of the chain rule and learn how to take derivatives of compositions of functions of more than one variable.

Chain Rules for One or Two Independent Variables

Recall that the chain rule for the derivative of a composite of two functions can be written in the form

$$\frac{d}{dx}\Big(f(g(x))\Big) = f'\big(g(x)\big)g'(x).$$

In this equation, both $f(x)$ and $g(x)$ are functions of one variable. Now suppose that f is a function of two variables and g is a function of one variable. Or perhaps they are both functions of two variables, or even more. How would we calculate the derivative in these cases? The following theorem gives us the answer for the case of one independent variable.

Theorem 14.5.1: Chain Rule for One Independent Variable

Suppose that $x = g(t)$ and $y = h(t)$ are differentiable functions of t and $z = f(x, y)$ is a differentiable function of x and y. Then $z = f(x(t), y(t))$ is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}, \tag{14.5.1}$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y).

Proof

The proof of this theorem uses the definition of differentiability of a function of two variables. Suppose that f is differentiable at the point $P(x_0, y_0)$, where $x_0 = g(t_0)$ and $y_0 = h(t_0)$ for a fixed value of t_0. We wish to prove that $z = f\big(x(t), y(t)\big)$ is differentiable at $t = t_0$ and that Equation 14.5.1 holds at that point as well.

Since f is differentiable at P, we know that

$$z(t) = f(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + E(x, y),$$

where

$$\lim_{(x,y)\to(x_0,y_0)} \frac{E(x, y)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} = 0.$$

We then subtract $z_0 = f(x_0, y_0)$ from both sides of this equation:

$$\begin{aligned} z(t) - z(t_0) &= f(x(t), y(t)) - f(x(t_0), y(t_0)) \\ &= f_x(x_0, y_0)(x(t) - x(t_0)) + f_y(x_0, y_0)(y(t) - y(t_0)) + E(x(t), y(t)). \end{aligned}$$

Next, we divide both sides by $t - t_0$:

$$\frac{z(t) - z(t_0)}{t - t_0} = f_x(x_0, y_0)\frac{x(t) - x(t_0)}{t - t_0} + f_y(x_0, y_0)\frac{y(t) - y(t_0)}{t - t_0} + \frac{E(x(t), y(t))}{t - t_0}.$$

Then we take the limit as t approaches t_0:

$$\lim_{t\to t_0}\frac{z(t)-z(t_0)}{t-t_0} = f_x(x_0,y_0)\lim_{t\to t_0}\left(\frac{x(t)-x(t_0)}{t-t_0}\right) + f_y(x_0,y_0)\lim_{t\to t_0}\left(\frac{y(t)-y(t_0)}{t-t_0}\right) + \lim_{t\to t_0}\frac{E(x(t),y(t))}{t-t_0}.$$

The left-hand side of this equation is equal to dz/dt, which leads to

$$\frac{dz}{dt} = f_x(x_0,y_0)\frac{dx}{dt} + f_y(x_0,y_0)\frac{dy}{dt} + \lim_{t\to t_0}\frac{E(x(t),y(t))}{t-t_0}.$$

The last term can be rewritten as

$$\begin{aligned}\lim_{t\to t_0}\frac{E(x(t),y(t))}{t-t_0} &= \lim_{t\to t_0}\frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}\frac{\sqrt{(x-x_0)^2+(y-y_0)^2}}{t-t_0})\\ &= \lim_{t\to t_0}\left(\frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}\right)\lim_{t\to t_0}\left(\frac{\sqrt{(x-x_0)^2+(y-y_0)^2}}{t-t_0}\right).\end{aligned}$$

As t approaches t_0, $\big(x(t),y(t)\big)$ approaches $\big(x(t_0),y(t_0)\big)$, so we can rewrite the last product as

$$\lim_{(x,y)\to(x_0,y_0)}\frac{E(x,y)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}\lim_{(x,y)\to(x_0,y_0)}\left(\frac{\sqrt{(x-x_0)^2+(y-y_0)^2}}{t-t_0}\right).$$

Since the first limit is equal to zero, we need only show that the second limit is finite:

$$\begin{aligned}\lim_{(x,y)\to(x_0,y_0)}\frac{\sqrt{(x-x_0)^2+(y-y_0)^2}}{t-t_0} &= \lim_{(x,y)\to(x_0,y_0)}\sqrt{\frac{(x-x_0)^2+(y-y_0)^2}{(t-t_0)^2}}\\ &= \lim_{(x,y)\to(x_0,y_0)}\sqrt{\left(\frac{x-x_0}{t-t_0}\right)^2+\left(\frac{y-y_0}{t-t_0}\right)^2}\\ &= \sqrt{\left[\lim_{(x,y)\to(x_0,y_0)}\left(\frac{x-x_0}{t-t_0}\right)\right]^2+\left[\lim_{(x,y)\to(x_0,y_0)}\left(\frac{y-y_0}{t-t_0}\right)\right]^2}.\end{aligned}$$

Since $x(t)$ and $y(t)$ are both differentiable functions of t, both limits inside the last radical exist. Therefore, this value is finite. This proves the chain rule at $t=t_0$; the rest of the theorem follows from the assumption that all functions are differentiable over their entire domains.

□

Closer examination of Equation 14.5.1 reveals an interesting pattern. The first term in the equation is $\frac{\partial f}{\partial x}\cdot\frac{dx}{dt}$ and the second term is $\frac{\partial f}{\partial y}\cdot\frac{dy}{dt}$. Recall that when multiplying fractions, cancelation can be used. If we treat these derivatives as fractions, then each product "simplifies" to something resembling $\partial f/dt$. The variables x and y that disappear in this simplification are often called **intermediate variables**: they are independent variables for the function f, but are dependent variables for the variable t. Two terms appear on the right-hand side of the formula, and f is a function of two variables. This pattern works with functions of more than two variables as well, as we see later in this section.

Example 14.5.1: Using the Chain Rule

Calculate dz/dt for each of the following functions:

a. $z=f(x,y)=4x^2+3y^2, \quad x=x(t)=\sin t, \quad y=y(t)=\cos t$
b. $z=f(x,y)=\sqrt{x^2-y^2}, \quad x=x(t)=e^{2t}, \quad y=y(t)=e^{-t}$

Solution

a. To use the chain rule, we need four quantities—$\partial z/\partial x,\ \partial z/\partial y,\ dx/dt$, and dy/dt:

- $\frac{\partial z}{\partial x} = 8x$
- $\frac{dx}{dt} = \cos t$
- $\frac{\partial z}{\partial y} = 6y$
- $\frac{dy}{dt} = -\sin t$

Now, we substitute each of these into Equation 14.5.1:

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \\ &= (8x)(\cos t) + (6y)(-\sin t) \\ &= 8x\cos t - 6y\sin t.\end{aligned}$$

This answer has three variables in it. To reduce it to one variable, use the fact that $x(t) = \sin t$ and $y(t) = \cos t$. We obtain

$$\begin{aligned}\frac{dz}{dt} &= 8x\cos t - 6y\sin t \\ &= 8(\sin t)\cos t - 6(\cos t)\sin t \\ &= 2\sin t\cos t.\end{aligned}$$

This derivative can also be calculated by first substituting $x(t)$ and $y(t)$ into $f(x, y)$, then differentiating with respect to t:

$$\begin{aligned}z = f(x, y) &= f\big(x(t), y(t)\big) \\ &= 4(x(t))^2 + 3(y(t))^2 \\ &= 4\sin^2 t + 3\cos^2 t.\end{aligned}$$

Then

$$\begin{aligned}\frac{dz}{dt} &= 2(4\sin t)(\cos t) + 2(3\cos t)(-\sin t) \\ &= 8\sin t\cos t - 6\sin t\cos t \\ &= 2\sin t\cos t,\end{aligned}$$

which is the same solution. However, it may not always be this easy to differentiate in this form.

b. To use the chain rule, we again need four quantities—$\partial z/\partial x, \partial z/dy, dx/dt$, and dy/dt :

- $\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 - y^2}}$
- $\frac{dx}{dt} = 2e^{2t}$
- $\frac{\partial z}{\partial y} = \frac{-y}{\sqrt{x^2 - y^2}}$
- $\frac{dy}{dt} = -e^{-t}.$

We substitute each of these into Equation 14.5.1:

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \\ &= \left(\frac{x}{\sqrt{x^2 - y^2}}\right)(2e^{2t}) + \left(\frac{-y}{\sqrt{x^2 - y^2}}\right)(-e^{-t}) \\ &= \frac{2xe^{2t} - ye^{-t}}{\sqrt{x^2 - y^2}}.\end{aligned}$$

To reduce this to one variable, we use the fact that $x(t) = e^{2t}$ and $y(t) = e^{-t}$. Therefore,

$$\begin{aligned}\frac{dz}{dt} &= \frac{2xe^2t + ye^{-t}}{\sqrt{x^2 - y^2}} \\ &= \frac{2(e^{2t})e^{2t} + (e^{-t})e^{-t}}{\sqrt{e^{4t} - e^{-2t}}} \\ &= \frac{2e^{4t} + e^{-2t}}{\sqrt{e^{4t} - e^{-2t}}}.\end{aligned}$$

To eliminate negative exponents, we multiply the top by e^{2t} and the bottom by $\sqrt{e^{4t}}$:

$$\begin{aligned}\frac{dz}{dt} &= \frac{2e^{4t} + e^{-2t}}{\sqrt{e^{4t} - e^{-2t}}} \cdot \frac{e^{2t}}{\sqrt{e^{4t}}} \\ &= \frac{2e^{6t} + 1}{\sqrt{e^{8t} - e^{2t}}} \\ &= \frac{2e^{6t} + 1}{\sqrt{e^{2t}(e^{6t} - 1)}} \\ &= \frac{2e^{6t} + 1}{e^t\sqrt{e^{6t} - 1}}.\end{aligned}$$

Again, this derivative can also be calculated by first substituting $x(t)$ and $y(t)$ into $f(x, y)$, then differentiating with respect to t:

$$\begin{aligned}z &= f(x, y) \\ &= f(x(t), y(t)) \\ &= \sqrt{(x(t))^2 - (y(t))^2} \\ &= \sqrt{e^{4t} - e^{-2t}} \\ &= (e^{4t} - e^{-2t})^{1/2}.\end{aligned}$$

Then

$$\begin{aligned}\frac{dz}{dt} &= \frac{1}{2}(e^{4t} - e^{-2t})^{-1/2}\left(4e^{4t} + 2e^{-2t}\right) \\ &= \frac{2e^{4t} + e^{-2t}}{\sqrt{e^{4t} - e^{-2t}}}.\end{aligned}$$

This is the same solution.

? Exercise 14.5.1

Calculate dz/dt given the following functions. Express the final answer in terms of t.

$$\begin{aligned}z = f(x, y) &= x^2 - 3xy + 2y^2, \\ x = x(t) &= 3\sin 2t, \\ y = y(t) &= 4\cos 2t\end{aligned}$$

Hint

Calculate $\partial z/\partial x, \partial z/dy, dx/dt$, and dy/dt, then use Equation 14.5.1.

Answer

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} \\ &= (2x - 3y)(6\cos 2t) + (-3x + 4y)(-8\sin 2t) \\ &= -92\sin 2t\cos 2t - 72(\cos^2 2t - \sin^2 2t) \\ &= -46\sin 4t - 72\cos 4t.\end{aligned}$$

It is often useful to create a visual representation of Equation 14.5.1 for the chain rule. This is called a **tree diagram** for the chain rule for functions of one variable and it provides a way to remember the formula (Figure 14.5.1). This diagram can be expanded for functions of more than one variable, as we shall see very shortly.

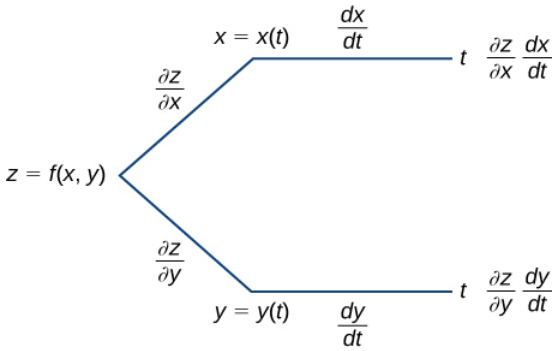


Figure 14.5.1: Tree diagram for the case $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$.

In this diagram, the leftmost corner corresponds to $z = f(x, y)$. Since f has two **independent variables**, there are two lines coming from this corner. The upper branch corresponds to the variable x and the lower branch corresponds to the variable y. Since each of these variables is then dependent on one variable t, one branch then comes from x and one branch comes from y. Last, each of the branches on the far right has a label that represents the path traveled to reach that branch. The top branch is reached by following the x branch, then the t branch; therefore, it is labeled $(\partial z/\partial x) \times (dx/dt)$. The bottom branch is similar: first the y branch, then the t branch. This branch is labeled $(\partial z/\partial y) \times (dy/dt)$. To get the formula for dz/dt, add all the terms that appear on the rightmost side of the diagram. This gives us Equation.

In Chain Rule for Two Independent Variables, $z = f(x, y)$ is a function of x and y, and both $x = g(u, v)$ and $y = h(u, v)$ are functions of the independent variables u and v.

Theorem 14.5.2: Chain Rule for Two Independent Variables

Suppose $x = g(u, v)$ and $y = h(u, v)$ are differentiable functions of u and v, and $z = f(x, y)$ is a differentiable function of x and y. Then, $z = f(g(u, v), h(u, v))$ is a differentiable function of u and v, and

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \tag{14.5.2}$$

and

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}. \tag{14.5.3}$$

We can draw a tree diagram for each of these formulas as well as follows.

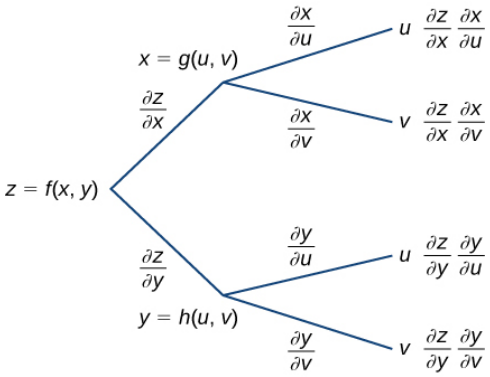


Figure 14.5.2: Tree diagram for $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$ and $\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$.

To derive the formula for $\partial z/\partial u$, start from the left side of the diagram, then follow only the branches that end with u and add the terms that appear at the end of those branches. For the formula for $\partial z/\partial v$, follow only the branches that end with v and add the terms that appear at the end of those branches.

There is an important difference between these two chain rule theorems. In Chain Rule for One Independent Variable, the left-hand side of the formula for the derivative is not a partial derivative, but in Chain Rule for Two Independent Variables it is. The reason is that, in Chain Rule for One Independent Variable, z is ultimately a function of t alone, whereas in Chain Rule for Two Independent Variables, z is a function of both u and v.

Example 14.5.2: Using the Chain Rule for Two Variables

Calculate $\partial z/\partial u$ and $\partial z/\partial v$ using the following functions:

$$z = f(x,y) = 3x^2 - 2xy + y^2,\ x = x(u,v) = 3u + 2v,\ y = y(u,v) = 4u - v.$$

Solution

To implement the chain rule for two variables, we need six partial derivatives—$\partial z/\partial x$, $\partial z/\partial y$, $\partial x/\partial u$, $\partial x/\partial v$, $\partial y/\partial u$, and $\partial y/\partial v$:

$$\frac{\partial z}{\partial x} = 6x - 2y \qquad \frac{\partial z}{\partial y} = -2x + 2y$$
$$\frac{\partial x}{\partial u} = 3 \qquad \frac{\partial x}{\partial v} = 2$$
$$\frac{\partial y}{\partial u} = 4 \qquad \frac{\partial y}{\partial v} = -1.$$

To find $\partial z/\partial u$, we use Equation 14.5.2:

$$\begin{aligned}\frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u} \\ &= 3(6x - 2y) + 4(-2x + 2y) \\ &= 10x + 2y.\end{aligned}$$

Next, we substitute $x(u,v) = 3u + 2v$ and $y(u,v) = 4u - v$:

$$\begin{aligned}\frac{\partial z}{\partial u} &= 10x + 2y \\ &= 10(3u + 2v) + 2(4u - v) \\ &= 38u + 18v.\end{aligned}$$

To find $\partial z/\partial v$, we use Equation 14.5.3:

$$\begin{aligned}\frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} \\ &= 2(6x - 2y) + (-1)(-2x + 2y) \\ &= 14x - 6y.\end{aligned}$$

Then we substitute $x(u,v) = 3u + 2v$ and $y(u,v) = 4u - v$:

$$\begin{aligned}\frac{\partial z}{\partial v} &= 14x - 6y \\ &= 14(3u + 2v) - 6(4u - v) \\ &= 18u + 34v\end{aligned}$$

Exercise 14.5.2

Calculate $\partial z/\partial u$ and $\partial z/\partial v$ given the following functions:

$$z = f(x,y) = \frac{2x-y}{x+3y}, \quad x(u,v) = e^{2u}\cos 3v, \quad y(u,v) = e^{2u}\sin 3v.$$

Hint

Calculate $\partial z/\partial x$, $\partial z/\partial y$, $\partial x/\partial u$, $\partial x/\partial v$, $\partial y/\partial u$, and $\partial y/\partial v$, then use Equation 14.5.2 and Equation 14.5.3.

Answer

$$\frac{\partial z}{\partial u} = 0, \quad \frac{\partial z}{\partial v} = \frac{-21}{(3\sin 3v + \cos 3v)^2}$$

The Generalized Chain Rule

Now that we've see how to extend the original chain rule to functions of two variables, it is natural to ask: Can we extend the rule to more than two variables? The answer is yes, as the *generalized chain rule* states.

Theorem 14.5.3: Generalized Chain Rule

Let $w = f(x_1, x_2, \ldots, x_m)$ be a differentiable function of m independent variables, and for each $i \in 1, \ldots, m$, let $x_i = x_i(t_1, t_2, \ldots, t_n)$ be a differentiable function of n independent variables. Then

$$\frac{\partial w}{\partial t_j} = \frac{\partial w}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial w}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial w}{\partial x_m}\frac{\partial x_m}{\partial t_j}$$

for any $j \in 1, 2, \ldots, n$.

In the next example we calculate the derivative of a function of three independent variables in which each of the three variables is dependent on two other variables.

Example 14.5.3: Using the Generalized Chain Rule

Calculate $\partial w/\partial u$ and $\partial w/\partial v$ using the following functions:

$$\begin{aligned} w &= f(x,y,z) = 3x^2 - 2xy + 4z^2 \\ x &= x(u,v) = e^u \sin v \\ y &= y(u,v) = e^u \cos v \\ z &= z(u,v) = e^u. \end{aligned}$$

Solution

The formulas for $\partial w/\partial u$ and $\partial w/\partial v$ are

$$\begin{aligned} \frac{\partial w}{\partial u} &= \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial u} \\ \frac{\partial w}{\partial v} &= \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial v}. \end{aligned}$$

Therefore, there are nine different partial derivatives that need to be calculated and substituted. We need to calculate each of them:

$$\begin{aligned} &\frac{\partial w}{\partial x} = 6x - 2y \quad \frac{\partial w}{\partial y} = -2x \quad \frac{\partial w}{\partial z} = 8z \\ &\frac{\partial x}{\partial u} = e^u \sin v \quad \frac{\partial y}{\partial u} = e^u \cos v \quad \frac{\partial z}{\partial u} = e^u \\ &\frac{\partial x}{\partial v} = e^u \cos v \quad \frac{\partial y}{\partial v} = -e^u \sin v \quad \frac{\partial z}{\partial v} = 0. \end{aligned}$$

Now, we substitute each of them into the first formula to calculate $\partial w/\partial u$:

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial u}$$
$$= (6x - 2y)e^u \sin v - 2xe^u \cos v + 8ze^u,$$

then substitute $x(u,v) = e^u \sin v$, $y(u,v) = e^u \cos v$, and $z(u,v) = e^u$ into this equation:

$$\begin{aligned}\frac{\partial w}{\partial u} &= (6x - 2y)e^u \sin v - 2xe^u \cos v + 8ze^u \\ &= (6e^u \sin v - 2eu \cos v)e^u \sin v - 2(e^u \sin v)e^u \cos v + 8e^{2u} \\ &= 6e^{2u}\sin^2 v - 4e^{2u}\sin v \cos v + 8e^{2u} \\ &= 2e^{2u}(3\sin^2 v - 2\sin v \cos v + 4).\end{aligned}$$

Next, we calculate $\partial w/\partial v$:

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial v}$$
$$= (6x - 2y)e^u \cos v - 2x(-e^u \sin v) + 8z(0),$$

then we substitute $x(u,v) = e^u \sin v$, $y(u,v) = e^u \cos v$, and $z(u,v) = e^u$ into this equation:

$$\begin{aligned}\frac{\partial w}{\partial v} &= (6x - 2y)e^u \cos v - 2x(-e^u \sin v) \\ &= (6e^u \sin v - 2e^u \cos v)e^u \cos v + 2(e^u \sin v)(e^u \sin v) \\ &= 2e^{2u}\sin^2 v + 6e^{2u}\sin v \cos v - 2e^{2u}\cos^2 v \\ &= 2e^{2u}(\sin^2 v + \sin v \cos v - \cos^2 v).\end{aligned}$$

? Exercise 14.5.3

Calculate $\partial w/\partial u$ and $\partial w/\partial v$ given the following functions:

$$w = f(x,y,z) = \frac{x + 2y - 4z}{2x - y + 3z}$$
$$x = x(u,v) = e^{2u}\cos 3v$$
$$y = y(u,v) = e^{2u}\sin 3v$$
$$z = z(u,v) = e^{2u}.$$

Hint

Calculate nine partial derivatives, then use the same formulas from Example 14.5.3.

Answer

$$\frac{\partial w}{\partial u} = 0$$
$$\frac{\partial w}{\partial v} = \frac{15 - 33\sin 3v + 6\cos 3v}{(3 + 2\cos 3v - \sin 3v)^2}$$

✓ Example 14.5.4: Drawing a Tree Diagram

Create a tree diagram for the case when

$$w = f(x,y,z), \quad x = x(t,u,v), \quad y = y(t,u,v), \quad z = z(t,u,v)$$

and write out the formulas for the three partial derivatives of w.

Solution

Starting from the left, the function f has three independent variables: x, y, and z. Therefore, three branches must be emanating from the first node. Each of these three branches also has three branches, for each of the variables t, u, and v.

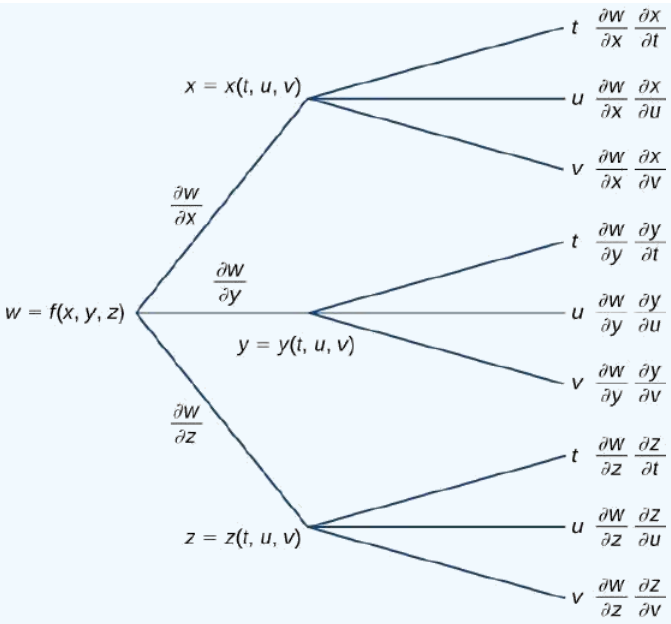


Figure 14.5.3: Tree diagram for a function of three variables, each of which is a function of three independent variables.

The three formulas are

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$
$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u}$$
$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v}.$$

? Exercise 14.5.4

Create a tree diagram for the case when

$$w = f(x, y), \quad x = x(t, u, v), \quad y = y(t, u, v)$$

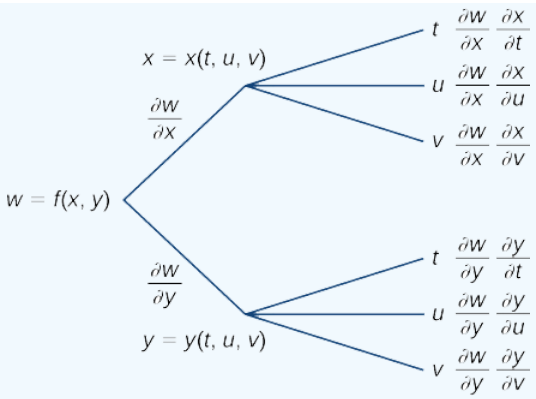
and write out the formulas for the three partial derivatives of w.

Hint

Determine the number of branches that emanate from each node in the tree.

Answer

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t}$$
$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u}$$
$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v}$$



Implicit Differentiation

Recall from implicit differentiation provides a method for finding dy/dx when y is defined implicitly as a function of x. The method involves differentiating both sides of the equation defining the function with respect to x, then solving for dy/dx. Partial derivatives provide an alternative to this method.

Consider the ellipse defined by the equation $x^2 + 3y^2 + 4y - 4 = 0$ as follows.

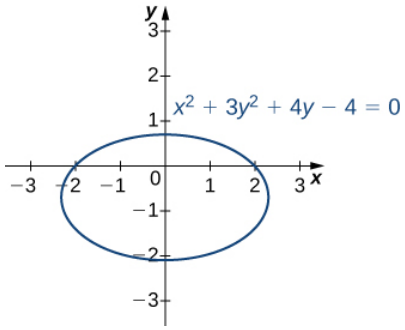


Figure 14.5.4: Graph of the ellipse defined by $x^2 + 3y^2 + 4y - 4 = 0$.

This equation implicitly defines y as a function of x. As such, we can find the derivative dy/dx using the method of implicit differentiation:

$$\begin{aligned} \frac{d}{dx}(x^2 + 3y^2 + 4y - 4) &= \frac{d}{dx}(0) \\ 2x + 6y\frac{dy}{dx} + 4\frac{dy}{dx} &= 0 \\ (6y + 4)\frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{x}{3y + 2} \end{aligned}$$

We can also define a function $z = f(x, y)$ by using the left-hand side of the equation defining the ellipse. Then $f(x, y) = x^2 + 3y^2 + 4y - 4$. The ellipse $x^2 + 3y^2 + 4y - 4 = 0$ can then be described by the equation $f(x, y) = 0$. Using this function and the following theorem gives us an alternative approach to calculating dy/dx.

Theorem 14.5.4: Implicit Differentiation of a Function of Two or More Variables

Suppose the function $z = f(x, y)$ defines y implicitly as a function $y = g(x)$ of x via the equation $f(x, y) = 0$. Then

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} \tag{14.5.4}$$

provided $f_y(x, y) \neq 0$.

If the equation $f(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y, then

$$\frac{dz}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial z} \quad \text{and} \quad \frac{dz}{dy} = -\frac{\partial f/\partial y}{\partial f/\partial z} \tag{14.5.5}$$

as long as $f_z(x, y, z) \neq 0$.

Equation 14.5.4 is a direct consequence of Equation 14.5.2. In particular, if we assume that y is defined implicitly as a function of x via the equation $f(x, y) = 0$, we can apply the chain rule to find dy/dx :

$$\begin{aligned} \frac{d}{dx} f(x, y) &= \frac{d}{dx}(0) \\ \frac{\partial f}{\partial x} \cdot \frac{dx}{dx} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} &= 0 \\ \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} &= 0. \end{aligned}$$

Solving this equation for dy/dx gives Equation 14.5.4. Equation 14.5.4 can be derived in a similar fashion.

Let's now return to the problem that we started before the previous theorem. Using Theorem 14.5.4 and the function $f(x, y) = x^2 + 3y^2 + 4y - 4$, we obtain

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2x \\ \frac{\partial f}{\partial y} &= 6y + 4. \end{aligned}$$

Then Equation 14.5.4 gives

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} = -\frac{2x}{6y + 4} = -\frac{x}{3y + 2},$$

which is the same result obtained by the earlier use of implicit differentiation.

Example 14.5.5: Implicit Differentiation by Partial Derivatives

a. Calculate dy/dx if y is defined implicitly as a function of x via the equation $3x^2 - 2xy + y^2 + 4x - 6y - 11 = 0$. What is the equation of the tangent line to the graph of this curve at point $(2, 1)$?

b. Calculate $\partial z/\partial x$ and $\partial z/\partial y$, given $x^2 e^y - yze^x = 0$.

Solution

a. Set $f(x, y) = 3x^2 - 2xy + y^2 + 4x - 6y - 11 = 0$, then calculate f_x and f_y : $f_x(x, y) = 6x - 2y + 4$ and $f_y(x, y) = -2x + 2y - 6$.

The derivative is given by

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y} = \frac{6x - 2y + 4}{-2x + 2y - 6} = \frac{3x - y + 2}{x - y + 3}.$$

The slope of the tangent line at point $(2, 1)$ is given by

$$\left.\frac{dy}{dx}\right|_{(x,y)=(2,1)} = \frac{3(2) - 1 + 2}{2 - 1 + 3} = \frac{7}{4}$$

To find the equation of the tangent line, we use the point-slope form (Figure 14.5.5):

$$\begin{aligned} y - y_0 &= m(x - x_0) \\ y - 1 &= \frac{7}{4}(x - 2) \\ y &= \frac{7}{4}x - \frac{7}{2} + 1 \\ y &= \frac{7}{4}x - \frac{5}{2}. \end{aligned}$$

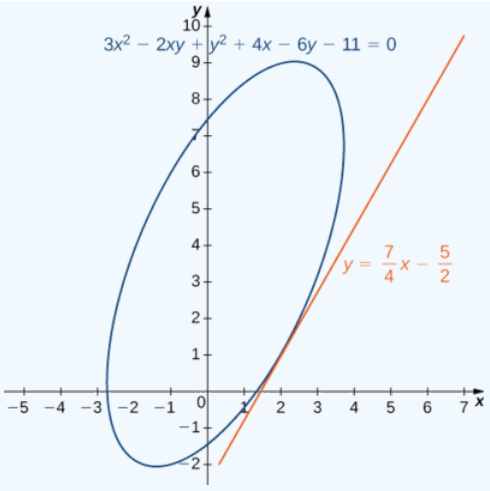


Figure 14.5.5: Graph of the rotated ellipse defined by $3x^2 - 2xy + y^2 + 4x - 6y - 11 = 0$.

b. We have $f(x, y, z) = x^2e^y - yze^x$. Therefore,

$$\frac{\partial f}{\partial x} = 2xe^y - yze^x$$
$$\frac{\partial f}{\partial y} = x^2e^y - ze^x$$
$$\frac{\partial f}{\partial z} = -ye^x$$

Using Equation 14.5.5,

$$\frac{\partial z}{\partial x} = -\frac{\partial f/\partial x}{\partial f/\partial y} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\partial f/\partial y}{\partial f/\partial z}$$
$$= -\frac{2xe^y - yze^x}{-ye^x} \qquad = -\frac{x^2e^y - ze^x}{-ye^x}$$
$$= \frac{2xe^y - yze^x}{ye^x} \qquad = \frac{x^2e^y - ze^x}{ye^x}$$

? Exercise 14.5.5

Find dy/dx if y is defined implicitly as a function of x by the equation $x^2 + xy - y^2 + 7x - 3y - 26 = 0$. What is the equation of the tangent line to the graph of this curve at point $(3, -2)$?

Hint

Calculate $\partial f/dx$ and $\partial f/dy$, then use Equation 14.5.4.

Solution

$$\frac{dy}{dx} = \frac{2x + y + 7}{2y - x + 3}\Big|_{(3,-2)} = \frac{2(3) + (-2) + 7}{2(-2) - (3) + 3} = -\frac{11}{4}$$

Equation of the tangent line: $y = -\frac{11}{4}x + \frac{25}{4}$

Key Concepts

- The chain rule for functions of more than one variable involves the partial derivatives with respect to all the independent variables.
- Tree diagrams are useful for deriving formulas for the chain rule for functions of more than one variable, where each independent variable also depends on other variables.

Key Equations

- **Chain rule, one independent variable**

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

- **Chain rule, two independent variables**

$$\frac{dz}{du} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \quad \frac{dz}{dv} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

- **Generalized chain rule**

$$\frac{\partial w}{\partial t_j} = \frac{\partial w}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial w}{\partial x_2} \frac{\partial x_1}{\partial t_j} + \cdots + \frac{\partial w}{\partial x_m} \frac{\partial x_m}{\partial t_j}$$

Glossary

generalized chain rule

the chain rule extended to functions of more than one independent variable, in which each independent variable may depend on one or more other variables

intermediate variable

given a composition of functions (e.g., $f(x(t), y(t))$), the intermediate variables are the variables that are independent in the outer function but dependent on other variables as well; in the function $f(x(t), y(t))$, the variables x and y are examples of intermediate variables

tree diagram

illustrates and derives formulas for the generalized chain rule, in which each independent variable is accounted for

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14.5E: Exercises for Section 14.5

In exercises 1 - 6, use the information provided to solve the problem.

1) Let $w(x,y,z) = xy\cos z$, where $x = t, y = t^2$, and $z = \arcsin t$. Find $\frac{dw}{dt}$.

Answer

$$\frac{dw}{dt} = y\cos z + x\cos z(2t) - \frac{xy\sin z}{\sqrt{1-t^2}}$$

2) Let $w(t,v) = e^{tv}$ where $t = r + s$ and $v = rs$. Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$.

3) If $w = 5x^2 + 2y^2, \quad x = -3u + v$, and $y = u - 4v$, find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$.

Answer

$$\frac{\partial w}{\partial u} = -30x + 4y \quad = \quad -30(-3u+v) + 4(u-4v) \quad = \quad 90u - 30v + 4u - 16v \quad = \quad 94u - 46v,$$
$$\frac{\partial w}{\partial v} = 10x - 16y \quad = \quad 10(-3u+v) - 16(u-4v) \quad = \quad -30u + 10v - 16u + 64v \quad = \quad -46u + 74v$$

4) If $w = xy^2, x = 5\cos(2t)$, and $y = 5\sin(2t)$, find $\frac{\partial w}{\partial t}$.

5) If $f(x,y) = xy, x = r\cos\theta$, and $y = r\sin\theta$, find $\frac{\partial f}{\partial r}$ and express the answer in terms of r and θ.

Answer

$$\frac{\partial f}{\partial r} = r\sin(2\theta)$$

6) Suppose $f(x,y) = x + y, u = e^x\sin y, \quad x = t^2$ and $y = \pi t$, where $x = r\cos\theta$ and $y = r\sin\theta$. Find $\frac{\partial f}{\partial \theta}$.

In exercises 7 - 12, find $\frac{dz}{dt}$ in two ways, first using the chain rule and then by direct substitution.

7) $z = x^2 + y^2, \quad x = t, y = t^2$

Answer

$$\frac{dz}{dt} = 2t + 4t^3$$

8) $z = \sqrt{x^2 + y^2}, \quad y = t^2, x = t$

9) $z = xy, \quad x = 1 - \sqrt{t}, y = 1 + \sqrt{t}$

Answer

$$\frac{dz}{dt} = -1$$

10) $z = \frac{x}{y}, \quad x = e^t, y = 2e^t$

11) $z = \ln(x + y), \quad x = e^t, y = e^t$

Answer

$$\frac{dz}{dt} = 1$$

12) $z = x^4, \quad x = t, y = t$

13) Let $w(x,y,z) = x^2 + y^2 + z^2, \quad x = cost, y = sint$, and $z = e^t$. Express w as a function of t and find $\frac{dw}{dt}$ directly. Then, find $\frac{dw}{dt}$ using the chain rule.

Answer

$\frac{dw}{dt} = 2e^{2t}$ in both cases

14) Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Find $\frac{dz}{dt}$.

15) Let $u = e^x \sin y$, where $x = -\ln 2t$ and $y = \pi t$. Find $\frac{du}{dt}$ when $x = \ln 2$ and $y = \frac{\pi}{4}$.

Answer

$\frac{du}{dt} = \sqrt{2}(\pi - 4)$

In exercises 16 - 33, find $\frac{dy}{dx}$ using partial derivatives.

16) $\sin(6x) + \tan(8y) + 5 = 0$

17) $x^3 + y^2x - 3 = 0$

Answer

$\frac{dy}{dx} = -\frac{3x^2 + y^2}{2xy}$

18) $\sin(x + y) + \cos(x - y) = 4$

19) $x^2 - 2xy + y^4 = 4$

Answer

$\frac{dy}{dx} = \frac{y - x}{-x + 2y^3}$

20) $xe^y + ye^x - 2x^2y = 0$

21) $x^{2/3} + y^{2/3} = a^{2/3}$

Answer

$\frac{dy}{dx} = -\sqrt[3]{\frac{y}{x}}$

22) $x\cos(xy) + y\cos x = 2$

23) $e^{xy} + ye^y = 1$

Answer

$\frac{dy}{dx} = -\frac{ye^{xy}}{xe^{xy} + e^y(1 + y)}$

24) $x^2y^3 + \cos y = 0$

25) Find $\frac{dz}{dt}$ using the chain rule where $z = 3x^2y^3$, $x = t^4$, and $y = t^2$.

Answer

$\frac{dz}{dt} = 42t^{13}$

26) Let $z = 3\cos x - \sin(xy), x = \frac{1}{t}$, and $y = 3t$. Find $\frac{dz}{dt}$.

27) Let $z = e^{1-xy}$, $x = t^{1/3}$, and $y = t^3$. Find $\frac{dz}{dt}$.

Answer

$\frac{dz}{dt} = -\frac{10}{3}t^{7/3} \times e^{1-t^{10/3}}$

28) Find $\frac{dz}{dt}$ by the chain rule where $z = \cosh^2(xy)$, $x = \frac{1}{2}t$, and $y = e^t$.

29) Let $z = \dfrac{x}{y},\ x = 2\cos u$, and $y = 3\sin v$. Find $\dfrac{\partial z}{\partial u}$ and $\dfrac{\partial z}{\partial v}$.

Answer

$\dfrac{\partial z}{\partial u} = \dfrac{-2\sin u}{3\sin v}$ and $\dfrac{\partial z}{\partial v} = \dfrac{-2\cos u\cos v}{3\sin^2 v}$

30) Let $z = e^{x^2y}$, where $x = \sqrt{uv}$ and $y = \frac{1}{v}$. Find $\dfrac{\partial z}{\partial u}$ and $\dfrac{\partial z}{\partial v}$.

31) If $z = xye^{x/y},\ x = r\cos\theta$, and $y = r\sin\theta$, find $\dfrac{\partial z}{\partial r}$ and $\dfrac{\partial z}{\partial \theta}$ when $r = 2$ and $\theta = \frac{\pi}{6}$.

Answer

$\dfrac{\partial z}{\partial r} = \sqrt{3}e^{\sqrt{3}}, \dfrac{\partial z}{\partial \theta} = (2 - 4\sqrt{3})e^{\sqrt{3}}$

32) Find $\dfrac{\partial w}{\partial s}$ if $w = 4x + y^2 + z^3,\ x = e^{rs^2},\ y = \ln(\frac{r+s}{t})$, and $z = rst^2$.

33) If $w = \sin(xyz),\ x = 1 - 3t,\ y = e^{1-t}$, and $z = 4t$, find $\dfrac{\partial w}{\partial t}$.

Answer

$\dfrac{\partial w}{\partial t} = -3yz\cos(xyz) - xze^{1-t}\cos(xyz) + 4xy\cos(xyz)$

In exercises 34 - 36, use this information: A function $f(x,y)$ is said to be homogeneous of degree n if $f(tx,ty) = t^n f(x,y)$. For all homogeneous functions of degree n, the following equation is true: $x\dfrac{\partial f}{\partial x} + y\dfrac{\partial f}{\partial y} = nf(x,y)$. Show that the given function is homogeneous and verify that $x\dfrac{\partial f}{\partial x} + y\dfrac{\partial f}{\partial y} = nf(x,y)$.

34) $f(x,y) = 3x^2 + y^2$

35) $f(x,y) = \sqrt{x^2 + y^2}$

Answer

$f(tx,ty) = \sqrt{t^2x^2 + t^2y^2} = t^1 f(x,y), \quad \dfrac{\partial f}{\partial y} = x\frac{1}{2}(x^2+y^2)^{-1/2}\times 2x + y\frac{1}{2}(x^2+y^2)^{-1/2}\times 2y = 1f(x,y)$

36) $f(x,y) = x^2y - 2y^3$

37) The volume of a right circular cylinder is given by $V(x,y) = \pi x^2 y$, where x is the radius of the cylinder and y is the cylinder height. Suppose x and y are functions of t given by $x = \frac{1}{2}t$ and $y = \frac{1}{3}t$ so that x and y are both increasing with time. How fast is the volume increasing when $x = 2$ and $y = 5$? Assume time is measured in seconds.

Answer

\ddfrac$dVdt = \frac{34\pi}{3}$ units^3/s

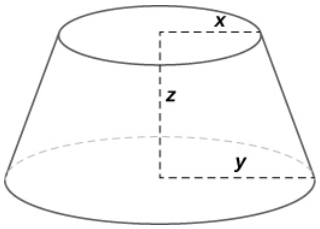
38) The pressure P of a gas is related to the volume and temperature by the formula $PV = kT$, where temperature is expressed in kelvins. Express the pressure of the gas as a function of both V and T. Find $\dfrac{dP}{dt}$ when $k = 1, \dfrac{dV}{dt} = 2$ cm^3/min, $\dfrac{dT}{dt} = 12$ K/min, $V = 20cm^3$, and $T = 20°F$.

39) The radius of a right circular cone is increasing at 3 cm/min whereas the height of the cone is decreasing at 2 cm/min. Find the rate of change of the volume of the cone when the radius is 13 cm and the height is 18 cm.

Answer

$\dfrac{dV}{dt} = \frac{1066\pi}{3}$ cm^3/min

40) The volume of a frustum of a cone is given by the formula $V = \frac{1}{3}\pi z(x^2 + y^2 + xy)$, where x is the radius of the smaller circle, y is the radius of the larger circle, and z is the height of the frustum (see figure). Find the rate of change of the volume of this frustum when $x = 10$ in., $y = 12$ in., and $z = 18$ in.



41) A closed box is in the shape of a rectangular solid with dimensions $x, y,$ and z. (Dimensions are in inches.) Suppose each dimension is changing at the rate of 0.5 in./min. Find the rate of change of the total surface area of the box when $x = 2$ in., $y = 3$ in., and $z = 1$ in.

Answer

$\frac{dA}{dt} = 12\,\text{in.}^2/\text{min}$

42) The total resistance in a circuit that has three individual resistances represented by $x, y,$ and z is given by the formula $R(x, y, z) = \frac{xyz}{yz + xz + xy}$. Suppose at a given time the x resistance is $100\,\Omega$, the y resistance is $200\,\Omega$, and the z resistance is $300\,\Omega$. Also, suppose the x resistance is changing at a rate of $2\,\Omega/\text{min}$, the y resistance is changing at the rate of $1\,\Omega/\text{min}$, and the z resistance has no change. Find the rate of change of the total resistance in this circuit at this time.

43) The temperature T at a point (x, y) is $T(x, y)$ and is measured using the Celsius scale. A fly crawls so that its position after t seconds is given by $x = \sqrt{1 + t}$ and $y = 2 + \frac{1}{3}t$, where x and y are measured in centimeters. The temperature function satisfies $T_x(2, 3) = 4$ and $T_y(2, 3) = 3$. How fast is the temperature increasing on the fly's path after 3 sec?

Answer

2°C/sec

44) The x and y components of a fluid moving in two dimensions are given by the following functions: $u(x, y) = 2y$ and $v(x, y) = -2x$ with $x \geq 0$ and $y \geq 0$. The speed of the fluid at the point (x, y) is $s(x, y) = \sqrt{u(x, y)^2 + v(x, y)^2}$. Find $\frac{\partial s}{\partial x}$ and $\frac{\partial s}{\partial y}$ using the chain rule.

45) Let $u = u(x, y, z)$, where $x = x(w, t)$, $y = y(w, t)$, $z = z(w, t)$, $w = w(r, s)$, and $t = t(r, s)$. Use a tree diagram and the chain rule to find an expression for $\frac{\partial u}{\partial r}$.

Answer

$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\left(\frac{\partial x}{\partial w}\frac{\partial w}{\partial r} + \frac{\partial x}{\partial t}\frac{\partial t}{\partial r}\right) + \frac{\partial u}{\partial y}\left(\frac{\partial y}{\partial w}\frac{\partial w}{\partial r} + \frac{\partial y}{\partial t}\frac{\partial t}{\partial r}\right) + \frac{\partial u}{\partial z}\left(\frac{\partial z}{\partial w}\frac{\partial w}{\partial r} + \frac{\partial z}{\partial t}\frac{\partial t}{\partial r}\right)$

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14.6: Directional Derivatives and the Gradient

Learning Objectives

- Determine the directional derivative in a given direction for a function of two variables.
- Determine the gradient vector of a given real-valued function.
- Explain the significance of the gradient vector with regard to direction of change along a surface.
- Use the gradient to find the tangent to a level curve of a given function.
- Calculate directional derivatives and gradients in three dimensions.

A function $z = f(x, y)$ has two partial derivatives: $\partial z/\partial x$ and $\partial z/\partial y$. These derivatives correspond to each of the independent variables and can be interpreted as instantaneous rates of change (that is, as slopes of a tangent line). For example, $\partial z/\partial x$ represents the slope of a tangent line passing through a given point on the surface defined by $z = f(x, y)$, assuming the tangent line is parallel to the x-**axis**. Similarly, $\partial z/\partial y$ represents the slope of the tangent line parallel to the y-**axis.** Now we consider the possibility of a tangent line parallel to neither axis.

Directional Derivatives

We start with the graph of a surface defined by the equation $z = f(x, y)$. Given a point (a, b) in the domain of f, we choose a direction to travel from that point. We measure the direction using an angle θ, which is measured counterclockwise in the xy-plane, starting at zero from the positive x-axis (Figure 14.6.1). The distance we travel is h and the direction we travel is given by the unit vector $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$. Therefore, the z-coordinate of the second point on the graph is given by $z = f(a + h\cos\theta, b + h\sin\theta)$.

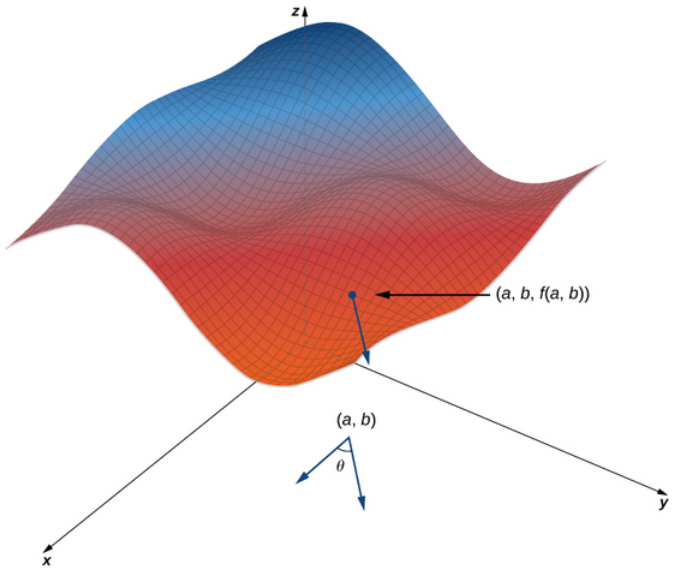


Figure 14.6.1: Finding the directional derivative at a point on the graph of $z = f(x, y)$. The slope of the blue arrow on the graph indicates the value of the directional derivative at that point.

We can calculate the slope of the secant line by dividing the difference in z-values by the length of the line segment connecting the two points in the domain. The length of the line segment is h. Therefore, the slope of the secant line is

$$m_{sec} = \frac{f(a + h\cos\theta, b + h\sin\theta) - f(a, b)}{h}$$

To find the slope of the tangent line in the same direction, we take the limit as h approaches zero.

Definition: Directional Derivatives

Suppose $z = f(x, y)$ is a function of two variables with a domain of D. Let $(a, b) \in D$ and define $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$. Then the directional derivative of f in the direction of $\vec{\mathbf{u}}$ is given by

$$D_{\vec{\mathbf{u}}} f(a,b) = \lim_{h \to 0} \frac{f(a + h\cos\theta, b + h\sin\theta) - f(a,b)}{h} \tag{14.6.1}$$

provided the limit exists.

Equation 14.6.1 provides a formal definition of the directional derivative that can be used in many cases to calculate a directional derivative.

Note that since the point (a,b) is chosen randomly from the domain D of the function f, we can use this definition to find the directional derivative as a function of x and y.

That is,

$$D_{\vec{\mathbf{u}}} f(x,y) = \lim_{h \to 0} \frac{f(x + h\cos\theta, y + h\sin\theta) - f(x,y)}{h} \tag{14.6.2}$$

Example 14.6.1: Finding a Directional Derivative from the Definition

Let $\theta = \arccos(3/5)$. Find the directional derivative $D_{\vec{\mathbf{u}}} f(x,y)$ of $f(x,y) = x^2 - xy + 3y^2$ in the direction of $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$.

Then determine $D_{\vec{\mathbf{u}}} f(-1,2)$.

Solution

First of all, since $\cos\theta = 3/5$ and θ is acute, this implies

$$\sin\theta = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25}} = \frac{4}{5}.$$

Using $f(x,y) = x^2 - xy + 3y^2$, we first calculate $f(x + h\cos\theta, y + h\sin\theta)$:

$$\begin{aligned} f(x + h\cos\theta, y + h\sin\theta) &= (x + h\cos\theta)^2 - (x + h\cos\theta)(y + h\sin\theta) + 3(y + h\sin\theta)^2 \\ &= x^2 + 2xh\cos\theta + h^2\cos^2\theta - xy - xh\sin\theta - yh\cos\theta - h^2\sin\theta\cos\theta \\ &\quad + 3y^2 + 6yh\sin\theta + 3h^2\sin^2\theta \\ &= x^2 + 2xh(\frac{3}{5}) + \frac{9h^2}{25} - xy - \frac{4xh}{5} - \frac{3yh}{5} - \frac{12h^2}{25} \\ &\quad + 3y^2 + 6yh(\frac{4}{5}) + 3h^2(\frac{16}{25}) \\ &= x^2 - xy + 3y^2 + \frac{2xh}{5} + \frac{9h^2}{5} + \frac{21yh}{5}. \end{aligned}$$

We substitute this expression into Equation 14.6.1 with $a = x$ and $b = y$:

$$\begin{aligned} D_{\vec{\mathbf{u}}} f(x,y) &= \lim_{h \to 0} \frac{f(x + h\cos\theta, y + h\sin\theta) - f(x,y)}{h} \\ &= \lim_{h \to 0} \frac{(x^2 - xy + 3y^2 + \frac{2xh}{5} + \frac{9h^2}{5} + \frac{21yh}{5}) - (x^2 - xy + 3y^2)}{h} \\ &= \lim_{h \to 0} \frac{\frac{2xh}{5} + \frac{9h^2}{5} + \frac{21yh}{5}}{h} \\ &= \lim_{h \to 0} \frac{2x}{5} + \frac{9h}{5} + \frac{21y}{5} \\ &= \frac{2x + 21y}{5}. \end{aligned}$$

To calculate $D_{\vec{\mathbf{u}}} f(-1,2)$, we substitute $x = -1$ and $y = 2$ into this answer (Figure 14.6.2):

$$D_{\vec{\mathbf{u}}} f(-1,2) = \frac{2(-1) + 21(2)}{5} = \frac{-2 + 42}{5} = 8.$$

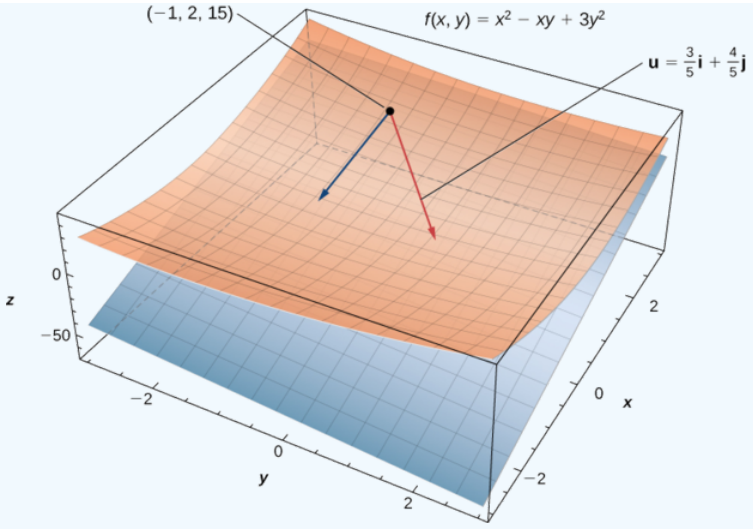


Figure 14.6.2: Finding the directional derivative in a given direction $\vec{\mathbf{u}}$ at a given point on a surface. The plane is tangent to the surface at the given point $(-1, 2, 15)$.

An easier approach to calculating directional derivatives that involves partial derivatives is outlined in the following theorem.

Theorem 14.6.1: Directional Derivative of a Function of Two Variables

Let $z = f(x, y)$ be a function of two variables x and y, and assume that f_x and f_y exist. Then the directional derivative of f in the direction of $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$ is given by

$$D_{\vec{\mathbf{u}}} f(x, y) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta. \tag{14.6.3}$$

Proof

Applying the definition of a directional derivative stated above in Equation 14.6.1, the directional derivative of f in the direction of $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$ at a point (x_0, y_0) in the domain of f can be written

$$D_{\vec{\mathbf{u}}} f((x_0, y_0)) = \lim_{t\to 0} \frac{f(x_0 + t\cos\theta, y_0 + t\sin\theta) - f(x_0, y_0)}{t}.$$

Let $x = x_0 + t\cos\theta$ and $y = y_0 + t\sin\theta$, and define $g(t) = f(x, y)$. Since f_x and f_y both exist, we can use the chain rule for functions of two variables to calculate $g'(t)$:

$$g'(t) = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta.$$

If $t = 0$, then $x = x_0$ and $y = y_0$, so

$$g'(0) = f_x(x_0, y_0)\cos\theta + f_y(x_0, y_0)\sin\theta$$

By the definition of $g'(t)$, it is also true that

$$g'(0) = \lim_{t\to 0}\frac{g(t) - g(0)}{t} = \lim_{t\to 0}\frac{f(x_0 + t\cos\theta, y_0 + t\sin\theta) - f(x_0, y_0)}{t}.$$

Therefore, $D_{\vec{\mathbf{u}}} f(x_0, y_0) = f_x(x_0, y_0)\cos\theta + f_y(x_0, y_0)\sin\theta$.

Since the point (x_0, y_0) is an arbitrary point from the domain of f, this result holds for all points in the domain of f for which the partials f_x and f_y exist.

Therefore,

$$D_{\vec{\mathbf{u}}} f(x, y) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta.$$

□

Example 14.6.2: Finding a Directional Derivative: Alternative Method

Let $\theta = \arccos(3/5)$. Find the directional derivative $D_{\vec{\mathbf{u}}} f(x,y)$ of $f(x,y) = x^2 - xy + 3y^2$ in the direction of $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$.

Then determine $D_{\vec{\mathbf{u}}} f(-1,2)$.

Solution

First, we must calculate the partial derivatives of f:

$$\begin{aligned} f_x(x,y) &= 2x - y \\ f_y(x,y) &= -x + 6y, \end{aligned}$$

Then we use Equation 14.6.3 with $\theta = \arccos(3/5)$:

$$\begin{aligned} D_{\vec{\mathbf{u}}} f(x,y) &= f_x(x,y)\cos\theta + f_y(x,y)\sin\theta \\ &= (2x-y)\frac{3}{5} + (-x+6y)\frac{4}{5} \\ &= \frac{6x}{5} - \frac{3y}{5} - \frac{4x}{5} + \frac{24y}{5} \\ &= \frac{2x+21y}{5}. \end{aligned}$$

To calculate $D_{\vec{\mathbf{u}}} f(-1,2)$, let $x = -1$ and $y = 2$:

$$D_{\vec{\mathbf{u}}} f(-1,2) = \frac{2(-1)+21(2)}{5} = \frac{-2+42}{5} = 8.$$

This is the same answer obtained in Example 14.6.1.

? Exercise 14.6.1:

Find the directional derivative $D_{\vec{\mathbf{u}}} f(x,y)$ of $f(x,y) = 3x^2y - 4xy^3 + 3y^2 - 4x$ in the direction of $\vec{\mathbf{u}} = (\cos\frac{\pi}{3})\,\hat{\mathbf{i}} + (\sin\frac{\pi}{3})\,\hat{\mathbf{j}}$ using Equation 14.6.3.

What is $D_{\vec{\mathbf{u}}} f(3,4)$?

Hint

Calculate the partial derivatives and determine the value of θ.

Answer

$$D_{\vec{\mathbf{u}}} f(x,y) = \frac{(6xy - 4y^3 - 4)(1)}{2} + \frac{(3x^2 - 12xy^2 + 6y)\sqrt{3}}{2}$$

$$D_{\vec{\mathbf{u}}} f(3,4) = \frac{72 - 256 - 4}{2} + \frac{(27 - 576 + 24)\sqrt{3}}{2} = -94 - \frac{525\sqrt{3}}{2}$$

If the vector that is given for the direction of the derivative is not a unit vector, then it is only necessary to divide by the norm of the vector. For example, if we wished to find the directional derivative of the function in Example 14.6.2 in the direction of the vector $\langle -5, 12\rangle$, we would first divide by its magnitude to get $\vec{\mathbf{u}}$. This gives us $\vec{\mathbf{u}} = \langle -\frac{5}{13}, \frac{12}{13}\rangle$.

Then

$$\begin{aligned} D_{\vec{\mathbf{u}}} f(x,y) &= f_x(x,y)\cos\theta + f_y(x,y)\sin\theta \\ &= -\frac{5}{13}(2x-y) + \frac{12}{13}(-x+6y) \\ &= -\frac{22}{13}x + \frac{17}{13}y \end{aligned}$$

Gradient

The right-hand side of Equation 14.6.3 is equal to $f_x(x,y)\cos\theta + f_y(x,y)\sin\theta$, which can be written as the dot product of two vectors. Define the first vector as $\vec{\nabla} f(x,y) = f_x(x,y)\,\hat{\mathbf{i}} + f_y(x,y)\,\hat{\mathbf{j}}$ and the second vector as $\vec{\mathbf{u}} = (\cos\theta)\,\hat{\mathbf{i}} + (\sin\theta)\,\hat{\mathbf{j}}$. Then the right-hand side of the equation can be written as the dot product of these two vectors:

$$D_{\vec{\mathbf{u}}} f(x,y) = \vec{\nabla} f(x,y) \cdot \vec{\mathbf{u}}. \tag{14.6.4}$$

The first vector in Equation 14.6.4 has a special name: the gradient of the function f. The symbol ∇ is called **nabla** and the vector $\vec{\nabla} f$ is read "**del** f."

Definition: The Gradient

Let $z = f(x,y)$ be a function of x and y such that f_x and f_y exist. The vector $\vec{\nabla} f(x,y)$ is called the **gradient** of f and is defined as

$$\vec{\nabla} f(x,y) = f_x(x,y)\,\hat{\mathbf{i}} + f_y(x,y)\,\hat{\mathbf{j}}. \tag{14.6.5}$$

The vector $\vec{\nabla} f(x,y)$ is also written as "**grad** f."

Example 14.6.3: Finding Gradients

Find the gradient $\vec{\nabla} f(x,y)$ of each of the following functions:

a. $f(x,y) = x^2 - xy + 3y^2$
b. $f(x,y) = \sin 3x \cos 3y$

Solution

For both parts a. and b., we first calculate the partial derivatives f_x and f_y, then use Equation 14.6.5.

a. $f_x(x,y) = 2x - y$ and $f_y(x,y) = -x + 6y$, so

$$\begin{aligned}\vec{\nabla} f(x,y) &= f_x(x,y)\,\hat{\mathbf{i}} + f_y(x,y)\,\hat{\mathbf{j}} \\ &= (2x - y)\,\hat{\mathbf{i}} + (-x + 6y)\,\hat{\mathbf{j}}.\end{aligned}$$

b. $f_x(x,y) = 3\cos 3x \cos 3y$ and $f_y(x,y) = -3\sin 3x \sin 3y$, so

$$\begin{aligned}\vec{\nabla} f(x,y) &= f_x(x,y)\,\hat{\mathbf{i}} + f_y(x,y)\,\hat{\mathbf{j}} \\ &= (3\cos 3x \cos 3y)\,\hat{\mathbf{i}} - (3\sin 3x \sin 3y)\,\hat{\mathbf{j}}.\end{aligned}$$

Exercise 14.6.2

Find the gradient $\vec{\nabla} f(x,y)$ of $f(x,y) = \dfrac{x^2 - 3y^2}{2x + y}$.

Hint

Calculate the partial derivatives, then use Equation 14.6.5.

Answer

$$\vec{\nabla} f(x,y) = \frac{2x^2 + 2xy + 6y^2}{(2x+y)^2}\,\hat{\mathbf{i}} - \frac{x^2 + 12xy + 3y^2}{(2x+y)^2}\,\hat{\mathbf{j}}$$

The gradient has some important properties. We have already seen one formula that uses the gradient: the formula for the directional derivative. Recall from The Dot Product that if the angle between two vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is φ, then $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = \|\vec{\mathbf{a}}\|\|\vec{\mathbf{b}}\|\cos\varphi$. Therefore, if the angle between $\vec{\nabla} f(x_0,y_0)$ and $\vec{\mathbf{u}} = (cos\theta)\,\hat{\mathbf{i}} + (sin\theta)\,\hat{\mathbf{j}}$ is φ, we have

$$D_{\vec{\mathbf{u}}} f(x_0,y_0) = \vec{\nabla} f(x_0,y_0) \cdot \vec{\mathbf{u}} = \|\vec{\nabla} f(x_0,y_0)\|\|\vec{\mathbf{u}}\|\cos\varphi = \|\vec{\nabla} f(x_0,y_0)\|\cos\varphi.$$

The $\|\vec{\mathbf{u}}\|$ disappears because $\vec{\mathbf{u}}$ is a unit vector. Therefore, the directional derivative is equal to the magnitude of the gradient evaluated at (x_0, y_0) multiplied by $\cos\varphi$. Recall that $\cos\varphi$ ranges from -1 to 1.

If $\varphi = 0$, then $\cos\varphi = 1$ and $\vec{\nabla} f(x_0, y_0)$ and $\vec{\mathbf{u}}$ both point in the same direction.

If $\varphi = \pi$, then $\cos\varphi = -1$ and $\vec{\nabla} f(x_0, y_0)$ and $\vec{\mathbf{u}}$ point in opposite directions.

In the first case, the value of $D_{\vec{\mathbf{u}}} f(x_0, y_0)$ is maximized; in the second case, the value of $D_{\vec{\mathbf{u}}} f(x_0, y_0)$ is minimized.

We can also see that if $\vec{\nabla} f(x_0, y_0) = \vec{0}$, then

$$D_{\vec{\mathbf{u}}} f(x_0, y_0) = \vec{\nabla} f(x_0, y_0) \cdot \vec{\mathbf{u}} = 0$$

for any vector $\vec{\mathbf{u}}$. These three cases are outlined in the following theorem.

Theorem 14.6.2: Properties of the Gradient

Suppose the function $z = f(x, y)$ is differentiable at (x_0, y_0) (Figure 14.6.3).

i. If $\vec{\nabla} f(x_0, y_0) = \vec{0}$, then $D_{\vec{\mathbf{u}}} f(x_0, y_0) = 0$ for any unit vector $\vec{\mathbf{u}}$.
ii. If $\vec{\nabla} f(x_0, y_0) \neq \vec{0}$, then $D_{\vec{\mathbf{u}}} f(x_0, y_0)$ is **maximized** when $\vec{\mathbf{u}}$ points in the same direction as $\vec{\nabla} f(x_0, y_0)$. The maximum value of $D_{\vec{\mathbf{u}}} f(x_0, y_0)$ is $\|\vec{\nabla} f(x_0, y_0)\|$.
iii. If $\vec{\nabla} f(x_0, y_0) \neq \vec{0}$, then $D_{\vec{\mathbf{u}}} f(x_0, y_0)$ is **minimized** when $\vec{\mathbf{u}}$ points in the opposite direction from $\vec{\nabla} f(x_0, y_0)$. The minimum value of $D_{\vec{\mathbf{u}}} f(x_0, y_0)$ is $-\|\vec{\nabla} f(x_0, y_0)\|$.

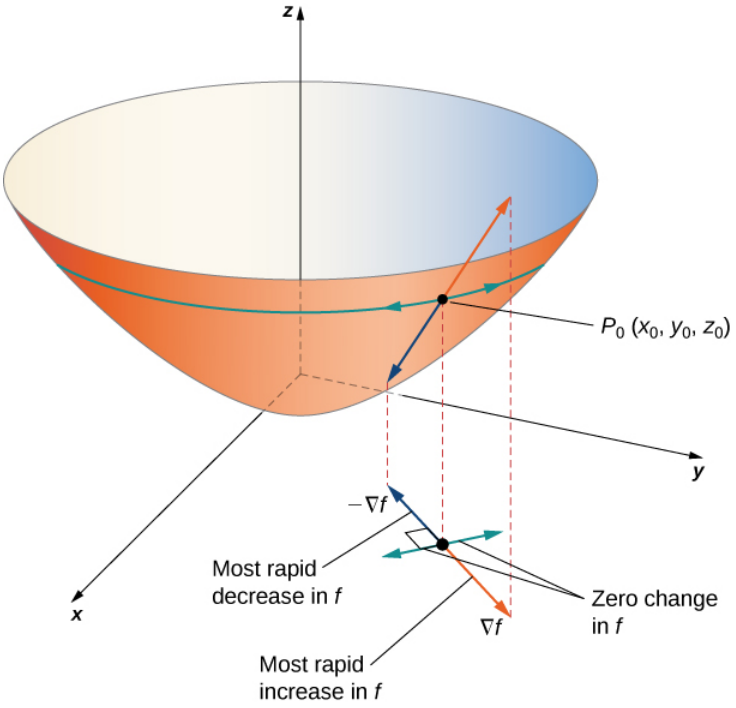


Figure 14.6.3: The gradient indicates the maximum and minimum values of the directional derivative at a point.

Example 14.6.4: Finding a Maximum Directional Derivative

Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value?

Solution

The maximum value of the directional derivative occurs when $\vec{\nabla} f$ and the unit vector point in the same direction. Therefore, we start by calculating $\vec{\nabla} f(x, y)$:

$$f_x(x,y) = 6x - 4y \text{ and } f_y(x,y) = -4x + 4y$$

so

$$\vec{\nabla} f(x,y) = f_x(x,y)\,\hat{\mathbf{i}} + f_y(x,y)\,\hat{\mathbf{j}} = (6x - 4y)\,\hat{\mathbf{i}} + (-4x + 4y)\,\hat{\mathbf{j}}.$$

Next, we evaluate the gradient at $(-2,3)$:

$$\vec{\nabla} f(-2,3) = (6(-2) - 4(3))\,\hat{\mathbf{i}} + (-4(-2) + 4(3))\,\hat{\mathbf{j}} = -24\,\hat{\mathbf{i}} + 20\,\hat{\mathbf{j}}.$$

We need to find a unit vector that points in the same direction as $\vec{\nabla} f(-2,3)$, so the next step is to divide $\vec{\nabla} f(-2,3)$ by its magnitude, which is $\sqrt{(-24)^2 + (20)^2} = \sqrt{976} = 4\sqrt{61}$. Therefore,

$$\frac{\vec{\nabla} f(-2,3)}{\|\vec{\nabla} f(-2,3)\|} = \frac{-24}{4\sqrt{61}}i + \frac{20}{4\sqrt{61}}j = -\frac{6\sqrt{61}}{61}\,\hat{\mathbf{i}} + \frac{5\sqrt{61}}{61}\,\hat{\mathbf{j}}.$$

This is the unit vector that points in the same direction as $\vec{\nabla} f(-2,3)$. To find the angle corresponding to this unit vector, we solve the equations

$$\cos\theta = \frac{-6\sqrt{61}}{61} \text{ and } \sin\theta = \frac{5\sqrt{61}}{61}$$

for θ. Since cosine is negative and sine is positive, the angle must be in the second quadrant. Therefore, $\theta = \pi - \arcsin((5\sqrt{61})/61) \approx 2.45$ rad.

The maximum value of the directional derivative at $(-2,3)$ is $\|\vec{\nabla} f(-2,3)\| = 4\sqrt{61}$ (Figure 14.6.4).

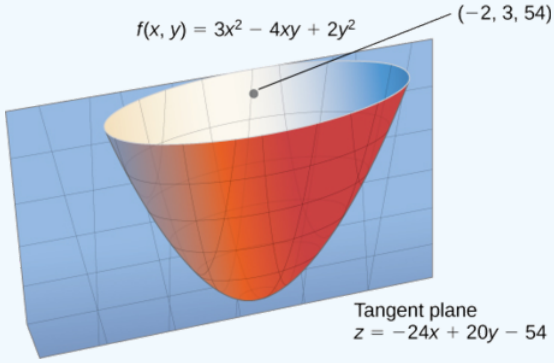


Figure 14.6.4: The maximum value of the directional derivative at $(-2,3)$ is in the direction of the gradient.

Exercise 14.6.3

Find the direction for which the directional derivative of $g(x,y) = 4x - xy + 2y^2$ at $(-2,3)$ is a maximum. What is the maximum value?

Hint

Evaluate the gradient of g at point $(-2,3)$.

Answer

The gradient of g at $(-2,3)$ is $\vec{\nabla} g(-2,3) = \hat{\mathbf{i}} + 14\,\hat{\mathbf{j}}$. The unit vector that points in the same direction as $\vec{\nabla} g(-2,3)$ is

$$\frac{\vec{\nabla} g(-2,3)}{\|\vec{\nabla} g(-2,3)\|} = \frac{1}{\sqrt{197}}\,\hat{\mathbf{i}} + \frac{14}{\sqrt{197}}\,\hat{\mathbf{j}} = \frac{\sqrt{197}}{197}\,\hat{\mathbf{i}} + \frac{14\sqrt{197}}{197}\,\hat{\mathbf{j}},$$

which gives an angle of $\theta = \arcsin((14\sqrt{197})/197) \approx 1.499$ rad.

The maximum value of the directional derivative is $\|\vec{\nabla} g(-2,3)\| = \sqrt{197}$.

Figure 14.6.5 shows a portion of the graph of the function $f(x, y) = 3 + \sin x \sin y$. Given a point (a, b) in the domain of f, the maximum value of the directional derivative at that point is given by $\|\vec{\nabla} f(a, b)\|$. This would equal the rate of greatest ascent if the surface represented a topographical map. If we went in the opposite direction, it would be the rate of greatest descent.

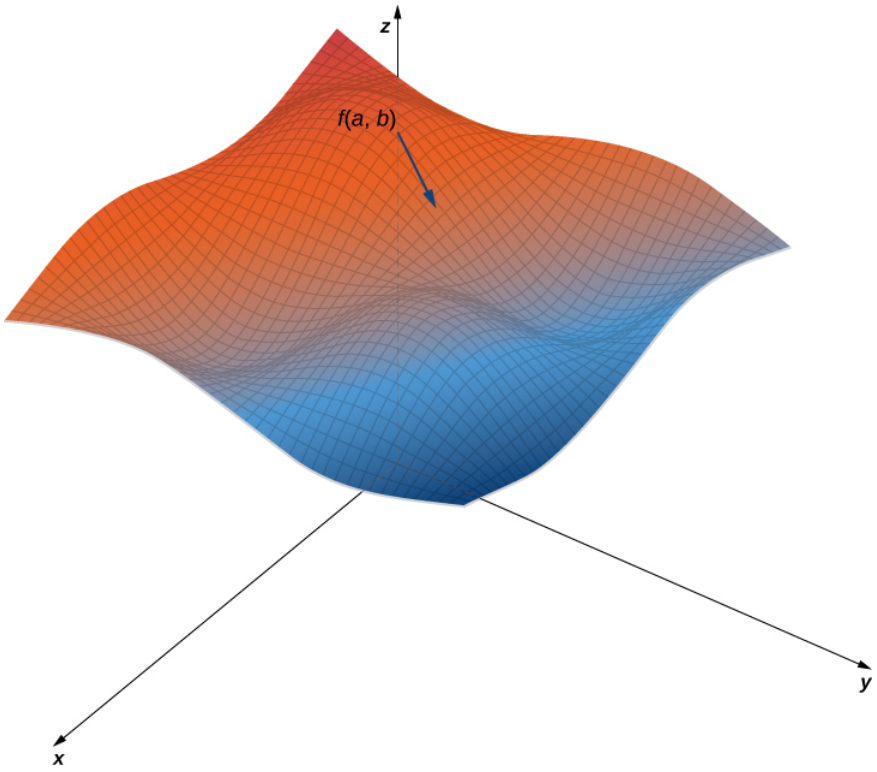


Figure 14.6.5: A typical surface in $\mathbb{R}^3$. Given a point on the surface, the directional derivative can be calculated using the gradient.

When using a topographical map, the steepest slope is always in the direction where the contour lines are closest together (Figure 14.6.6). This is analogous to the contour map of a function, assuming the level curves are obtained for equally spaced values throughout the range of that function.

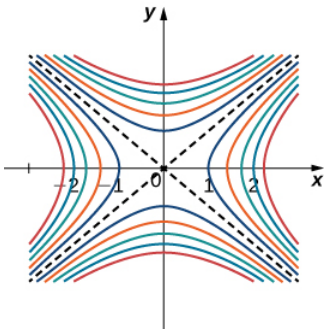


Figure 14.6.6: Contour map for the function $f(x, y) = x^2 - y^2$ using level values between -5 and 5.

Gradients and Level Curves

Recall that if a curve is defined parametrically by the function pair $(x(t), y(t))$, then the vector $x'(t)\,\hat{\mathbf{i}} + y'(t)\,\hat{\mathbf{j}}$ is tangent to the curve for every value of t in the domain. Now let's assume $z = f(x, y)$ is a differentiable function of x and y, and (x_0, y_0) is in its domain. Let's suppose further that $x_0 = x(t_0)$ and $y_0 = y(t_0)$ for some value of t, and consider the level curve $f(x, y) = k$. Define $g(t) = f(x(t), y(t))$ and calculate $g'(t)$ on the level curve. By the chain Rule,

$$g'(t) = f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t).$$

But $g'(t) = 0$ because $g(t) = k$ for all t. Therefore, on the one hand,

$$f_x(x(t), y(t))x'(t) + f_y(x(t), y(t))y'(t) = 0;$$

on the other hand,

$$f_x(x(t),y(t))x'(t)+f_y(x(t),y(t))y'(t)=\vec{\nabla}f(x,y)\cdot\langle x'(t),y'(t)\rangle.$$

Therefore,

$$\vec{\nabla}f(x,y)\cdot\langle x'(t),y'(t)\rangle=0.$$

Thus, the dot product of these vectors is equal to zero, which implies they are orthogonal. However, the second vector is tangent to the level curve, which implies the gradient must be normal to the level curve, which gives rise to the following theorem.

Theorem 14.6.3: Gradient Is Normal to the Level Curve

Suppose the function $z=f(x,y)$ has continuous first-order partial derivatives in an open disk centered at a point (x_0,y_0). If $\vec{\nabla}f(x_0,y_0)\neq 0$, then $\vec{\nabla}f(x_0,y_0)$ is normal to the level curve of f at (x_0,y_0).

We can use this theorem to find tangent and normal vectors to level curves of a function.

Example 14.6.5: Finding Tangents to Level Curves

For the function $f(x,y)=2x^2-3xy+8y^2+2x-4y+4$, find a tangent vector to the level curve at point $(-2,1)$. Graph the level curve corresponding to $f(x,y)=18$ and draw in $\vec{\nabla}f(-2,1)$ and a tangent vector.

Solution

First, we must calculate $\vec{\nabla}f(x,y)$:

$$f_x(x,y)=4x-3y+2 \text{ and } f_y=-3x+16y-4 \text{ so}$$
$$\vec{\nabla}f(x,y)=(4x-3y+2)\,\hat{\mathbf{i}}+(-3x+16y-4)\,\hat{\mathbf{j}}.$$

Next, we evaluate $\vec{\nabla}f(x,y)$ at $(-2,1)$:

$$\vec{\nabla}f(-2,1)=(4(-2)-3(1)+2)\,\hat{\mathbf{i}}+(-3(-2)+16(1)-4)\,\hat{\mathbf{j}}=-9\,\hat{\mathbf{i}}+18\,\hat{\mathbf{j}}.$$

This vector is orthogonal to the curve at point $(-2,1)$. We can obtain a tangent vector by reversing the components and multiplying either one by -1. Thus, for example, $-18\,\hat{\mathbf{i}}-9\,\hat{\mathbf{j}}$ is a tangent vector (Figure 14.6.7).

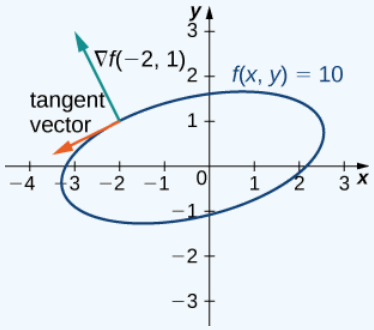


Figure 14.6.7: Tangent and normal vectors to $2x^2-3xy+8y^2+2x-4y+4=18$ at point $(-2,1)$.

Exercise 14.6.4

For the function $f(x,y)=x^2-2xy+5y^2+3x-2y+3$, find the tangent to the level curve at point $(1,1)$. Draw the graph of the level curve corresponding to $f(x,y)=8$ and draw $\vec{\nabla}f(1,1)$ and a tangent vector.

Hint

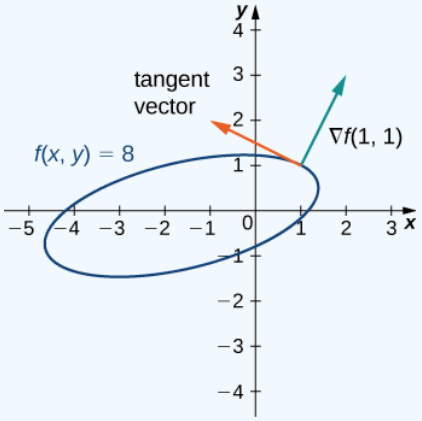
Calculate the gradient at point $(1,1)$.

Answer

$\vec{\nabla} f(x,y) = (2x - 2y + 3)\,\hat{\mathbf{i}} + (-2x + 10y - 2)\,\hat{\mathbf{j}}$

$\vec{\nabla} f(1,1) = 3\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{j}}$

Tangent vector: $6\,\hat{\mathbf{i}} - 3\,\hat{\mathbf{j}}$ or $-6\,\hat{\mathbf{i}} + 3\,\hat{\mathbf{j}}$



Three-Dimensional Gradients and Directional Derivatives

The definition of a gradient can be extended to functions of more than two variables.

Definition: Gradients in 3D

Let $w = f(x,y,z)$ be a function of three variables such that f_x, f_y, and f_z exist. The vector $\vec{\nabla} f(x,y,z)$ is called the gradient of f and is defined as

$$\vec{\nabla} f(x,y,z) = f_x(x,y,z)\,\hat{\mathbf{i}} + f_y(x,y,z)\,\hat{\mathbf{j}} + f_z(x,y,z)\,\hat{\mathbf{k}}. \tag{14.6.6}$$

$\vec{\nabla} f(x,y,z)$ can also be written as **grad** $f(x,y,z)$.

Calculating the gradient of a function in three variables is very similar to calculating the gradient of a function in two variables. First, we calculate the partial derivatives f_x, f_y, and f_z, and then we use Equation 14.6.6.

Example 14.6.6: Finding Gradients in Three Dimensions

Find the gradient $\vec{\nabla} f(x,y,z)$ of each of the following functions:

a. $f(x,y,z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz$
b. $f(x,y,z) = e^{-2z}\sin 2x \cos 2y$

Solution

For both parts a. and b., we first calculate the partial derivatives f_x, f_y, and f_z, then use Equation 14.6.6.

a. $f_x(x,y,z) = 10x - 2y + 3z$, $f_y(x,y,z) = -2x + 2y - 4z$, and $f_z(x,y,z) = 3x - 4y + 2z$, so

$$\begin{aligned}\vec{\nabla} f(x,y,z) &= f_x(x,y,z)\,\hat{\mathbf{i}} + f_y(x,y,z)\,\hat{\mathbf{j}} + f_z(x,y,z)\,\hat{\mathbf{k}} \\ &= (10x - 2y + 3z)\,\hat{\mathbf{i}} + (-2x + 2y - 4z)\,\hat{\mathbf{j}} + (3x - 4y + 2z)\,\hat{\mathbf{k}}.\end{aligned}$$

b. $f_x(x,y,z) = 2e^{-2z}\cos 2x \cos 2y$, $f_y(x,y,z) = -2e^{-2z}\sin 2x \sin 2y$, and $f_z(x,y,z) = -2e^{-2z}\sin 2x \cos 2y$, so

$$\begin{aligned}\vec{\nabla} f(x,y,z) &= f_x(x,y,z)\,\hat{\mathbf{i}} + f_y(x,y,z)\,\hat{\mathbf{j}} + f_z(x,y,z)\,\hat{\mathbf{k}} \\ &= (2e^{-2z}\cos 2x \cos 2y)\,\hat{\mathbf{i}} + (-2e^{-2z}\sin 2x \sin 2y)\,\hat{\mathbf{j}} + (-2e^{-2z}\sin 2x \cos 2y)\,\hat{\mathbf{k}} \\ &= 2e^{-2z}(\cos 2x \cos 2y\,\hat{\mathbf{i}} - \sin 2x \sin 2y\,\hat{\mathbf{j}} - \sin 2x \cos 2y\,\hat{\mathbf{k}}).\end{aligned}$$

Exercise 14.6.5:

Find the gradient $\vec{\nabla} f(x,y,z)$ of $f(x,y,z) = \dfrac{x^2 - 3y^2 + z^2}{2x + y - 4z.}$

Answer

$$\vec{\nabla} f(x,y,z) = \frac{2x^2 + 2xy + 6y^2 - 8xz - 2z^2}{(2x + y - 4z)^2}\,\hat{\mathbf{i}} - \frac{x^2 + 12xy + 3y^2 - 24yz + z^2}{(2x + y - 4z)^2}\,\hat{\mathbf{j}} + \frac{4x^2 - 12y^2 - 4z^2 + 4xz + 2yz}{(2x + y - 4z)^2}\,\hat{\mathbf{k}}$$

The directional derivative can also be generalized to functions of three variables. To determine a direction in three dimensions, a vector with three components is needed. This vector is a unit vector, and the components of the unit vector are called *directional cosines*. Given a three-dimensional unit vector $\vec{\mathbf{u}}$ in standard form (i.e., the initial point is at the origin), this vector forms three different angles with the positive x-, y-, and z-axes. Let's call these angles $\alpha, \beta,$ and γ. Then the directional cosines are given by $\cos\alpha, \cos\beta,$ and $\cos\gamma$. These are the components of the unit vector $\vec{\mathbf{u}}$; since $\vec{\mathbf{u}}$ is a unit vector, it is true that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

Definition: Directional Derivative of a Function of Three variables

Suppose $w = f(x,y,z)$ is a function of three variables with a domain of D. Let $(x_0, y_0, z_0) \in D$ and let $\vec{\mathbf{u}} = \cos\alpha\,\hat{\mathbf{i}} + \cos\beta\,\hat{\mathbf{j}} + \cos\gamma\,\hat{\mathbf{k}}$ be a unit vector. Then, the directional derivative of f in the direction of u is given by

$$D_{\vec{\mathbf{u}}} f(x_0, y_0, z_0) = \lim_{t \to 0} \frac{f(x_0 + t\cos\alpha, y_0 + t\cos\beta, z_0 + t\cos\gamma) - f(x_0, y_0, z_0)}{t}$$

provided the limit exists.

We can calculate the directional derivative of a function of three variables by using the gradient, leading to a formula that is analogous to Equation 14.6.3.

Theorem 14.6.4: Directional Derivative of a Function of Three Variables

Let $f(x,y,z)$ be a differentiable function of three variables and let $\vec{\mathbf{u}} = \cos\alpha\,\hat{\mathbf{i}} + \cos\beta\,\hat{\mathbf{j}} + \cos\gamma\,\hat{\mathbf{k}}$ be a unit vector. Then, the directional derivative of f in the direction of $\vec{\mathbf{u}}$ is given by

$$\begin{aligned} D_{\vec{\mathbf{u}}} f(x,y,z) &= \vec{\nabla} f(x,y,z) \cdot \vec{\mathbf{u}} \\ &= f_x(x,y,z)\cos\alpha + f_y(x,y,z)\cos\beta + f_z(x,y,z)\cos\gamma. \end{aligned} \tag{14.6.7}$$

The three angles $\alpha, \beta,$ and γ determine the unit vector $\vec{\mathbf{u}}$. In practice, we can use an arbitrary (nonunit) vector, then divide by its magnitude to obtain a unit vector in the desired direction.

Example 14.6.7: Finding a Directional Derivative in Three Dimensions

Calculate $D_{\vec{\mathbf{v}}} f(1,-2,3)$ in the direction of $\vec{\mathbf{v}} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ for the function

$$f(x,y,z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$

Solution:

First, we find the magnitude of v:

$$\|\vec{\mathbf{v}}\| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = \sqrt{9} = 3.$$

Therefore, $\dfrac{\vec{\mathbf{v}}}{\|\vec{\mathbf{v}}\|} = \dfrac{-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}}{3} = -\dfrac{1}{3}\,\hat{\mathbf{i}} + \dfrac{2}{3}\,\hat{\mathbf{j}} + \dfrac{2}{3}\,\hat{\mathbf{k}}$ is a unit vector in the direction of $\vec{\mathbf{v}}$, so $\cos\alpha = -\dfrac{1}{3}, \cos\beta = \dfrac{2}{3},$ and $\cos\gamma = \dfrac{2}{3}$. Next, we calculate the partial derivatives of f:

$$f_x(x,y,z) = 10x - 2y + 3z$$
$$f_y(x,y,z) = -2x + 2y - 4z$$
$$f_z(x,y,z) = -4y + 2z + 3x,$$

then substitute them into Equation 14.6.7:

$$\begin{aligned} D_{\vec{\mathbf{v}}} f(x,y,z) &= f_x(x,y,z)\cos\alpha + f_y(x,y,z)\cos\beta + f_z(x,y,z)\cos\gamma \\ &= (10x - 2y + 3z)(-\frac{1}{3}) + (-2x + 2y - 4z)(\frac{2}{3}) + (-4y + 2z + 3x)(\frac{2}{3}) \\ &= -\frac{10x}{3} + \frac{2y}{3} - \frac{3z}{3} - \frac{4x}{3} + \frac{4y}{3} - \frac{8z}{3} - \frac{8y}{3} + \frac{4z}{3} + \frac{6x}{3} \\ &= -\frac{8x}{3} - \frac{2y}{3} - \frac{7z}{3}. \end{aligned}$$

Last, to find $D_{\vec{\mathbf{v}}} f(1,-2,3)$, we substitute $x = 1$, $y = -2$, and $z = 3$:

$$\begin{aligned} D_{\vec{\mathbf{v}}} f(1,-2,3) &= -\frac{8(1)}{3} - \frac{2(-2)}{3} - \frac{7(3)}{3} \\ &= -\frac{8}{3} + \frac{4}{3} - \frac{21}{3} \\ &= -\frac{25}{3}. \end{aligned}$$

? Exercise 14.6.6:

Calculate $D_{\vec{\mathbf{v}}} f(x,y,z)$ and $D_{\vec{\mathbf{v}}} f(0,-2,5)$ in the direction of $\vec{\mathbf{v}} = -3\,\hat{\mathbf{i}} + 12\,\hat{\mathbf{j}} - 4\,\hat{\mathbf{k}}$ for the function

$$f(x,y,z) = 3x^2 + xy - 2y^2 + 4yz - z^2 + 2xz.$$

Hint

First, divide $\vec{\mathbf{v}}$ by its magnitude, calculate the partial derivatives of f, then use Equation 14.6.7.

Answer

$$D_{\vec{\mathbf{v}}} f(x,y,z) = -\frac{3}{13}(6x + y + 2z) + \frac{12}{13}(x - 4y + 4z) - \frac{4}{13}(2x + 4y - 2z)$$

$$D_{\vec{\mathbf{v}}} f(0,-2,5) = \frac{384}{13}$$

Summary

- A directional derivative represents a rate of change of a function in any given direction.
- The gradient can be used in a formula to calculate the directional derivative.
- The gradient indicates the direction of greatest change of a function of more than one variable.

Key Equations

- **directional derivative (two dimensions)**

$$D_{\vec{\mathbf{u}}} f(a,b) = \lim_{h \to 0} \frac{f(a + h\cos\theta, b + h\sin\theta) - f(a,b)}{h}$$

or

$$D_{\vec{\mathbf{u}}} f(x,y) = f_x(x,y)\cos\theta + f_y(x,y)\sin\theta$$

- **gradient (two dimensions)**

$$\vec{\nabla} f(x,y) = f_x(x,y)\,\hat{\mathbf{i}} + f_y(x,y)\,\hat{\mathbf{j}}$$

- **gradient (three dimensions)**

$$\vec{\nabla} f(x,y,z) = f_x(x,y,z)\,\hat{\mathbf{i}} + f_y(x,y,z)\,\hat{\mathbf{j}} + f_z(x,y,z)\,\hat{\mathbf{k}}$$

- **directional derivative (three dimensions)**

$$D_{\vec{\mathbf{u}}} f(x,y,z) = \vec{\nabla} f(x,y,z) \cdot \vec{\mathbf{u}} = f_x(x,y,z)\cos\alpha + f_y(x,y,z)\cos\beta + f_x(x,y,z)\cos\gamma$$

Glossary

directional derivative

the derivative of a function in the direction of a given unit vector

gradient

the gradient of the function $f(x,y)$ is defined to be $\vec{\nabla} f(x,y) = (\partial f/\partial x)\,\hat{\mathbf{i}} + (\partial f/\partial y)\,\hat{\mathbf{j}}$, which can be generalized to a function of any number of independent variables

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14.6E: Exercises for Section 14.6

In exercise 1, find the directional derivative using the limit definition only.

1) a. $f(x,y) = 5 - 2x^2 - \frac{1}{2}y^2$ at point $P(3,4)$ in the direction of $\vec{\mathbf{u}} = (\cos\frac{\pi}{4})\,\hat{\mathbf{i}} + (\sin\frac{\pi}{4})\,\hat{\mathbf{j}}$

Answer

a. $D_{\vec{\mathbf{u}}} f(3,4) = -8\sqrt{2}$

1) b. $f(x,y) = y^2\cos(2x)$ at point $P(\frac{\pi}{3},2)$ in the direction of $\vec{\mathbf{u}} = (\cos\frac{\pi}{4})\,\hat{\mathbf{i}} + (\sin\frac{\pi}{4})\,\hat{\mathbf{j}}$

Answer

b. $D_{\vec{\mathbf{u}}} f(\frac{\pi}{3},2) = -2\sqrt{6} - \sqrt{2}$

2) Find the directional derivative of $f(x,y) = y^2\sin(2x)$ at point $P(\frac{\pi}{4},2)$ in the direction of $\vec{\mathbf{u}} = 5\,\hat{\mathbf{i}} + 12\,\hat{\mathbf{j}}$.

In exercises 3 - 13, find the directional derivative of the function in the direction of $\vec{\mathbf{v}}$ as a function of x and y. Remember that you first need to find a unit vector in the direction of the direction vector. Then find the value of the directional derivative at point P.

3) $f(x,y) = xy, \quad P(-2,0), \quad \vec{\mathbf{v}} = \frac{1}{2}\,\hat{\mathbf{i}} + \frac{\sqrt{3}}{2}\,\hat{\mathbf{j}}$

Answer

$D_{\vec{\mathbf{v}}} f(x,y) = \frac{1}{2}y + \frac{\sqrt{3}}{2}x$
$D_{\vec{\mathbf{v}}} f(-2,0) = -\sqrt{3}$

4) $h(x,y) = e^x\sin y, \quad P(1,\frac{\pi}{2}), \quad \vec{\mathbf{v}} = -\hat{\mathbf{i}}$

5) $f(x,y) = x^2y, \quad P(-5,5), \quad \vec{\mathbf{v}} = 3\,\hat{\mathbf{i}} - 4\,\hat{\mathbf{j}}$

Answer

$D_{\vec{\mathbf{v}}} f(x,y) = \frac{6}{5}xy - \frac{4}{5}x^2$
$D_{\vec{\mathbf{v}}} f(-5,5) = -50$

6) $f(x,y) = xy, \quad P(1,1), \quad \vec{\mathbf{u}} = \langle\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2}\rangle$

7) $f(x,y) = x^2 - y^2, \quad P(1,0), \quad \vec{\mathbf{u}} = \langle\frac{\sqrt{3}}{2},\frac{1}{2}\rangle$

Answer

$D_{\vec{\mathbf{u}}} f(x,y) = x\sqrt{3} - y$
$D_{\vec{\mathbf{u}}} f(1,0) = \sqrt{3}$

8) $f(x,y) = 3x + 4y + 7, \quad P(0,\frac{\pi}{2}), \quad \vec{\mathbf{u}} = \langle\frac{3}{5},\frac{4}{5}\rangle$

9) $f(x,y) = e^x\cos y, \quad P = (0,\frac{\pi}{2}), \quad \vec{\mathbf{u}} = \langle 0,5\rangle$

Answer

$D_{\vec{\mathbf{u}}} f(x,y) = -e^x\sin y$
$D_{\vec{\mathbf{u}}} f(0,\frac{\pi}{2}) = -1$

10) $f(x,y) = y^{10}, \quad \vec{\mathbf{u}} = \langle 0,-3\rangle, \quad P = (1,-1)$

11) $f(x,y) = \ln(x^2+y^2), \quad \vec{\mathbf{u}} = \langle 2,-5\rangle, \quad P(1,2)$

Answer

$D_{\vec{\mathbf{u}}} f(x,y) = \frac{\sqrt{29}}{29}\left(\frac{4x - 10y}{x^2+y^2}\right)$
$D_{\vec{\mathbf{u}}} f(1,2) = -\frac{16\sqrt{29}}{145}$

12) $h(x,y,z) = xyz, \quad P(2,1,1), \quad \vec{\mathbf{v}} = 2\,\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$

Answer

$D_{\vec{\mathbf{v}}}h(x,y,z) = \frac{\sqrt{6}}{6}(2yz + xz - xy)$
$D_{\vec{\mathbf{v}}}h(2,1,1) = \frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}$

13) $f(x,y,z) = y^2 + xz, \quad P(1,2,2), \quad \vec{\mathbf{v}} = \langle 2,-1,2\rangle$

Answer

$D_{\vec{\mathbf{v}}}f(x,y,z) = \frac{2}{3}(z - y + x)$
$D_{\vec{\mathbf{v}}}f(1,2,2) = \frac{2}{3}$

In exercises 14 - 19, find the directional derivative of the function in the direction of the unit vector $\vec{\mathbf{u}} = \cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}$.

14) $f(x,y) = x^2 + 2y^2, \quad \theta = \frac{\pi}{6}$

15) $f(x,y) = \dfrac{y}{x+2y}, \quad \theta = -\frac{\pi}{4}$

Answer

$$D_{\vec{\mathbf{u}}}f(x,y) = \frac{-\sqrt{2}(x+y)}{2(x+2y)^2}$$

16) $f(x,y) = \cos(3x+y), \quad \theta = \frac{\pi}{4}$

17) $w(x,y) = ye^x, \quad \theta = \frac{\pi}{3}$

Answer

$$D_{\vec{\mathbf{u}}}f(x,y) = \frac{e^x(y+\sqrt{3})}{2}$$

18) $f(x,y) = x\arctan(y), \quad \theta = \frac{\pi}{2}$

19) $f(x,y) = \ln(x+2y), \quad \theta = \frac{\pi}{3}$

Answer

$$D_{\vec{\mathbf{u}}}f(x,y) = \frac{1+2\sqrt{3}}{2(x+2y)}$$

In exercises 20 - 23, find the gradient $\vec{\nabla}f$.

20) Find the gradient of $f(x,y) = 3x^2 + y^3 - 3x + y$. Then find it's value at the point $P(2,3)$.

21) Find the gradient of $f(x,y) = \dfrac{14 - x^2 - y^2}{3}$. Then, find the gradient at point $P(1,2)$.

Answer

$\vec{\nabla}f(x,y) = -\frac{2}{3}x\,\hat{\mathbf{i}} - \frac{2}{3}y\hat{\mathbf{j}}$
$\vec{\nabla}f(1,2) = -\frac{2}{3}\,\hat{\mathbf{i}} - \frac{4}{3}\,\hat{\mathbf{j}}$

22) Find the gradient of $f(x,y) = \ln(4x^3 - 3y)$. Then, find the gradient at point $P(1,1)$.

23) Find the gradient of $f(x,y,z) = xy + yz + xz$. Then find the gradient at point $P(1,2,3)$.

Answer

$\vec{\nabla}f(x,y,z) = \langle y+z, x+z, y+x\rangle$
$\vec{\nabla}f(1,2,3) = \langle 5,4,3\rangle$

In exercises 24 - 25, find the directional derivative of the function at point P in the direction of Q.

24) $f(x,y) = x^2 + 3y^2, \quad P(1,1), \quad Q(4,5)$

25) $f(x,y,z)=\dfrac{y}{x+z}, \quad P(2,1,-1), \quad Q(-1,2,0)$

Answer

$D_{\overrightarrow{PQ}}f(x,y)=\frac{3}{\sqrt{11}}$

26) Find the directional derivative of $f(x,y,z))$ at P and in the direction of $\vec{\mathbf{u}}$: $\quad f(x,y,z)=\ln(x^2+2y^2+3z^2), \quad P(2,1,4), \quad \vec{\mathbf{u}}=\frac{-3}{13}\,\hat{\mathbf{i}}-\frac{4}{13}\,\hat{\mathbf{j}}-\frac{12}{13}\,\hat{\mathbf{k}}$.

In exercises 27 - 29, find the directional derivative of the function at P in the direction of $\vec{\mathbf{u}}$.

27) $f(x,y)=\ln(5x+4y), \quad P(3,9), \quad \vec{\mathbf{u}}=6\,\hat{\mathbf{i}}+8\,\hat{\mathbf{j}}$

Answer

$D_{\vec{\mathbf{u}}}f(3,9)=\frac{31}{255}$

28) $f(x,y)=-7x+2y, \quad P(2,-4), \quad \vec{\mathbf{u}}=4\,\hat{\mathbf{i}}-3\,\hat{\mathbf{j}}$

29) $f(x,y,z)=4x^5y^2z^3, \quad P(2,-1,1), \quad \vec{\mathbf{u}}=\frac{1}{3}\,\hat{\mathbf{i}}+\frac{2}{3}\,\hat{\mathbf{j}}-\frac{2}{3}\,\hat{\mathbf{k}}$

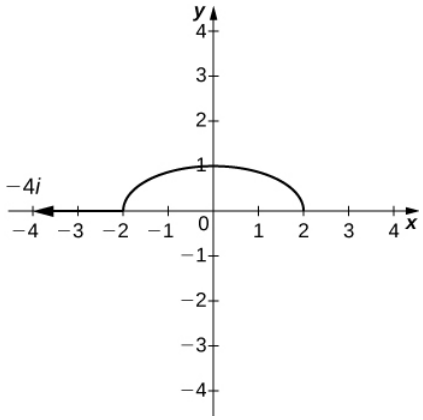
Answer

$D_{\vec{\mathbf{u}}}f(2,-1,1)=-320$

30) [T] Use technology to sketch the level curve of $f(x,y)=4x-2y+3$ that passes through $P(1,2)$ and draw the gradient vector at P.

31) [T] Use technology to sketch the level curve of $f(x,y)=x^2+4y^2$ that passes through $P(-2,0)$ and draw the gradient vector at P.

Answer



In exercises 32 - 35, find the gradient vector at the indicated point.

32) $f(x,y)=xy^2-yx^2, \quad P(-1,1)$

33) $f(x,y)=xe^y-\ln(x), \quad P(-3,0)$

Answer

$\vec{\nabla}f(-3,0)=\frac{4}{3}\,\hat{\mathbf{i}}-3\,\hat{\mathbf{j}}$

34) $f(x,y,z)=xy-\ln(z), \quad P(2,-2,2)$

35) $f(x,y,z)=x\sqrt{y^2+z^2}, \quad P(-2,-1,-1)$

Answer

$\vec{\nabla}f(-2,-1,-1)=\sqrt{2}\,\hat{\mathbf{i}}+\sqrt{2}\,\hat{\mathbf{j}}+\sqrt{2}\,\hat{\mathbf{k}}$

In exercises 36 - 40, find the indicated directional derivative of the function.

36) $f(x,y) = x^2 + xy + y^2$ at point $(-5,-4)$ in the direction the function increases most rapidly.

37) $f(x,y) = e^{xy}$ at point $(6,7)$ in the direction the function increases most rapidly.

Answer

$1.6(10^{19})$

38) $f(x,y) = \arctan\left(\frac{y}{x}\right)$ at point $(-9,9)$ in the direction the function increases most rapidly.

39) $f(x,y,z) = \ln(xy + yz + zx)$ at point $(-9,-18,-27)$ in the direction the function increases most rapidly.

Answer

$\frac{5\sqrt{2}}{99}$

40) $f(x,y,z) = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$ at point $(5,-5,5)$ in the direction the function increases most rapidly.

In exercises 41 - 43, find the maximum rate of change of f at the given point and the direction in which it occurs.

41) $f(x,y) = xe^{-y}, \quad (-2,0)$

Answer

$\max\{D_{\vec{u}} f(-2,0)\} = \sqrt{5}, \quad \langle 1,2\rangle$

Solution:

$\vec{\nabla} f(x,y) = e^{-y}\hat{\mathbf{i}} - xe^{-y}\hat{\mathbf{j}}$. So $\max\{D_{\vec{u}} f(-2,0)\} = \|\vec{\nabla} f(-2,0)\| = \sqrt{5}$.
The direction in which it occurs will just be the direction of the gradient vector: $\vec{\nabla} f(-2,0) = \langle 1,2\rangle$.

42) $f(x,y) = \sqrt{x^2 + 2y}, \quad (4,10)$

43) $f(x,y) = \cos(3x + 2y), \quad (\frac{\pi}{6}, -\frac{\pi}{8})$

Answer

$\max\{D_{\vec{u}} f\left(\frac{\pi}{6}, -\frac{\pi}{8}\right)\} = \sqrt{\frac{13}{2}} = \frac{\sqrt{26}}{2}, \quad \langle -3,-2\rangle$

In exercises 44 - 47, find equations of

a. the tangent plane and

b. the normal line to the given surface at the given point.

44) The level curve $f(x,y,z) = 12$ for $f(x,y,z) = 4x^2 - 2y^2 + z^2$ at point $(2,2,2)$.

45) $f(x,y,z) = xy + yz + xz = 3$ at point $(1,1,1)$

Answer

a. tangent plane equation: $x + y + z = 3$,
b. normal line equations: $x - 1 = y - 1 = z - 1$

46) $f(x,y,z) = xyz = 6$ at point $(1,2,3)$

47) $f(x,y,z) = xe^y \cos z - z = 1$ at point $(1,0,0)$

Answer

a. tangent plane equation: $x + y - z = 1$,
b. normal line equations: $x - 1 = y = -z$

In exercises 48 - 51, solve the stated problem.

48) The temperature T in a metal sphere is inversely proportional to the distance from the center of the sphere (the origin: $(0,0,0)$). The temperature at point $(1,2,2)$ is $120°$C.

a. Find the rate of change of the temperature at point $(1,2,2)$ in the direction toward point $(2,1,3)$.

b. Show that, at any point in the sphere, the direction of greatest increase in temperature is given by a vector that points toward the origin.

49) The **electrical potential** (voltage) in a certain region of space is given by the function $V(x,y,z) = 5x^2 - 3xy + xyz$.

a. Find the rate of change of the voltage at point $(3,4,5)$ in the direction of the vector $\langle 1,1,-1\rangle$.

b. In which direction does the voltage change most rapidly at point $(3,4,5)$?

c. What is the maximum rate of change of the voltage at point $(3,4,5)$?

Answer

a. $\frac{32}{\sqrt{3}}$,
b. $\langle 38,6,12\rangle$,
c. $2\sqrt{406}$

50) If the electric potential at a point (x,y) in the xy-plane is $V(x,y) = e^{-2x}\cos(2y)$, then the electric intensity vector at (x,y) is $E = -\vec{\nabla}V(x,y)$.

a. Find the electric intensity vector at $(\frac{\pi}{4},0)$.

b. Show that, at each point in the plane, the electric potential decreases most rapidly in the direction of the vector E.

51) In two dimensions, the motion of an ideal fluid is governed by a velocity potential φ. The velocity components of the fluid u in the x-direction and v in the y-direction, are given by $\langle u,v\rangle = \vec{\nabla}\varphi$. Find the velocity components associated with the velocity potential $\varphi(x,y) = \sin\pi x \sin 2\pi y$.

Answer

$\langle u,v\rangle = \langle \pi\cos(\pi x)\sin(2\pi y), 2\pi\sin(\pi x)\cos(2\pi y)\rangle$

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- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) created problems 20 and 22 and added a solution to problem 41.

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14.7: Maxima/Minima Problems

Learning Objectives

- Use partial derivatives to locate critical points for a function of two variables.
- Apply a second derivative test to identify a critical point as a local maximum, local minimum, or saddle point for a function of two variables.
- Examine critical points and boundary points to find absolute maximum and minimum values for a function of two variables.

One of the most useful applications for derivatives of a function of one variable is the determination of maximum and/or minimum values. This application is also important for functions of two or more variables, but as we have seen in earlier sections of this chapter, the introduction of more independent variables leads to more possible outcomes for the calculations. The main ideas of finding critical points and using derivative tests are still valid, but new wrinkles appear when assessing the results.

Critical Points

For functions of a single variable, we defined critical points as the values of the variable at which the function's derivative equals zero or does not exist. For functions of two or more variables, the concept is essentially the same, except for the fact that we are now working with partial derivatives.

Definition: Critical Points

Let $z = f(x, y)$ be a function of two variables that is differentiable on an open set containing the point (x_0, y_0). The point (x_0, y_0) is called a *critical point* of a function of two variables f if one of the two following conditions holds:

1. $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$
2. Either $f_x(x_0, y_0)$ or $f_y(x_0, y_0)$ does not exist.

Example 14.7.1: Finding Critical Points

Find the critical points of each of the following functions:

a. $f(x, y) = \sqrt{4y^2 - 9x^2 + 24y + 36x + 36}$
b. $g(x, y) = x^2 + 2xy - 4y^2 + 4x - 6y + 4$

Solution

a. First, we calculate $f_x(x, y)$ and $f_y(x, y)$:

$$f_x(x, y) = \frac{1}{2}(-18x + 36)(4y^2 - 9x^2 + 24y + 36x + 36)^{-1/2}$$
$$= \frac{-9x + 18}{\sqrt{4y^2 - 9x^2 + 24y + 36x + 36}}$$

$$f_y(x, y) = \frac{1}{2}(8y + 24)(4y^2 - 9x^2 + 24y + 36x + 36)^{-1/2}$$
$$= \frac{4y + 12}{\sqrt{4y^2 - 9x^2 + 24y + 36x + 36}}.$$

Next, we set each of these expressions equal to zero:

$$\frac{-9x + 18}{\sqrt{4y^2 - 9x^2 + 24y + 36x + 36}} = 0$$
$$\frac{4y + 12}{\sqrt{4y^2 - 9x^2 + 24y + 36x + 36}} = 0.$$

Then, multiply each equation by its common denominator:

$$-9x + 18 = 0$$
$$4y + 12 = 0.$$

Therefore, $x = 2$ and $y = -3$, so $(2, -3)$ is a critical point of f.

We must also check for the possibility that the denominator of each partial derivative can equal zero, thus causing the partial derivative not to exist. Since the denominator is the same in each partial derivative, we need only do this once:

$$4y^2 - 9x^2 + 24y + 36x + 36 = 0. \tag{14.7.1}$$

Equation 14.7.1 represents a hyperbola. We should also note that the domain of f consists of points satisfying the inequality

$$4y^2 - 9x^2 + 24y + 36x + 36 \geq 0.$$

Therefore, any points on the hyperbola are not only critical points, they are also on the boundary of the domain. To put the hyperbola in standard form, we use the method of completing the square:

$$\begin{aligned} 4y^2 - 9x^2 + 24y + 36x + 36 &= 0 \\ 4y^2 - 9x^2 + 24y + 36x &= -36 \\ 4y^2 + 24y - 9x^2 + 36x &= -36 \\ 4(y^2 + 6y) - 9(x^2 - 4x) &= -36 \\ 4(y^2 + 6y + 9) - 9(x^2 - 4x + 4) &= -36 - 36 + 36 \\ 4(y + 3)^2 - 9(x - 2)^2 &= -36. \end{aligned}$$

Dividing both sides by -36 puts the equation in standard form:

$$\begin{aligned} \frac{4(y+3)^2}{-36} - \frac{9(x-2)^2}{-36} &= 1 \\ \frac{(x-2)^2}{4} - \frac{(y+3)^2}{9} &= 1. \end{aligned}$$

Notice that point $(2, -3)$ is the center of the hyperbola.

Thus, the critical points of the function f are $(2, -3)$ and all points on the hyperbola, $\frac{(x-2)^2}{4} - \frac{(y+3)^2}{9} = 1$.

b. First, we calculate $g_x(x, y)$ and $g_y(x, y)$:

$$\begin{aligned} g_x(x, y) &= 2x + 2y + 4 \\ g_y(x, y) &= 2x - 8y - 6. \end{aligned}$$

Next, we set each of these expressions equal to zero, which gives a system of equations in x and y:

$$\begin{aligned} 2x + 2y + 4 &= 0 \\ 2x - 8y - 6 &= 0. \end{aligned}$$

Subtracting the second equation from the first gives $10y + 10 = 0$, so $y = -1$. Substituting this into the first equation gives $2x + 2(-1) + 4 = 0$, so $x = -1$.

Therefore $(-1, -1)$ is a critical point of g. There are no points in $\mathbb{R}^2$ that make either partial derivative not exist.

Figure 14.7.1 shows the behavior of the surface at the critical point.

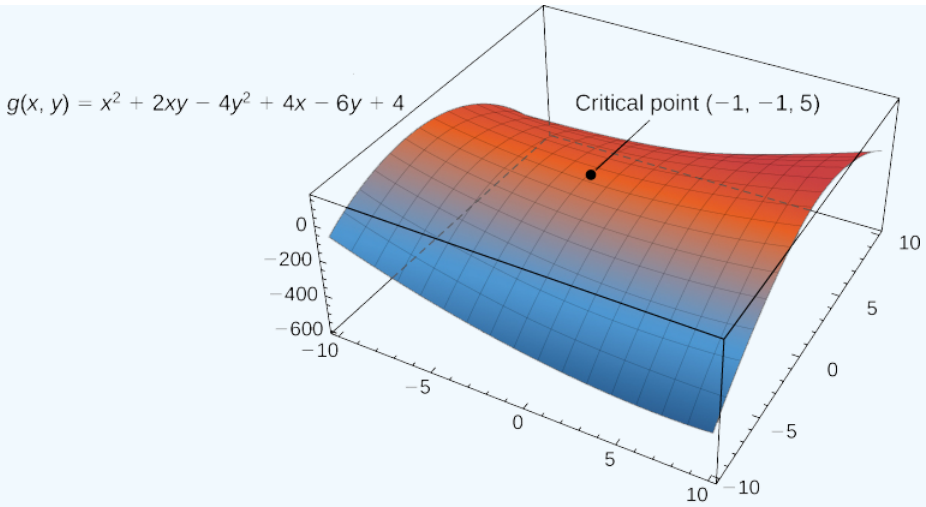


Figure 14.7.1: The function $g(x,y)$ has a critical point at $(-1,-1,5)$.

Exercise 14.7.1

Find the critical point of the function $f(x,y)=x^3+2xy-2x-4y$.

Hint

Calculate $f_x(x,y)$ and $f_y(x,y)$, then set them equal to zero.

Answer

The only critical point of f is $(2,-5)$.

The main purpose for determining critical points is to locate relative maxima and minima, as in single-variable calculus. When working with a function of one variable, the definition of a local extremum involves finding an interval around the critical point such that the function value is either greater than or less than all the other function values in that interval. When working with a function of two or more variables, we work with an open disk around the point.

Definition: Global and Local Extrema

Let $z=f(x,y)$ be a function of two variables that is defined and continuous on an open set containing the point (x_0,y_0). Then f has a **local maximum** at (x_0,y_0) if

$$f(x_0,y_0)\geq f(x,y)$$

for all points (x,y) within some disk centered at (x_0,y_0). The number $f(x_0,y_0)$ is called a local maximum value. If the preceding inequality holds for every point (x,y) in the domain of f, then f has a **global maximum** (also called an absolute maximum) at (x_0,y_0).

The function f has a local minimum at (x_0,y_0) if

$$f(x_0,y_0)\leq f(x,y)$$

for all points (x,y) within some disk centered at (x_0,y_0). The number $f(x_0,y_0)$ is called a local minimum value. If the preceding inequality holds for every point (x,y) in the domain of f, then f has a **global minimum** (also called an absolute minimum) at (x_0,y_0).

If $f(x_0,y_0)$ is either a local maximum or local minimum value, then it is called a **local extremum** (see the following figure).

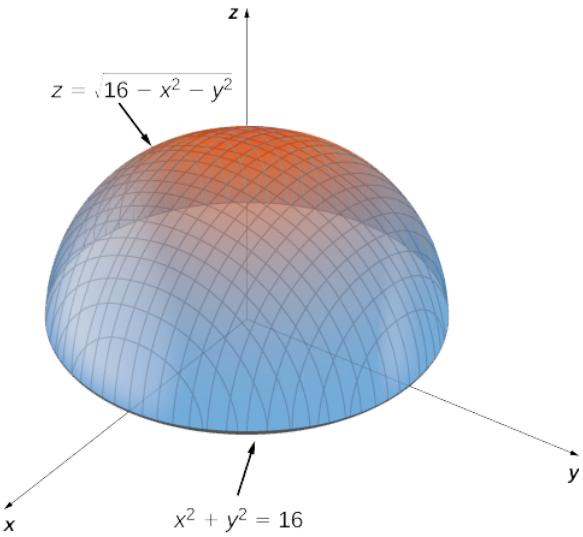


Figure 14.7.2: The graph of $z = \sqrt{16 - x^2 - y^2}$ has a maximum value when $(x, y) = (0, 0)$. It attains its minimum value at the boundary of its domain, which is the circle $x^2 + y^2 = 16$.

In Calculus 1, we showed that extrema of functions of one variable occur at critical points. The same is true for functions of more than one variable, as stated in the following theorem.

Fermat's Theorem for Functions of Two Variables

Let $z = f(x, y)$ be a function of two variables that is defined and continuous on an open set containing the point (x_0, y_0). Suppose f_x and f_y each exist at (x_0, y_0). If f has a local extremum at (x_0, y_0), then (x_0, y_0) is a critical point of f.

Second Derivative Test

Consider the function $f(x) = x^3$. This function has a critical point at $x = 0$, since $f'(0) = 3(0)^2 = 0$. However, f does not have an extreme value at $x = 0$. Therefore, the existence of a critical value at $x = x_0$ does not guarantee a local extremum at $x = x_0$. The same is true for a function of two or more variables. One way this can happen is at a **saddle point**. An example of a saddle point appears in the following figure.

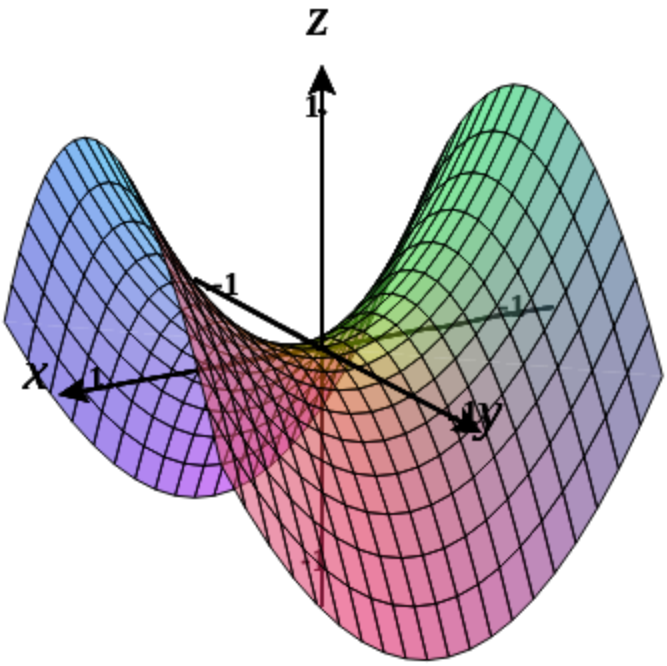


Figure 14.7.3**:** Graph of the function $z = x^2 - y^2$. This graph has a saddle point at the origin.

In this graph, the origin is a saddle point. This is because the first partial derivatives of $f(x, y) = x^2 - y^2$ are both equal to zero at this point, but it is neither a maximum nor a minimum for the function. Furthermore the vertical trace corresponding to $y = 0$ is $z = x^2$ (a parabola opening upward), but the vertical trace corresponding to $x = 0$ is $z = -y^2$ (a parabola opening downward). Therefore, it is both a global maximum for one trace and a global minimum for another.

Definition: Saddle Point

Given the function $z = f(x, y)$, the point $\big(x_0, y_0, f(x_0, y_0)\big)$ is a saddle point if both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$, but f does not have a local extremum at (x_0, y_0).

The second derivative test for a function of one variable provides a method for determining whether an extremum occurs at a critical point of a function. When extending this result to a function of two variables, an issue arises related to the fact that there are, in fact, four different second-order partial derivatives, although equality of mixed partials reduces this to three. The second derivative test for a function of two variables, stated in the following theorem, uses a **discriminant** D that replaces $f''(x_0)$ in the second derivative test for a function of one variable.

Second Derivative Test

Let $z = f(x, y)$ be a function of two variables for which the first- and second-order partial derivatives are continuous on some disk containing the point (x_0, y_0). Suppose $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$. Define the quantity

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - \big(f_{xy}(x_0, y_0)\big)^2.$$

Then:

i. If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a local minimum at (x_0, y_0).
ii. If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a local maximum at (x_0, y_0).
iii. If $D < 0$, then f has a saddle point at (x_0, y_0).

iv. If $D = 0$, then the test is inconclusive.

See Figure 14.7.4.

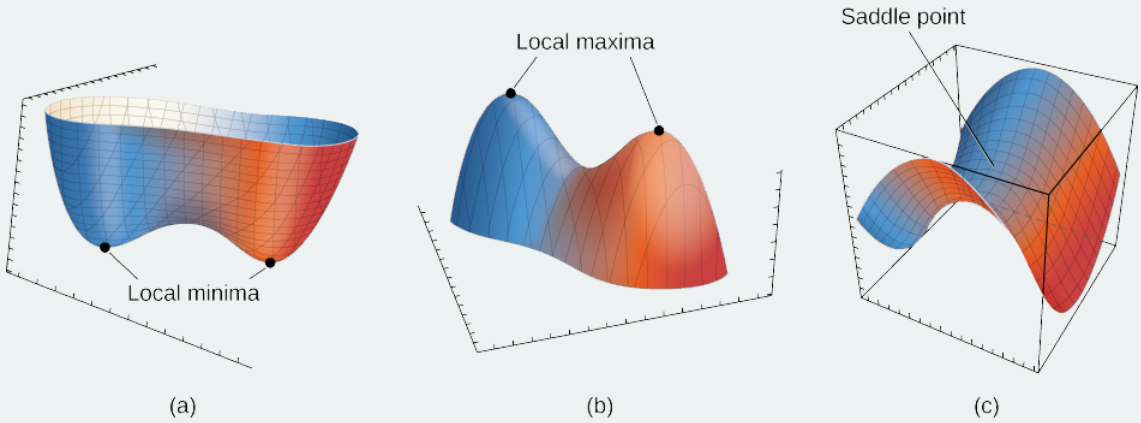


Figure 14.7.4: The second derivative test can often determine whether a function of two variables has local minima (a), local maxima (b), or a saddle point (c).

To apply the second derivative test, it is necessary that we first find the critical points of the function. There are several steps involved in the entire procedure, which are outlined in a problem-solving strategy.

Problem-Solving Strategy: Using the Second Derivative Test for Functions of Two Variables

Let $z = f(x, y)$ be a function of two variables for which the first- and second-order partial derivatives are continuous on some disk containing the point (x_0, y_0). To apply the second derivative test to find local extrema, use the following steps:

1. Determine the critical points (x_0, y_0) of the function f where $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$. Discard any points where at least one of the partial derivatives does not exist.
2. Calculate the discriminant $D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - \big(f_{xy}(x_0, y_0)\big)^2$ for each critical point of f.
3. Apply the four cases of the test to determine whether each critical point is a local maximum, local minimum, or saddle point, or whether the theorem is inconclusive.

Example 14.7.2: Using the Second Derivative Test

Find the critical points for each of the following functions, and use the second derivative test to find the local extrema:

a. $f(x, y) = 4x^2 + 9y^2 + 8x - 36y + 24$

b. $g(x, y) = \frac{1}{3}x^3 + y^2 + 2xy - 6x - 3y + 4$

Solution

a. **Step 1** of the problem-solving strategy involves finding the critical points of f. To do this, we first calculate $f_x(x, y)$ and $f_y(x, y)$, then set each of them equal to zero:

$$f_x(x, y) = 8x + 8$$
$$f_y(x, y) = 18y - 36.$$

Setting them equal to zero yields the system of equations

$$8x + 8 = 0$$
$$18y - 36 = 0.$$

The solution to this system is $x = -1$ and $y = 2$. Therefore $(-1, 2)$ is a critical point of f.

Step 2 of the problem-solving strategy involves calculating D. To do this, we first calculate the second partial derivatives of f:

$$f_{xx}(x,y) = 8$$
$$f_{xy}(x,y) = 0$$
$$f_{yy}(x,y) = 18.$$

Therefore, $D = f_{xx}(-1,2)f_{yy}(-1,2) - \left(f_{xy}(-1,2)\right)^2 = (8)(18) - (0)^2 = 144$.

Step 3 states to apply the four cases of the test to classify the function's behavior at this critical point.

Since $D > 0$ and $f_{xx}(-1,2) > 0$, this corresponds to case 1. Therefore, f has a local minimum at $(-1,2)$ as shown in the following figure.

$z = 4x^2 + 9y^2 + 8x - 36y + 24$

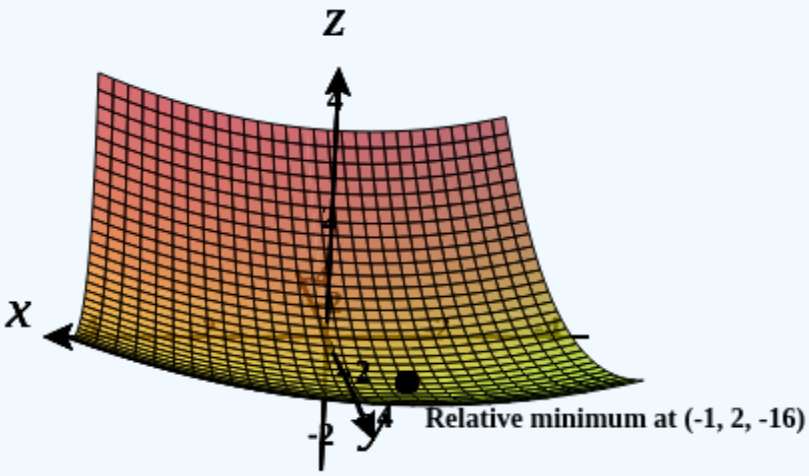


Figure 14.7.5**:** The function $f(x,y)$ has a local minimum at $(-1,2,-16)$. Note the scale on the y-axis in this plot is in thousands.

b. For **step 1**, we first calculate $g_x(x,y)$ and $g_y(x,y)$, then set each of them equal to zero:

$$g_x(x,y) = x^2 + 2y - 6$$
$$g_y(x,y) = 2y + 2x - 3.$$

Setting them equal to zero yields the system of equations

$$x^2 + 2y - 6 = 0$$
$$2y + 2x - 3 = 0.$$

To solve this system, first solve the second equation for y. This gives $y = \dfrac{3-2x}{2}$. Substituting this into the first equation gives

$$x^2 + 3 - 2x - 6 = 0$$
$$x^2 - 2x - 3 = 0$$
$$(x-3)(x+1) = 0.$$

Therefore, $x = -1$ or $x = 3$. Substituting these values into the equation $y = \dfrac{3-2x}{2}$ yields the critical points $\left(-1, \frac{5}{2}\right)$ and $\left(3, -\frac{3}{2}\right)$.

Step 2 involves calculating the second partial derivatives of g:

$$g_{xx}(x,y) = 2x$$
$$g_{xy}(x,y) = 2$$
$$g_{yy}(x,y) = 2.$$

Then, we find a general formula for D:

$$\begin{aligned} D(x_0,y_0) &= g_{xx}(x_0,y_0)g_{yy}(x_0,y_0) - \left(g_{xy}(x_0,y_0)\right)^2 \\ &= (2x_0)(2) - 2^2 \\ &= 4x_0 - 4. \end{aligned}$$

Next, we substitute each critical point into this formula:

$$D\left(-1, \tfrac{5}{2}\right) = (2(-1))(2) - (2)^2 = -4 - 4 = -8$$
$$D\left(3, -\tfrac{3}{2}\right) = (2(3))(2) - (2)^2 = 12 - 4 = 8.$$

In step 3, we note that, applying Note to point $\left(-1, \frac{5}{2}\right)$ leads to case 3, which means that $\left(-1, \frac{5}{2}\right)$ is a saddle point. Applying the theorem to point $\left(3, -\frac{3}{2}\right)$ leads to case 1, which means that $\left(3, -\frac{3}{2}\right)$ corresponds to a local minimum as shown in the following figure.

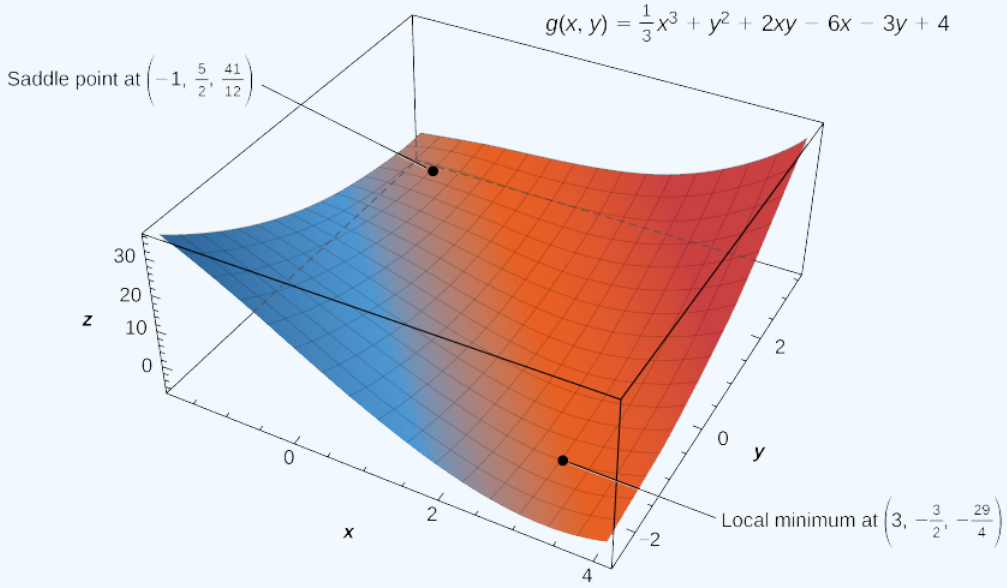


Figure 14.7.6: The function $g(x,y)$ has a local minimum and a saddle point.

Exercise 14.7.2

Use the second derivative test to find the local extrema of the function

$$f(x,y) = x^3 + 2xy - 6x - 4y^2.$$

Hint

Follow the problem-solving strategy for applying the second derivative test.

Answer

$\left(\frac{4}{3}, \frac{1}{3}\right)$ is a saddle point, $\left(-\frac{3}{2}, -\frac{3}{8}\right)$ is a local maximum.

Absolute Maxima and Minima

When finding global extrema of functions of one variable on a closed interval, we start by checking the critical values over that interval and then evaluate the function at the endpoints of the interval. When working with a function of two variables, the closed interval is replaced by a closed, bounded set. A set is bounded if all the points in that set can be contained within a ball (or disk) of finite radius. First, we need to find the critical points inside the set and calculate the corresponding critical values. Then, it is necessary to find the maximum and minimum value of the function on the boundary of the set. When we have all these values, the largest function value corresponds to the global maximum and the smallest function value corresponds to the absolute minimum. First, however, we need to be assured that such values exist. The following theorem does this.

Extreme Value Theorem

A continuous function $f(x, y)$ on a closed and bounded set D in the plane attains an absolute maximum value at some point of D and an absolute minimum value at some point of D.

Now that we know any continuous function f defined on a closed, bounded set attains its extreme values, we need to know how to find them.

Finding Extreme Values of a Function of Two Variables

Assume $z = f(x, y)$ is a differentiable function of two variables defined on a closed, bounded set D. Then f will attain the absolute maximum value and the absolute minimum value, which are, respectively, the largest and smallest values found among the following:

1. The values of f at the critical points of f in D.
2. The values of f on the boundary of D.

The proof of this theorem is a direct consequence of the extreme value theorem and Fermat's theorem. In particular, if either extremum is not located on the boundary of D, then it is located at an interior point of D. But an interior point (x_0, y_0) of D that's an absolute extremum is also a local extremum; hence, (x_0, y_0) is a critical point of f by Fermat's theorem. Therefore the only possible values for the global extrema of f on D are the extreme values of f on the interior or boundary of D.

Problem-Solving Strategy: Finding Absolute Maximum and Minimum Values

Let $z = f(x, y)$ be a continuous function of two variables defined on a closed, bounded set D, and assume f is differentiable on D. To find the absolute maximum and minimum values of f on D, do the following:

1. Determine the critical points of f in D.
2. Calculate f at each of these critical points.
3. Determine the maximum and minimum values of f on the boundary of its domain.
4. The maximum and minimum values of f will occur at one of the values obtained in steps 2 and 3.

Finding the maximum and minimum values of f on the boundary of D can be challenging. If the boundary is a rectangle or set of straight lines, then it is possible to parameterize the line segments and determine the maxima on each of these segments, as seen in Example 14.7.3. The same approach can be used for other shapes such as circles and ellipses.

If the boundary of the set D is a more complicated curve defined by a function $g(x, y) = c$ for some constant c, and the first-order partial derivatives of g exist, then the method of Lagrange multipliers can prove useful for determining the extrema of f on the boundary which is introduced in Lagrange Multipliers.

Example 14.7.3: Finding Absolute Extrema

Use the problem-solving strategy for finding absolute extrema of a function to determine the absolute extrema of each of the following functions:

a. $f(x,y) = x^2 - 2xy + 4y^2 - 4x - 2y + 24$ on the domain defined by $0 \le x \le 4$ and $0 \le y \le 2$
b. $g(x,y) = x^2 + y^2 + 4x - 6y$ on the domain defined by $x^2 + y^2 \le 16$

Solution

a. Using the problem-solving strategy, step 1 involves finding the critical points of f on its domain. Therefore, we first calculate $f_x(x,y)$ and $f_y(x,y)$, then set them each equal to zero:

$$f_x(x,y) = 2x - 2y - 4$$
$$f_y(x,y) = -2x + 8y - 2.$$

Setting them equal to zero yields the system of equations

$$2x - 2y - 4 = 0$$
$$-2x + 8y - 2 = 0.$$

The solution to this system is $x = 3$ and $y = 1$. Therefore $(3,1)$ is a critical point of f. Calculating $f(3,1)$ gives $f(3,1) = 17$.

The next step involves finding the extrema of f on the boundary of its domain. The boundary of its domain consists of four line segments as shown in the following graph:

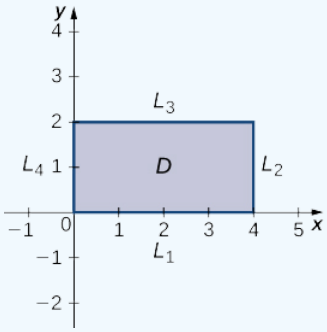


Figure 14.7.7: Graph of the domain of the function $f(x,y) = x^2 - 2xy + 4y^2 - 4x - 2y + 24$.

L_1 is the line segment connecting $(0,0)$ and $(4,0)$, and it can be parameterized by the equations $x(t) = t, y(t) = 0$ for $0 \le t \le 4$. Define $g(t) = f\big(x(t), y(t)\big)$. This gives $g(t) = t^2 - 4t + 24$. Differentiating g leads to $g'(t) = 2t - 4$. Therefore, g has a critical value at $t = 2$, which corresponds to the point $(2,0)$. Calculating $f(2,0)$ gives the z-value 20.

L_2 is the line segment connecting $(4,0)$ and $(4,2)$, and it can be parameterized by the equations $x(t) = 4, y(t) = t$ for $0 \le t \le 2$. Again, define $g(t) = f\big(x(t), y(t)\big)$. This gives $g(t) = 4t^2 - 10t + 24$. Then, $g'(t) = 8t - 10$. g has a critical value at $t = \frac{5}{4}$, which corresponds to the point $\left(4, \frac{5}{4}\right)$. Calculating $f\left(4, \frac{5}{4}\right)$ gives the z-value 17.75.

L_3 is the line segment connecting $(0,2)$ and $(4,2)$, and it can be parameterized by the equations $x(t) = t, y(t) = 2$ for $0 \le t \le 4$. Again, define $g(t) = f\big(x(t), y(t)\big)$. This gives $g(t) = t^2 - 8t + 36$. The critical value corresponds to the point $(4,2)$. So, calculating $f(4,2)$ gives the z-value 20.

L_4 is the line segment connecting $(0,0)$ and $(0,2)$, and it can be parameterized by the equations $x(t) = 0, y(t) = t$ for $0 \le t \le 2$. This time, $g(t) = 4t^2 - 2t + 24$ and the critical value $t = \frac{1}{4}$ correspond to the point $\left(0, \frac{1}{4}\right)$. Calculating $f\left(0, \frac{1}{4}\right)$ gives the z-value 23.75.

We also need to find the values of $f(x,y)$ at the corners of its domain. These corners are located at $(0,0), (4,0), (4,2)$ and $(0,2)$:

$$f(0,0) = (0)^2 - 2(0)(0) + 4(0)^2 - 4(0) - 2(0) + 24 = 24$$
$$f(4,0) = (4)^2 - 2(4)(0) + 4(0)^2 - 4(4) - 2(0) + 24 = 24$$
$$f(4,2) = (4)^2 - 2(4)(2) + 4(2)^2 - 4(4) - 2(2) + 24 = 20$$
$$f(0,2) = (0)^2 - 2(0)(2) + 4(2)^2 - 4(0) - 2(2) + 24 = 36.$$

The absolute maximum value is 36, which occurs at $(0,2)$, and the global minimum value is 20, which occurs at both $(4,2)$ and $(2,0)$ as shown in the following figure.

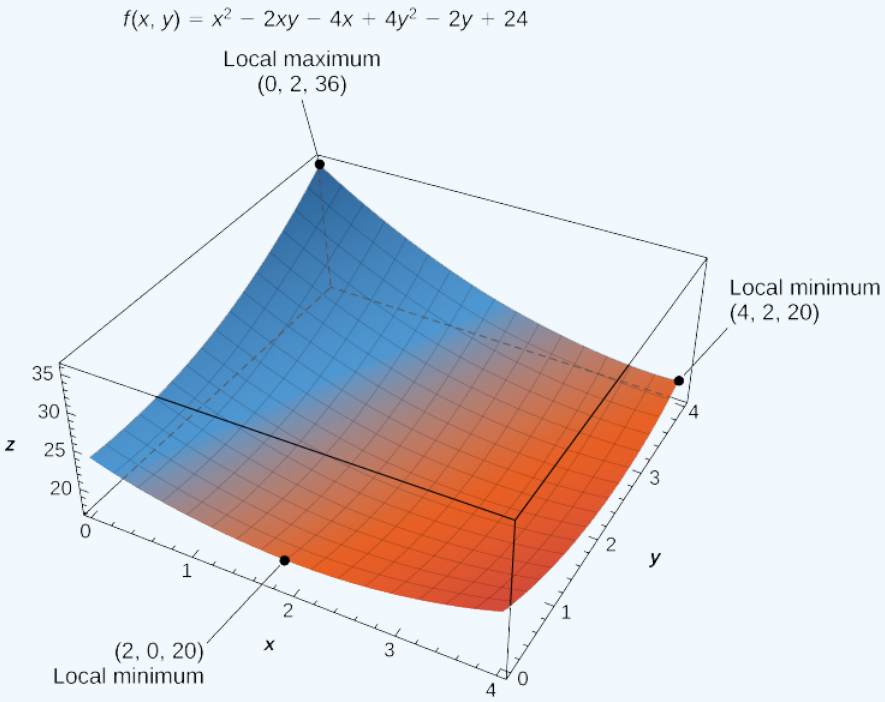


Figure 14.7.8: The function $f(x, y)$ has two global minima and one global maximum over its domain.

b. Using the problem-solving strategy, step 1 involves finding the critical points of g on its domain. Therefore, we first calculate $g_x(x, y)$ and $g_y(x, y)$, then set them each equal to zero:

$$g_x(x, y) = 2x + 4$$
$$g_y(x, y) = 2y - 6.$$

Setting them equal to zero yields the system of equations

$$2x + 4 = 0$$
$$2y - 6 = 0.$$

The solution to this system is $x = -2$ and $y = 3$. Therefore, $(-2, 3)$ is a critical point of g. Calculating $g(-2, 3)$, we get

$$g(-2, 3) = (-2)^2 + 3^2 + 4(-2) - 6(3) = 4 + 9 - 8 - 18 = -13.$$

The next step involves finding the extrema of g on the boundary of its domain. The boundary of its domain consists of a circle of radius 4 centered at the origin as shown in the following graph.

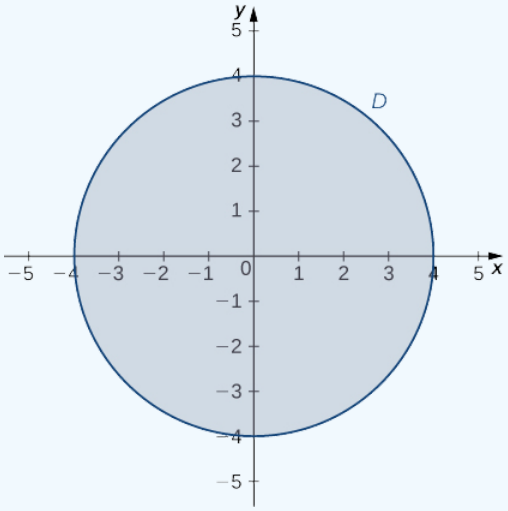


Figure 14.7.9: Graph of the restricted domain of the function $g(x,y)=x^2+y^2+4x-6y$.

The boundary of the domain of g can be parameterized using the functions $x(t)=4\cos t,\ y(t)=4\sin t$ for $0\le t\le 2\pi$. Define $h(t)=g\big(x(t),y(t)\big)$:

$$\begin{aligned} h(t) &= g\big(x(t),y(t)\big) \\ &= (4\cos t)^2+(4\sin t)^2+4(4\cos t)-6(4\sin t) \\ &= 16\cos^2 t+16\sin^2 t+16\cos t-24\sin t \\ &= 16+16\cos t-24\sin t. \end{aligned}$$

Setting $h'(t)=0$ leads to

$$\begin{aligned} -16\sin t-24\cos t &= 0 \\ -16\sin t &= 24\cos t \\ \frac{-16\sin t}{-16\cos t} &= \frac{24\cos t}{-16\cos t} \\ \tan t &= -\frac{3}{2}. \end{aligned}$$

This equation has two solutions over the interval $0\le t\le 2\pi$. One is $t=\pi-\arctan(\frac{3}{2})$ and the other is $t=2\pi-\arctan(\frac{3}{2})$. For the first angle,

$$\sin t=\sin(\pi-\arctan(\tfrac{3}{2}))=\sin(\arctan(\tfrac{3}{2}))=\frac{3\sqrt{13}}{13}$$

$$\cos t=\cos(\pi-\arctan(\tfrac{3}{2}))=-\cos(\arctan(\tfrac{3}{2}))=-\frac{2\sqrt{13}}{13}.$$

Therefore, $x(t)=4\cos t=-\frac{8\sqrt{13}}{13}$ and $y(t)=4\sin t=\frac{12\sqrt{13}}{13}$, so $\left(-\frac{8\sqrt{13}}{13},\frac{12\sqrt{13}}{13}\right)$ is a critical point on the boundary and

$$\begin{aligned} g\left(-\tfrac{8\sqrt{13}}{13},\tfrac{12\sqrt{13}}{13}\right) &= \left(-\tfrac{8\sqrt{13}}{13}\right)^2+\left(\tfrac{12\sqrt{13}}{13}\right)^2+4\left(-\tfrac{8\sqrt{13}}{13}\right)-6\left(\tfrac{12\sqrt{13}}{13}\right) \\ &= \frac{144}{13}+\frac{64}{13}-\frac{32\sqrt{13}}{13}-\frac{72\sqrt{13}}{13} \\ &= \frac{208-104\sqrt{13}}{13}\approx -12.844. \end{aligned}$$

For the second angle,

$$\sin t=\sin(2\pi-\arctan(\tfrac{3}{2}))=-\sin(\arctan(\tfrac{3}{2}))=-\frac{3\sqrt{13}}{13}$$

$$\cos t=\cos(2\pi-\arctan(\tfrac{3}{2}))=\cos(\arctan(\tfrac{3}{2}))=\frac{2\sqrt{13}}{13}.$$

Therefore, $x(t) = 4\cos t = \frac{8\sqrt{13}}{13}$ and $y(t) = 4\sin t = -\frac{12\sqrt{13}}{13}$, so $\left(\frac{8\sqrt{13}}{13}, -\frac{12\sqrt{13}}{13}\right)$ is a critical point on the boundary and

$$\begin{aligned} g\left(\tfrac{8\sqrt{13}}{13}, -\tfrac{12\sqrt{13}}{13}\right) &= \left(\tfrac{8\sqrt{13}}{13}\right)^2 + \left(-\tfrac{12\sqrt{13}}{13}\right)^2 + 4\left(\tfrac{8\sqrt{13}}{13}\right) - 6\left(-\tfrac{12\sqrt{13}}{13}\right) \\ &= \frac{144}{13} + \frac{64}{13} + \frac{32\sqrt{13}}{13} + \frac{72\sqrt{13}}{13} \\ &= \frac{208 + 104\sqrt{13}}{13} \approx 44.844. \end{aligned}$$

The absolute minimum of g is -13, which is attained at the point $(-2, 3)$, which is an interior point of D. The absolute maximum of g is approximately equal to 44.844, which is attained at the boundary point $\left(\frac{8\sqrt{13}}{13}, -\frac{12\sqrt{13}}{13}\right)$. These are the absolute extrema of g on D as shown in the following figure.

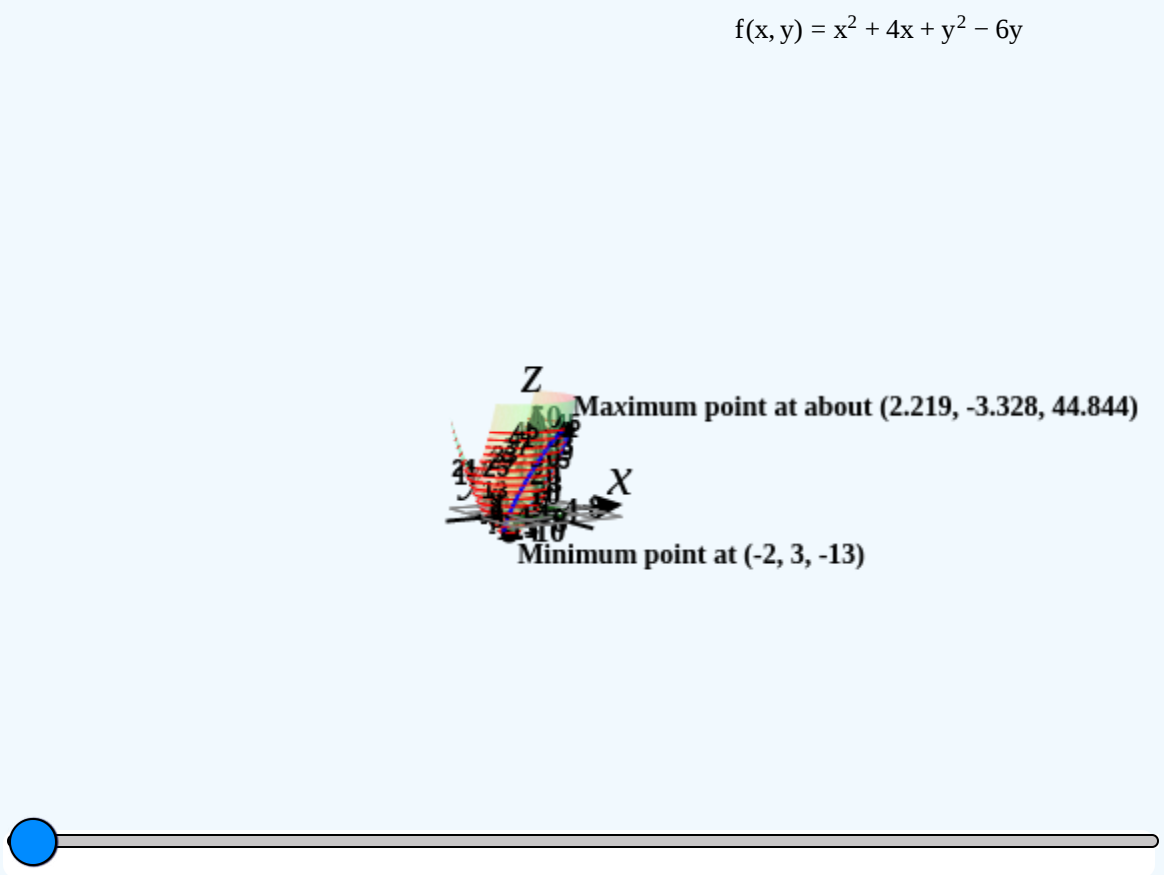


Figure 14.7.10: The function $f(x, y)$ has a local minimum and a local maximum.

Exercise 14.7.3:

Use the problem-solving strategy for finding absolute extrema of a function to find the absolute extrema of the function

$$f(x, y) = 4x^2 - 2xy + 6y^2 - 8x + 2y + 3$$

on the domain defined by $0 \le x \le 2$ and $-1 \le y \le 3$.

Hint

Calculate $f_x(x, y)$ and $f_y(x, y)$, and set them equal to zero. Then, calculate f for each critical point and find the extrema of f on the boundary of D.

Answer

The absolute minimum occurs at $(1,0)$: $f(1,0) = -1$.

The absolute maximum occurs at $(0,3)$: $f(0,3) = 63$.

Example 14.7.4: Profitable Golf Balls

Pro-T company has developed a profit model that depends on the number x of golf balls sold per month (measured in thousands), and the number of hours per month of advertising y, according to the function

$$z = f(x,y) = 48x + 96y - x^2 - 2xy - 9y^2,$$

where z is measured in thousands of dollars. The maximum number of golf balls that can be produced and sold is $50,000$, and the maximum number of hours of advertising that can be purchased is 25. Find the values of x and y that maximize profit, and find the maximum profit.



Figure 14.7.11: (credit: modification of work by oatsy40, Flickr)

Solution

Using the problem-solving strategy, step 1 involves finding the critical points of f on its domain. Therefore, we first calculate $f_x(x,y)$ and $f_y(x,y)$, then set them each equal to zero:

$$f_x(x,y) = 48 - 2x - 2y$$
$$f_y(x,y) = 96 - 2x - 18y.$$

Setting them equal to zero yields the system of equations

$$48 - 2x - 2y = 0$$
$$96 - 2x - 18y = 0.$$

The solution to this system is $x = 21$ and $y = 3$. Therefore $(21,3)$ is a critical point of f. Calculating $f(21,3)$ gives $f(21,3) = 48(21) + 96(3) - 21^2 - 2(21)(3) - 9(3)^2 = 648$.

The domain of this function is $0 \le x \le 50$ and $0 \le y \le 25$ as shown in the following graph.

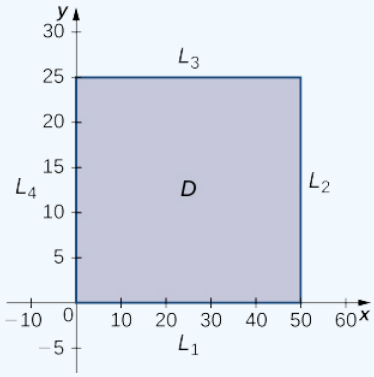


Figure 14.7.12: Graph of the domain of the function $f(x,y) = 48x + 96y - x^2 - 2xy - 9y^2$.

L_1 is the line segment connecting $(0,0)$ and $(50,0)$, and it can be parameterized by the equations $x(t)=t, y(t)=0$ for $0 \le t \le 50$. We then define $g(t)=f(x(t),y(t))$:

$$\begin{aligned} g(t) &= f(x(t),y(t)) \\ &= f(t,0) \\ &= 48t+96(0)-y^2-2(t)(0)-9(0)^2 \\ &= 48t-t^2. \end{aligned}$$

Setting $g'(t)=0$ yields the critical point $t=24$, which corresponds to the point $(24,0)$ in the domain of f. Calculating $f(24,0)$ gives 576.

L_2 is the line segment connecting $(50,0)$ and $(50,25)$, and it can be parameterized by the equations $x(t)=50, y(t)=t$ for $0 \le t \le 25$. Once again, we define $g(t)=f\big(x(t),y(t)\big)$:

$$\begin{aligned} g(t) &= f\big(x(t),y(t)\big) \\ &= f(50,t) \\ &= 48(50)+96t-50^2-2(50)t-9t^2 \\ &= -9t^2-4t-100. \end{aligned}$$

This function has a critical point at $t=-\frac{2}{9}$, which corresponds to the point $(50,-29)$. This point is not in the domain of f.

L_3 is the line segment connecting $(0,25)$ and $(50,25)$, and it can be parameterized by the equations $x(t)=t, y(t)=25$ for $0 \le t \le 50$. We define $g(t)=f\big(x(t),y(t)\big)$:

$$\begin{aligned} g(t) &= f\big(x(t),y(t)\big) \\ &= f(t,25) \\ &= 48t+96(25)-t^2-2t(25)-9(25^2) \\ &= -t^2-2t-3225. \end{aligned}$$

This function has a critical point at $t=-1$, which corresponds to the point $(-1,25)$, which is not in the domain.

L_4 is the line segment connecting $(0,0)$ to $(0,25)$, and it can be parameterized by the equations $x(t)=0, y(t)=t$ for $0 \le t \le 25$. We define $g(t)=f\big(x(t),y(t)\big)$:

$$\begin{aligned} g(t) &= f\big(x(t),y(t)\big) \\ &= f(0,t) \\ &= 48(0)+96t-(0)^2-2(0)t-9t^2 \\ &= 96t-9t^2. \end{aligned}$$

This function has a critical point at $t=\frac{16}{3}$, which corresponds to the point $\left(0,\frac{16}{3}\right)$, which is on the boundary of the domain. Calculating $f\left(0,\frac{16}{3}\right)$ gives 256.

We also need to find the values of $f(x,y)$ at the corners of its domain. These corners are located at $(0,0),(50,0),(50,25)$ and $(0,25)$:

$$\begin{aligned} f(0,0) &= 48(0)+96(0)-(0)^2-2(0)(0)-9(0)^2=0 \\ f(50,0) &= 48(50)+96(0)-(50)^2-2(50)(0)-9(0)^2=-100 \\ f(50,25) &= 48(50)+96(25)-(50)^2-2(50)(25)-9(25)^2=-5825 \\ f(0,25) &= 48(0)+96(25)-(0)^2-2(0)(25)-9(25)^2=-3225. \end{aligned}$$

The maximum value is 648, which occurs at $(21,3)$. Therefore, a maximum profit of $\$648,000$ is realized when $21,000$ golf balls are sold and 3 hours of advertising are purchased per month as shown in the following figure.

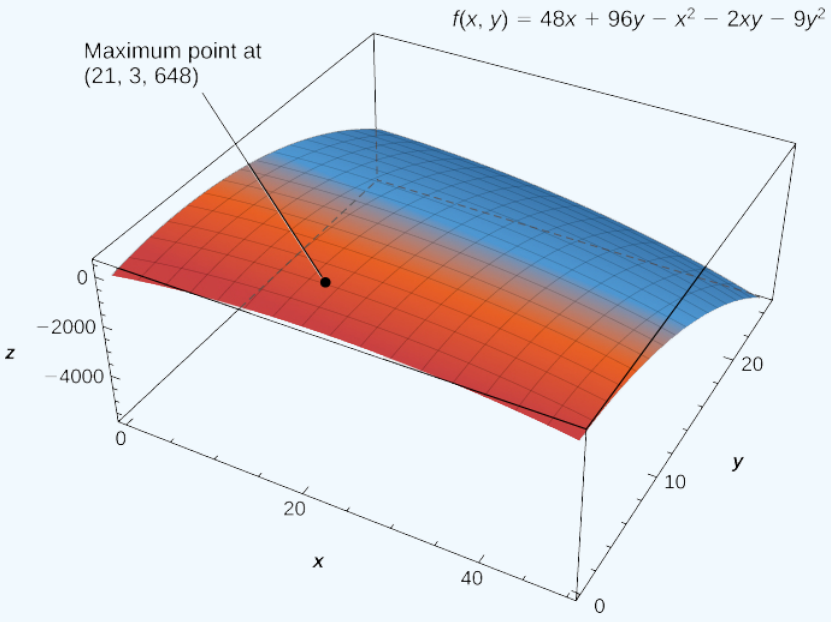


Figure 14.7.13: The profit function $f(x, y)$ has a maximum at $(21, 3, 648)$.

Key Concepts

- A critical point of the function $f(x, y)$ is any point (x_0, y_0) where either $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$, or at least one of $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ do not exist.
- A saddle point is a point (x_0, y_0) where $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$, but $f(x_0, y_0)$ is neither a maximum nor a minimum at that point.
- To find extrema of functions of two variables, first find the critical points, then calculate the discriminant and apply the second derivative test.

Key Equations

- **Discriminant**

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

Glossary

critical point of a function of two variables

the point (x_0, y_0) is called a critical point of $f(x, y)$ if one of the two following conditions holds:

1. $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$
2. At least one of $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ do not exist

discriminant

the discriminant of the function $f(x, y)$ is given by the formula $D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$

saddle point

given the function $z = f(x, y)$, the point $(x_0, y_0, f(x_0, y_0))$ is a saddle point if both $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$, but f does not have a local extremum at (x_0, y_0)

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14.7E: Exercises for Section 14.7

Finding Critical Points

In exercises 1 - 5, find all critical points.

1) $f(x,y) = 1 + x^2 + y^2$

Answer

$(0,0)$

2) $f(x,y) = 1 - (x-2)^2 + (y+3)^2$

3) $f(x,y) = (3x-2)^2 + (y-4)^2$

Answer

$\left(\frac{2}{3}, 4\right)$

4) $f(x,y) = x^4 + y^4 - 16xy$

Answer

$(0,0), \quad (-2,-2), \quad (2,2)$

5) $f(x,y) = 15x^3 - 3xy + 15y^3$

Answer

$(0,0), \quad \left(\frac{1}{15}, \frac{1}{15}\right)$

Finding Extrema & the Second Partials Test

In exercises 6 - 9, find the critical points of the function and test for extrema or saddle points by using algebraic techniques (completing the square) or by examining the form of the equation. Where possible, verify your results using the Second Partials Test.

6) $f(x,y) = -\sqrt{x^2 + y^2}$

Answer

Crit. pts.: $(0,0)$

Extrema: f has a relative maximum of 0 at $(0,0)$.

To justify this, consider the fact that the square root function cannot give a negative value, so this function cannot possibly return a positive value. Since it's value is 0 at the critical point $(0,0)$, we know it must be the function's **absolute maximum** value.

7) $f(x,y) = -x^2 - 5y^2 + 8x - 10y - 13$

Answer

Crit. pts.: $(4,-1)$

Extrema: f has a relative maximum of 8 at $(4,-1)$.

To justify this, we complete the square on this function, being careful to factor out the coefficient of the squared terms before completing the square.

$$\begin{aligned} f(x,y) &= -x^2 - 5y^2 + 8x - 10y - 13 \\ &= -(x^2 - 8x \quad\quad) - 5(y^2 + 2y \quad\quad) - 13 \\ &= -(x^2 - 8x + 16) - 5(y^2 + 2y + 1) - 13 + 16 + 5 \\ &= -(x-4)^2 - 5(y+1)^2 + 8 \end{aligned}$$